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Transport Parallèle pour l'Estimation de Monte-Carlo sur des Graphes Fibrés

Parallel Transport for Monte-Carlo Estimation on Graph Bundles

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Parallel Transport for Monte-Carlo Estimation on Graph Bundles

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Hugo Jaquard: *Parallel Transport for Monte-Carlo Estimation on Graph Bundles*

Having finally begun, how does one stop?
I suppose I can simply wait to be interrupted
as in my parents' case by a large tree–
the barge, so to speak, will have passed
for the last time between the mountains.
Something, they say, like falling asleep,
which I proceeded to do.

[...]

I was content with my brooding.
I spent my days with the colored pencils
(I soon used up the darker colors)
though what I saw, as I told my aunt,
was less a factual account of the world
than a vision of its transformation
subsequent to passage through the void of myself.

— Louise Glück[1]

A famous excerpt from the 1830's exchange between Analysts and Geometers Carl Gustav Jakob Jacobi and Adrien-Marie Legendre goes as follows (in french).

Il est vrai que M. Fourier avait l'opinion que le but principal des mathématiques était l'utilité publique et l'explication des phénomènes naturels ; mais un philsophe comme lui aurait dû savoir que le but unique de la science, c'est l'honneur de l'esprit humain, et que sous ce titre, une question des nombres vaut autant qu'une question du système du monde.

— Carl Gustav Jakob Jacobi[2]

This thesis happily sits in-between, and seeks pleasure from both perspectives. I sincerely hope you find enjoyment in these pages.

1 From [Glück, 2014].
2 For an extensive record of the correspondence, see [Pieper, 2013].

A B S T R A C T

Problems such as intra-class denoising in cryogenic electron microscopy, phase-reconstruction in ptychography, or vector-field extension in computational geometry all involve graphs with edges endowed with *rotations*. In these settings, one wishes to process data associated to the nodes of the graph (*e.g.*, vectors or complex numbers), called a graph signal, while taking into account these rotations.

A frequent task in these pipelines is that of (Tikhonov) smoothing of a given graph signal, a linear-algebraic computation too expensive to be performed exactly when the graph is too large. Approximations become necessary, and a modern methodology consists in introducing *randomization* in numerical linear-algebraic algorithms to speed-up the computation. We explore one such randomization strategy for rotation-aware smoothing of graph signals.

Our main contribution consists in the introduction of a distribution on random decompositions of graph, called *multi-type spanning forests*, and of associated Monte-Carlo estimators that can be used to perform rotation-aware smoothing. This decomposition is obtained as a rotation-aware generalization of the classical uniform spanning-tree distribution, by leveraging the theory of *determinantal point procesess*. We propose a Julia implementation of these estimators, that competes with standard iterative state-of-the-art algorithms, and outperforms them in some cases depending on the topology of the graph. We apply these estimators to the problems of *angular synchronization* and *rotation-aware interpolation*.

RÉSUMÉ

Les problèmes comme ceux du débruitage intra-classe en cryomicroscopie électronique, de la reconstruction de phase en ptychographie, ou d'extension de champs de vecteurs en géométrie algorithmique impliquent tous des graphes dont les arcs sont équipés de *rotations*. On souhaite dans ces situations traiter des données associées aux nœuds du graphe, ce que l'on appelle un signal sur graphe, tout en prenant en compte ces rotations.

Une tâche fréquente dans ces traitements est celle du lissage (de Tikhonov) d'un signal sur graphe donné, qui est une opération d'algèbre linéaire trop chère pour être calculée exactement en pratique. Il devient nécessaire d'employer des approximations, et une méthodologie moderne d'accélération de calculs consiste à *introduire de l'aléatoire* dans les algorithmes d'algèbre linéaire. Nous explorons une telle stratégie pour un problème de lissage, prenant en compte les rotations, de signal sur graphe.

Notre contribution principale consiste en l'introduction d'une distribution de décompositions aléatoires de graphes, appelées des *forêts couvrantes multi-types*, et d'estimateurs de Monte-Carlo associés pour le problème de lissage avec rotations. Cette décomposition est obtenue comme une généralisation de la classique distribution uniforme des arbres couvrants d'un graphe, en s'appuyant sur la théorie des *processus ponctuels déterminantaux*. Nous proposons une implémentation en Julia de ces estimateurs, qui rivalisent avec des algorithmes itératifs standards de l'état de l'art, et sont plus performants dans certains cas, selon la topologie du graphe. Nous appliquons ces estimateurs aux problèmes de *synchronisation angulaire* et d'*interpolation avec rotations*.

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Simon, Nicolas : c'est avant tout avec vous que j'ai appris à transformer mes élucubrations en véritable travail de recherche ; à mener un travail expérimental rigoureux ; à valoriser et à présenter tout ce travail ! Pour le cœur scientifique et pour tout ce qui va autour, merci ; j'espère que vous savez, *chacun*, combien je vous dois. Bidou, dans ton bureau on parle science, mais aussi *de science* : c'est émancipateur, toujours stimulant, et toujours un plaisir. Merci d'être toujours si doux et attentionné ; que l'on n'entende personne grogner le contraire !
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Arthur, je ne te dis évidemment pas « merci ». =)

Tsiky, je ne te le dirai jamais assez.

GLOSSARY OF NOTATION

We describe below the principal notational conventions we adopt.

Sets. $\mathbf{N}$, $\mathbf{Z}$, $\mathbf{R}$ and $\mathbf{C}$ denote respectively the sets of non-negative integers, integer, real and complex numbers. $\mathbf{R}_+$ denotes the set on non-negative real numbers, and $\mathbf{R}_+^*$ of positive real numbers. The conjugate of a complex number $z \in \mathbf{C}$ is denoted by z^*, and its modulus by $|z|$. $\arg(z)$ denotes the argument of z. The imaginary complex unit is denoted by ι; the real and imaginary parts of z are denoted by $\mathfrak{Re}(z)$ and $\mathfrak{Im}(z)$ respectively.
$[n]$ denotes the set of integers from 1 to n. Given a finite set S, $|S|$ (in some cases, $\#S$) is its cardinality, 2^S is the set of subsets of S, and $\binom{S}{k}$ is the set of subsets of S of cardinality $k \in \mathbf{N}$. S^k is the set of k-tuples of S, and S^ω the set of ordered "infinite" sequences of elements of S.

Graphs and Linear Algebra. Combinatorial objects related to graphs are denoted by capital letters in calligraphic fonts: for instance, $\mathcal{V}$ and $\mathcal{E}$ are the sets of vertices and edges of the graph. Matrices are denoted using capital letters in "sans serif" fonts: for instance, L is the Laplacian of a graph. Identity matrices are in turn denoted by I, the dimension of the matrix being implicit from context: for instance, in $\mathsf{L} + \mathsf{I}$, we have $\mathsf{I} \in M_{\mathcal{V}}(\mathbf{C})$. The transpose of A is denoted by A^t, and its conjugate-transpose by A^*. The same convention applies to vectors.
If $W \subseteq V$ is a subspace of a linear space V, $W^\perp$ is the orthogonal complement of W in V.
Given a matrix $A \in M_n(\mathbf{C})$, we denote by $A_{R,C}$ its restrictions to the rows and columns indexed by $R, C \subseteq [n]$, and by A_R its restrictions to the rows and columns indexed by R.
A vector $f \in \mathbf{C}^{\mathcal{V}}$ (always denoted in lowercase) is frequently regarded as a linear map from $\mathbf{C}^{\mathcal{V}}$ to $\mathbf{C}$, and its entries denoted either by f_i or $f(i)$ depending on context.
For two vectors $f, g \in \mathbf{C}^{\mathcal{V}}$, we denote the standard l_2 inner product by

$$\langle f, g \rangle = \sum_{i \in \mathcal{V}} f(i) g(i)^*,$$

and by $\|f\|_2$ the associated norm.
The trace and determinant of a (square) matrix A are denoted by $\mathrm{tr}(A)$ and $\det(A)$ respectively.
If $i \in \mathcal{V}$, $\delta_i \in \mathbf{C}^{\mathcal{V}}$ is the vector such that $\delta_i(i) = 1$ and 0 on all other entries.

Probabilities. For a probability distribution $\mathcal{D}$ defined over (the measurable subsets of) a set S, we denote by $\mathbf{P}_{\mathcal{D}}(X)$ the probability of drawing $X \subseteq S$. As we mostly work with discrete probabilities, we often write $\mathbf{P}_{\mathcal{D}}(a) = \mathbf{P}_{\mathcal{D}}(\{a\})$ instead. The expectation of a random variable $f : S \to \mathbf{C}$ is denoted by $\mathbf{E}_{\mathcal{D}}(f)$.
We often define a probability distribution by $\mathbf{P}_{\mathcal{D}}(a) \propto p_a$ (for $p_A \in \mathbf{R}$), which means that

$$\mathbf{P}_{\mathcal{D}}(a) = \frac{p_a}{\sum_{a \in S} p_a}.$$

We denote by $\mathcal{N}_{\mathbf{C}}(0,1)$ the law of a complex normal random variable, which is such that $\mathfrak{Re}(z) \sim \mathcal{N}(0, 1/2)$ and $\mathfrak{Im}(z) \sim \mathcal{N}(0, 1/2)$. $\mathcal{U}(I)$ is the uniform law over an interval $I \subseteq \mathbf{R}$.

CONTENTS

Part I

PROCESSING DATA OVER GRAPHS: PROBABILITY AND GEOMETRIES

1

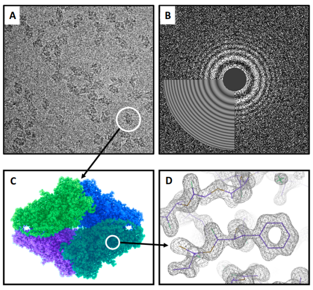
A TAILORED INTRODUCTION TO GRAPHS

Now I will have less distraction.
— Leonhard Euler[1]

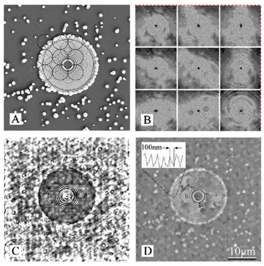
A *graph* $\mathcal{G}$ is a set of nodes $\mathcal{V}$ pairwise interconnected by a set of edges $\mathcal{E}$ [Bollobás, 1998]. Modern graph theory is a thriving domain of research, cross-fertilized by ideas coming from fundamental mathematics, informatics, and physical sciences. Our work also feeds on interdisciplinarity, and draws on more recently-established links with *graph signal processing* (GSP) [Shuman et al., 2013, Ortega et al., 2018]. Let us start by introducing a number of (arbitrarily selected) exciting potential applications for of our (theoretical) work.

1. *Cryogenic Electron Microscopy.* Cryo-EM is a Nobel-prize-winning technology[2], allowing to infer with high resolution the structure of biological macro-molecules *in solution*, which has revolutionized structural biology and biochemistry [Bendory et al., 2020]. In particular, these techniques involve numerous measurements of very noisy 2-dimensional projections of the 3-dimensional molecule, at unknown angles. The very poor signal-to-noise-ratio of the resulting images impedes further processing, and many approaches proceed by rotationally-registering (some pairs of) these images, thereby describing a graph, before denoising them and constructing a 3-dimensional model of the molecule (see, *e.g.*, [Singer and Wu, 2012, Zhao and Singer, 2014]).

1 See [Eves, 2003].
2 https://www.nobelprize.org/prizes/chemistry/2017/press-release/



(a) Cryo-EM-based 3D-imaging of a β-galactosidase enzyme. Picture from [Bendory et al., 2020].



(b) Ptychography-based imaging of a 1 μm-thick gold Fresnel zone plate. Picture from [Rodenburg et al., 2007, Rodenburg, 2008].

(c) Unicorn model from the *Onward* Pixar movie, with fur and feathers automatically generated from a few guiding strands. Picture from [De Goes et al., 2020].

Figure 1.1: Examples of results for the three problems listed above.

2. *Ptychography*, a powerful and flexible diffractive-imaging method, with wide-spread application in crystallography, microscopic imaging, or cell-imaging [Rodenburg, 2008]. Ptychographic imaging involves a phase-retrieval problem of the entries of a complex-valued vector $x \in \mathbf{C}^n$, obtained from diffraction patterns captured when illumimating different regions of the object of interest. A modern approach to this problem consists in approximating some pairwise phase-offsets $x_i(x_j)^*$, which describe a graph topology, and using these values to estimate x (see, *e.g.*, [Iwen et al., 2017]).

3. *Grooming*, that is, helping graphic artists' work by automatically generating fur on beautiful 3-dimensional creatures [De Goes et al., 2020]. This is an example of a common situation in computational geometry involving a vector-field defined over the nodes of a discrete surface (which, in particular, defines a graph) that has to be processed while taking into account the geometry and curvature of the model.

We refer the reader to Figure 1.1 for illustrations pertaining to these three applications, the processing pipelines of which all share a common structure at some point. These problems require processing potentially massive data-sets, and consequently, computational bottlenecks are a harsh reality that practitioners have to face; this issue is a driving motivation for the active research on developing more efficient processing algorithms. Our work in this thesis follows this course, and we strive to develop efficient, widely-applicable, graph-theory-based algorithms to solve challenging data-processing tasks.

The common ideas underlying these applications will be introduced in the course of this manuscript, and we now begin with definitions and notations.

Formally, we denote by $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ a graph with finite set of nodes $\mathcal{V}$ and *non-oriented* edges $\mathcal{E} \subseteq \mathcal{V}^{(2)}$ (the set of *non-ordered* pairs in $\mathcal{V}$). A *weighted graph* is a graph whose edges $e \in \mathcal{E}$ are associated to weights $w_e \in \mathbf{R}^+$. See Figure 1.2 for an example of a well-known graph. The results we obtain in this thesis apply to both weighted and unweighted graphs, and we may omit the weights to make the notation less cumbersome. We are specifically interested in *randomized algorithms* to process data supported *over* graphs: an elementary instance of such graph-supported data is that of real values attached to the nodes of the graph, usually formalized as a vector

$f \in \mathbf{R}^{\mathcal{V}}$, and called a *graph signal*[3]. Established GSP applications range from digital image processing, where each pixel is a node and the signal represents the corresponding intensity at this pixel (which is a classical setting of discrete signal processing; see Section 1.1.1), to, *e.g.*, the analysis of functional and effective connectivity between regions of the human brain based on fMRI[4] data [Huang et al., 2018, Farahani et al., 2019], and more modern developments further include the problems we mentioned above.
Practical instances of these problems may involve large graphs with hundreds of thousands of nodes, and solving these problems translates to linear-algebraic computations involving large matrices, so that the runtime of *exact* solvers becomes prohibitive. This is in particular the case of a smoothing problem we consider in **Part** ii, for which exact solvers have $\mathcal{O}(|\mathcal{V}|^3)$ time-complexity. In turn, one has to rely on very efficient approximate solvers. For some problems, including the smoothing problem, a number of different techniques can be used, including the following.

- A classical solution is to use (deterministic) Krylov-subspaces-based iterative algorithms [Saad, 2003, Saad, 2011][5]; the main drawback of which are that their convergence depends in a complex manner on the spectrum of the matrix at hand (conjugate-gradient algorithms for instance require good-quality preconditioners to ensure fast convergence), and that these methods require matrix-vector multiplications involving large matrices, which become computationally more expensive as the graph gets denser. These observations are exemplified in the following generic result (see, *e.g.*, [Vishnoi et al., 2013]), pertaining to a general form of the smoothing problem mentioned above.

  Theorem 1.0.1. *Given a positive definite $n \times n$ symmetric matrix A and a vector b, consider the linear least-squares problem*

  $$\operatorname*{argmin}_{w} \|Aw - y\|_2^2, \tag{1.1}$$

  where $\|x\|_2 = \langle x, x\rangle$ denotes the usual squared l_2-norm. Denoting by w_* the optimal solution to this problem, the conjugate-gradient algorithm outputs an approximation $\widetilde{w}$ of w_* satisfying

  $$\frac{\|\widetilde{w} - w_*\|_A^2}{\|w_*\|_A^2} \leq \varepsilon, \tag{1.2}$$

  where $\varepsilon > 0$, and $\|x\|_A^2 = \langle x, Ax\rangle$, in time $\mathcal{O}\left(t_A \sqrt{\kappa_A} \log\left(\frac{1}{\varepsilon}\right)\right)$, with κ_A the condition number[6] *of A, and t_A the cost of a matrix-vector product involving A.*

- A second, more modern option is to rely on *Randomized Numerical Linear Algebra* routines (RandNLA) [Drineas and Mahoney, 2016, Martinsson and Tropp, 2020] to either build randomized preconditioners (to be used with conjugate-gradient algorithms for instance) or to perform *Monte-Carlo estimation* [Hammersley, 2013]. Even for Monte-Carlo estimators with convergence rates in $\mathcal{O}\left(\frac{\sigma}{\sqrt{m}}\right)$ (for σ^2 the variance of the estimator and m the number of samples), RandNLA techniques can reach state-of-the-art practical performance, and

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3 Such a signal can also be formalized as a *discrete differential form* on $\mathcal{G}$, which may at times be a more natural concept [Hansen and Ghrist, 2019]. Both formalisms are equivalent for signals taking values in a field, and we stick to the vector representation for simplicity. The difference will be important when considering simplicial complexes in Chapter 7.
4 Functional Magnetic Resonance Imaging.
5 Named after Russian engineer and mathematician Aleksey Nikolaevich Krylov, 1863−1945.
6 In this case, the condition (for the 2-norm) of the matrix A is the ratio of its extremal eigenvalues: $\kappa(A) = \frac{\lambda_{\max}}{\lambda_{\min}}$.

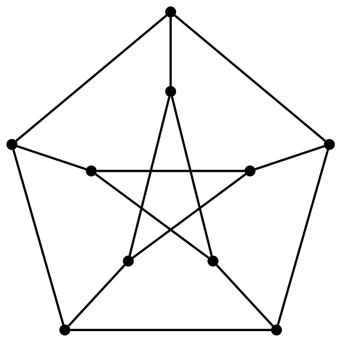


Figure 1.2: The Petersen graph [Holton and Sheehan, 1993].

allow for flexible schemes of computation, with low memory footprints. The most famous line of research is certainly that initiated in [Spielman and Teng, 2011] in order to sparsify and solve "Laplacian" systems; the techniques developed in subsequent works are already the topic of doctoral courses [Tropp, 2019] and have matured enough to be compared with high-performance implementations of Krylov-subspace-based methods [Gao et al., 2023].

Our work explores the second option, and expands on the range of RandNLA techniques available to solve graph-based problems. In particular, these will apply to some *smoothing and interpolation problems* (encountered for instance in computational geometry), and to angular synchronization (encountered in, *e.g.*, phase retrieval).

The study of data defined over graphs largely pre-dates GSP, and the randomized algorithms we develop are variations of techniques first studied in the work of Kirchhoff [Kirchhoff, 1847][7]. One natural and pervasive tool in this setting is a *discrete differential* operator $\nabla : \mathbf{R}^{\mathcal{V}} \to \mathbf{R}^{\mathcal{E}}$. Its definition hinges on a choice of a reference *orientation* of the edges: to each edge $e = \{s_e, t_e\} \in \mathcal{E}$, one can associate two oriented edges (s_e, t_e) and (t_e, s_e) in $\mathcal{V}^2$, and we arbitrarily choose one such reference orientation for each edge. We abuse notation and denote by $e = (s_e, t_e) \in \overrightarrow{\mathcal{E}}$ the corresponding oriented edge (whether we mean the oriented or non-oriented edge will be clear depending on context), by $e^* = (t_e, s_e)$ the reversely-oriented edge, and by $\overrightarrow{\mathcal{E}} \subseteq \mathcal{V}^2$ the set of *oriented* edges of $\mathcal{G}$, which contains the two oriented versions $e, e^* \in \overrightarrow{\mathcal{E}}$ for each $e \in \mathcal{E}$. *All graphs throughout this manuscript are endowed with one such reference orientation.* The discrete differential ∇ is then defined as the linear map satisfying

$$(\nabla f)(e) = \sqrt{w_e}(f(t_e) - f(s_e)) \tag{1.3}$$

for all edges $e \in \mathcal{E}$, and $f \in \mathbf{R}^{\mathcal{V}}$. We abusively identify ∇ with its matrix in the natural bases[8] of $\mathbf{R}^{\mathcal{V}}$ and $\mathbf{R}^{\mathcal{E}}$, which is exactly the $|\mathcal{V}| \times |\mathcal{E}|$ edge-vertex incidence matrix of $\mathcal{G}$, with

$$\nabla_{v,e} = \begin{cases} -\sqrt{w_e} & \text{if } v = s_e \\ \sqrt{w_e} & \text{if } v = t_e \\ 0 & \text{otherwise.} \end{cases} \tag{1.4}$$

We abuse notation in the same manner for all operators encountered throughout the manuscript.

7 German physicist and mathematician Gustav Robert Kirchhoff, 1924−1887.
8 Specifically, the bases generated by the nodes and the edges respectively.

(a) Electric potential f on $\mathcal{G}$.

(b) Induced voltage ∇f on $\mathcal{G}$.

Figure 1.3: Representation of the action of ∇, on the 2×3 grid graph $\mathcal{G}$.

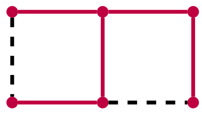


Figure 1.4: A spanning tree (in purple) on the 2×3 grid graph.

As an illustration of these notions, let us informally expose the result of Kirchhoff, which pertains to the analysis of electrical networks (notions important for our developments will be more precisely re-introduced). For simplicity, graphs are unweighted throughout the following paragraph, and references to electrical-circuits theory only serve as a motivation: Kirchhoff's result is of a purely mathematical nature (for a more exhaustive exposition of the links between electrical-circuits theory, linear algebra, optimization and probability, see [Bollobás, 1998]).

Electrical circuits and spanning trees. Graphs have long since been recognized as a model for idealized electrical networks [Kirchhoff, 1847], the branches of which can be represented as the edges $\mathcal{E}$ of a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. In this representation, an electrical potential f is nothing but a real-valued function on its nodes $f \in \mathbf{R}^{\mathcal{V}}$, and the potential difference $f(t_e) - f(s_e)$ between the two endpoints s_e and t_e of an (oriented) edge $e \in \mathcal{E}$ is the induced voltage on this branch of the circuit [Bollobás, 1998][9]. The vector of voltages $g \in \mathbf{R}^{\mathcal{E}}$ induced by a potential f is conveniently obtained as the image of f by the discrete differential ∇:

$$g = \nabla f, \tag{1.5}$$

as we illustrate in Figure 1.3, for $f = (-0.91, 1.21, 0.59, 0.39, 0.81, -0.69)^t$. In this specific example, the matrix of ∇ reads:

$$\nabla = \begin{array}{c} \\ (1,2) \\ (1,4) \\ (2,3) \\ (2,5) \\ (3,6) \\ (4,5) \\ (5,6) \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \end{array}, \tag{1.6}$$

where the rows (resp. columns) are explicitly indexed by the edges (resp. nodes) of $\mathcal{G}$.
If a potential difference $g(e) = f(t_e) - f(s_e)$ between two nodes s_e and t_e is observed on some

9 Note that this signal is naturally oriented from s_e to t_e.

edge e, an electrical current $c(e) \in \mathbf{R}$ also flows along this edge. According to Ohm's law[10], one has

$$c(e) = \frac{g(e)}{r_e}, \tag{1.7}$$

where $r_e \in \mathbf{R}$ is the resistance of edge e, *i.e.*, the current along an edge is always proportional to the voltage on that edge. In the following, we denote by R the $|\mathcal{E}| \times |\mathcal{E}|$ diagonal matrix with $\mathsf{R}_{e,e} = r_e$, so that the current $c \in \mathbf{R}^{\mathcal{E}}$ satisfies $\mathsf{R}c = g$ for some potential difference g.

A basic problem in electrical engineering then consists in determining, from a given voltage c imposed on some branches of the circuit[11], the resulting current $c \in \mathbf{R}^{\mathcal{E}}$ over the entire network. For the problem to be well-posed, and to be physically meaningful, one must impose additional constraints on the current c, namely, to follow Kirchhoff's voltage and current laws [Tipler and Mosca, 2007][12], so that, in the end, we can obtain the following formulation. Given a network $\mathcal{G}$, with resistance matrix R, and a voltage $g \in \mathbf{R}^{\mathcal{E}}$ imposed on a part of the network, find a current $c \in \mathbf{R}^{\mathcal{E}}$ on $\mathcal{G}$ satisfying both:

$$c \in \ker \nabla^* \text{ and } \mathsf{R}c - g \in \mathrm{im}\nabla, \tag{1.8}$$

which expresses both Kirchhoff's and Ohm's laws, and where ∇^* denotes the adjoint of ∇ for the standard inner products on $\mathbf{R}^{\mathcal{V}}$ and $\mathbf{R}^{\mathcal{E}}$ (*i.e.*, $\nabla^* = \nabla^t$, its transpose).

One can show that such a current c exists and is unique [Weyl, 1923, Bollobás, 1998], and Kirchhoff's network theorem allows to express c as an average over the *spanning trees* of $\mathcal{G}$ of simple currents associated to these trees. A spanning tree is a subset of edges that is connected to all the nodes in the graph, but does not contain a cycle; see Section 1.2 for a more in-depth discussion, and Figure 1.4 for an illustration of a spanning tree [Kirchhoff, 1847, Nerode and Shank, 1961]. Mathematically, Kirchhoff's network theorem reads

$$z = \frac{1}{\#\text{spanning trees}} \sum_T \left(\mathsf{R}^{-1} \overline{T}^* g \right), \tag{1.9}$$

where $\overline{T}^* g$ denotes the previously mentioned current associated to a tree T (see [Nerode and Shank, 1961] for a precise definition). What should be noted here is that Equation (1.9) expresses the current resulting on *all the edges of the graph*, and not just one isolated part of the graph (which is more common, as we will discuss in this chapter through a number of examples). This result has been influential for two main reasons.

- It is an early example of a more general result called a *mean projection theorem* [Maurer, 1976, Lyons, 2003, Catanzaro et al., 2012, Kassel and Lévy, 2022b], that pertains to a special type of probability distribution called a *Determinantal Point Process*, discussed in Chapter 3.
- It gives an alternative way to compute the current z going through a circuit, which is a non-trivial electrical-engineering problem [MacWilliams, 1958].

Importantly, Kirchhoff's theorem can be understood as a probabilistic statement: if one considers the uniform distribution over the spanning trees of $\mathcal{G}$, then

$$c = \mathbf{E}(\mathsf{R}^{-1} \overline{T}^* g). \tag{1.10}$$

10 Named after german physicist Georg Ohm, 1789−1854.
11 Obtained by plugging a generator on these branches.
12 The two laws respectively state that the sum of the potential differences along a cycle should sum to zero, and that the sum of the currents on the edges incident to any given nodes should be zero (that is, the total currents incoming and coming out of this node should be equal).

Equation (1.10) is computationally useful because it can be leveraged in Monte-Carlo estimators, by repeating the following m times:

1. sample a spanning tree T uniformly among all the spanning trees of $\mathcal{G}$;
2. compute the current $\overline{T}^* c$ induced by the tree T and use it for the estimation.

Averaging the result over a number m of samples then yields an approximation[13] of z.
The uniform distribution over spanning trees has later applied as a tool to build spectral sparsifiers of graphs [Kyng and Song, 2018, Kaufman et al., 2022] that can be used to efficiently solve some application-relevant linear systems appearing in, *e.g.*, discrete geometry and combinatorial optimization [Levcopoulos and Lingas, 1992]. It turns out that these methods are *practically* useful because of efficient tree-sampling algorithms that have been developed, such as those of Wilson [Wilson, 1996] and of Aldous and Broder [Aldous, 1990, Broder, 1989] (see Section 1.2).

Our work introduces Monte-Carlo estimators, in the spirit of Kirchhoff's theorem, to solve a number of problems that are the subject of growing interest in GSP, in the specific setting of edges describing additional *rotations*. In **Part** i, we introduce some preliminary background.

- Chapter 1 contains an introduction to graphs and classical GSP problems, as well as a presentation of existing Kirchhoff-like estimators for some of these problems.
- Chapter 2 focuses on graphs whose edges are endowed with rotations, which is known as a *connection*. Interest in these structures has flourished in the past few years, and we present some important practical problems on these graphs, which we frame in the context of GSP.
- Chapter 3 finally introduces important technical tools that we need to develop our estimators: determinantal point processes.

Part ii contains our principal contribution: a theoretical analysis of Kirchhoff-like estimators for rotation-aware GSP problems, and a numerical evaluation of these estimators as compared to standard, deterministic methods [Saad, 2003].

Chapter 1 is more specifically organized as follows.

- We introduce in Section 1.1 the *combinatorial graph Laplacian*, a linear operator that serves as a basis to formulate many GSP problems. We further discuss some specific GSP problems and related properties of the Laplacian. The analysis of these problems and their solution largely depends on the topology of the graph, and we recall some associated generative models of random graphs.
- We discuss in Section 1.2 existing Kirchhoff-like techniques, leveraging generalizations of the uniform distribution over spanning trees, that can be used to solve some of these problems. These techniques are meant to provide approximate, cheaply-computable solutions, and we invite the reader to question how these methods are affected by the structure of the graph.

13 Early algorithms, such as in [MacWilliams, 1958], were inefficient, and used to generate *all* the spanning trees of $\mathcal{G}$ to get an exact result, instead of relying on Monte-Carlo estimations.

1.1 THE GRAPH LAPLACIAN

In the design of processing pipelines for graph-based problems, one may leverage information coming from a number of different perspectives, pertaining to:

- variational properties of signals on $\mathcal{G}$;
- probabilistic properties of random processes on $\mathcal{G}$;
- structural and combinatorial properties of $\mathcal{G}$.

It turns out that many of these properties can be jointly captured by algebraic operators (*i.e.*, matrices) representing the graph, such as the graph Laplacian[14] [Chung, 1997].
The *combinatorial Laplacian* $\mathsf{L} : \mathbf{R}^{\mathcal{V}} \to \mathbf{R}^{\mathcal{V}}$ of a graph $\mathcal{G}$ is a symmetric, *semi-definite positive* operator, defined by

$$\mathsf{L} = \nabla^t \nabla \tag{1.11}$$
$$= \mathsf{D} - \mathsf{A}, \tag{1.12}$$

where D is the diagonal matrix of the degree (where each node i has degree $d_i = \sum_{j\in\mathcal{V}} w_{i,j}$), and A is the (possibly weighted) adjacency matrix of the graph (with $\mathsf{A}_{(i,j)} = w_{i,j}$ if $(i,j) \in \overrightarrow{\mathcal{E}}$ and 0 otherwise). Let us note that combinatorial Laplacian is also called Kirchhoff's matrix (see, *e.g.*, [Bollobás, 1998]), and that Equation (1.12) above is obtained from an explicit computation of the entries of $\mathsf{L} = \nabla^t \nabla$.

We explore in this section some properties that the combinatorial Laplacian captures, and how those can be used in practical GSP problems.

1.1.1 *Smoothing Graph Signals*

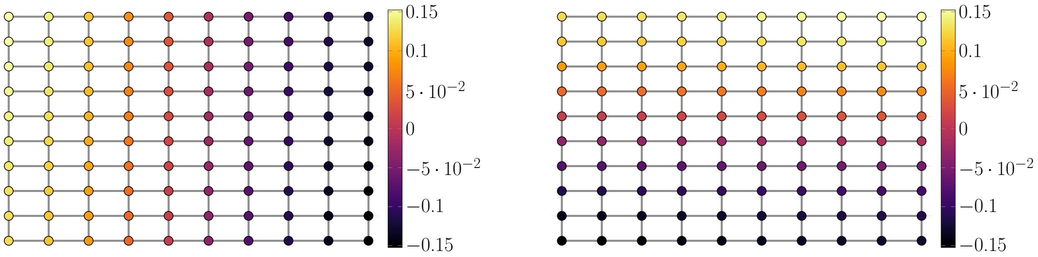
One common hypothesis about graph signals is that they are *smooth*: the values of a signal on two adjacent nodes should be close. Examples of smooth signals may include sensor data over a network (see *e.g.* [Narang and Ortega, 2010, Narang and Ortega, 2012]), or opinions in social networks. The quadratic form associated to L provides a natural way to quantify this smoothness [Shuman et al., 2013], as we immediately obtain from Equation (1.11) that it acts as:

$$\langle f, \mathsf{L} f\rangle = \frac{1}{2} \sum_{e\in\overrightarrow{\mathcal{E}}} w_e |f(t_e) - f(s_e)|^2, \tag{1.13}$$

which associates a high value to functions with important local variations between adjacent vertices. Conversely, a signal f is smooth if it has small local variations between adjacent nodes. Note that $\langle f, \mathsf{L} f\rangle$ is always non-negative, hence, L is semi-definite positive.

Another complementary way to conceptualize the smoothness of a graph signal is in terms of its *frequency components*. Denoting by $0 \le \lambda_1 \le ... \le \lambda_{|\mathcal{V}|}$ and $u_1, ..., u_{|\mathcal{V}|} \in \mathbf{R}^{\mathcal{V}}$ the eigenvalues and an associated orthonormal basis of eigenvectors of L, we say that a signal f is *B-bandlimited* if $f \in \operatorname{span}(u_1, ..., u_B)$, that is, if it is a linear combination of the B first eigenvectors of L. Let us provide more insight into this definition.

14 Named after French mathematician, physicist and astronomer Pierre-Simon de Laplace, 1749−1827.



(a) u_2

(b) u_3

Figure 1.5: Second and third eigenfunctions of the 10×10 grid graph.

(I1) First, since L is symmetric semi-definite positive, the eigenpairs (λ_i, u_i) can be recovered as the successive optimizers of the Rayleigh quotient [Horn and Johnson, 2012][15]:

$$u_k = \underset{f \in \text{span}(u_1,\ldots,u_{k-1})^\perp}{\text{argmin}} \frac{\langle f, \mathsf{L} f \rangle}{\langle f, f \rangle}, \tag{1.14}$$

which allows for both an intuitive understanding of the eigenvectors of L, and for simple proofs of classical results, such as the following. A graph is *connected* if there exists a path between any two nodes i and j (*i.e.*, an ordered sequence of edges $((i = v_1, v_2), (v_2, v_3), \ldots, (v_{l-1}, v_l = j))$; if a graph is not connected, it is separated into multiple maximal *connected components* (*i.e.*, maximal subsets of nodes such that the graph restricted to these nodes *is* connected). Then, it is clear from Equation (1.14) that 1-bandlimited signals are connected-component-wise *constant* signals, since $\langle f, \mathsf{L} f \rangle = 0$ for such signals (and the same holds true for k-bandlimited signals, where k is the number of connected components of $\mathcal{G}$).

(I2) Second, when $\mathcal{G}$ is a d-dimensional torus[16] $\mathbb{T}^d$, the eigenvectors of L are the d-dimensional discrete Fourier modes [Gray et al., 2006, Fracastoro et al., 2016]. In particular, when $d = 1$, $\mathcal{G}$ is a cycle, and we have

$$u_k(i) = \frac{1}{\sqrt{n}} \mathfrak{Re} \left(e^{-\frac{2i\pi(k-1)(i-1)}{n}} \right) \tag{1.15}$$

for a cycle with $n = |\mathcal{V}|$ vertices. In analogy with this classical situation, we more generally say that u_k is a frequency component with frequency λ_k (for any given graph $\mathcal{G}$).

Overall, a B-bandlimited signal is a linear combination of the B smoothest orthogonal signals on the graph. We illustrate in Figure 1.5 the second and third eigenvectors of L on a 2D grid graph, which are the two smoothest, orthogonal, non-constant signals on this graph.

Like regular (discrete) signals, one can apply filters to graph signals to, *e.g.*, denoise them. Formally, we call a filter a function $h : \mathbf{R} \to \mathbf{R}$ and, if a signal $f \in \mathbf{R}^{\mathcal{V}}$ decomposes as a sum of eigenfunctions $f = \hat{f}_1 u_1 + \ldots + \hat{f}_{|\mathcal{V}|} u_{|\mathcal{V}|}$, the filtered signal f^h writes:

$$f^h = \sum_{k=1}^{|\mathcal{V}|} h(\lambda_k) \hat{f}_k u_k, \tag{1.16}$$

15 Where $\text{span}(u_1, \ldots, u_{k-1})^\perp$ is the orthogonal complement of $\text{span}(u_1, \ldots, u_{k-1})^\perp$ in $\mathbf{R}^{\mathcal{V}}$.
16 More precisely, a Cartesian product of cycle graphs (which are graphs consisting of a single cycle).

where $\hat{f}_k = \langle f, u_k \rangle$. In general, applying a filter as in Equation (1.16) requires computing a full eigendecomposition of L, at $\mathcal{O}(|\mathcal{V}|^3)$ cost, but some filters may be less expensive to compute. In particular, a classical choice of cheaply-computable filters are *polynomial filters*, *i.e.*, filters for which h is a polynomial, in which case applying the filter h only requires computing matrix-vector products (see, *e.g.*, [Hammond et al., 2011]):

$$f^h = h(\mathsf{L})f, \tag{1.17}$$

where $h(\mathsf{L})$ abusively denotes a matrix polynomial in L.

Let us now describe two classical low-pass filters.

Smoothing. Consider a noisy measurement $f_\varepsilon = f_\top + \varepsilon$ of a smooth signal $f_\top \in \mathbf{R}^{\mathcal{V}}$, where $\varepsilon \in \mathbf{R}^{\mathcal{V}}$ represents some noise (*e.g.* Gaussian[17] noise). Different filters may be used to estimate $f_\top$ from f_ε [Shuman et al., 2013]; we give two examples below. See Figure 1.6

- An ideal low-pass filter is a map $h_\tau : \mathbf{R} \to \{0, 1\}$, with $h_\tau(\lambda) = 1$ if $\lambda \leq \tau$ and $h_\tau(\lambda) = 0$ otherwise (for a fixed threshold τ). Applying a low-pass filter simply amounts to filtering out the high-frequency components of a signal:

  $$f_\varepsilon^{h_\tau} = \sum_{k=1}^{K} (\hat{f}_\varepsilon)_k u_k \tag{1.18}$$

  for $\lambda_K \leq \lambda < \lambda_{K+1}$. Note that this is not a polynomial filter, and that computing the K first eigenpairs (λ_i, u_i) is required in order to apply this filter.

- Instead of completely getting rid of the high-frequency components, one can consider a soft filter of the form

  $$h_q(\lambda) = \frac{q}{q + \lambda} \tag{1.19}$$

  for some positive constant $q \in \mathbf{R}_+^*$. One reason this filter is especially interesting is because it corresponds to the solution of the graph Tikhonov smoothing problem[18]:

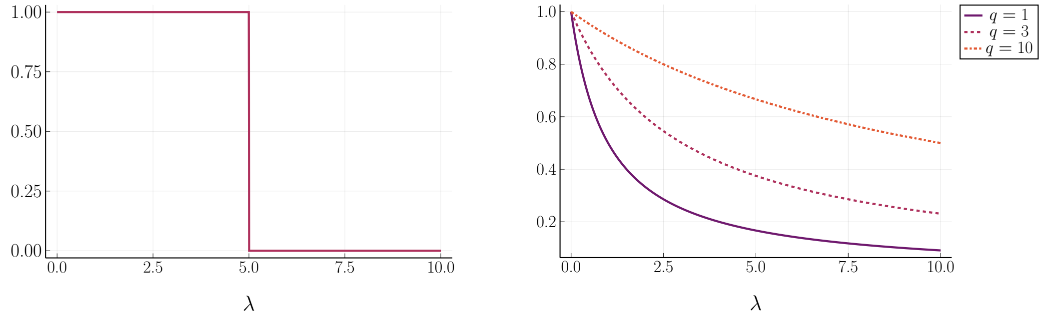
  $$\underset{f \in \mathbf{R}^{\mathcal{V}}}{\operatorname{argmin}} \; q\|f - f_\varepsilon\|_2^2 + \langle f, \mathsf{L}f \rangle, \tag{1.20}$$

  whose solution $f_* = f_\varepsilon^{h_q}$ simply reads

  $$f_* = (\mathsf{L} + q\mathsf{I})^{-1} q\mathsf{I} \; f_\varepsilon. \tag{1.21}$$

  In particular, applying the corresponding filter simply amounts to solving the linear system $q^{-1}(\mathsf{L} + q\mathsf{I})f = f_\varepsilon$ instead of computing an eigendecomposition[19] of L. Note that, while computing f_* *exactly* has cost $\mathcal{O}(|\mathcal{V}|^3)$, iterative, approximate, algorithms such as the conjugate-gradient method can provide significant speed-ups [Saad, 2003] (see Chapter 6 for a comparison between our estimators and the conjugate-gradient method, on a similar problem), and we present in Section 1.2 an existing Kirchhoff-like estimator to compute f_*.

17 After German mathematician, physicist and astronomer Carl Friedrich Gauss, 1777−1855.
18 Named after Russian mathematician and geophysicist Andry Nikolayevich Tikhonov, 1906−1993.
19 Tikhonov regularization is not a polynomial filter, but a *rational* filter (and more specifically corresponds to an ARMA filter) [Loukas et al., 2015], and one may notice that it is expressible as a power-series in L.



(a) Ideal hard-pass filter with $\tau = 5$. (b) Tikhonov-based low-pass filters for $q = 1, 3, 10$.

Figure 1.6: Low-pass filters.

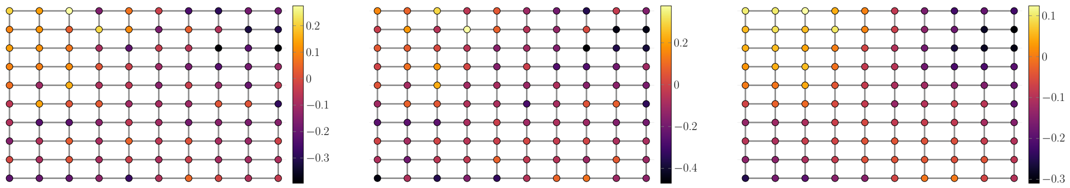


Figure 1.7: Tikhonov-based moothing of a 5-bandlimited signal on the 2D grid. Left: ground-truth signal. Center: noisy signal. Right: smoothed signal.

If the filter is well-tuned (τ or q chosen so that $h(\lambda) \simeq 1$ for the first B frequencies $\lambda_1 \leq \ldots \leq \lambda_B$), f^h should give a good approximation of $f_\top$.
We illustrate Tikhonov-based smoothing in Figure 1.7, for a 5-bandlimited ground-truth signal $f_\top$:

$$f_\top = \alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3 + \alpha_4 u_4 + \alpha_5 u_5, \tag{1.22}$$

with randomly generated coefficients $\alpha_i \sim \mathcal{N}(0, 1)$, and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, where σ^2 is selected so as to obtain an signal-to-noise ratio[20] of 1.5. We take $q = 1$ as a regularization parameter. As compared to the noisy signal, the filtered signal is perceptually smoother.

Note that the previous smoothing methodology relies on the analogy between classical signals on the torus and graph signals (recall Point (I2) above), but this analogy may not always be valid. We illustrate in Figure 1.8 the relative error $\frac{\|f_\top - f_*\|_2}{\|f_\top\|_2}$ for different values of q, on two randomly-generated graphs.

- An Erdös-Rényi graph[21], with a set of n nodes and an edge between each pair of nodes $\{i, j\}$ appearing with probability p (independently for each pair). Erdös-Rényi graphs are among the most simple random graph models, but do not exhibit structures often encountered in real-world graphs, such as spatial correlations (*e.g.* sensor networks), or clustering (*e.g.* social networks). Here, we take $n = 10000 = 10^5$ and $p = \frac{40}{10000} = 4 \times 10^{-4}$.
- A random geometric graph (ε-graph), built from n points uniformly sampled in $[0, 1]^d$, with an edge between x_i and x_j whenever $\|x_i - x_j\|_2 < \varepsilon$. This type of graph describes an

20 That is, we set $\sigma^2 = \frac{1}{SNR \times n}$ with $SNR = 1.5$.
21 After Hungarian mathematicians Alfréd Rényi (1920−1970) and Paul Erdös (1913−1996).

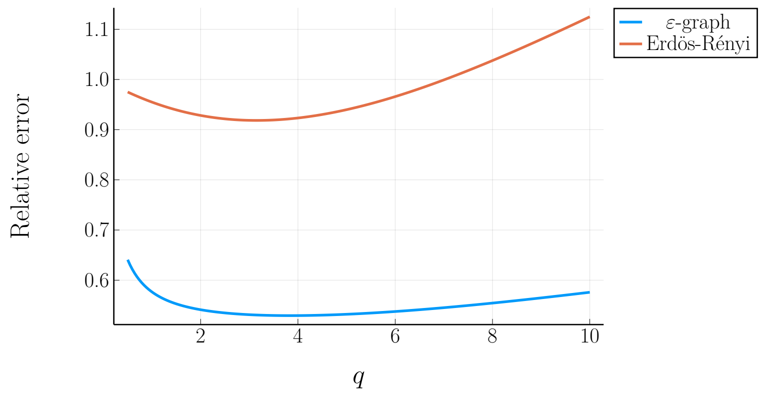


Figure 1.8: Relative errors between $f_\top$ and f_* on the Erdös-Rényi-graph and ε-graph respectively, for a range of values of $q \in [0.5, 10]$.

irregular mesh in $\mathbf{R}^d$, similarly to a grid-graph, and also typically exhibits some clustering of its nodes [Dall and Christensen, 2002]. In our case, we take $n = 10000$, $d = 3$ and $\varepsilon = 0.1$.

We follow the same signal-generation scheme as in Figure 1.7, with a SNR of 3. For all $q \in [0.5, 10]$, the resulting relative error is much lower for the ε-graph than for the Erdös-Rényi graph: the lower-energy eigenvectors of the ε-graph capture a meaningful notion of smoothness, with higher-energy eigenvectors behaving similarly to noise. On the other hand, the eigenvectors of the Erdös-Rényi graph behave similarly to noise across the entire spectrum.

1.1.2 *Interpolation and Semi-Supervised learning*

Beyond globally smooth signals, another class of signals with important applications is that of signals $f \in \mathbf{R}^{\mathcal{V}}$ whose values are fixed on a subset $\mathcal{A} \subseteq \mathcal{V}$ of nodes, called a *boundary*, and maximally smooth when disregarding this boundary, that is, satisfying:

$$\sum_{j \in \mathcal{V}} (f(i) - f(j)) \mathbf{1}_{\mathcal{E}} (\{i, j\}) = 0, \tag{1.23}$$

for all nodes $i \in \mathcal{V} \setminus \mathcal{A}$, where $\mathbf{1}_{\mathcal{E}} (\{i, j\})$ is the indicator that $\{i, j\} \in \mathcal{E}$. Such signals are called *harmonic* on $\mathcal{V} \setminus \mathcal{A}$, in reference to the solution of Laplace's equation that asks to satisfy a similar condition [Axler et al., 2013]. More precisely, a signal $f \in \mathbf{R}^{\mathcal{V}}$ is harmonic with boundary condition $g : \mathcal{A} \to \mathbf{R}$ if it satisfies the two conditions:

$$\begin{cases} f(a) = g(a) \text{ if } i \in \mathcal{A}, \\ (\mathsf{L}f)(i) = 0 \text{ if } i \in \mathcal{V} \setminus \mathcal{A}. \end{cases} \tag{1.24}$$

Harmonic interpolation consists in finding an harmonic signal $f \in \mathbf{R}^{\mathcal{V}}$ from a given boundary condition g, and finds applications in *e.g.* image inpainting [Bertalmio et al., 2000], or semi-supervised node classification (discussed below).

One shows both the existence and unicity of such a signal from a straightforward linear-algebraic argument (see *e.g.* [Rosenthal, 2014] for a proof in a much more general situation, or Appendix B.1).

Proposition 1.1.1. *Suppose that $\mathcal{A}$ intersects all the connected components of $\mathcal{G}$. Then, there exists a unique harmonic signal f on $\mathcal{A}$ with boundary condition g, and we have*

$$f_{|\mathcal{V}\setminus\mathcal{A}} = -(\mathsf{L}_{\mathcal{V}\setminus\mathcal{A}})^{-1}\mathsf{L}_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\; g, \tag{1.25}$$

where $M_{R,C}$ denotes the restriction of a matrix M to its rows (resp. columns) indexed by R (resp. C), and $M_R = M_{R,R}$ to its rows and colums indexed by R. We call f the harmonic interpolation *of g.*

In the following, we denote by $\mathrm{h}(g)$ this unique harmonic interpolation of a signal g. Note that, like the smoothing operation, computing harmonic interpolations *exactly* is relatively expensive, and requires solving a linear system, in $\mathcal{O}(|\mathcal{V}\setminus\mathcal{A}|^3)$ time.

A well-known and very useful property of harmonic interpolations is that we can express them as expectations of random processes [Lawler and Limic, 2010]. Consider the natural random walk on $\mathcal{G}$, starting from a given node i and transitioning from a node u to v with probability $\frac{w_{u,v}}{d_u}$, where we recall that d_u is the (weighted) degree of node u. Denote by a the first node in $\mathcal{A}$ reached by a realization of this walk, and by ρ_i the resulting distribution on $\mathcal{A}$. Then, we have the following.

Proposition 1.1.2. *For $i \in \mathcal{V}\setminus\mathcal{A}$, the signal defined by*

$$\begin{aligned} f(i) &= \mathbf{E}_{a\sim\rho_i}(g(a)) && (1.26)\\ &= \sum_{a\in\mathcal{A}} \mathbf{P}_{\rho_i}(a)g(a) && (1.27)\end{aligned}$$

is the unique harmonic interpolation of $g : \mathcal{A} \to \mathbf{R}$.

Proposition 1.1.2 is a famous example of a variational property (harmonicity) being captured by a random process. Its classical proof leverages the Markov property of the random walk [Lawler and Limic, 2010], and we will also obtain it as a corollary of a more general lemma we derive in **Part** ii of this manuscript (Lemma 4.1.6).

As a by-product, Proposition 1.1.2 yields a simple algorithmic procedure to compute $f(i)$, by repeating the following:

- Run a random walk from i until it reaches $\mathcal{A}$.
- Set the value $f(i) = g(a)$ on node i, where a is the node encountered when first reaching $\mathcal{A}$.

Averaging the values $f(i)$ obtained from a number of different runs, one obtains an unbiased estimate of $\mathrm{h}(g)$. This strategy can also be employed to solve (partial) differential equations defined on differential manifolds, resulting in the so-called *walk-on-spheres* algorithms [Muller, 1956, Sawhney and Crane, 2020]. An interesting property of these algorithms is that no centralized knowledge of the graph is necessary to estimate a given $f(i)$. On the other hand, unlike Kirchhoff-like estimators, the estimation remains localized on a single node, that is, on node i.

Semi-Supervised Classification. Let us give an intuitive explanation of the link between harmonic interpolation and the previously-mentioned graph-signal smoothing operation, on a practical semi-supervised classification problem [Zhu et al., 2003, Avrachenkov et al., 2017]. Suppose that each node of the graph belongs to one of finitely many disjoint classes C_k (*i.e.*,

$\bigcup_k C_k$ partitions $\mathcal{V}$), that can represent, for instance, political opinion [Garimella et al., 2018]. The semi-supervised node-classification problem asks, given a subset of vertices $\mathcal{A}$ whose classes are known (or estimated), to infer the classes of the other nodes $\mathcal{V} \setminus \mathcal{A}$. For each class C_k, we build the *a priori* labeling function $g_k : \mathcal{V} \to \mathbf{R}$ satisfying:

$$g_k(i) = \begin{cases} 1 \text{ if } i \in C_k \\ 0 \text{ otherwise.} \end{cases} \tag{1.28}$$

A common strategy consists in estimating *classification functions* $f_k : \mathcal{V} \to \mathbf{R}$ for each class C_k, that represent how likely a point is to belong to class C_k. The resulting estimated class of point i is then $\operatorname{argmax}_k \; f_k(i)$. Multiple strategies are possible to build the classification functions.

1. Giving absolute confidence to the *a priori* classification functions, in which case one can choose the f_k's to be the harmonic interpolations of the g_k's [Zhu et al., 2003].

2. Giving more importance to the hypothesis of homophily which, in network theory [McPherson et al., 2001, Borgatti and Halgin, 2011], states that nodes sharing the same label are more likely to be connected to each other. Under this hypothesis, one expects classification functions to be smooth *over the whole graph*, and may set, *e.g.*, $f_k = g_k^{h_q}$ the Tikhonov regularization of the *a priori* labeling function, by solving the problem in Equation (1.20) (see [Avrachenkov et al., 2017]). Here, the parameter q controls the confidence-level on the priors f_k.

One potential drawback of the smoothing-based approach is that the confidence level q given to the *a priori* labels is the same for all nodes. We can more generally try to estimate the classification functions as

$$f_k = \underset{f \in \mathbf{R}^{\mathcal{V}}}{\operatorname{argmin}} \sum_{i \in \mathcal{V}} q_i |f(i) - g_k(i)|^2, \tag{1.29}$$

where the non-homogeneous q_i's associate different confidence levels to different nodes. In fact, one can show that by setting $q_i = 0$ for $i \in \mathcal{V} \setminus \mathcal{A}$ and $q_i = \alpha$ for $i \in \mathcal{A}$, we obtain [Pilavcı et al., 2021]:

$$\lim_{\alpha \to \infty} (f_k^{h_\alpha})_{|\mathcal{V}\setminus\mathcal{A}} = (\mathrm{h}(g_k))_{|\mathcal{V}\setminus\mathcal{A}}, \tag{1.30}$$

where $f_k^{h_\alpha}$ denotes the optimal solution of Equation (1.29) with the previous choice of q_i's.

Remark 1.1.3. *There exist random-walk interpretations for the smoothing operator* $(\mathsf{L} + q\mathsf{I})^{-1} q\mathsf{I}$, *such as the one in [Avrachenkov et al., 2017], based on random-walks with* a priori *geometrically distributed length. Following the previously-mentioned Lemma 4.1.6, we obtain in Chapter 4 a random-walk interpretation of the solutions of the problems in Equations* (1.20) *and* (1.29), *completely analogous to the one in Proposition 1.1.2. Our interpretation also relies on interrupted random walks, where the probability of interruption at each node depends on both the degree of the node and the parameter q.*

Finally, is it reasonable to expect that $g_k^{h_q}$ is smooth when the graph presents a clear community structure? We will see in the next section that this is indeed the case, for an appropriate choice of the bandlimited-ness parameter B.

1.1.3 *Community Detection*

Community detection is a completely unsupervised classification problem, which asks to recover a partition of the nodes from the topology of the graph itself (following, for instance, the homophily assumption). It is not straightforward to define what a community structure is, but we can rely on generative procedures to model some types of community structures. One prominent such model is the *Stochastic Block Model* (SBM), which generates graphs with a community structure as follows (we restrict our attention to graphs containing *two* communities) [Karrer and Newman, 2011]:

1. Each node belongs to one of two classes: $\mathcal{V} = C_1 \cup C_2$, which can be built deterministically or at random.
2. An edge $\{i, j\}$ is included with probability[22]

$$\begin{cases} \frac{c_{1,1}}{n} \text{ if } i, j \in C_1, \\ \frac{c_{2,2}}{n} \text{ if } i, j \in C_2, \\ \frac{c_{1,2}}{n} \text{ otherwise,} \end{cases} \tag{1.31}$$

   where $c_{1,1}, c_{2,2}, c_{1,2} \in \mathbf{R}^+$.

When $c_{1,1}, c_{2,2} \gg c_{1,2}$, we expect to recover two clearly-defined communities in the resulting graph (see Figure 1.9 below for an illustration)[23]. One can for instance show that, when $|C_1| = |C_2|$, $c_{1,1} = c_{2,2} = a$ and $|\mathcal{V}| \to \infty$, it is (statistically) possible to recover a partition that is non-trivially correlated with the true partition $\mathcal{V} = C_1 \cup C_2$ if and only if [Decelle et al., 2011, Mossel et al., 2012, Massoulié, 2014, Mossel et al., 2018]:

$$(a - c_{1,2})^2 > 2(a + c_{1,2}). \tag{1.32}$$

In words, for large graphs, the SBM displays an increasingly meaningful community structure as:

- the intra-class connectivity increases;
- the inter-class connectivity decreases.

An extreme instance of this scenario is observed when the graph has two connected components, corresponding to the two communities C_1 and C_2. In this situation, recovering the partition is trivial: the community structure is reflected in the kernel $\ker L$ (generated by the constant signals on each connected component).

One drawback of the SBM is that it exhibits homogeneous intra-class degree distributions, which is not commonly observed in real-word datasets [Barabási, 2013]. An alternative model is the *degree-corrected SBM* (DC-SBM), which replaces step 2. to introduce degree-heterogeneity:

2'. An edge $\{i, j\}$ is included with probability

$$\begin{cases} \xi_i \xi_j \frac{c_{1,1}}{n} \text{ if } i, j \in C_1, \\ \xi_i \xi_j \frac{c_{2,2}}{n} \text{ if } i, j \in C_2, \\ \xi_i \xi_j \frac{c_{1,2}}{n} \text{ otherwise,} \end{cases} \tag{1.33}$$

22 It is usually assumed that $n > c_{k,l}$ in Equation (1.31), so that we always have $\frac{c_{1,1}}{n}, \frac{c_{2,2}}{n}, \frac{c_{1,2}}{n} \in (0, 1)$. Actual implementations may use probabilities of the form $\min\left(\frac{c_{k,l}}{n}, 1\right)$ instead.

23 On the other hand, when $c_{1,1} = c_{2,2} = c_{1,2}$, we recover the Erdös-Rényi model.

where each ξ_i is a (iid) randomly sampled positive real value, representing the intrinsic *connectivity* of node i, with $\mathbf{E}(\xi_i) = 1$ and finite second moment Ξ [Karrer and Newman, 2011].

Similarly to the SBM, DC-SBMs admit a detectability threshold [Mossel et al., 2015, Gulikers et al., 2016], which reads:

$$(a - c_{1,2})^2 \Xi > 2(a + c_{1,2}). \tag{1.34}$$

Given a graph with a community structure, how can we detect these two communities? It turns out that, much like the trivial case of a disconnected graph, they are reflected in the spectrum of the Laplacian [von Luxburg, 2007, Dall'Amico et al., 2021].

Spectral Clustering. Spectral clustering is a popular community detection algorithm on graphs, and dates back to [Fiedler, 1973]. In its simplest variant, the procedure goes as follows.

- First, compute the eigenvector u_2 associated to the second smallest eigenvalue λ_2 of L.
- Then, partition $\mathcal{V}$ into $\widetilde{C_1} \cup \widetilde{C_2}$ according to the sign of the entries of u_2:

$$\begin{cases} i \in \widetilde{C_1} \text{ if } u_2(i) > 0, \\ i \in \widetilde{C_2} \text{ otherwise.} \end{cases} \tag{1.35}$$

A heuristic explanation as to why spectral clustering works is the following [von Luxburg, 2007]. Suppose that the graph is connected, and that $|C_1| = |C_2|$. Consider further an ideal classification function f_* such that $f_*(i) = 1$ if $i \in C_1$, and $f_*(i) = -1$ if $i \in C_2$. Equivalently, the condition $|C_1| = |C_2|$ can be re-written as:

$$\langle f_*, \mathbf{1} \rangle = 0, \tag{1.36}$$

where $\mathbf{1}$ denodes the all-one vector in $\mathbf{R}^{\mathcal{V}}$.
One condition that f_* should intuitively satisfy is to define a *minimum cut* [Schrijver et al., 2003], *i.e.*:

$$f_* \in \underset{f \in \{-1,1\}^{\mathcal{V}}}{\operatorname{argmin}} \frac{1}{2} \sum_{e \in \vec{\mathcal{E}}} |f(t_e) - f(s_e)|^2 \tag{1.37}$$

or, equivalently

$$f_* \in \underset{f \in \{-1,1\}^{\mathcal{V}}}{\operatorname{argmin}} \langle f, \mathsf{L} f \rangle. \tag{1.38}$$

Plugging-in the orthogonality condition, we obtain:

$$f_* \in \underset{\substack{f \in \{-1,1\}^{\mathcal{V}} \\ \langle f, \mathbf{1} \rangle = 0}}{\operatorname{argmin}} \langle f, \mathsf{L} f \rangle. \tag{1.39}$$

While Problem (1.38) can be solved efficiently (using the randomized Stoer-Wagner algorithm [Stoer and Wagner, 1997]), this is no longer the case for Problem (1.39). Instead, a spectral relaxation reads

$$f_* \in \underset{\substack{f \in \mathbf{R}^{\mathcal{V}} \\ \langle f, \mathbf{1} \rangle = 0 \\ \|f\|_2^2 = n}}{\operatorname{argmin}} \langle f, \mathsf{L} f \rangle, \tag{1.40}$$

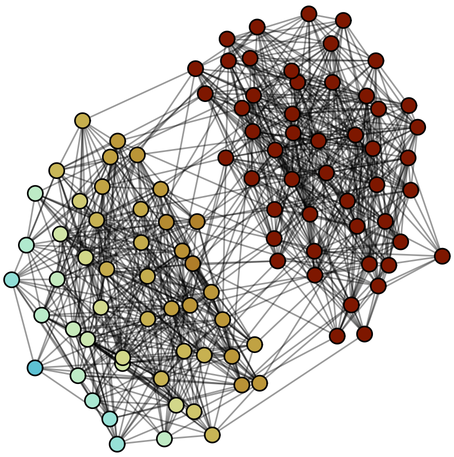


Figure 1.9: Second eigenvector u_2 on a SBM with 100 nodes.

the (in general unique) solution of which is $f_* = \sqrt{|\mathcal{V}|}u_2$, where u_2 again denotes the second eigenvector of L (this result follows directly from the Rayleigh-quotient minimization from Equation (1.14).

Practically, spectral clustering is observed to work well on graphs with two clearly-identifiable communities [von Luxburg, 2007], and some variants of the algorithm even *provably* recover classes non-trivially correlated with the ground-truth partition [Coja-Oghlan, 2010, Qin and Rohe, 2013, Krzakala et al., 2013, Dall'Amico et al., 2021]. From a GSP perspective, this has an interesting consequence: 2-bandlimited signals are meaningfully smooth, in that their intra-class fluctuations remain small. We plot in Figure 1.9 the second eigenvector u_2 of a SBM-graph, with two communities of size 50 ($n = 100$), $c_{1,1} = c_{2,2} = 19$ and $c_{1,2} = 1$. The community structured can be clearly identified, and u_2 is smooth on each community.

In practice, the size of the classes may not be equal, and the quality of the results may be degraded in the presence of high-degree nodes, so that other variants of spectral clustering are usually preferred. Common modifications include (see [von Luxburg, 2007] and references therein):

- Using the eigenvector of another operator instead of the combinatorial Laplacian L.
- Replacing the sign-based class-assignment by an embedding of the nodes into $\mathbf{R}^2$ (based on the values of u_1 and u_2), and applying a spatial clustering algorithm such as k-means.

In particular, two classical choices of operator are the *random-walk Laplacian*

$$\Delta = \mathsf{D}^{-1}\mathsf{L}, \tag{1.41}$$

and the similar *normalized Laplacian*

$$\widetilde{\mathsf{L}} = \mathsf{D}^{-\frac{1}{2}}\mathsf{L}\mathsf{D}^{-\frac{1}{2}}. \tag{1.42}$$

Remark 1.1.4. *Interestingly, other results show that better performance can be obtained by further replacing* D *with* $\mathsf{D} + q\mathsf{I}$ *[Qin and Rohe, 2013, Dall'Amico et al., 2020], resulting in a regularized normalized Laplacian* $(\mathsf{D}+q\mathsf{I})^{-\frac{1}{2}}\mathsf{L}(\mathsf{D}+q\mathsf{I})^{-\frac{1}{2}}$. *We will also encounter in **Part** ii a regularized random-walk Laplacian* $\Delta_q = (\mathsf{D}+q\mathsf{I})^{-1}\mathsf{L}$, *which is closely related to the regularized Laplacian* $(\mathsf{L}+q\mathsf{I})^{-1}q\mathsf{I}$ *from Section 1.1.1. Specifically, we will explain how* Δ_q *serves to obtain a random-walk interpretation of the operator* $(\mathsf{L}+q\mathsf{I})^{-1}q\mathsf{I}$.

This concludes our introduction of the graph Laplacian. In the following, we go over existing work on RandNLA estimators for GSP, in the spirit of Kirchhoff's seminal network theorem.

1.2 FOREST ESTIMATORS

Random processes evolving on a graph capture many of its properties; some examples we already mentioned:

- Kirchhoff's network theorem expresses the current in an electrical network resulting from a voltage source as the expectation of elementary currents built from uniformly sampled spanning trees;
- random walks allow to express the result of many variational problems in graph signal processing (*e.g.*, harmonic interpolation, or semi-supervised classification [Avrachenkov et al., 2017]).

These two examples however differ in a crucial manner: the random-walk-based expressions rely on random processes that are *dependent on the starting point of the random walk*, which enables estimation on that specific node; on the other hand Kirchhoff's theorem gives a *global* expression of the solution over the entire network. The goal of this section is to review existing work on *global* estimators for GSP quantities, in the spirit of Kirchhoff's theorem, based on (random) *rooted spanning forests*.

Formally, a spanning tree is a subset of edges $\mathsf{t} \subseteq \mathcal{E}$ that is *connected*, cycle-free, and spans all the nodes in $\mathcal{G}$ [Bollobás, 1998][24]. Equivalently, it is a cycle-free subset of edges of size $|\mathcal{V}| - 1$. See Figures 1.4 and 1.10a for illustrations of spanning trees.
A spanning forest is a subset of edges that is cycle-free and spans all the nodes in $\mathcal{G}$, but needs not be connected. In particular, it describes a family of disjoint connected components that must be trees. A *rooted spanning forest* (RSF) is a subset of edges and nodes $\phi \subseteq \mathcal{V} \cup \mathcal{E}$ satisfying the following [Avena and Gaudillière, 2013].

1. The edges in $\phi \cap \mathcal{E}$ form a spanning forest, that is, a disjoint set of tree.
2. Each node r in $\phi \cap \mathcal{V}$ is connected to exactly one tree $\mathsf{t} \subseteq \phi \cap \mathcal{E}$. In this case, r is called the *root* of the tree t.

Note that a rooted spanning forest always has cardinality $|\mathcal{V}|$. See Figure 1.10b for an illustration of a RSF on the 4×4 grid graph. *In the following, we will only consider rooted spanning forests.*

Consider the following distribution $\mathcal{D}_\mathcal{F}$ over rooted spanning forests, discussed in [Avena and Gaudillière, 2013]:

$$\mathbf{P}_{\mathcal{D}_\mathcal{F}}(\phi) \propto q^{|\phi \cap \mathcal{V}|} \prod_{e \in \phi \cap \mathcal{E}} w_e, \tag{1.43}$$

24 Note that spanning trees only exist for connected graphs.

where $q \in \mathbf{R}_+^*$ is an *a priori* parameter. Note that large values of q tend to favor sampling rooted spanning forests with a large number of components, whereas low values tend to reduce the number of components to only one tree.

One important property of the distribution $\mathcal{D}_\mathcal{F}$ that makes it suitable for applications is that there exists an associated efficient sampling algorithm. This algorithm is a variant of Wilson's algorithm [Wilson, 1996], that samples from the following distribution $\mathcal{D}_{\mathcal{T}_r}$ over spanning trees *rooted in some node* r:

$$\mathbf{P}_{\mathcal{D}_{\mathcal{T}_r}}(\mathsf{t}) \propto \prod_{e \in \mathsf{t}} w_e. \tag{1.44}$$

Note that, when the graph is non-weighted (*i.e.*, $w_e = 1$ for all $e \in \mathcal{E}$), $\mathcal{D}_{\mathcal{T}_r}$ is the uniform distribution over the spanning trees of $\mathcal{G}$. Further, note that the choice of root bears little relevance[25] to the distribution $\mathcal{D}_{\mathcal{T}_r}$: for any two roots $r, r' \in \mathcal{V}$, we have $\mathbf{P}_{\mathcal{D}_{\mathcal{T}_r}}(\mathsf{t}) = \mathbf{P}_{\mathcal{D}_{\mathcal{T}_{r'}}}(\mathsf{t})$.

Wilson's Algorithm. Wilson's algorithm exactly samples from $\mathcal{D}_{\mathcal{T}_r}$ by running multiple random walks on $\mathcal{G}$, and iteratively constructing a tree t from the resulting random paths based on a loop-erasure procedure. Specifically, suppose that the nodes of $\mathcal{G}$ are arbitrarily ordered in a queue, and say that a node i is spanned by t if i is an endpoint of some edge in t. Wilson's algorithm goes as follows.

1. Remove the first node i in the queue that is not spanned by t, and start a random walk from i while keeping track of the resulting path p in $\mathcal{G}$.
2. If p self-intersects, erase the resulting loop.
3. If p reaches the root r or a node already spanned by t, add p to the tree t.

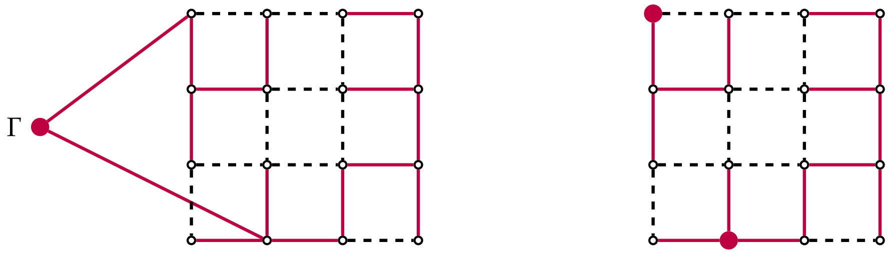
Repeat this procedure for the next node in the queue, until the queue is exhausted. See Algorithm 1 for a more detailed statement, where `random_successor(u)` denotes the function which, at each call, randomly outputs a neighbor of u with probability $\frac{w_{u,v}}{d_u}$. While other theoretically more efficient algorithms sampling from $\mathcal{D}_{\mathcal{T}_r}$ exist, *e.g.* [Schild, 2018, Anari et al., 2022][26], Wilson's algorithm has two practical advantages:

- it is easy to implement and yields practically good performance;
- it can be modified to sample from similar distributions.

In particular, $\mathcal{D}_\mathcal{F}$ can be sampled from by running Wilson's algorithm on an extended version of the graph $\mathcal{G}$ [Wilson, 1996, Avena et al., 2018]. Precisely, consider the extended graph $\mathcal{G}^\Gamma$, with nodes $\mathcal{V} \cup \{\Gamma\}$ and edges $\mathcal{E} \cup \bigcup_{v \in \mathcal{V}} \{\{v, \Gamma\}\}$, so that all the nodes of $\mathcal{V}$ are connected to Γ, with weight $w_{(v,\Gamma)} = q$. Here, Γ acts as an additional *boundary* node, similar to the boundaries encountered in Section 1.1.2. Sampling a tree rooted in Γ by running Wilson's algorithm on $\mathcal{G}^\Gamma$, and replacing the edges of the form (v, Γ) by the roots $\{v\}$, yields a rooted spanning forest distributed according to $\mathcal{D}_\mathcal{F}$. See Figure 1.10 for an illustration of this correspondence.

In this case, the expected time-complexity to sample a rooted spanning forest is in $\mathcal{O}\left(|\mathcal{V}| + \frac{|\mathcal{E}|}{q}\right)$ for non-weighted graphs, and $\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q}\right)$ for weighted graphs, where $\mathrm{Vol}(\mathcal{G}) = 2\sum_{e \in \mathcal{E}} w_e$ denotes the "volume" of the graph, *i.e.*, the sum of its degrees [Tremblay, 2024].

25 Let us temper this observation a bit. One can study similar distributions for directed graphs, in which case the choice of root *is* relevant. Further, we will see that the choice of a root is very convenient for sampling algorithms.
26 Note that algorithms such as that of [Anari et al., 2022] do not perform *exact* sampling, but rely on a MarkovChain Monte-Carlo strategy.



(a) A spanning tree on $\mathcal{G}^\Gamma$.

(b) A RSF on $\mathcal{G}$ with two trees.

Figure 1.10: Rooted spanning forest and spanning tree on the (extended) 4×4 grid graph $\mathcal{G}$. Edges belonging to the tree (resp. RSF) are represented in purple, and roots as large purple nodes. We only represent the edges of the form (v, Γ) in $\mathcal{G}^\Gamma$ belonging to the tree.

Algorithm 1 Wilson's uniform-spanning-tree sampling algorithm, with root r [Wilson, 1996].

```
t = {r}
while t not spanning do
    Let i ∈ V be the first node in the queue not spanned by t
    u = i                                   ▷ u the current node of the random walk
    p = ϵ                                   ▷ With ϵ the empty path
    while u ≠ r and p does not intersect t do
        u′ = random_successor(u)
        Add the edge (u, u′) to the path p
        if p contains a cycle then
            Remove this cycle from p
        end if
        u = u′
    end while
    t = t ∪ p
end while
Output t
```

The study of random forests distributed according to Equation (1.43), and of related combinatorial properties, goes back to at least [Chebotarev and Shamis, 2006], in the special case of $q = 1$. Random forests were more recently re-introduced in [Avena and Gaudillière, 2013, Avena and Gaudillière, 2018] to study well-distributed points in networks, with applications to the study of metastable dynamics [Avena et al., 2022], and multi-resolution analysis on graphs [Avena et al., 2020, Avena et al., 2021] (see also [Avena et al., 2018] for a survey). Closer to our work are the applications of random forests to GSP and randomized algorithms, pertaining to node-sampling [Tremblay et al., 2017], trace-estimation [Barthelmé et al., 2019], and graph Tikhonov smoothing and harmonic interpolation [Pilavcı et al., 2021], recalled below.

Graph Tikhonov Regularization. Recall the graph Tikhonov smoothing problem from Equation (1.20):

$$\operatorname*{argmin}_{f \in \mathbf{R}^{\mathcal{V}}} \; q\|f - g\|^2 + \langle f, \mathsf{L} f \rangle,$$

Figure 1.11: Illustration of the estimator $\widetilde{f}$ on the 4×4 grid graph $\mathcal{G}$. Left: a signal g on $\mathcal{G}$. Right: the estimation on all the nodes of $\mathcal{G}$ for the RSF ϕ represented in Figure 1.10b.

the solution of which reads $f_* = (\mathsf{L} + q\mathsf{I})^{-1} q\, g$, and which aims at inferring a smooth signal $f \in \mathbf{R}^{\mathcal{V}}$ from a noisy measured signal $g \in \mathbf{R}^{\mathcal{V}}$. A network-theorem-inspired estimator of the solution f_*, based on rooted spanning forests, was proposed in [Pilavcı et al., 2021].

Theorem 1.2.1 (from [Pilavcı et al., 2021])**.** *For a given RSF ϕ, denote by $r_\phi(v)$ the root of the tree containing node v. Then, we have:*

$$f_*(i) = \mathbf{E}_{\mathcal{D}_{\mathcal{F}}}(g(r_\phi(i))). \tag{1.45}$$

That is, setting $\widetilde{f}(i, \phi, g) = g(r_\phi(i))$ (*i.e.* the value at the root is propagated to all the nodes in the tree) yields an unbiased estimator of f_*, and is computed as follows.

- Sample a rooted spanning forest ϕ using Algorithm 1 on the graph $\mathcal{G}^\Gamma$.
- Set $\widetilde{f}(i) = g(r_\phi(i))$ for all nodes $i \in \mathcal{V}$.

See Figure 1.11 for an illustration of the estimator on a sampled rooted spanning forest.

For $q = 1$, Theorem 1.2.1 also appears in another guise in [Chebotarev and Shamis, 2006]. The connection between the result of [Chebotarev and Shamis, 2006] and graph Tikhonov regularization was first observed in [Avrachenkov et al., 2017], and interpreted in terms of information dissemination for semi-supervised node classification.
From a computational perspective, it was observed on synthetic problems that forest-based estimators such as $\widetilde{f}$ obtain similar, but not better, performance in terms of runtime as compared to iterative solvers [Saad, 2003], but may have interesting advantages in specific pipelines [Pilavcı et al., 2021]. Further improvements on these techniques were also described in [Pilavcı et al., 2022b], but not thoroughly benchmarked.
These approaches were later adapted to personalized PageRank computations, a problem analogous to semi-supervised classification, for which significant speed-ups on some practical problems were reported [Liao et al., 2022, Yang et al., 2024].

Trace Estimation. An associated problem to the graph Tikhonov smoothing problem consists in determining the optimal q that should be used to recover the ground-truth signal $f_\top$ from $f_\varepsilon = f_\top + \varepsilon$. Assuming homoscedastic Gaussian noise ε, a number of model selection criteria such as, *e.g.*, Akaike's Information Criterion require computing the *effective degrees of freedom* of the model [Akaike, 1969, Akaike, 1987] (see also [Hastie et al., 2009] for other examples), in this case

$$s(q) = \operatorname{tr}\left((\mathsf{L} + q\mathsf{I})^{-1} q\mathsf{I}\right), \tag{1.46}$$

across a range of values of q. It turns out that a simple RSF-based estimator can be used to estimate $s(q)$, and reaches good performance on this problem [Barthelmé et al., 2019].

Theorem 1.2.2 (from [Barthelmé et al., 2019])**.**

$$s(q) = \mathbf{E}_{\mathcal{D}_{\mathcal{F}}}(|\phi \cap \mathcal{V}|). \tag{1.47}$$

In words, the number of roots in a random forest ϕ distributed according to $\mathcal{D}_{\mathcal{F}}$ is an unbiased estimator of $s(q)$. Further improvements on this estimator, leveraging variance-reduction techniques, allow to reach state-of-the-art performance [Pilavcı et al., 2022a].

This concludes our presentation of RSF-based estimators for GSP problems.

1.3 SUMMARY

We covered in this chapter a number methodological and mathematical notions pertaining to graphs that shall serve as a basis for our developments in **Part** ii. Let us give a brief overview.

- First, many graph problems are based on the combinatorial Laplacian L, which serves to capture a number of graph-properties. One important problem we will generalize in **Part** ii is the graph Tikhonov smoothing problem, which aims at smoothing a noisy signal $f_\varepsilon = f_\top + \varepsilon$ by computing:

$$\underset{f \in \mathbf{R}^{\mathcal{V}}}{\operatorname{argmin}} \; q\|f - f_\varepsilon\|_2^2 + \langle f, \mathsf{L} f\rangle. \tag{1.48}$$

  One should keep in mind that the resulting signal will only be meaningfully smooth in specific settings, such as when $f_\top$ is bandlimited and the graph presents an appropriate structure (*e.g.* built from spatially sampled points, or with a community structure).

- A second important idea we reviewed is that the solution to graph problems can be expressed using random processes. It is helpful to distinguish two types of approaches.

  1. Techniques based on random walks (*e.g.*, Proposition 1.1.2), the most classical in the literature, which allow for node-wise estimation obtained from random-walks started at that node. Since the estimation remains localized on a a single node at a time, only a small part of the graph has to be processed for each node, which is useful when the quantity of interest need only be computed for a few nodes.
  2. Techniques based on random decompositions of the graph (*e.g.*, Kirchhoff's network theorem, or the RSF-based estimator for the graph Tikhonov smoothing problem), which allow to estimate the solution of a specific problem *on the whole graph* at once. While the sampling process may be more complicated and costly than for localized methods, *e.g.*, sampling a rooted spanning forest distributed according to

$$\mathbf{P}_{\mathcal{M}_{\mathcal{F}}}(\phi) \propto q^{|\phi \cap \mathcal{V}|} \prod_{e \in \phi \cap \mathcal{E}} w_e \tag{1.49}$$

     using Wilson's algorithm as compared to running a simple random-walk, the global estimation allows for more aggressive variance-reduction techniques to apply, resulting in high-quality estimations from fewer samples [Pilavcı et al., 2022b, Pilavcı et al., 2022a].

Let us finally mention a well-known companion result to Kirchhoff's network theorem: the "matrix-tree" theorem [Kirchhoff, 1847, Harary, 1966, Kirby et al., 2016, Lyons and Peres, 2017], a remarkable identity that relates the number of spanning trees in a (non-weighted) graph to the Laplacian matrix L[27].

Theorem 1.3.1. *Matrix-Tree theorem*

$$\#\texttt{spanning trees} = |m_{i,j}|, \text{ where } m_{i,j} \text{ denotes any first minor of } \mathsf{L}. \tag{1.50}$$

In words, the number of spanning trees of a graph is given, up to the sign, by any first minor of its Laplacian matrix[28]. It turns out that this type of relation between determinants of (submatrices of) the Laplacian decompositions of the graph $\mathcal{G}$ underlies Theorems 1.2.1 and 1.2.2, and will use them to derive generalized RSF-based estimators for a class of problems we introduce in Chapter 2.

The remainder of **Part** i is organized as follows.

- We discuss in Chapter 2 the notion of a connection on graphs, which associates linear maps to the edges of the graph. The specific instance of maps acting by 2-dimensional rotations will be our main focus in **Part** ii.
- We cover in Chapter 3 the theory of determinantal point processes. In particular, we provide a proof of a probabilistic version of Theorem 1.3.1, and introduce the necessary tool to extend RSF-based estimators to graphs with a connection.

27 A more general formula involving the weights is obtained instead for weighted graphs.
28 Recall that the first minor of a matrix A is the determinant of this matrix obtained when removing one of its rows and one of its columns.

2

GEOMETRIES ON GRAPHS

There appears a fundamental principle which can serve to characterize all possible geometries...

— Felix Klein[1]

Graphs serve as a simple model for topological and geometrical situations, and were introduced by Euler[2], for this very reason, asking whether it was possible to walk along a loop in the city of Königsberg[3] while only crossing each of its seven bridges once [Euler, 1741]. The answer to this question is no, but many other notions from geometry have found their way into graph theory[4]. The applications we consider in **Part** ii pertain to one such notion, called a *connection* [Cartan, 1926].

Let us begin by discussing the specific case of *signed graphs*, for which a number of general notions and phenomena can be introduced in a simple setting.

Signed Graphs. More specifically, an elementary instance of a connection corresponds to signed graphs [Zaslavsky, 1982], whose edges are additionally endowed with a $+$ or $-$ sign; and we denote by $\sigma_e \in \{-1, 1\}$ the sign associated to edge e. Processing pipelines have to account for this additional labelling, and GSP tools like the low-pass filters we mentioned in Chapter 1 can

1 See [Klein, 2004].
2 Leonhard Euler, Swiss mathematician, 1707−1783.
3 Now Kaliningrad, Russia. Königsberg also happens to be the birthplace and city of residence of young Gustav Kirchhoff.
4 The study of graphs actually predates that of surfaces in geometry, first studied by Gauss, and of more general manifolds as introduced by German mathematician Bernhard Riemann (1826−1866).

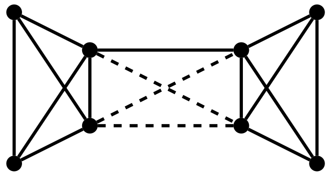


Figure 2.1: A signed graph with 8 vertice with an implicit community structures (left and right). $+$-edges are full, and $-$-edges are dashed. Intra-community edges are $+$-edges, and inter-community edges are $-$-edges, except for one $+$-edge between the two communities (as may occur in real-world datasets).

be readily adapted to process signals on a signed graph $\mathcal{G}$, by considering the *signed Laplacian* $\mathsf{L}_\pm$ [Dittrich and Matz, 2020]:

$$\mathsf{L}_\pm = \mathsf{D} - \mathsf{A}_\pm, \tag{2.1}$$

where $A_\pm$ is a signed adjacency matrix, such that $(\mathsf{A}_\pm)_{i,j} = w_{i,j}\sigma_{i,j}$ if $\{i,j\} \in \mathcal{E}$, and $(\mathsf{A}_\pm)_{i,j} = 0$ otherwise. In this case, the smoothness of a signal $f \in \mathbf{R}^{\mathcal{V}}$ is quantified by

$$\langle f, \mathsf{L}_\pm f \rangle = \frac{1}{2} \sum_{e \in \overrightarrow{\mathcal{E}}} w_e |f(t_e) - \sigma_e f(s_e)|^2 \tag{2.2}$$

which, when the graph further presents a community structure in its topology that is coherent with the connection (*e.g.*, a SBM with positive intra-community edges and negative inter-communities edges), assigns lower values to signals that are in agreement with this community structure. For real-world networks, the sign of the edges may not be perfectly coherent with the underlying community, as we illustrate in Figure 2.1; in this case, one can show that the spectrum of the signed Laplacian $\mathsf{L}_\pm$ exhibits a key difference from that of L: its smallest eigenvalue is strictly positive [Dittrich and Matz, 2020].
Applications of signed graphs do not suffer from this difference, and may typically concern social networks, with the sign of the edges representing the friendly/hostile nature of the relationship between two different persons (stemming from, *e.g.*, political opinion) [Neumann and Peng, 2022]. In this context, a natural question is to infer the community structure from the connection [Lelarge et al., 2013], which can be formalized in a number of manners.

1. One approach consists in framing community-detection as a *synchronization problem* [Cucuringu, 2015][5]. If there are two communities to be inferred, $\mathbf{Z}_2$-synchronization[6] associates a label $f_*(i) \in \mathbf{Z}_2 = \{-1, 1\}$ to each node $i \in \mathcal{V}$ by finding

$$f_* \in \operatorname*{argmin}_{f \in \mathbf{Z}_2^{\mathcal{V}}} \langle f, \mathsf{L}_\pm f \rangle. \tag{2.3}$$

   Solving the $\mathbf{Z}_2$-synchronization problem in Equation (2.3) amounts to finding a *maximum cut* in $\mathcal{G}$, which is NP-complete in general [Karp, 2010][7]. Instead, one relies on spectral

5 The term “synchronization” was dubbed in analogy to the time-synchronization problem in (sensor) networks, as one tries to best satisfy a number of constraints over a graph, *i.e.*, to synchronize the data over a number of nodes (see, *e.g.*, [Karp et al., 2003, Sivrikaya and Yener, 2004]).
6 Here, $\mathbf{Z}_2$ denotes the multiplicative group with two elements.
7 The maximum-cut problem alluded to here considers edge-weights that can take negative values, and in particular of the form $\sigma_e w_e$.

relaxations similar to the spectral clustering algorithm from Chapter 1, and looks for a maximally-smooth signal

$$f_* = \underset{\|f\|_2^2=n}{\operatorname{argmin}} \langle f, \mathsf{L}_\pm f\rangle. \tag{2.4}$$

Note that, like spectral clustering algorithms, this approach can be modified by considering variants of $\mathsf{L}_\pm$, such as $\widetilde{\mathsf{L}_\pm} = \mathsf{I} - \mathsf{D}^{\frac{1}{2}}\mathsf{A}_\pm\mathsf{D}^{\frac{1}{2}}$. In this setting, introducing regularization once again helps to achieve better performance (*e.g.*, [Cucuringu et al., 2021])[8].

2. Another line of work explores the use of fixed-length random walks [Neumann and Peng, 2022]. Informally, a sparse embedding $f(i)$ of node $i \in \mathcal{V}$ into $\mathbf{R}^{\mathcal{V}}$ is constructed based on statistics extracted from the multiplication of the signs of the edges encountered by random walks $v^k = (v_0^k, ..., v_l^k)$ started at i:

$$\prod_{i=0}^{l-1} \sigma_{(i,i+1)}. \tag{2.5}$$

The resulting embeddings of the nodes are then used to recover the communities. Interestingly, the computed statistics can be related to the spectrum of $\widetilde{\mathsf{L}_\pm}$, and leveraged to obtain *formal guarantees* on the clustering; similar sign-aware random walks have also been used to perform personalized ranking in signed graphs [Jung et al., 2016, Jung et al., 2020], without establishing theoretical guarantees.

To our knowledge, unlike for combinatorial Laplacians, the relations between signed Laplacians and random-walks on graphs have received relatively little interest[9], and it is noteworthy that formal links can be established between the following [Neumann and Peng, 2022]:

- the topology of $\mathcal{G}$ (*e.g.*, the community structure);
- statistics of connection-aware random walks on $\mathcal{G}$;
- the spectrum of $\mathsf{L}_\pm$ and variational properties of signals on $\mathcal{G}$.

Investigating this type of properties for random processes on graphs endowed with more general edge-supported information, and Laplacian-like operators associated to these graphs, is one of our main concerns in **Part** ii of this manuscript.

Let us now describe this generalization.

It is convenient to to adopt a slightly different perspective on graph signals, *i.e.*, data defined *over* a graph [Kenyon, 2011, Hansen and Ghrist, 2019]. Given a vector space[10] $\mathbf{V}$, a *vector bundle* over a graph $\mathcal{G}$ (also, $\mathbf{V}$-bundle) is the assignment of an isomorphic copy $\mathbf{V}_i$ of $\mathbf{V}$ to each vertex[11]

8 One may also notice that $\langle f, \mathsf{L}_\pm f\rangle$ is a variant of the Hamiltonian classically considered in the Ising model from statistical mechanics, which defines a Gibbs probability measure over configurations in $\mathbf{Z}_2^{\mathcal{V}}$. The mode of this distribution is the solution f_* to the $\mathbf{Z}_2$-synchronization problem. Another classical problem consists in minimizing the *free energy* associated to the Gibbs measure, which is the solution of an energy-entropy competition, and describes the distribution of the configurations that are the most likely to emerge from the model [Mezard and Montanari, 2009].

9 The spectral theory of the signed Laplacian, on the other hand, is the subject of active interest. See, *e.g.*, [Belardo et al., 2019] for a review.

10 More generally, $\mathbf{V}$ can be replaced by a module over some ring or algebra, such as a matrix group.

11 Formally, one requests that each $\mathbf{V}_i$ be isomorphic to $\mathbf{V}$ as a vector space.

$i \in \mathcal{V}$. A $\mathbf{V}$-signal f over $\mathcal{G}$ is the assignment of an element of $\mathbf{V}_i$ to each node i, so that $f \in \mathbf{V}^{\mathcal{V}} = \bigoplus_{i \in \mathcal{V}} \mathbf{V}_i$. In particular, the signals we have considered so far are $\mathbf{R}$-signals in this sense, defined over $\mathbf{R}$-bundles.
From this perspective, it is natural to expect that signal-processing pipelines for $\mathbf{V}$-signals should account for the different copies $\mathbf{V}_i$ each data-point $f(i)$ belongs to: edges of $\mathcal{G}$ should describe *how* two data-points defined on adjacent nodes are to be compared, and thus processed. In geometrical parlance, this information is known as a *connection* [Cartan, 1926]. Specifically, a connection Ψ is a collection of linear maps $\Psi = (\psi_e)_{e \in \overrightarrow{\mathcal{E}}}$, which associates to each directed edge $e = (s_e, t_e) \in \overrightarrow{\mathcal{E}}$ an *invertible* map $\psi_e : \mathbf{V}_{s_e} \to \mathbf{V}_{t_e}$, called the *connection map* from s_e to t_e, such that $\psi^{-1}_{(s_e,t_e)} = \psi_{(t_e,s_e)}$.

Remark 2.0.1. *In the case of signed graphs, the connection maps $\psi_e : \mathbf{R} \to \mathbf{R}$ for each edge e are defined by multiplication with σ_e:*

$$\psi_e(x) = \sigma_e \cdot x. \tag{2.6}$$

The GSP pipelines described in Chapter 1, on the other hand, did not account for this kind of additional structure, and all the connection maps were all trivial: $\psi_e = \mathrm{id}_{\mathbf{R}_{s_e}, \mathbf{R}_{t_e}}$.

The problems we considered on signed graphs can be cast in this more general framework, and are of practical relevance in diverse settings. In particular, the last decade has witnessed a growing interest in *synchronization problems*, in which the connection maps arise from group actions (*e.g.*, that of $\mathbf{Z}_2$ on $\mathbf{R}$ in Remark 2.0.1).

Synchronization is a completely non-supervised inference problem, generalizing the community-detection problems on regular and signed graphs, in which there is no *a priori* signal to be processed[12]. More specifically, for a group $\mathfrak{g}$, the *group-synchronization* problem asks to recover a set of n group elements $g_i \in \mathfrak{g}$ from a given subset of noisy measurements [Lerman and Shi, 2022]:

$$g_{i,j} \simeq g_i g_j^{-1}. \tag{2.7}$$

From this perspective, in $\mathbf{Z}_2$-synchronization for instance, we have $\mathfrak{g} = \mathbf{Z}_2$ (where the group multiplication is defined as the composition of the connections maps, following Remark 2.0.1), and the strategy proposed above to recover the g_i's (in Equation (2.4)) is to minimize $\langle f, \mathsf{L}_\pm f \rangle$ over all $f \in \mathbf{Z}_2^{\mathcal{V}}$ (but note that this is only one possible strategy). More general synchronization problems can be attacked from this GSP perspective as well: if $\mathfrak{g} \subseteq \mathbf{V}$ a Hilbert space[13], synchronization

12 The problem of synchronization we consider is different from the cross-disciplinary notion of synchronization more commonly studied in network-related sciences for, *e.g.*, phase oscillators [Dörfler and Bullo, 2014]: we are not interested in the process of coupled dynamical systems evolving towards a synchronized dynamic (dubbed the "dynamic-synchronization" problem in the following). From this classical dynamical-system perspective, *and in this classical setting of phase oscillators*, our problem is rather trivial: a solution of the synchronization problem we consider would correspond to any possible state of the coupled dynamical system once dynamical-synchronization has been reached; that is, any state constant over all nodes of the network would be a solution. For the specific problem of angular synchronization (which we introduce in the remainder of this section, and which is one of the main problems we consider in this manuscript), we instead ask the following: suppose that we modify slightly our definition of what a "synchronized state" is, and say that the coupled system is in a synchronized state when all phase-differences between adjacent nodes *agree with some* a priori *prescribed phase-offsets*; computing a solution to this problem is no longer trivial. Note that a dynamical-synchronization counterpart to this setting has also been considered in, *e.g.*, the statistical-physics literature [Zdeborová and Krzakala, 2016, Lupo and Ricci-Tersenghi, 2017].

13 Named after German mathematician David Hilbert, 1862−1943.

can be achieved by finding some signal $g \in \mathfrak{g}^{\mathcal{V}}$ that is *maximally smooth*[14], and we sketch in the remainder of this introduction how this smoothness can be quantified for more general connections using Laplacian-like operators (we focus on $U(\mathbf{C})$-connections, but the discussion generalizes to general $O(\mathbf{R}^d)$-connections as-is). Applications of synchronization range from solving jigsaw puzzles (with $\mathfrak{g} = \mathbf{Z}_4$) [Huroyan et al., 2020], to sensor localization [Cucuringu et al., 2012], cryo-electron microscopy [Bandeira et al., 2020], simultaneous localization and mapping (SLAM) [Rosen et al., 2019], or structure-from-motion problems (with − a subgroup of − $\mathfrak{g} = SE(\mathbf{R}^d)$, the special Euclidean group in $\mathbf{R}^d$) [Tron et al., 2016].

Our principal interest in the remainder of this manuscript lies in synchronization and smoothing on $\mathbf{C}$-bundles endowed with a $U(\mathbf{C})$-connection (also called *unitary connection*), in which case the connection maps represent 2D-rotations attached to the edges of the graph. More precisely, $\mathbf{C}$-bundles associate to each node i a copy $\mathbf{C}_i \simeq \mathbf{C}$ of the complex plane $\mathbf{C}$, and each connection map $\psi_e : \mathbf{C}_{s_e} \to \mathbf{C}_{t_e}$ in a $U(\mathbf{C})$-connection acts as multiplication by a unitary complex number [Kenyon, 2011] $e^{\iota\theta_e}$, so that

$$\psi_e(z) = e^{\iota\theta_e} \cdot z, \tag{2.8}$$

with ι the complex imaginary unit (we sometimes abuse notation and write $\psi_e = e^{\iota\theta_e}$ when there is no risk of confusion)[15]. In this case, the condition $\psi_e^{-1} = \psi_{e^*}$ reads:

$$\psi_{e^*} = \psi_e^*, \tag{2.9}$$

or, equivalently, $\psi_{(t_e,s_e)} = \psi^*_{(s_e,t_e)}$, and the transformation associated to an edge traversed in one direction or the other differs only by conjugation. This notion is illustrated in Figure 2.2, where we represent the application of the maps ψ_e to a vector $z_1 \in \mathbf{C}_{v_1}$ along a cycle, which translates the geometrical inspiration behind a connection. Unitary connections appear in a number of problems, such as *angular synchronization* [Singer, 2011, Yu, 2012], signal processing over directed graphs [Furutani et al., 2019, Zhang et al., 2021], discrete geometry processing [Knöppel et al., 2013, Sharp et al., 2019], or even direction of arrival estimation [Moreira et al., 2019, Raia et al., 2020], and will be our main object of interest in **Part** ii of this manuscript. To set the idea of what a typical pipeline looks like, the reader can keep in mind that, in most cases, connections are built as a pre-processing step, before defining a $\mathbf{C}$-signal on the graph; see the following remark for an explicit construction in classical computational-geometry pipelines.

Remark 2.0.2. *Vector-fields over a (discretized) surface $\mathcal{M}$ can be understood as 2-dimensional,* i.e. *$\mathbf{R}^2$-signals or, equivalently, complex-valued $\mathbf{C}$-signals, each vector in the field belonging to a tangent plane, that is, copy of $\mathbf{R}^2$ (resp. $\mathbf{C}$) associated to one node of the discretization. See Figure 2.3. In this context, processing pipelines have to take into account the curvature of the surface, and the "natural" connection to consider is (a discretization of) the Levi-Civita connection of the manifold[16], a practical and common construction of which (for triangulated surfaces) goes as follows [Knöppel et al., 2013].*

14 Natural choices of $\mathbf{V}$, with respect to synchronization problems, include algebras over a vector-space/module, and one can for instance consider connections acting on matrix-valued signals, with $\mathbf{V} = M_n(\mathbf{R})$, or even operator-valued signals. These models are still be amenable to GSP-like variational analyses based, for instance, on the theory of W^* algebras.

15 This association of a copy of $\mathbf{C}$ is analogous to a fiber bundle over a manifold (*e.g.* its tangent bundle), and is known as a *complex line bundle* over $\mathcal{G}$.

16 In Riemannian geometry, the Levi-Civita connection, named after Italian mathematician Tullio Levi-Civita (1873−1941), is a connection (in a sense adapted to differentiable manifolds, which is usually introduced as a generalization of directional derivatives, see *e.g.* [Sylvestre Gallot, 2004, Faure, 2022]) that is uniquely associated

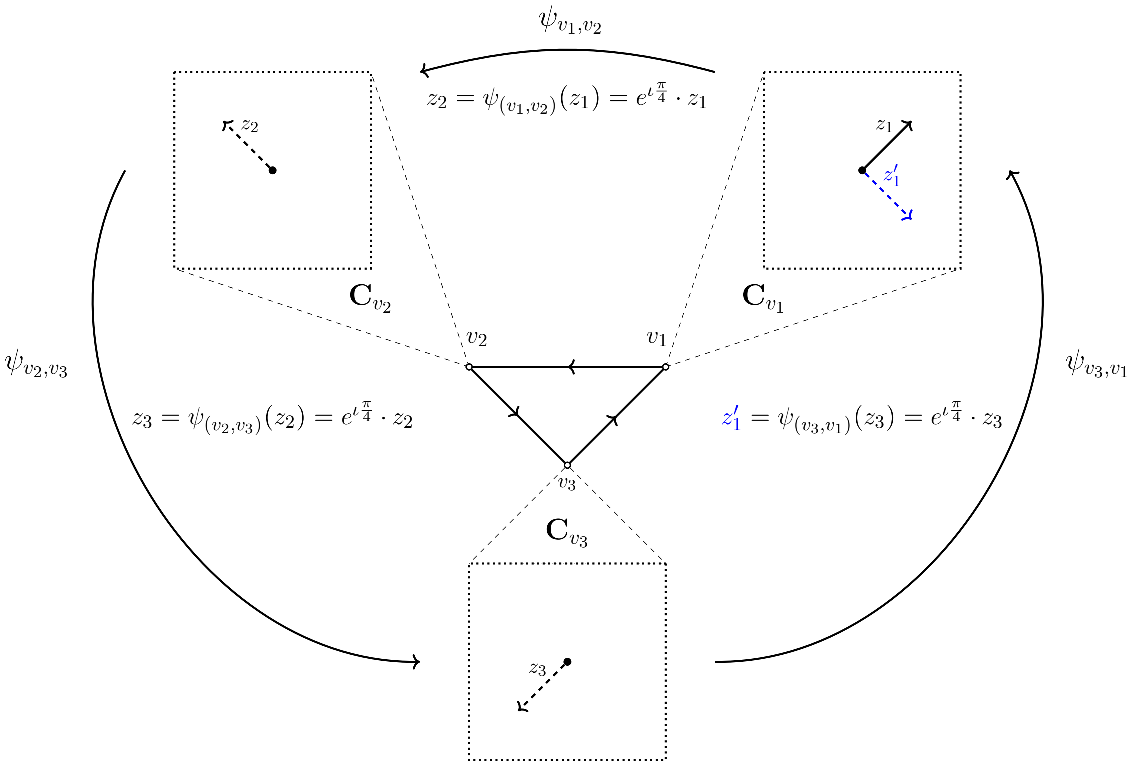


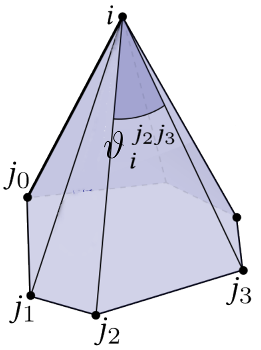
Figure 2.2: A $U(\mathbf{C})$-connection Ψ on the triangle graph K_3 (in the center). The connections maps ψ_e are rotations with associated angles $\theta_{(v_1,v_2)} = \theta_{(v_2,v_3)} = \theta_{(v_3,v_1)} = \frac{\pi}{4}$, and we represent their action on some $z_1 \in \mathbf{C}_{v_1}$. A cyclic path $C = ((v_1, v_2), (v_2, v_3), (v_3, v_1))$ is depicted using directed arrows, we denote by $\psi_C = \psi_{(v_3,v_1)} \circ \psi_{(v_2,v_3)} \circ \psi_{(v_1,v_2)}$ the composition of the connection maps along this cycle (a rotation of $\theta_C = \frac{3\pi}{4}$). Top right: $z_1 \in \mathbf{C}_{v_1}$ (bold arrow) and $z_1' = \psi_C(z_1) \in \mathbf{C}_{v_1}$ (blue dotted arrow). Top left, bottom: $z_2 = \psi_{(v_1,v_2)}(z_2) \in \mathbf{C}_{v_2}$ and $z_3 = \psi_{(v_2,v_3)}(z_2) \in \mathbf{C}_{v_3}$ (dotted arrows).

1. *Choose, for each node i on the surface, an adjacent node j_0. Starting from $\{i, j_0\}$, order the other nodes adjacent to i in counter-clockwise order along a neighborhood of i, and denote by $\vartheta_i^{j_k,j_{k+1}}$ the interior angle at node i between edges $\{i, j_k\}$ and $\{i, j_{k+1}\}$.*

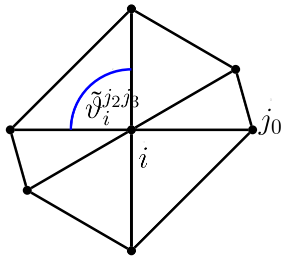
2. *Define the normalized angles*

$$\widetilde{\vartheta}_i^{j_k,j_{k+1}} = 2\pi \frac{\vartheta_i^{j_k,j_{k+1}}}{\sum_k \vartheta_i^{j_k,j_{k+1}}}. \tag{2.10}$$

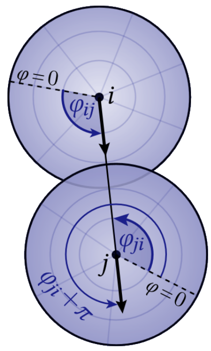
to a manifold $\mathcal{M}$, depending on its metric (*i.e.*, the choice of an inner-product on each tangent space). Here, the connection considered is that of the surface $\mathcal{M}$ understood as a sub-manifold of $\mathbf{R}^3$, and endowed with the metric induced by the usual l_2 inner-product.



(a) 3D neighborhood of node i.



(b) Flattened neighborhood of i.



(c) Connection map from i to j.

Figure 2.3: Construction of the connection for a triangulated surface. From left to right: 3-dimensional representation of the neighborhood of i, with a choice of reference edge $\{i, j_0\}$; 2-dimensional representation of the flattened neighborhood (normalized angles are represented approximately); explicit description of the connection map, so that the transported vectors remain parallel in the flattened neighborhoods. Figures on the left and right are from [Sharp et al., 2019]. The angle $\varphi = 0$ in the figure on the right represents the direction of the reference edge $\{i, j_0\}$.

Note that the sum of normalized angles satisfies $\sum_k \widetilde{\vartheta}_i^{j_k, j_{k+1}} = 2\pi$ *at each node* i*, and that one can use this construction to assign a "direction"* $\varphi_{i,j}$ *to each outgoing edge* $\{i, j\}$*:*

$$\varphi_{i,j} = \sum_{l=0}^{k-1} \widetilde{\vartheta}_i^{j_k, j_{k+1}}. \tag{2.11}$$

Practically, the angles $\widetilde{\vartheta}_i^{j_k, j_{k+1}}$ *correspond to angle-measurements in a "flattened out" neighborhood of* i *(since, in that case, the interior angles always sum to* 2π*), and* $\varphi_{i,j}$ *simply describes the directed angle from* $\{i, j_0\}$ *to* $\{i, j\}$ *(traversed in a counter-clockwise fashion).*

3. *Finally, define the connection map* $\psi_{i,j} : \mathbf{C} \to \mathbf{C}$ *such that*

$$\psi_e(z) = e^{\iota \theta_{i,j}} \cdot z, \tag{2.12}$$

where $\theta_{i,j} = (\varphi_{j,i} + \pi) - \varphi_{i,j}$*, so that the displacement of the vector* z *through the connection map results in parallel vector* in the flattened neighborhood.

When the surface is flat, this construction results in the trivial connection. See also [Sharp et al., 2019] for variants adapted to other discretizations.

Much like $\mathbf{R}$-signals on signed graphs, the analysis of $\mathbf{C}$-signals on graphs endowed with a $U(\mathbf{C})$-connection allows for a variational interpretation, based on the *connection Laplacian* $\mathsf{L}_\theta : \mathbf{C}^{\mathcal{V}} \to \mathbf{C}^{\mathcal{V}}$, defined by $\mathsf{L}_\theta = \mathsf{D} - \mathsf{A}_\theta$, with D the diagonal degree matrix with $\mathsf{D}_{i,i} = \sum_j w_{i,j}$, and A_θ a connection-aware adjacency matrix such that $(\mathsf{A}_\theta)_{i,j} = w_e e^{-\iota \theta_{(i,j)}}$ if $\{i, j\} \in \mathcal{E}$, and $\mathsf{A}_{i,j} = 0$ otherwise. This operator generalizes the signed Laplacian to more diverse connections, and it will be useful to keep in mind the two following properties.

1. The connection Laplacian is associated to the quadratic form[17]

$$\langle f, \mathsf{L}_\theta f \rangle = \frac{1}{2} \sum_{e \in \overrightarrow{\mathcal{E}}} |f(t_e) - e^{\iota\theta_e} f(s_e)|^2. \tag{2.13}$$

   For the trivial connection, with $\psi_e = \mathrm{id}_{\mathbf{C}_{s_e}, \mathbf{C}_{t_e}}$ (*i.e.* $\theta_e = 0$) for all edges $e \in \overrightarrow{\mathcal{E}}$, we recover $\mathsf{L}_\theta = \mathsf{L}$ and the expression in Equation (1.13), and we recover $\mathsf{L}_\theta = \mathsf{L}_\pm$ and Equation (2.2) for the sign-aware connections $\psi_e = \sigma_e \mathrm{id}_{\mathbf{C}_{s_e}, \mathbf{C}_{t_e}}$.

2. Unlike the combinatorial Laplacian, and similarly to the signed Laplacian, $\langle f, \mathsf{L}_\theta f \rangle$ also penalizes functions that are incoherent with respect to the connection. Further, L_θ is only *semi-definite positive*, and not positive-definite, *i.e.*, there exists no non-identically-zero signal $f \in \mathbf{C}^{\mathcal{V}}$ such that $\mathsf{L}_\theta f = 0$ (equivalently, such that $\langle f, \mathsf{L}_\theta f \rangle = 0$), and it follows that *the smallest eigenvalue* λ_0 *of* L_θ *is non-zero.* See Example 2.0.3 below for an illustration pertaining to the graph-bundle in Figure 2.2.

Let us in particular stress that constant-signals are in general *not* maximally-smooth for $U(\mathbf{C})$-connections.

Example 2.0.3. *Consider the signal $f = (1\ 1\ 1)^t$ defined over the cyclic graph represented in Figure 2.2. Here, the connection is constant, and we simply have:*

$$\langle f, \mathsf{L}_\theta f \rangle = 3|1 - e^{\iota\frac{\pi}{4}}|^2 > 0. \tag{2.14}$$

One can go a step further, and show that there exists no non-zero signal $f \in \mathbf{C}^{\mathcal{V}}$ such that $\langle f, \mathsf{L}_\theta f \rangle = 0$. Indeed, in that case, one would have $f(v_1) = e^{\iota\frac{\pi}{4}} f(v_3) = e^{\iota\frac{2\pi}{2}} f(v_2) = e^{\iota\frac{3\pi}{4}} f(v_1)$; a contradiction.

Our goal in this chapter is two-fold. First, to present other examples of $\mathbf{C}$-signals and $U(\mathbf{C})$-connections on graphs, along with specific connection-Laplacian-based problems and associated GSP-like pipelines. Second, to provide a rough but clear idea of how properties of the spectrum of L_θ can be leveraged to understand and solve these problems; we deliberately emphasize qualitative discussions over precise quantitative statements, and encourage the reader to go through the many references for more details. Specifically, we discuss the two following problems, for which we later provide Kirchhoff-like forest-based estimators in **Part** ii[18].

- A connection-aware Tikhonov smoothing. This is the topic of Sections 2.1, in which we explore the analogy between the uses of L_θ and L for GSP-inspired pipelines.
- The $U(\mathbf{C})$-synchronization problem, also called the *angular synchronization* problem, discussed in Sections 2.3 and 2.4.

17 The expression in Equation (2.13) can be derived directly from the definition of L_θ. An alternative proof consists in re-writing the connection Laplacian using a connection-aware discrete differential $\nabla_\theta : \mathbf{C}^{\mathcal{V}} \to \mathbf{C}^{\mathcal{E}}$, so that $\mathsf{L}_\theta = \nabla_\theta^* \nabla_\theta$. See Chapters 5 and 6 for more details on this decomposition.

18 Some of our developments in Chapter 4 will apply to more general connections over unitary matrix groups as well.

2.1 TIKHONOV SMOOTHING ON **C**-BUNDLES

C-signals are complex-valued functions defined over the nodes of a graph $\mathcal{G}$, that can be used to represent 2-dimensional data over the nodes of a graph $\mathcal{G}$. In **C**-signal processing pipelines, such as in computational geometry (recall Remark 2.0.2), one often has access to a number of different types of information that can be leveraged.

- The graph $\mathcal{G}$ itself, which gives a combinatorial representation of an underlying space.
- A $U(\mathbf{C})$-connection Ψ, which represents additional information on the underlying space that is not captured in the topology of $\mathcal{G}$.
- A signal f to be processed, *e.g.*, smoothed out accross the whole space, or interpolated.

In turn, filtering operations have to account for the additional structure provided by the connection Ψ, and models adapted to this setting have to be defined. In particular, it is no longer appropriate to quantify the smoothness of a signal in terms of its projection on the spectrum of L (like in Chapter 1), and our first development focuses on how graph-signal-processing tools should be translated to the setting of a $U(\mathbf{C})$-connection. Our first development consists is a description of a connection-aware Tikhonov smoothing problem, along with a number of existing applications.

Connection-aware graph Tikhonov smoothing mimicks the soft filter used the classical graph Tikhonov smoothing problem, and amounts to solving:

$$\underset{f \in \mathbf{C}^n}{\operatorname{argmin}} \; q\|f-g\|_2^2 + \langle f, \mathsf{L}_\theta f \rangle, \tag{2.15}$$

where $g \in \mathbf{C}^n$, and $q \in \mathbf{R}_+^*$ is an *a priori* known regularization parameter. This specific problem has been considered in a number in different contexts, such as vector-field extension in computational geometry [Sharp et al., 2019], or signal processing over directed graphs [Furutani et al., 2019], as detailed below.

Proposition 2.1.1. *The optimal solution f_* of the problem in Equation (2.15) takes the form*

$$(\mathsf{L}_\theta + q\mathsf{I})^{-1} q\mathsf{I}\; g. \tag{2.16}$$

Proof. The proof simply proceeds by finding the unique critical point in **C** of the cost function $f \mapsto q\|f-g\|_2^2 + \langle f, \mathsf{L}_\theta f \rangle$. The slight subtlety is that we need to consider functions from **C** to **R**, and consider Wirtinger derivatives[19]. See Appendix A.1 for details. ■

The variant of Tikhonov smoothing considered in Equation (2.15) is adapted to **C**-signals g degraded by homoscedastic noise (*i.e.*, the strength of the noise is the same on all nodes). Other variants adapted to heteroscedastic noise are also considered in practice, in which case one associates different regularization values q_i to different nodes in the graph (this is completely analogous to our discussion on semi-supervised node-classification in Section 1.1.2 of Chapter 1, and is not specific to **C**-signals; in fact, the RSF-based estimators of Section 1.2 have been

19 Named after Austrian mathematician Wilhelm Wirtinger, 1865−1945.

extended to this setting [Pilavcı et al., 2021]). Aggregating these values into a diagonal matrix Q with $\mathsf{Q}_{i,i} = q_i$, the solution to the resulting regularization problem simply reads

$$(\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}\, g. \tag{2.17}$$

Let us warn the reader that, while connection-aware Tikhonov smoothing is encountered in applications, it has not, to our knowledge, been systematically studied. In particular, the interplay between the topology of $\mathcal{G}$ and the connection Ψ can become quite intricate, and it is not *a priori* clear for which graphs the resulting signal f_* is meaningfully smooth (recall the comparison in Chapter 1 between Tikhonov smoothing on Erdös-Rényi and random geometric graphs). Empirical experiments should be designed with this consideration in mind.

2.2 EXAMPLES

Let us discuss two practical examples of connection-aware smoothing problems. Along the way, we discuss an important parameter called the *incoherence* of the connection. Though this parameter lacks a formal definition (but can be quantified in a number of different ways), it plays an important role in most problems involving $U(\mathbf{C})$-connections.

Vector-field Extension on Surfaces. Discrete versions of computational geometry problems are naturally encoded using the connection Laplacian. Given a triangulated surface $\mathcal{M}$, that can be represented as a graph $\mathcal{G}$, typical problems seek to infer a vector field on the surface. For instance, given a vector field g defined on $\mathcal{A} \subseteq \mathcal{V}$, the data of the graph itself is not enough to infer a vector field that is consistent with the underlying (smooth) surface, as it fails to encode for:

- the position of the nodes in space (if the surface is embedded into $\mathbf{R}^3$);
- the curvature of the surface.

The connection, and associated connection Laplacian, serve to encode the missing curvature information[20]. A concrete example of this problem is that of "grooming" in discrete geometry processing, in which one aims to add fur or hair on a 3D model, from the data of a few guiding strands of hair; see, Figure 2.4. If one applies techniques that do not account for the curvature of the mesh, the resulting hairy model ends up looking very unnatural.
Why the curvature matters is better understood on a toy example[21]. Consider computing the "parallel transport" of a guiding strand of hair on the surface, that is, of a 2-dimensional vector z defined in the tangent plane T_A at some point A of the surface to other tangent planes. Parallel transport simply refers to the application of the connection maps ψ_e encountered along a path.

- For the trivial connection, all resulting strands of hair will point in the same direction (in which case we say that the connection is *flat*). Non-trivial connections properly account for the curvature, and the result is usually much more convincing.

20 For discretization of Riemannian (sub-)manifolds, one typically defines the connection as a discretization of the Levi-Civita connection, following, *e.g.*, the construction in Remark 2.0.2. If the discretized graph is built as a random geometric graph using points uniformly sampled from the manifold, and the connection is connection is built in a precise manner, one can further show that the connection Laplacian actually converges to the Laplace-Beltrami operator [Singer and Wu, 2012].

21 We roughly follow the nice introductory exposition of Mathieu Desbrun during the Journées Nationales de l'Informatique Mathématique, held in Grenoble in early 2024. See also [Crane et al., 2013].

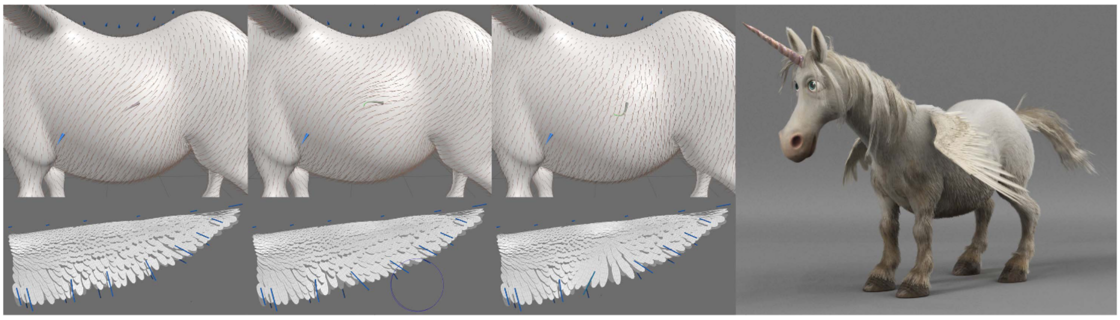


Figure 2.4: Figure from [De Goes et al., 2020]. Grooming from a few guiding strands (in blue), as applied to the fur and feathers of Unicorns from the *Onward* Pixar movie. Three images on the left: guiding strands for the tangent direction and shapes of the fur and feathers. Right: resulting model after grooming, further processing and rendering.

- One issue that comes up in the non-trivial-connection setting is that the resulting direction depends on the specific path: two paths will, in general, not yield the same result. In particular, transporting a vector along a loop C on the surface will result in a vector $z' = \left(\prod_{e \in C} \psi_e\right) z \in T_A$ that differs from z; in mathematical terminology, the map ψ_C is called the *holonomy* of the connection along the cycle C [Joyce, 2000][22]. See Figure 2.5 for an illustration on the sphere[23]. This same phenomenon is represented in Figure 2.2 on the triangle graph K_3 endowed with a $U(\mathbf{C})$-connection: the vector z_1' recovered after transport along the cycle C is a rotation of the original vector z_1 by an angle of $\theta_C = \frac{3\pi}{4}$. We say that the connection is *incoherent* along the cycle C.

Let us go back to the problem of vector-field extension on surfaces, which can be formulated as a $\mathbf{C}$-signal processing problem. More precisely, one wants to infer a signal $f \in \mathbf{C}^V$ from a signal $g : \mathcal{A} \to \mathcal{V}$ defined on some subset $\mathcal{A} \subseteq \mathcal{V}$ of nodes. Informally, one strategy to compute this extension f consists in accumulating the result of the parallel transports of the signal defined on $\mathcal{A}$ to all the other nodes in $\mathcal{V} \setminus \mathcal{A}$, along all paths of minimal length[24] [Sharp et al., 2019]. It can further be argued that these parallel transport maps are well-approximated by short-time-horizon heat diffusion on the surface [Sharp et al., 2019][25], in which case one is led to compute solutions of the form:

$$(\mathsf{Q} + \mathsf{L}_\theta)^{-1}\overline{g}, \tag{2.18}$$

where $\overline{g} \in \mathbf{C}^{\mathcal{V}}$ is such that $\overline{g}(v) = g(v)$ if $v \in \mathcal{A}$, and $\overline{g}(v) = 0$ otherwise. Substituting $\overline{g} \mapsto \mathsf{Q}^{-1}\overline{g}$ in Equation (2.18), we exactly recover the solution of the connection-aware Tikhonov smoothing problem for heteroscedastic noise (Equation (2.17)): in this approximation, vector-field extension over surfaces amounts to a connection-aware smoothing of the *a priori* signal $\overline{g}$.

22 In physics, the terminology "non-holonomic" refers to systems for which any state may be achieved by going along cycles in its parameter space. One famous example is the the "geometric" Pancharatnam-Berry phase [Berry, 1987].
23 Figure from Wikipedia, by Luca Antonelli (`https://commons.wikimedia.org/wiki/User:Luca_Antonelli`), distributed under CC BY-SA 3.0 License (`https://creativecommons.org/licenses/by-sa/3.0/`). `https://fr.wikipedia.org/wiki/Holonomie`
24 Also called *geodesics*.
25 The solution of heat-diffusion equations is often expressed using Green's functions. For flat connections, Green's functions are known to be related to random walks (see, *e.g.*, [Avena and Gaudillière, 2013]), and we explore connection-aware versions of these relations in **Part** ii.

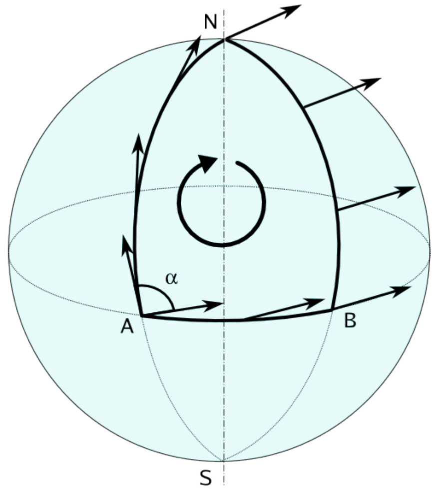


Figure 2.5: Parallel transport of a vector on the (tangent bundle of) the sphere, along a loop based in A (from Wikipedia). The transport results in a rotation of angle α, and the holonomy is non-trivial.

Another strategy for vector-field-extension algorithm consists in finding a connection-aware harmonic interpolant, and computing [De Goes et al., 2020]:

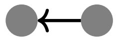
$$-\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\; g, \tag{2.19}$$

which is the solution to the interpolation problem in Equation (1.24) from Chapter 1, where L has been replaced by L_θ to account for the connection.
A number of other problems in discrete geometry processing also involve computing expressions of the form in Equation (2.18); one famous example is an implicit discretization of the "mean curvature" flow, which serves to smooth-out surfaces [Crane et al., 2013][26].

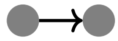
The previous example is rooted in geometry, and clearly outlines the importance of connections in practical problems, where graphs appear as a powerful data-representation tool. Let us now introduce, by way of a graph-specific example, a common methodology in which one controls the *incoherence* of the connection by way of an additional hyper-parameter.

Signal Processing on Directed Graphs. A directed graph $\mathcal{G}^d$ is given by a set on nodes $\mathcal{V}^d$ and a set of directed edges $\overrightarrow{\mathcal{E}}^d \subseteq \mathcal{V}^2$; if the directed graph is weighted, a weight w_e is associated to each *directed* edge $e \in \overrightarrow{\mathcal{E}}$. Note that we do not impose that $w_e = w_{e^*}$, and that we set $w_e = 0$ if $e \notin \overrightarrow{\mathcal{E}}^d$. Let us also stress that directed graphs differ from the graphs we have considered so far which are non-directed graphs that, for technical convenience, we have *endowed with a reference orientation*; in particular, the set of directed edges $\overrightarrow{\mathcal{E}}$ is closed under orientation-reversal of edges. This is not the case for the set of edges $\overrightarrow{\mathcal{E}}^d$ in directed graphs.
Directed graphs are natural models for a number of network-structured data such as, for instance, scientific citation networks or more general social networks [Marques et al., 2020], and it is natural

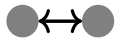
26 The mean curvature flow is a type of geometric flow, which describes the deformation in time of an embedding $f : \mathcal{M} \to \mathbf{R}^3$ into $\mathbf{R}^3$ as controlled by a differential equation; in this case, a heat equation [Andrews et al., 2022]. Note that, in this case, one considers the trivial connection.



(a) $\theta_e = -\gamma 2\pi$



(b) $\theta_e = \gamma 2\pi$



(c) $\theta_e = 0$

Figure 2.6: Connections on directed graphs with two vertices, according to Equation (2.20). The graphs are trivially weighted for simplicity ($w \in \{0, 1\}$).

to apply a GSP methodology to process signals defined over directed graphs. One obstacle in this endeavor is that there is a number of competing notions for what the Laplacian of directed graphs should be, with different drawbacks and advantages (see, *e.g.*, [Chung, 2005, Sevi et al., 2023]). One strategy consists in regarding the Laplacian of a directed graph as the connection Laplacian of an associated non-directed graph $\mathcal{G}$, for a hand-crafted $U(\mathbf{C})$-connection [Furutani et al., 2019]. One advantage of this construction is that the resulting operator remains Hermitian[27] and positive semi-definite (in most cases, even positive definite). Specifically, one takes $\mathcal{G} = (\mathcal{V}^d, \mathcal{E})$ with $\mathcal{E} = \left\{ \{i, j\};\ (i, j) \in \overrightarrow{\mathcal{E}}^d \text{ or } (j, i) \in \overrightarrow{\mathcal{E}}^d \right\}$, and considers the connection with

$$\theta_e = \gamma 2\pi (w_{s_e, t_e} - w_{t_e, s_e}), \tag{2.20}$$

where $\gamma \in \mathbf{R}_+^*$ is a (small) scale parameter, which controls the incoherence of the connection. The weight of the (undirected) edge e is set to $w_e = \frac{1}{2}(w_{s_e,t_e} + w_{t_e,s_e})$. This choice of connection encodes the direction of the edge $e \in \overrightarrow{\mathcal{E}}^d$ in the phase of the connection map $\psi_e = e^{\iota \theta_e}$. See Figure 2.6 for an illustration of this phenomenon. The scale parameter γ is usually taken to be small, so that there can be no ambiguity on the preponderant direction of each edge in the angular representation[28].
To filter a $\mathbf{R}$-signal $g \in \mathbf{R}^{\mathcal{V}}$ defined over $\mathcal{G}^d$, one then proceeds as follows.

1. Construct the associated non-directed graph $\mathcal{G}$.

2. Apply a connection-aware filter to g (*e.g.* Tikhonov smoothing [Furutani et al., 2019]), resulting in a $\mathbf{C}$-signal f.

3. Compute the real part $\mathfrak{Re}(f)$ of the recovered signal as the final result.

This type of strategy comes with no guarantee on the estimated solution, but encodings of directed graphs using connection Laplacians have been used in machine-learning applications to reach competitive performance on, *e.g.*, link-prediction tasks in directed networks [Zhang et al., 2021].

The reader should now have a rough idea of how the incoherence of a connection impacts processing on $\mathbf{C}$-bundles. We discuss in the following sections how this incoherence impacts our second problem of interest: angular synchronization.

27 Named after French mathematician Charles Hermite, 1822–1901.
28 This further ensures that the eigenvectors of L_θ are "close" to those of L, and that both can be used similarly, as we discuss in Appendix D.4 for more general $U(\mathbf{C})$-connections. In particular, for low values of γ, L_θ-based Tikhonov smoothing results in meaningfully smooth signals whenever L-based Tikhonov smoothing does.

2.3 ANGULAR SYNCHRONIZATION

The angular-synchronization problem is an instance of group synchronization, which asks to recover a set of n unknown angles $\omega = (\omega)_i \subseteq [0, 2\pi)^n$ from measured pairwise offset measurements $\{\theta_{i,j}\}_{i,j}$ [Singer, 2011]:

$$\theta_{i,j} = \omega_j - \omega_i + \varepsilon_{i,j} \mod 2\pi, \tag{2.21}$$

where $\varepsilon_{i,j}$ represents some unknown degradation of the measurement (*e.g.*, measurement noise, or corruption of a subset of measurements). In this case, the group $\mathfrak{g}$ in question may be either the additive group over $[0, 2\pi)$ (mod 2π), or the multiplicative group $U(\mathbf{C})$; we focus on this second perspective[29,30].

Angular synchronization appears in many structured signal processing problems, where it is often a key component in state-of-the-art recovery methods. Examples include perceived-luminance reconstruction [Yu, 2009], ptychography [Filbir et al., 2021] (and more generally phase reconstruction problems, see *e.g.* [Alexeev et al., 2014, Iwen et al., 2017]), ranking [Cucuringu, 2016], clock synchronization [Giridhar and Kumar, 2006], or plenty others in computational geometry (see, *e.g.*, [Knöppel et al., 2013] and references therein[31]). From a different perspective, the problem of angular synchronization also appears in the statistical physics literature, as a model for spin-systems (*e.g.* [Lupo and Ricci-Tersenghi, 2017, Chen et al., 2022]).

In most applications, we only observe a subset of all the measurements $\theta_{i,j}$, and the problem is naturally formulated on a graph $\mathcal{G}$ with n nodes whose set of edges $\mathcal{E}$ is indexed by the number of measurements, and where edge (i, j) (resp. (j, i)) carries the offset $\theta_{i,j}$ (resp. $(\theta_{j,i} = -\theta_{i,j})$). *For simplicity, we assume that this graph is connected.*

When noiseless measurements $\theta_{i,j} = \omega_j - \omega_i$ are available, the connection is *coherent*, and exact recovery can be trivially performed up to a global phase shift, by *propagating* values according to the offset measurements along a spanning tree of $\mathcal{G}$ as follows.

1. Fix an arbitrary value s_r on some root node r.
2. Recursively set $s_j = s_i + \theta_{i,j}$ for all edges $\{i, j\}$ in the tree, starting from edges adjacent to the root $i = r$.

See Figures 2.7a and 2.7b for illustrations. In this setting, the angles ω_i can be recovered exactly (up to a global shift) from the first eigenvector v_1 of L_θ. We make a more explicit development on this statement below, but let us first describe what happens when the connection is incoherent.

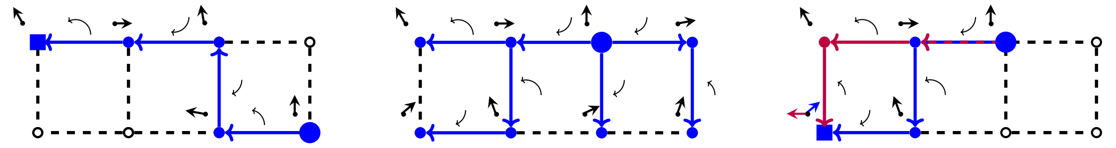
The problem is more involved when considering imperfect measurements $\theta_{i,j}$, with non-zero noise $\varepsilon_{i,j}$ (*incoherent* connections). In general, *exact* angular synchronization can no longer be performed in this noisy regime, as long as even one *incoherent* cycle is present in the graph (recall the incoherence along loops in the vector-field example from Section 2.2, and in Example 2.0.3). Indeed, consider the following.

---

29 Note that the two groups are isomorphic, *e.g.*, through the isomorphism $x \mapsto e^{\iota x}$. In fact, one may consider a family of such isomorphisms $x \mapsto e^{\iota \gamma x}$, depending on a scale parameter γ (recall the representation of directed graphs using connection Laplacians, or see also example [Cucuringu, 2016] for another example). Group-synchronization algorithms often consider one such $U(\mathbf{C})$-representation, with scale-parameter γ small enough so that the connection remains coherent.

30 The reader familiar with differential geometry may recognize this problem as that of looking for a *global section* of a vector bundle [Lafontaine, 2021].

31 Angular synchronization has been in use for a long time in this literature, but the emphasis is most often put on practical results rather than theoretical analyses.



(a) Propagation from the large blue node (bottom left) to the large blue square (top right) along the path in blue. A rotation is applied when traversing each edge.

(b) Synchronization by propagation from the large blue node to all other nodes in the graph, along the branches of the blue spanning tree (only possible if there is no incoherence).

(c) Propagation from the large blue node to the large blue rectangle (bottom left), along two different paths (blue and purple). Here, the rotations are incoherent and the result of the propagation depends on the specific path considered.

Figure 2.7: Propagations on the 4×2 grid graph, along different paths. The rotation angles $\theta_{i,j}$'s associated to each edge are represented as circular rotating arrows (only drawn along the paths we consider). Propagations always start from the large blue node, and propagated angles are represented as straight arrows (top left of each node). Left 2.7a: propagation along an arbitrary path. Center 2.7b: exact synchronization achieved by propagation along a spanning tree, in the absence of noise. Right 2.7c: synchronization impossible due to noise and incoherence along cycles.

- If $i, j \in \mathcal{V}$ are spanned by some incoherent cycle $C \subseteq \mathcal{E}$ (that is, $\theta_C = \sum_{e \in \mathcal{C}} \theta_e \neq 0$), set an arbitrary value s_i on node i, and propagate this value to node j along the two possible paths in the cycle. In general, we find two different solutions. See Figure 2.7c for an illustration.
- In the special case $i = j$, the propagation occurs along the complete loop C, and we find two different values $s_i \neq s_i + \theta_C$. It is impossible to satisfy all the pairwise constraints $s_j = s_i + \theta_{i,j}$, and this shows that exact synchronization is impossible in general.

A common workaround, first introduced in [Yu, 2009], consists in quantifying the incoherence of an angular assignment $s \in [0, 2\pi)^n$ as

$$\mathcal{I}(s) = \sum_{\{i,j\} \in \mathcal{E}} \left(2 - 2\cos\left((s_j - s_i) - \theta_{i,j}\right)\right), \tag{2.22}$$

before minimizing this incoherence over all possible assignments. In case the connection is coherent, the minimizer of $\mathcal{I}(s)$ is an exact solution to the angular-synchronization problem. While this problem is non-convex and NP-hard in general [Zhang and Huang, 2006] (see also [Boumal, 2016] for a discussion), different techniques allowing to (approximately) recover a solution have been proposed, *e.g.*, [Yu, 2012, Singer, 2011, Boumal, 2016, Perry et al., 2018, He et al., 2023]. For now, let us discuss some links between angular synchronization and the connection Laplacian L_θ, which appears as a key ingredient in most estimation strategies.

We further develop our earlier observation concerning coherent connections. Consider a vector $f \in U(\mathbf{C})^{\mathcal{V}}$, the entries f_i of which are *unitary* complex numbers $f_i = e^{\iota s_i}$ representing some angular assignment. Applying to f the quadratic form associated to L_θ (Equation (2.13)), we recover exactly the expression in Equation (2.22). Indeed, in this case, $|f(j) - e^{\iota\theta_{(i,j)}} f(i)|^2 = 2 - 2\mathfrak{Re}((e^{\iota s_j})^* e^{\iota\theta_{(i,j)}} e^{\iota s_i})$, which, thanks to conjugation-invariance of the real part, results in:

$$\langle f, \mathsf{L}_\theta f\rangle = \sum_{\{i,j\} \in E} 2 - 2\cos((s_j - s_i) - \theta_{(i,j)}). \tag{2.23}$$

It follows that $\langle f, \mathsf{L}_\theta f\rangle = 0$ if and only if $f_i = e^{\iota\theta_{(j,i)}} f_j$ for all $(i,j) \in \overrightarrow{\mathcal{E}}$, that is, if $s_i = s_j + \theta_{(j,i)}$. This has one nice consequence: L_θ is not only semi-definite positive (as seen from Equation (2.13)), but *positive* definite, *unless* there is an *exact solution* $x \in U(\mathbf{C})^{\mathcal{V}}$ to the associated angular-synchronization problem. Further, in this case, the kernel $\ker \mathsf{L}_\theta$ is generated by x, which is nothing but an eigenvector of L_θ associated to the eigenvalue $\lambda_1 = 0$.

There is actually no need to consider the incoherence measurement $\mathcal{I}(s)$ in the previous argument, as one could reach the conclusion by inspecting the quadratic form $\langle f, \mathsf{L}_\theta f\rangle$ directly. On the other hand, it serves as a useful motivation to build a spectral algorithm for angular synchronization. Consider finding an optimal angular assignment $f_* \in U(\mathbf{C})$ satisfying

$$f_* \in \underset{f \in U(\mathbf{C})^{\mathcal{V}}}{\operatorname{argmin}} \ \langle f, \mathsf{L}_\theta f\rangle, \tag{2.24}$$

which corresponds to minimizing $\mathcal{I}(s)$, with $f(i) = e^{\iota s_i}$. Relaxing the condition $f_* \in U(\mathbf{C})$ to $\|f\|_2^2 = n$, the problem re-formulates as

$$f_* \in \underset{\substack{f \in \mathbf{C}^{\mathcal{V}} \\ \|f\|_2^2 = n}}{\operatorname{argmin}} \langle f, \mathsf{L}_\theta f\rangle, \tag{2.25}$$

with optimal solution $f_* = \sqrt{n} v_1$, where v_1 is a (normalized) eigenvector of L_θ associated to the smallest eigenvalue λ_1. In particular, λ_1 can be understood as quantifying the quality of the spectral solution of the angular-synchronization problem and, by extrapolation, as a measure of the coherence of the connection[32].

Experimental results suggest that the spectral method for angular synchronization described in Equation (2.25), first studied in [Yu, 2009] and [Singer, 2011] (and later refined in, *e.g.*, [Cucuringu et al., 2012, Yu, 2012] based on *normalized* connection Laplacians), can reach competitive performance on practical angular-synchronization problems [He et al., 2023]. In addition, it is straightforward to solve, and we propose fast forest-based methods that can quickly compute an approximation of the solution in **Part** ii (Chapter 6). On the other hand, the literature concerning theoretical guarantees for angular synchronization can be fairly confusing. In particular, it is not clear how the spectral method compares to other angular-synchronization solvers: guarantees are often quite weak as compared to other techniques, but are available in a wide range of situations (in contrast to other methods). Our goal in the following section is to roughly survey this situation by way of examples, and to make evident what the important (spectral) parameters are, with respect to the (statistical) difficulty of the problem.

2.4 MODELS OF ANGULAR SYNCHRONIZATION: FEASIBILITY AND HARDNESS

The hardness of the angular-synchronization problem depends on two main factors:

1. the incoherence of the connection;

2. the topology of the graph.

32 When the connection stems from pairwise phase offsets of the form $x_i(x_j)^* e^{\iota\omega_i - \omega_j}$ of an underlying vector $x \in U(\mathbf{C})$ with $x_i = e^{\iota\omega_i}$, that have been degraded by additive Gaussian noise, the solution to the linear-least-squares problem in Equation (2.24) is also the maximum likelihood estimator of the $e^{\iota\omega_i}$ for this model [Filbir et al., 2021].

Let us first consider the performance of spectral estimators, for which the influence of these two parameters is nicely illustrated in the following result from [Singer, 2011], pertaining to the eigenvectors of the connection-aware adjacency matrix A_θ. The model considered is a *corruption model*, for which a subset of the edges provide the exact offsets $\omega_i - \omega_j$, and where the others carry erroneous information. Specifically, the graph and the connection are built as follows.

- First, a set of "good" edges is selected as an Erdös-Rényi graph on n vertices: each edge in $\mathcal{V}^{(2)}$ is selected with probability p, and we denote by $\mathcal{E}_g \subseteq \mathcal{V}^{(2)}$ the resulting set of good edges. For these edges, the measured offset $\theta_{i,j}$ is the exact angular difference $\theta_{i,j} = \omega_j - \omega_i$.
- Second, a subset of "bad" edges is added to the graph, by uniformly drawing a subset of edges from the remaining non-good edges $\mathcal{V}^{(2)} \setminus \mathcal{E}_g$; we denote by $\mathcal{E}_b$ this subset of bad edges. For bad edges, the measurements are corrupted, and taken uniformly in $[0, 2\pi)$, that is, $\theta_{i,j} \sim \mathcal{U}\left([0, 2\pi)\right)$.

This results in a graph with $m = |\mathcal{E}_g| + |\mathcal{E}_b|$ edges, for which the measured offsets are either exact ($\theta_{i,j} = \omega_j - \omega_i$ for $\{i,j\} \in \mathcal{E}_g$), corrupted ($\theta_{i,j} \sim \mathcal{U}\left([0, 2\pi)\right)$ for $\{i,j\} \in \mathcal{E}_b$), or missing ($\{i,j\} \notin \mathcal{E}$). Note that, much like the analysis carried-out in Section 2.3, exact angular-synchronization can be trivially performed as long as $\mathcal{E}_g$ contains a spanning tree of $\mathcal{G}$ (provided that we are able to distinguish this tree from the bad edges, which we are not).
On the other hand, one may still be able to exploit the preponderance of good edges to perform the estimation. In particular, one can then show that, in the large n limit, the first eigenvector v_1 of the connection-aware adjacency matrix A_θ associated to this graph is positively correlated with the true solution x (with $x_i = e^{\iota\omega_i}$) as soon as [Singer, 2011][33]

$$p > \sqrt{\frac{n^5}{8m^3}}. \tag{2.26}$$

The left-hand-side in Equation (2.26) is the probability p of an edge to be good, and controls the (in)coherence of the connection; the right-hand-side is a measure of the density of the graph, that is, a very crude descriptor of its topology, and gets larger the amount of both good and bad edges increases. In words, Equation (2.26) expresses that the threshold on the proportion of good measurements p for v_1 to be correlated with the ground-truth solution x gets higher as the proportion of bad edges increases, the exact threshold depending on the overall density of the graph. There are a few things that one should note.

- First, this bound is not sharp, as noted in [Singer, 2011]. v_1 may be positively correlated with x even when the condition from Equation (2.26) is not met.
- Second, this result pertains to the first eigenvector v_1 of A_θ, and does not apply to the first eigenvector v_1 of L_θ. On this front, one should keep in mind that spectral methods based on normalized operators of the form $\widetilde{\mathsf{L}_\theta} = \mathsf{D}^{-\frac{1}{2}}\mathsf{L}_\theta\mathsf{D}^{-\frac{1}{2}}$ (and $\Delta_\theta = \mathsf{D}^{-1}\mathsf{L}_\theta$) have been found to reach better practical performance (see, *e.g.*, [He et al., 2023]), and that $\widetilde{\mathsf{L}_\theta}$ and $\widetilde{\mathsf{A}_\theta} = \mathsf{D}^{-\frac{1}{2}}\mathsf{A}_\theta\mathsf{D}^{-\frac{1}{2}}$ have the same eigenvectors.

33 Results of this type usually pertain to large graphs, and are proved using tools from asymptotic random-matrix theory [Couillet and Liao, 2022, Tao, 2023], where $n \to \infty$. The result from [Singer, 2011] states that $\frac{1}{n}|\langle \mathsf{v}_1, x\rangle|^2 \to c$ where c is some *positive* constant, that is, v_1 and x are *not* orthogonal in the large n limit, and are correlated in this sense. In contrast, $\frac{1}{n}|\langle y, x\rangle|^2 \to 0$ when y yields an estimate that is not correlated to the ground-truth x.

The result of [Singer, 2011] is reminiscent of the detectability threshold for community detection recalled in Chapter 1, the condition in Equation (2.26) speaking on the ability of the spectral method to be *correlated* to the underlying ω_i's, but note that there is no guarantee of *exact recovery*. One may further remark that, unlike the community-detection threshold from Chapter 1, the condition in Equation (2.26) is method-specific. In relation with these observations, there are two points we would like to raise.

1. There exists methods for which exact-recovery may be certified (depending on the topology of the graph at hand), if one makes the assumption that the exact offsets have not been degraded too much [Wang and Singer, 2013, Zhong and Boumal, 2018] (in this case, the solution to the problem in Equation (2.24) is unique)[34]. To our knowledge, no such guarantees apply to spectral methods, and existing theorems are "only" able to establish Cheeger-type[35] bounds on the norm of the error between the estimate and the ground-truth x, depending on the amount on degradation (see, *e.g.*, [Filbir et al., 2021] and references therein).

2. There exists *method-independent* statistical thresholds for non-trivial recovery, that apply to the complete graph. These results are obtained by framing the angular-synchronization problem as a special case of a more classical low-rank-recovery problem for spiked Wigner matrices [Couillet and Liao, 2022][36]. Specifically, one considers an observation matrix of the form
$$Y = \frac{\sigma}{\sqrt{n}} x x^* + W, \tag{2.27}$$
where W is an Hermitian matrix, the off-diagonal entries of which are drawn according to a complex normal distribution ($W_{i,j} \sim \mathcal{N}_{\mathbf{C}}(0,1)$ for all $\{i,j\}$), and with $W_{i,i} = 0$. σ is a signal-to-noise-ratio parameter, and controls the amount of noise that is applied. In the end Y can be understood as a weighted adjacency matrix of a complete graph, and represents a global degradation of the measured offsets $(xx^*)_{i,j} = e^{\iota(\omega_j - \omega_i)}$ by homoscedastic noise, from which we wish to recover the ground-truth vector x.
When $\sigma > 1$, v_1 is positively correlated with x [Féral and Péché, 2007, Benaych-Georges and Nadakuditi, 2011]:
$$\frac{1}{n} |\langle \mathsf{v}_1, x \rangle|^2 \to 1 - \frac{1}{\sigma^2}. \tag{2.28}$$
In contrast, as soon as $\sigma < 1$, no estimator can have a positive correlation with x [Deshpande et al., 2016, Perry et al., 2016]. It is remarkable that v_1 is provably correlated with x down

34 These algorithms are based on two different techniques: one of them hinges on a formulation on angular-synchronization as a semi-definite programming problem (for which a number of existing solvers can be used), and the second implements a power-method-like iteration $(f)_i \mapsto \frac{(A_\theta f)_i}{|(A_\theta f)_i|}$.

35 The Cheeger inequality, named after American mathematician Jeff Cheeger (born 1943), bounds the isoperimetric constant of a Riemannien manifold using the first eigenvalue of the Laplace-Beltrami operator [Cheeger, 1970]. For graphs, the Cheeger inequality is a similar spectral bound on the connectivity of the graph and, in the specific case of connection Laplacians, on the quality of the angular synchronization problem [Bandeira et al., 2013] (see also Remark 2.4.1).

36 Named after Hungarian-American theoretical physicist Eugene Paul Wigner, 1902−1995, who introduced random matrices to study the nuclei of atoms (in this setting, eigenvalues represent (discrete) energy levels).

to the threshold $\sigma = 1$: to our knowledge, this has not been established for any other hyper-parameter-free estimator computable in polynomial time[37,38].

The results we have exposed so far suggest that the spectral class of methods provides robust angular-synchronization solvers, but only consider complete (trivial topology) or Erdös-Rényi (topology controlled by density) graphs, and do not yield much insight into the importance of the graph's topology for these problems. To our knowledge, the only mathematically rigorous work on the feasibility of angular synchronization that considered non-complete and non-Erdös-Rényi graphs is [Abbe et al., 2018], where the case of the *infinite* grid is considered, as a model for statistical-physics problems.

On the other hand, Cramér-Rao bounds[39] for general angular-synchronization have been derived, with a more explicit dependence on the topology of the graph [Boumal et al., 2014]. Cramér-Rao bounds are *method-independent* results, that lower-bound the variance of unbiased estimators for given problems (here, the variance depends on the generative model used in the set-up of the problem, and in particular on the generation of the offset measurements $\theta_{i,j}$) [Rao, 1992]. More precisely, the results in [Boumal et al., 2014] bound the expected squared geodesic distance (in $U(\mathbf{C})$) between ground-truth offsets $x_i(x_j)^*$ (that may or may not have been observed) and recovered offsets $f(i)(f(j))^*$, for small-noise regimes and connected graphs:

$$\mathbf{E}_\theta \left(d^2 \left(x_i(x_j)^*, f(i)(f(j))^* \right) \right) \gtrsim \tau_{i,j}^2, \tag{2.29}$$

where $\gtrsim$ represents inequality up to multiplicative constants, and $\tau_{i,j}$ denotes the commute time between nodes i and j (*i.e.*, the expected number of steps it takes for a random-walker to go from i to j, and then to i again)[40].

The commute time $\tau_{i,j}$ measures how "well-connected" two nodes i and j are, and provides an intuitive understanding of when an exact relative offset $\theta_{j,i} = \arg(x_i(x_j)^*)$ will be hard to recover, as is the case for the following examples.

- Regular grids (*e.g.*, 2-dimensional grids), in which case the phase-offset between two nodes that are far apart (for instance, top-left and bottom-right of the grid) will be hard to

37 Provided one has access to the value of σ, it is conjectured that the approximate-message-passing (AMP) algorithm from [Perry et al., 2018] also achieves this threshold. For a bad choice of hyper-parameter, statistical-physics heuristics suggest that this AMP algorithm can output an estimate that is not correlated with x [Chen et al., 2022]. Similar heuristics also suggest that the algorithm from [Zhong and Boumal, 2018], which *provably* achieves exact recovery for low-enough noise levels, can output results that are not correlated with x either if the noise-level is too high [Perry et al., 2018]. To our knowledge, it has not even been established that the solution to the non-relaxed angular-synchronization problem (Equation (2.24)) is unique in high-noise regimes (up to a global rotation). On this matter, analyses from statistical-physics suggest that angular-synchronization is not *replica-symmetric* (nor is it *full-replica-symmetry-breaking*), and instead likely behaves in a 1-*replica-symmetry-breaking* (1RSB) fashion [Lupo and Ricci-Tersenghi, 2017, Chen et al., 2022]: concretely a class of problems from which there are examples with a number of solutions *provably* larger than one [Nam et al., 2022, Nam et al., 2024] (in a precise technical sense, in the large n limit).

38 In analogy with the community-detection literature, one would expect algorithms based on regularized variants of the matrices L_θ and A_θ may reach better performance for angular-synchronization problems on graphs with heterogeneous degree-distribution, but we are not aware of any work exploring this direction. Another avenue for improvement would be to explore *survey-propagation* algorithms (which are generalizations of belief-propagation algorithms), which are better suited to 1RSB systems [Chen et al., 2022].

39 Named after Swedish mathematician and statistician Harard Cramér (1893−1985), and Indian statistician Calyampudi Radhakrishna Rao (1920−2023).

40 In this case, $\tau_{i,j}$ more precisely corresponds the commute time *for an appropriately re-weighted graph*, with weights depending on the connection. See [Boumal et al., 2014] for more details.

estimate. The same observation holds for graphs with homogeneous spatial correlations, such as ε-graphs or nearest-neigbors-graphs.

- Graphs with community structures such as SBMs, with bottlenecks (and poor expansion properties). In this case, the phase-offset between two nodes belonging to different communities will be hard to recover. In particular, this also holds for nodes belonging to different communities, but still connected by an edge.

As a corollary observation, the quality of angular synchronization (as measured by, *e.g.*, Equation (2.24)) will degrade when there are numerous edges between pairs of nodes belonging to groups that are *poorly connected* (*e.g.*, weak community-structure, which typically happens in practical situations when the graph and the connection both have to be estimated from some other underlying data[41]). For connections that are not very incoherent (low-noise regimes), this phenomenon is also reflected in the spectrum of L_θ: the larger the spectral gap between λ_1 (measure of the quality of the solution to the spectral formulation of angular-synchronization) and λ_2 (measure of the connectivity and expansion of the graph; larger values of λ_2 corresponding to better connectivity[42]), the better the estimation will be.

Remark 2.4.1. *The previous spectral interpretation is not completely rigorous: the interpretation of λ_2 as a measure of connectivity of the graph has only been established for combinatorial Laplacians [Chung, 1997], and it is unknown whether it can be applied to general connection Laplacians (though we argue in Appendix D.4 that it can for weakly incoherent connections). One can, on the other hand, derive explicit bounds on the quality of angular-synchronization solutions, depending only on the spectral ratio $\frac{\widetilde{\lambda}_1}{\widetilde{\lambda}_2}$ [Bandeira et al., 2013], where $\widetilde{\lambda}_1$ and $\widetilde{\lambda}_2$ are the two smallest eigenvalues of the normalized connection Laplacian $\widetilde{L_\theta}$.*

Overall, spectral methods for angular synchronization are simple agnostic solvers, that can be employed in a wide range of situations: spectral solutions are correlated with the ground-truth underlying angular assignment for all noise regimes (including critically high noise), with accuracy increasing as the noise decreases. These observations are backed empirically (*e.g.*, [He et al., 2023]), and there is no *a priori* reason to prefer another solver, except when the noise level is "reasonably" high, in which case alternatives with exact-recovery guarantees can be used (at least, on the complete graph)[43]. Moreover, the spectrum of L_θ, and in particular its spectral gap, can further be regarded as a measure of the difficulty of the problem, which allows to identify statistically challenging instances of the problem, regardless of the method considered.

2.5 SUMMARY

We discussed in this chapter an alternative perspective on complex-valued graph signals, taking inspiration from constructions commonly studied in differential geometry, as signals defined on

41 Instances of this problem may for example appear in intra-class denoising for cryo-EM. See, *e.g.*, the constructions in [Singer and Wu, 2012] and [Fan and Zhao, 2019b, Fan and Zhao, 2019a]. Let us stress that, in those applications, there is no way to circumvent this situation by considering a graph with a different topology.

42 This interpretation stems from the classical Cheeger inequality on graphs [Chung, 1997].

43 This affirmation should be nuanced: depending on the specific instance of the problem, other algorithms may very well outperform the spectral methods, but there is, to our knowledge, no reliable predictor of when one algorithm or the other should be used (though it has been observed that spectral methods tend to underperform on very sparse graphs).

a $\mathbf{C}$-bundle. One particular feature that emerges from this perspective is the notion of *unitary connection* on a graph $\mathcal{G}$, which associates 2-dimensional rotations to its edges. In particular, a *semi-definite* positive operator known as the *connection Laplacian* L_θ can be naturally associated to such structures, and acts as a generalization of the combinatorial Laplacian to this setting. Connections and connection-Laplacians are a flexible tool, and have found a growing number of applications in the last decade, such as the analysis of signed and directed graphs (and, in particular signal processing over such graphs), vector-field processing problems in computational geometry, and angular-synchronization problems. In these problems, the connection is obtained as part of a pre-processing, and serves as a basis for further processing.
There are two specific properties that the reader should take away from this chapter.

1. The quadratic form associated to L_θ reads
$$\langle f, \mathsf{L}_\theta f \rangle = \frac{1}{2} \sum_{e \in \overrightarrow{\mathcal{E}}} |f(t_e) - e^{\iota \theta_e} f(s_e)|^2, \tag{2.30}$$
which penalizes $\mathbf{C}$-signals that are incoherent with respect to the connection. In turn, it can be used to penalizes signals for connection-aware Tikhonov regularization, which takes the form
$$\underset{f \in \mathbf{C}^{\mathcal{V}}}{\operatorname{argmin}} \; q\|f-g\|^2 + \langle f, \mathsf{L}_\theta f \rangle, \tag{2.31}$$
with solution $f_* = (\mathsf{L}_\theta + q\mathsf{I})^{-1}(q\mathsf{I})g$.

2. An approximate solution to the angular-synchronization problem can be extracted from the eigenvector v_1 associated to the smallest eigenvalue λ_1 of L_θ. Further, λ_1 can be regarded as a measure of the coherence of the connection, and the spectral gap with λ_2 as a measure of the difficulty of the problem.

Let us note though that the the interplay between the topology of the graph and the connection is complex, and one can no-longer *a priori* rely on simple interpretations available for trivial connections (for instance, the fact that the eigenvectors of L are nearly-constant over communities in a graph, or that Tikhonov smoothing results in meaningfully smooth signals on *a-priori*-selected topologies)[44].

One may further notice that we did not discuss the last part of the triptych from Chapter 1: *how do probabilities relate to variational and structural properties of the graph* and, in particular, how randomized algorithms can be used to solve connection-aware problems on $\mathcal{G}$.
In the case of signed graphs, there is some existing work, such as the signed-random-walk-based community-detection algorithm of [Neumann and Peng, 2022]. On the other hand, to our knowledge, no connection-aware randomized algorithms for general unitary connections have found their way into applications yet. Theoretical investigations on these questions also remain very sparse: a connection-aware generalization of the uniform-spanning-tree distribution was for instance introduced in [Kenyon, 2011], and links between connection-aware Gaussian free fields (which can be understood as a type of random signal on the graph[45]) have been studied in [Kassel

44 As we will discuss in Chapter 6, it turns out that these observations *do* carry over to non-trivial connections, as long as the incoherence remains low.

45 The Gaussian free field is a classical model in probability, that enjoys many links with other processes, such as the loop-erased random walks in Wilson's algorithm [Le Jan, 2011].

and Lévy, 2021]. We are not aware of prior work investigating, *e.g.*, the algorithmics of graphs endowed with general unitary connections.

Notwithstanding, the computational challenges mentioned in Chapter 1 remain: large graphs may be too expensive to process using exact deterministic algorithms, and loading these graphs into memory may not even be an option. The aim of our contribution in **Part** ii is to fill this gap. We discuss in Chapter 3 the theory of peterminantal point processes, which is an important tool we use in **Part** ii to derive connection-aware forest-based estimators for Tikhonov regularization and angular synchronization.

3

DETERMINANTAL POINT PROCESSES

Je suis passée aux maths impures.
— Odile Macchi[1]

A point process on some space $\mathcal{S}$ (*e.g.*, $\mathbf{R}^n$)[2] describes random configurations of points in $\mathcal{S}$. Point processes can be useful computational tools, and are a crucial tool to our developments in **Part** ii. With no regard for a precise definition, let us present two well-known examples.

- CCD cameras reconstruct an image by measuring the amount of photons incident on a discretized (array-like) surface [Healey and Kondepudy, 1994]. The intensity of the image on a given pixel depends on the *number of photons* hitting the corresponding cell of the array over a fixed exposition time, which follows a Poisson distribution[3]. The *spatial distribution of the photons* on the surface is an instance of a *Poisson point process* [Cox and Isham, 1980], and can be observed in characteristic shot noise patterns in the resulting image[4].
- A number of point processes exhibit *repulsivity*. For instance, trees in a forest do not lie too close to each other. This situation can be modeled using the *hard-core point processes* described by Matérn[5], obtained by sampling points from a Poisson point process before applying a thinning rule to enforce a *minimum distance between two points* [Cox and Isham, 1980].

1 See https://interstices.info/odile-macchi-au-bonheur-des-maths/.
2 More generally, $\mathcal{S}$ can be a Polish space [Guyon, 1995].
3 Named after French mathematician and physicist Siméon Denis Poisson, 1781−1840.
4 See for instance https://en.wikipedia.org/wiki/Shot_noise.
5 Swedish statistician Bertil Matérn, 1917−2007.

Applying a thinning rule is a practical way to enforce additional structure in the sample (in this case, repulsivity between points), but results in hard-to-analyze models. Another fruitful approach, introduced by Macchi[6] (to model the position of fermions "in a trap", among other phenomena[7]), consists in formally prescribing the *co-incidence probabilities* of subsets of points in a sample [Macchi, 2017]: depending on these probabilities, models both repulsive and mathematically tractable can be described, such as determinantal point processes.

A (discrete) *Determinantal Point Process* (DPP) on a finite set $\mathcal{S}$ is a probability distribution over $2^{\mathcal{S}}$, satisfying a condition on the co-incidence probabilities of points in each random sample $X \subseteq \mathcal{S}$, which is parametrized by its *marginal kernel*: an Hermitian matrix $K \in M_{|\mathcal{S}|}(\mathbf{C})$ whose eigenvalues must all lie in $[0, 1]$ [Hough et al., 2006][8]. Precisely, we say that $X \sim \mathrm{DPP}(K)$ if

$$\mathbf{P}_{\mathrm{DPP}(K)}(A \subseteq X) = \det(K)_{A,A} \tag{3.1}$$

for all $A \subseteq \mathcal{S}$, where $\det(K)_{R,C}$ is the minor of K restricted to its rows and columns indexed respectively by R and C. We write $\det(K)_{:,C}$ (resp. $\det(K)_{R,:}$) in case $R = \mathcal{S}$ (resp. $C = \mathcal{S}$). Such distributions encode for a straightforward notion of repulsivity: for $i, j \in \mathcal{S}$, we have

$$\begin{aligned} \mathbf{P}_{\mathrm{DPP}(K)}(\{i,j\} \subseteq X) &= \det(A)_{\{i,j\},\{i,j\}} && (3.2) \\ &= \mathbf{P}_{\mathrm{DPP}(K)}(\{i\} \subseteq X)\mathbf{P}_{\mathrm{DPP}(K)}(\{j\} \subseteq X) - |K_{i,j}|^2, && (3.3) \end{aligned}$$

so that $\mathbf{P}_{\mathrm{DPP}(K)}(\{i,j\} \subseteq X) \leq \mathbf{P}_{\mathrm{DPP}(K)}(\{i\} \subseteq X)\mathbf{P}_{\mathrm{DPP}(K)}(\{j\} \subseteq X)$, the probability of independently drawing i and j with probabilities $\mathbf{P}_{\mathrm{DPP}(K)}(\{i\} \subseteq X)$ and $\mathbf{P}_{\mathrm{DPP}(K)}(\{j\} \subseteq X)$ respectively. When $K_{i,j}$ depends on the distance of i and j in $\mathcal{S}$, this repulsivity is clearly visible in pictures, see for instance (E3) below.

First, one should rest assured that DPPs do exist. We list below a few examples (some of which are spectacular, but pertain to *non-discrete* DPPs, that we do not define properly).

(E1) If $K = \Lambda$ a diagonal marginal kernel, $\mathrm{DPP}(K)$ draws a subset $\{i_1, ..., i_r\} \subseteq \mathcal{S}$ of size $r \leq |\mathcal{S}|$ with probability proportional to $\prod_{k=1}^{r} \lambda_{i_k}$ (*a.k.a.* independent sampling).

(E2) Consider a finite, linearly ordered, set $\mathcal{S}$ (we take $\mathcal{S} = \{1, ..., n\}$ for simplicity), and a sequence $(x_1, x_2, ..., x_m)$ of m elements sampled uniformly and independently in $\mathcal{S}$. The set of indices i such that $x_{i+1} < x_i$ is a DPP over $\{2, ..., m\}$ [Borodin et al., 2010].

(E3) There are two famous results about the eigenvalues of the complex Ginibre Ensemble[9] (*i.e.*, $n \times n$ random matrices with entries drawn from the complex normal distribution $\mathcal{N}_{\mathbf{C}}\left(0, \frac{1}{n}\right)$): the first is their distribution converges to a circle-law in $\mathbf{C}$; the second is that *they are distributed as a (non-discrete) DPP* [Ginibre, 1965, Tao, 2023]. See Figure 3.1.
The distributions of eigenvalues in Gaussian Ensembles (Hermitian matrices with normal random entries) also follow a DPP, and converge to semi-circle laws [Tao, 2023].

6 French physicist and mathematician Odile Macchi, born 1943.
7 And still used in this context to this day, see *e.g.* [Dean et al., 2019].
8 More generally K need not be Hermitian, but one can show that it should remain a contraction operator for Equation (3.1) to hold. Further, under this condition, there always exists a DPP parametrized by K [Macchi, 2017].
9 Named after French mathematical physicist Jean Ginibre, 1938−2020.

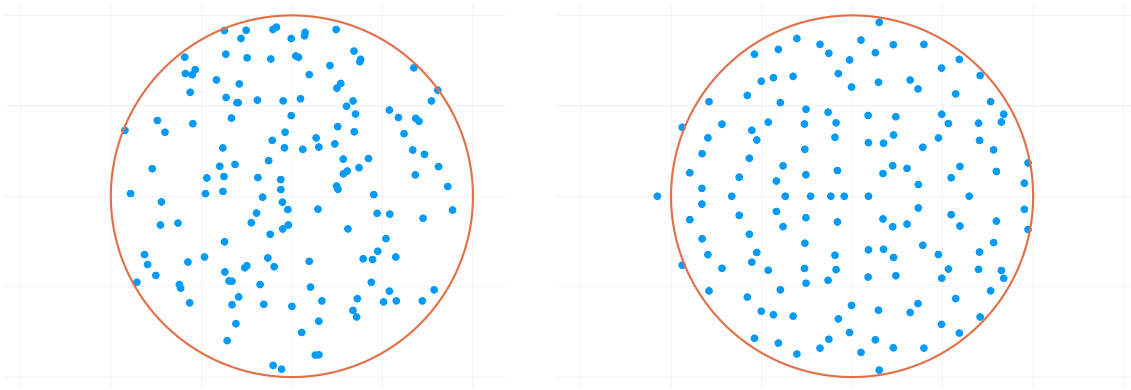


Figure 3.1: The complex unit circle (in orange), and randomly sampled points in $\mathbf{C}$. Left: i.i.d. uniform points in the complex unit disk. Right: eigenvalues sampled from a Ginibre Ensemble ($n = 150$).

(E4) The zeros of a randomly drawn power series $f : \mathbf{C} \to \mathbf{C}$ with coefficients $a_i(f) \sim \mathcal{N}_{\mathbf{C}}(0, 1)$ are distributed according to (non-discrete) DPP [Peres and Virág, 2005]. In this case, the (generalized) marginal kernel is the Bergman kernel[10] on $\mathbf{C}$.

The DPPs in Examples (E2), (E3) and (E4) above exhibit repulsivity of points in their samples but, if the underlying set $\mathcal{S}$ carries a combinatorial structure, some DPPs can encode for even more striking features, such as in the following example.

Theorem 3.0.1. *The uniform distribution over the spanning trees of a connected graph (*i.e.*, cycle-free subsets of edges* $\mathsf{t} \subseteq \mathcal{E} = \mathcal{S}$ *of size* $|\mathsf{t}| = |\mathcal{V}| - 1$*), denoted* $\mathcal{D}_{\mathcal{T}}$*, is a DPP with marginal kernel*

$$K_{\mathcal{T}} = \nabla(\nabla^t \nabla)^\dagger \nabla^t, \tag{3.4}$$

where ∇ *is the discrete differential operator from Chapter 1, and* $A^\dagger$ *denotes the Moore-Penrose*[11] *pseudo-inverse of a matrix* A *[Horn and Johnson, 2012].*

Theorem 3.0.1 provides a different perspective on the uniform distribution over spanning trees and is, in essence, a probabilistic re-formulation of Theorem 1.3.1 (and, arguably, the first non-trivial DPP to have been systematically studied). Its proof is instructive, and we present it in Section 3.2.

Second, beyond their appearance in many mathematical models, DPPs can also be *practically useful*. We mention some use-cases below.

- DPPs have been proposed as a tool to draw representative samples in larger datasets [Kulesza and Taskar, 2012], with applications such as document-summarization, or pose-estimation of multiple subjects in the same picture. Many inference tasks that are prohibitively expensive for alternative models (*e.g.*, Markov Random Fields[12]) remain tractable for DPPs (*e.g.*, computing marginals; see also Section 3.1).

10 The Bergman kernel, named after Polish-American mathematician Stefan Bergman (1895–1977), is the reproducing kernel in the RKHS of square-integrable holomorphic functions, for the "standard" L^2 inner product.

11 After American mathematician Eliakum Hastings Moore (1862–1932) and British mathematician and physicist Roger Penrose (born 1931).

12 Named after Russian mathematician Andreï Andreïevitch Markov, 1856–1922.

- The repulsivity of points in samples from DPPs can also be leveraged for Monte-Carlo integration. This idea was first investigated in [Bardenet and Hardy, 2020] (for continuously differentiable functions), and extended in [Coeurjolly et al., 2021][13].
- DPPs have also been used to build "coresets" [Tremblay et al., 2019], subsets of a whole dataset with the property that (up to a controlled error) a cost function only needs to be evaluated on the coreset instead of the complete dataset (a construction especially interesting for machine-learning applications on large datasets).

Finally, DPPs can also be applied to RandNLA. One such application is studied in [Derezinski and Warmuth, 2018, Derezinski and Mahoney, 2021], re-visiting an earlier algebraic result attributed to Jacobi [Jacobi, 1841][14] (see, *e.g.*, [Ben-Israel and Greville, 2003] and references therein[15]), which aims at solving linear least-squares problems of the form

$$\underset{w\in\mathbf{R}^p}{\operatorname{argmin}}\ \|Aw-y\|_2^2, \tag{3.5}$$

where $y\in\mathbf{R}^n$ (with $p\leq n$) and A is a full-rank $n\times p$ matrix with coefficients in $\mathbf{R}$. Denoting by $w_*\in\mathbf{R}^p$ the solution to the optimization problem in Equation (3.5) (which can be expressed as $(A^tA)^{-1}A^ty$), they show the following.

Theorem 3.0.2. *Consider a DPP with kernel $K_A=A(A^tA)^{-1}A^t$, then*

$$\mathbf{E}_{X\sim\mathrm{DPP}(K_A)}\left((A_{X,:})^{-1}y_{|X}\right)=w_*. \tag{3.6}$$

Theorem 3.0.2 holds more generally when replacing $\mathbf{R}$ with $\mathbf{C}$, and we re-enact in Section 3.3 an earlier proof from [Ben-Tal and Teboulle, 1990], which will be instructive for our developments in **Part** ii. Let us for now note that solving the linear least-squares problem from Equation (3.5) amounts to solving a linear system, an operation for which there is a number of existing efficient, high-quality approximate algorithms. As such, one should keep in mind that, in order for Theorem 3.0.2 to be practically useful, a number of technical conditions must be satisfied.

(C1) First, drawing a sample from $\mathrm{DPP}(K_A)$ should be inexpensive.

(C2) Second, computing $(A_{X,:})^{-1}y_{|X}$ should also be cheap.

(C3) Finally, the empirical mean should converge to the expectation quickly enough.

We will see that none of these conditions are satisfied *in general*, and we discuss Conditions (C1), (C2) and (C3) in Sections 3.1.2 and 3.3. On the other hand, *there are* practical situations in which the estimator from Equation (3.6) can largely outperform well-established solvers. Describing such a situation is one of the aims of **Part** ii of this manuscript, and Tikhonov smoothing of graph signals using RSFs (Section 1.2) is actually a special instance of the more general situation we describe.

In this chapter, we introduce some general background on DPPs that will be useful in the rest of the manuscript. We postpone the presentation of the specific DPPs we will be interested in in **Part** ii to Chapter 5.

13 Interestingly, these two works use *different* (non-discrete) kernels.
14 Carl Gustav Jakob Jacobi, Prussian mathematician, 1804−1851.
15 Interestingly, similar estimators had already been used in applications such as arc-measurements adjustment since at least the 18^{th} century, *e.g.*, in the work of Ragusan mathematician, physicist, astronomer, philosopher, poet, theologian and Jesuit priest Roger Joseph Boscovich (in english french; Ruggiero Giuseppe Boscovich in italian; Rogerius Iosephus Boscovicius in latin; Ruđer Josip Bošković in croatian), 1711−1787 [Sheynin, 1973].

3.1 GENERALITIES

Determinantal point processes satisfy a number of interesting properties. We begin with two elementary observations [Kulesza and Taskar, 2012].

Restriction. Let $S \subseteq \mathcal{S}$ and $X \sim \mathrm{DPP}(K)$. It immediately follows from the definition that the random subset $X \cap S$ also follows a DPP, with marginal kernel $K_{S,S}$.

Cardinality. The size of a DPP sample $|\mathcal{X}|$, which is itself a random quantity, is also tractable. Its first two moments are as follows:

$$\mathbf{E}_{X\sim\mathrm{DPP}(K)}(|X|) = \mathrm{tr}(K), \qquad \mathbf{Var}_{X\sim\mathrm{DPP}(K)}(|X|) = \mathrm{tr}(K - K^2). \tag{3.7}$$

See [Kulesza and Taskar, 2012] for a proof.

Remark 3.1.1. *We actually already encountered a consequence of these results In Chapter 1. One can show that the distribution $\mathcal{D}_{\mathcal{F}}$ over RSF (Section 1.2) is a DPP [Avena and Gaudillière, 2013], with marginal kernel*

$$K_{\mathcal{F}} = \begin{bmatrix} \nabla \\ \sqrt{q}\mathsf{I} \end{bmatrix} (\mathsf{L} + q\mathsf{I})^{-1} \begin{bmatrix} \nabla^* & \sqrt{q}\mathsf{I} \end{bmatrix}. \tag{3.8}$$

The restriction of $K_{\mathcal{F}}$ to its rows and columns indexed by $\mathcal{V}$ reads $(\mathsf{L}+q\mathsf{I})^{-1}q\mathsf{I}$, and describes a DPP over the nodes of RSFs (the distribution of the roots). Theorem 1.2.2 immediately follows: the number of roots $|\phi \cap \mathcal{V}|$ in a RSF drawn according to $\mathcal{D}_{\mathcal{F}}$ is an unbiased estimator of $\mathrm{tr}((\mathsf{L}+q\mathsf{I})^{-1}q\mathsf{I})$.

For now, we focus on an important special class of DPPs, with fixed-size samples.

3.1.1 *Projection Kernels*

The two most fundamental types of marginal kernels for DPPs are diagonal kernels (recall Example (E1) above) and *projection* kernels [Hough et al., 2006]. The goal of this short section is to give an introduction to DPPs defined from projection kernels.

Projection DPP. A linear operator $P : \mathbf{C}^n \to \mathbf{C}^n$ is a projection operator if it is idempotent, *i.e.* if $P^2 = P$. If P is Hermitian, $\ker P = \mathrm{im}P^\perp$ and P is the *orthogonal projection* on $\mathrm{im}P$, in which case its eigenvalues lie in $\{0, 1\}$ [Horn and Johnson, 2012]. When the marginal kernel K of a DPP is a projection operator, it is called a *projection DPP.*

There is one property of projection DPPs that sets them apart from more general DPPs.

Proposition 3.1.2. *If K is a projection operator and $X \sim \mathrm{DPP}(K)$, $|X| = \mathrm{rk}(K)$ almost surely, where $\mathrm{rk}(K)$ denotes the rank of K.*

Proposition 3.1.2 is a direct consequence of Equation (3.7). In words, it states that samples from a projection DPP have fixed cardinality.

This has a very concrete consequence: a projection DPP is a distribution over $\binom{\mathcal{S}}{\mathrm{rk}(K)}$, the subsets of $\mathcal{S}$ of size $\mathrm{rk}(K)$ (abusively extending the usual notation for combinations), *defined* by

$$\mathbf{P}_{\mathrm{DPP}(K)}(X) = \det(K)_{X,X}. \tag{3.9}$$

In particular, there exists a projection DPP associated to *any* projection kernel K.

Let us describe a practical way of building projection kernels.

Decomposition of projection kernels. Consider a linear operator $A : \mathbf{C}^p \to \mathbf{C}^n$, that can be represented as a matrix in $M_{n,p}(\mathbf{C})$ in a given basis. Denoting by A^* the adjoint of A, we can express the orthogonal projection on $\mathrm{im}A$ in $\mathbf{C}^n$ as [Horn and Johnson, 2012]:

$$P_{\mathrm{im}A} = A(A^*A)^{\dagger}A^*. \tag{3.10}$$

If the columns of $Q \in M_{n,p}(\mathbf{C})$ form an orthonormal basis of $\mathrm{im}A$ (*i.e.*, $Q^*Q = I$ and $\mathrm{im}A = \mathrm{im}Q$), we can even write:

$$P_{\mathrm{im}A} = QQ^*, \tag{3.11}$$

in which case a DPP with kernel $K = P_{\mathrm{im}A}$ will draw a sample $X \in \binom{\mathcal{S}}{\mathrm{rk}(K)}$ with probability

$$\mathbf{P}_{\mathrm{DPP}(K)}(X) = \det(Q)^2_{X,:}. \tag{3.12}$$

Note that the orthonormal basis Q may be constructed from the eigenvectors of K, but any other choice of basis would work, for instance obtained from a QR-decomposition [Horn and Johnson, 2012].

Remark 3.1.3. *Notice that, when A is full-rank, $P_{\mathrm{im}A}$ exactly takes the form of the kernel K_A in Theorem 3.0.2, and that the marginal kernel in Theorem 3.0.1 is actually the orthogonal projection on* $\mathrm{im}\nabla$.

Our interest in the rest of this manuscript mainly lies in projection DPPs. However, discussing properties of general DPPs is a natural way to introduce a pervasive tool in the theory of DPPs, and puts our work in a broader perspective. We take this opportunity to present a fundamental result [Hough et al., 2006], and its practical consequences.

3.1.2 *Mixture, Existence and Sampling*

The definition of DPPs we considered so far is frustrating: it does not provide much probabilistic intuition, and it is not clear whether a DPP associated to a given kernel K even exists (unless the kernel is diagonal, or a projection operator).

It turns out that general DPPs can be expressed as mixtures of projection DPPs, practically solving both of these problems [Hough et al., 2006, Kulesza and Taskar, 2012].

Theorem 3.1.4. *Fix a projection kernel $K = U\Lambda U^*$, with Λ the matrix of eigenvalues $\lambda_1, ..., \lambda_{|\mathcal{S}|}$ and U its matrix of eigenvectors. Consider the point process on $\mathcal{S}$ described as follows.*

(M1) First, draw a subset of indices $I = \{i_1, ..., i_r\} \subseteq \mathcal{S}$ *with probability* $\prod_{k=1}^{r} \lambda_k$.

(M2) Then, sample Y *from* $\mathrm{DPP}(K_{(I)})$ *where* $K_{(I)} = U_{:,I}(U_{:,I})^*$ *is a projection kernel.*

Then, Y *follows a DPP with marginal kernel* K.

The key tool in the proof of Theorem 3.1.4, and many other results concerning DPPs, is the Cauchy-Binet formula[16], which re-expresses the determinant of the product of two *non-square* matrices [Horn and Johnson, 2012, Muir, 1890].

Proposition 3.1.5. *For two matrices* $A \in M_{m,n}(\mathbf{C})$ *and* $B \in M_{n,m}(\mathbf{C})$ *with* $m < n$*, we have:*

$$\det(AB) = \sum_{T \in \binom{[n]}{m}} \det(A)_{:,T} \det(B)_{T,:}, \tag{3.13}$$

where $T \in \binom{[n]}{m}$ *ranges over all subsets of* $[n] = \{1, ..., n\}$ *of size* m.

From two successive applications of the Cauchy-Binet formula to the eigendecomposition $K = U\Lambda U^*$, one readily shows that, for all $A \subseteq \mathcal{S}$:

$$\det(U\Lambda U^*)_{A,A} = \sum_{I \in \binom{\mathcal{S}}{|A|}} \left(\prod_{i \in I} \lambda_i \right) \det(U)^2_{A,I}. \tag{3.14}$$

Recognizing in the left-hand-side (resp. right-hand-side) the probability that $A \subseteq X$ for $X \sim \mathrm{DPP}(K)$ (resp. $A \subseteq Y$ for Y the point process in Theorem 3.1.4), we obtain Theorem 3.1.4.

Let us stress once again the consequences of this result. First, $\mathrm{DPP}(K)$ exists for all Hermitian kernels K, and sampling from $\mathrm{DPP}(K)$ essentially boils down to sampling from a projection DPP (item (M2) in Theorem 3.1.4), with probabilities described in Equations (3.9) and (3.12). Further, we observe that $|X|$ is distributed as a sum of Bernoulli[17] variables $\mathcal{B}(\lambda_k)$ (see item (M1)).

In general, sampling from a DPP can be relatively expensive, and requires computing an eigendecomposition of K, in $\mathcal{O}\left(|\mathcal{S}| \times \mathrm{rk}(K)^2\right)$ operations, before sampling from $\mathrm{DPP}(K_{(I)})$ [Hough et al., 2006, Kulesza and Taskar, 2012]. Some applications only require sampling from a projection DPP (*e.g.*, Theorem 3.0.2): the algorithm from [Hough et al., 2006] specializes to this situation, but the computationally-expensive eigendecomposition must still be performed in this case, which can be prohibitive for runtime-sensitive applications (see, *e.g.*, [Barthelmé et al., 2023] for a self-contained analysis of the complexity).
Other DPP-sampling algorithms exist, such as those of [Barthelmé et al., 2023] (rejection-sampling-based), [Anari et al., 2022] (MCMC-based), or [Poulson, 2020] (for sparse kernels), with similar computational profiles. In general, DPPs are too expensive to sample for RandNLA applications.

This unfortunate observation only holds *in general*, and much faster sampling algorithms may exist for specific DPPs. We now discuss the important example of uniform spanning trees.

16 After French mathematicians Augustain Louis Cauchy (1789–1857) and Jacques Philippe Marie Binet (1786–1856).
17 Named after Swiss mathematician Jacob Bernoulli, 1655–1705.

3.2 A DETERMINANTAL PERSPECTIVE ON KIRCHHOFF'S MATRIX-TREE THEOREM

Our main goal in the following pages is to familiarize the reader with the systematic tools to study projection DPPs on graphs. We go through the proof of Theorem 3.0.1, and proceed to show that $\mathrm{DPP}(K_{\mathcal{T}})$ coincide with the uniform distribution over the spanning trees of $\mathcal{G}$. We assume to simplify notations that the graph is non-weighted, and outline key steps in the analysis that will be important in **Part** ii of this manuscript (multiple DPPs generalizing the uniform-spanning-tree distribution have been studied using similar tools, see *e.g.* [Lyons, 2009, Kassel and Lévy, 2022a]). We also briefly contrast the complexity of sampling algorithms for the uniform-spanning-tree distribution with the generic projection-DPP-samplers mentioned in Section 3.1.2.

Let us once again re-phrase Theorem 3.0.1.

Theorem 3.2.1. *Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a connected graph, and $K_{\mathcal{T}}$ and the orthogonal projection on* $\mathrm{im}\nabla$ *(∇ the discrete differential operator from Chapter 1). Then,* $\mathrm{DPP}(K_{\mathcal{T}})$ *(a DPP on $\mathcal{S} = \mathcal{E}$) is the uniform-spanning-tree distribution over $\mathcal{G}$.*

We always take $X \sim \mathrm{DPP}(K_{\mathcal{T}})$ in the remainder of this Section, and work with unweighted graphs for simplicity (all results generalize to the weighted case). We begin by computing $\mathrm{rk}(K_{\mathcal{T}})$:

$$\mathrm{tr}(K_{\mathcal{T}}) = \mathrm{tr}(\mathsf{L}\mathsf{L}^{\dagger}) = \mathrm{tr}(U\Lambda\Lambda^{\dagger}U^*) = |\mathcal{V}| - 1, \tag{3.15}$$

where we write $\mathsf{L} = U\Lambda U^*$ the eigendecomposition of L. The last equality holds because $\mathcal{G}$ is connected. It follows that, almost surely, $|X| = |\mathcal{V}| - 1$.

Our second step is a general lemma regarding projection DPPs.

Lemma 3.2.2. *Let $K = A(A^*A)^{\dagger}A^*$ be the orthogonal projection kernel onto* $\mathrm{im}A$*, where $A : \mathbf{C}^p \to \mathbf{C}^n$ is a linear operator, and $p \leq n$. We also denote by $A \in M_{n,p}(\mathbf{C})$ its matrix in a fixed basis. Note that, in this case, $\mathcal{S} = [n]$ and $\mathrm{rk}(K) \leq p$.*
Then, for all $X \in \binom{[n]}{\mathrm{rk}}$

$$\mathbf{P}_{\mathrm{DPP}(K)}(X) \propto \det(AA^*)_{X,X}. \tag{3.16}$$

Moreover, the normalizing constant is given by the product of non-zero eigenvalues of A^*A (if A^*A is invertible, this is $\det(A^*A)$).

Proof. First, write the eigendecomposition $A^*A = U\Lambda U^*$ (hence, $(A^*A)^{\dagger} = U\Lambda^{\dagger}U^*$). From two successive applications of the Cauchy-Binet formula, we then obtain, for $X \in \binom{[n]}{\mathrm{rk}}$:

$$\begin{aligned}\mathbf{P}_{\mathrm{DPP}(K)}(X) &= \sum_{T \in \binom{[n]}{\mathrm{rk}}} \sum_{S \in \binom{[n]}{\mathrm{rk}}} \det(AU)_{X,T} \, \det(\Lambda^{\dagger})_{T,S} \, \det((AU)^*)_{S,X} && (3.17)\\ &= \sum_{T \in \binom{[n]}{\mathrm{rk}}} \det(\Lambda^{\dagger})_{T,T} \, \det(A((U_{:,T})(U_{:,T})^*)A^*)_{X,X}, && (3.18)\end{aligned}$$

where the second line holds since, if $\Lambda^{\dagger}$ is diagonal, $\det(\Lambda^{\dagger})_{T,S} = 0$ whenever $T \neq S$.
Finally, note that $\det(\Lambda^{\dagger})_{T,T}$ is only non-zero when T indexes the non-zero eigenvalues, in which case the columns of $U_{:,T}$ form an orthonormal basis of $\mathrm{im}A$, and we obtain

$$\mathbf{P}_{\mathrm{DPP}(K)}(X) = \det(\Lambda^{\dagger})_{T,T} \, \det(AA^*)_{X,X}. \tag{3.19}$$

(a) An edge-signal $f \in \mathbf{C}^{\mathcal{E}}$.

(b) The associated signal $\sigma^v f \in \mathbf{C}^{\mathcal{V}}$.

Figure 3.2: Construction of the signal $\sigma^v f \in \mathbf{C}^{\mathcal{V}}$ (right) from a base signal $f \in \mathbf{C}^{\mathcal{V}}$ (both integer-valued for simplicity). The reference node v is depicted as a large black node (right).

■

Remark 3.2.3. *Remark that, when* rk $= p$*, we are dealing with a product of square* $p \times p$ *matrices, and we simply have:*

$$\mathbf{P}_{\mathrm{DPP}(K)}(X) = \frac{\det(AA^*)_{X,X}}{\det(A^*A)}, \tag{3.20}$$

by multiplicativity of the determinant. In our case, ∇ *is rank-deficient, and we need the more general result from Lemma 3.2.2*

Since $\det(AA^*)_{X,X} = \det(A_{X,:}(A^*)_{:,X})$, it follows from Lemma 3.2.2 that $\mathrm{im} A_{X,:} = \mathbf{C}^n$ almost surely. When $A = \nabla$, this has a strong consequence on the structure of $X \subseteq \mathcal{E}$.

Structure of the Samples: Hearing Cycles in a Graph. Let us first note that the restricted differential operator $\nabla_{X,:}$ is nothing but the differential operator associated to a restricted graph $\mathcal{G}_X = (\mathcal{V}, X)$, that we denote ∇_X for simplicity. We argue in the following that, much like the number of connected components of a graph corresponds to the number of zero eigenvalues of its Laplacian, the absence of zero eigenvalues in the spectrum of $\nabla_X \nabla_X^*$ is characteristic of X being a tree[18]. We rely on classical arguments; see, *e.g.* [Biggs, 1993].
Like we previously mentioned, we only need derive a condition under which $\mathrm{im}\nabla_X = \mathbf{C}^{\mathcal{E}}$. We proceed in two steps.

1. *If X is cycle-free, then* $\mathrm{im}\nabla_X = \mathbf{C}^{\mathcal{E}}$. Indeed, consider for any vertex $v \in \mathcal{V}$ the map $\sigma_X^v : \mathbf{C}^{\mathcal{E}} \to \mathbf{C}^{\mathcal{V}}$, defined by its action on signals $f \in \mathbf{C}^{\mathcal{E}}$:

$$(\sigma_X^v f)(i) = \sum_{e \in v \to i} f(e), \tag{3.21}$$

where $v \to i \subseteq X$ denotes the unique path in X between v and i. The value $(\sigma_X^v f)(i)$ on any node i is nothing but the sum of the values of f on the edges encountered on the path from v to i, as illustrated on a concrete example in Figure 3.2. Clearly, $\nabla_X \sigma_X^v f = f$ for all signals $f \in \mathbf{C}^{\mathcal{E}}$, hence, $\mathrm{im}\nabla_X = \mathbf{C}^{\mathcal{E}}$.

18 The study of topological and geometrical properties of manifolds by way of linear operators associated to these manifolds, spectral geometry, is a classical subject in physics and mathematics [Kac, 1966, Berger et al., 1971]. The spectral theory of graphs has also received significant attention [Chung, 1997], and we present but one very specific result.

2. *If X contains a cycle,* $\ker \nabla_X$ *is non-trivial, and* $\mathrm{im}\nabla_X \neq \mathbf{C}^{\mathcal{V}}$. For simplicity[19], let us further assume that $|X| = |\mathcal{V}| - 1$ (as we have already shown for the DPP we consider). In this case, the result follows from a simple cardinality constraint: since X contains a cycle, it can only span a subset of nodes $\mathcal{V}_X \subseteq \mathcal{V}$ of size at most $|\mathcal{V}| - 1$ nodes, and the function $f \in \mathbf{C}^{\mathcal{V}}$ defined by

$$f(i) = \begin{cases} a \text{ for some } a \in \mathcal{C} \setminus \{0\} \text{ if } i \in \mathcal{V} \setminus \mathcal{V}_X \\ 0 \text{ otherwise} \end{cases} \tag{3.22}$$

is such that $\nabla_x f = 0$, that is, it is a non-trivial element of $\ker \nabla_X$.

It follows that $\mathbf{P}_{\mathrm{DPP}(K_{\mathcal{T}})}(X) > 0$ if and only if the graph $\mathcal{G}_X = (\mathcal{V}, X)$ is cycle-free. In particular, if $\mathcal{G}_X$ is connected, almost surely, it must be a tree.

Remark 3.2.4. *Since* $\det(AA^*)_{X,X} = \det(A_{X,:}(A^*)_{:,X})$ *being strictly positive is also equivalent to the kernel* $\ker(A^*)_{:,X}$ *being trivial almost surely, the tree structure of the samples from* $\mathrm{DPP}(K_{\mathcal{T}})$ *can similarly be inferred from a dual argument, by leveraging ideas from combinatorial topology [Eckmann, 1944, Lyons and Peres, 2017]*[20]*. So far, we emphasized that ∇ captures variational properties of signals defined on $\mathcal{V}$; it turns out its adjoint ∇^* is a natural tool to reason about combinatorial properties such as the presence of cycles in $\mathcal{G}$.*
Given an orientation of the edges of $\mathcal{E}$, one can consider $\mathbf{C}$*-linear combinations of edges in $\overrightarrow{\mathcal{E}}$, called* 1*-chains, where we impose that the sum of two reversely-oriented versions of the same edge $e, \overline{e} \in \overrightarrow{\mathcal{E}}$ is zero:*

$$e + \overline{e} = 0.$$

Formally, a 1*-chain c_1 is an element of* $\left(\bigoplus_{e \in \overrightarrow{\mathcal{E}}} \mathbf{C}e\right)\big/\!\sim$*, where $\sim$ represents the relation defined by $e \sim e'$ if and only if $e' = e^*$.*
One can also write $\mathbf{C}$*-linear combinations c_0 of nodes in $\mathcal{V}$, called* 0*-chains, such that* $c_0 \in \bigoplus_{v \in \mathcal{V}} \mathbf{C}v$*. The crux of the argument then lies in relating* 1*-chains to* 0*-chains by way a* boundary *operator* B*, namely* $\mathsf{B} = \nabla^*$*. The boundary operator* B *maps* 1*-chains to* 0*-chains, and satisfies* $\mathsf{B}e = v' - v$ *for all edges* $e = (v, v')$*. As a consequence, for a path* $p = \sum_{i=1}^{m} e_i$ *with* $e_i = (v_i, v_{i+1})$*, we have* $\mathsf{B}p = v_{m+1} - v_1$*. If p is a directed cycle, it follows that* $\mathsf{B}p = v_1 - v_1 = 0$*. See Figure 3.3 for a visual representation of* 1*-chains and their boundaries.*
In words, if a 1*-chain c describes a cycle,* $\mathbf{C}c \subseteq \ker \mathsf{B}$*. One can further show that, if a graph $\mathcal{G}$ does not contain any cycle, then* $\ker \mathsf{B} = \{0\}$ *(see* e.g. *[Lyons and Peres, 2017, Lim, 2020] for more details). In words, the kernel of* B *is trivial as long $\mathcal{G}$ does not contain a cycle, and we can conclude in the same manner as before.*
Note that $\mathrm{rk}(K_{\mathcal{T}})$ *can also be computed using similar arguments (see Chapter 7), and that this perspective is the one more commonly employed to analyze combinatorially meaningful DPPs [Lyons, 2009, Lyons and Peres, 2017, Kassel and Lévy, 2022a].*

Together with the cardinality contsraint $|X| = |\mathcal{V}| - 1$, this shows that, almost surely, samples from $\mathrm{DPP}(K_{\mathcal{T}})$ are *spanning* trees. The last step of the proof consists in computing $\det(\nabla\nabla^*)_{X,X}$.

19 The result can be proved in greater generality, without relying on this hypothesis. Further, one can show that the dimension of the kernel $\ker \nabla_X$ is equal to the number of *algebraic cycles*. See Remark 3.2.4.

20 In the specific setting we consider, the theory was mostly developed by Kirchhoff to reason about currents in electrical networks.

(a) $\mathsf{B}(e_1 + e_2) = v_5 - v_1$.

(b) $\mathsf{B}(e_1 + e_2 - e_3 - e_4) = 0$.

Figure 3.3: Boundary of 1-chains on the 2×3 grid graph. Edges belonging to the 1-chain are represented in purple, and the associated boundaries (0-chains) as large purple nodes.

Lemma 3.2.5. *Let $X \subseteq E$ be a spanning tree. Then*

$$\det(\nabla\nabla^*)_{X,X} = |\mathcal{V}|. \tag{3.23}$$

Proof. From the Cauchy-Binet formula, we have

$$\det(\nabla\nabla^*)_{X,X} = \sum_{T \in \binom{\mathcal{V}}{|\mathcal{V}|-1}} \det(\nabla)^2_{X,T}, \tag{3.24}$$

and we finally conclude by computing $\det(\nabla)_{X,T}$: by the pigeonhole principle there can only be one non-zero permutation in the determinant

$$\det(\nabla)_{X,T} = \sum_{\sigma \in \mathcal{S}_{|\mathcal{V}|-1}} \prod_{1 \le i \le m} \operatorname{sgn}(\sigma)(\nabla_{X,T})_{i,\sigma(i)}, \tag{3.25}$$

where $\operatorname{sgn}(\sigma)$ denotes the signature of the permutation σ (note that, in our case, this permutation is actually a bijection between nodes and edges of $\mathcal{G}$), and $\det(\nabla)_{X,T} = \pm 1$. ■

Remark 3.2.6. *As a corollary, we also obtain that the distribution $\mathcal{D}_{\mathcal{F}}$ over RSFs is a DPP with kernel $K_{\mathcal{F}}$ (from Equation (3.8)). This is an example of a DPP with a fast sampling algorithm, Wilson's algorithm, which samples forests in $\mathcal{O}\left(|\mathcal{V}| + \frac{|\mathcal{E}|}{q}\right)$ time in expectation. In contrast, the classical DPP sampler from [Hough et al., 2006] requires $\mathcal{O}((|\mathcal{V}| + |\mathcal{E}|)^3)$ operations.*

The analysis we presented in this Section can be generalized to study other DPPs on graphs, and we describe in **Part** ii a generalization of the RSF distribution to graphs with a $U(\mathbf{C})$-connection. These distributions also admit efficient sampling algorithms, which makes them suitable candidates to apply Theorem 3.0.2 (recall Condition (C1)).

3.3 LEAST-SQUARES PROBLEMS

We further discuss Theorem 3.0.2 in this section. We start with a concise proof, originally due to [Ben-Tal and Teboulle, 1990]. This proof significantly differs from the one in [Derezinski and Warmuth, 2018], and reflects the methodology we use in **Part** ii.

We first re-state Theorem 3.0.2, and consider more general complex coefficients in $\mathbf{C}$.

Theorem 3.3.1. *Consider the linear least-squares problem*

$$\operatorname*{argmin}_{w \in \mathbf{C}^p} \|Aw - y\|_2^2, \tag{3.26}$$

with $y \in \mathbf{C}^n$, and A a full-rank $n \times p$ ($p \leq n$) matrix with coefficients in $\mathbf{C}$. Denote by w_* the solution of the problem in Equation (3.26), and consider the projection DPP associated to the orthogonal projection K_A over $\mathrm{im}A$. Then, we have

$$\mathbf{E}_{X \sim \mathrm{DPP}(K_A)} \left((A_{X,:})^{-1} y_{|X} \right) = w_*. \tag{3.27}$$

The proof only requires two tools: the Cauchy-Binet formula, and a general form of Cramer's rule, recalled below [Horn and Johnson, 2012].

Proposition 3.3.2. *Let $B \in M_{n,n}(\mathbf{C})$ be an invertible matrix, and $W, Y \in M_{n,m}(\mathbf{C})$ ($n, m \geq 1$). Then, the solution $W_* \in M_{n,m}$ to the matrix equation $BW = Y$ satisfies*

$$\det(W_*)_{I,J} = \frac{B[Y, I, J]}{\det(B)} \tag{3.28}$$

for all $I \in \binom{[n]}{k}$, $J \in \binom{[m]}{k}$, where $B[Y, I, J]$ denotes the matrix obtained from B by replacing its columns indexed by I by those of Y indexed by J.

In the specific instance that $w^* \in \mathbf{C}^n$ is the solution to the system $Bw = y$, with $B \in M_{n,n}(\mathbf{C})$ an invertible matrix and $y \in \mathbf{C}^n$, we have:

$$w_*(i) = \frac{\det(B[y, i])}{\det(B)}, \tag{3.29}$$

where $B[y, i] = B[y, \{i\}, \{1\}]$ denotes the matrix obtained from B by replacing its i-th column with y.
As a corollary, the entries of B^{-1} can be expressed as:

$$(B^{-1})_{i,j} = (-1)^{i+j} \det(B)_{[n]\setminus\{j\},[n]\setminus\{i\}}. \tag{3.30}$$

By "Cramer's rule", we refer to any of these three results, and will refer to the precise equation. Proving Theorem 3.0.2 is then very straightforward.

Proof of Theorem 3.0.2. First, note that w_* must satisfy the equation[21]:

$$A^* A w_* = A^* y. \tag{3.31}$$

21 This is a classical result for matrices with real coefficients [Horn and Johnson, 2012]. For matrices with entries in $\mathbf{C}$, this can be shown from standard **CR**-calculus arguments. We provide an explicit derivation in Chapter 2 in the case of the connection-aware graph Tikhonov smoothing problem.

Writing Cramer's rule (from Equation (3.29)), it follows that:

$$w_*(i) = \frac{\det((A^*A)[A^*y, i])}{\det(A^*A)} \tag{3.32}$$

$$= \frac{\det(A^*(A[y, i]))}{\det(A^*A)} \tag{3.33}$$

$$= \frac{1}{\det(A^*A)} \sum_{X \in \binom{[n]}{p}} \det(A^*)_{:,X} \det(A_{X,:}[y_{|X}]) \tag{3.34}$$

$$= \sum_{X \in \binom{[n]}{p}} \mathbf{P}_{\mathrm{DPP}(P_A)}(X)(A_{X,:})^{-1} y_{|X}, \tag{3.35}$$

where the last equation is again a consequence of Cramer's rule (Equation (3.28), and of Proposition 3.2.2. Equation (3.34) is obtained from the Cauchy-Binet formula. ∎

Generic algorithms to compute $((A_{X,:})^{-1}y'_{X,:}$ have $\mathcal{O}(n^3)$ cost, which is too expensive for Theorem 3.3.1 to be practically useful (recall Condition (C2)). We present in the following remark a noteworthy instance of cheaply-invertible system, pertaining to graph Tikhonov-regularization.

Remark 3.3.3. *If we consider a graph $\mathcal{G}$ together with a graph signal y embedded into $\mathbf{R}^{\mathcal{V}\cup\mathcal{E}}$ (that is, $y' = [y\ 0]^t$), and take $A = \begin{bmatrix} \nabla \\ \sqrt{q} \end{bmatrix}$, with $q \in \mathbf{R}_+^*$, Theorem 3.3.1 reads.*

Corollary 3.3.4. *Let ϕ denote a spanning forest sampled according to $\mathcal{D}_{\mathcal{F}}$. Then,*

$$\mathbf{E}_{\phi \sim \mathcal{D}_{\mathcal{F}}}((A_{\phi,:})^{-1}y'_{\phi,:}) = \operatorname*{argmin}_{f \in \mathbf{R}^{\mathcal{V}}} \|Af - y'\|^2 \tag{3.36}$$

$$= \operatorname*{argmin}_{f \in \mathbf{R}^{\mathcal{V}}} q\|f - g\|^2 + \langle f, \mathsf{L}f \rangle, \tag{3.37}$$

where $g = \frac{1}{\sqrt{q}}y$.

In words, Tikhonov-regularization of a graph-signal amounts to a linear least-squares problem of the type considered in Theorem 3.0.2, and is thus amenable to DPP-based Monte-Carlo estimation. In order to compute $((A_{\phi,:})^{-1}y'_{\phi,:}$, one can proceed much more efficiently than in the general case: expliciting the entries of $(A_{\phi,:})^{-1}$ using Cramer's rule (Equation (3.30)), we exactly recover Theorem 1.2.1, and an estimation can be computed in time $\mathcal{O}(|\mathcal{V}|)$ for each forest ϕ[22]*. Note that the same expression can be obtained by applying an adequate permutation of the entries of $A_{\phi,:}$ instead of Cramer's rule.*

Let us now briefly discuss Condition (C3): how soon is the empirical mean

$$\mu(X_1, ..., X_m) = \frac{1}{m} \sum_{X_i} \mathbf{P}_{\mathrm{DPP}(P_A)}(X_i)(A_{X,:})^{-1} y_{X,:}, \tag{3.38}$$

a good approximation of its mean $\mathbf{E}_{X \sim \mathrm{DPP}(P_A)}((A_{X,:})^{-1}y_{X,:}) = w_*$? Here, the sum is over m i.i.d. DPP samples $X_1, ..., X_m \sim \mathrm{DPP}(K_A)$. The generic answer is given, for real-valued random

22 This is a roundabout way of proving Theorem 3.0.2. See Chapter 6 for a more direct argument, in the case of graphs with a $U(\mathbf{C})$-connection.

variables, by the Central Limit Theorem [Fischer, 2011][23], and is quite unsatisfying (for simplicity, we assume that all our variables are real in the following lines): $\sqrt{m}(\mu_m(X_1, ..., X_m)(i) - w_*(i))$ converges in distribution to $\mathcal{N}_{\mathbf{R}}(0, \sigma_i^2)$, where σ_i^2 is the variance of $((A_{X,:})^{-1} y_{X,:})(i)$.
In other words, the error in the estimation of each entry $w_*(i)$ is a random variable centered in 0, whose concentration depends on $\mathcal{O}\left(\frac{\sigma}{\sqrt{m}}\right)$, the standard convergence rate of Monte-Carlo estimators [Hammersley, 2013]. Consequently, depending on the variance of the estimator, a large number of samples may be required to reach a desired precision. For the methods we propose in **Part** ii, as for RSF-based methods, the variance of the estimator is too large, and the convergence rate is too slow to compete with state-of-the-art deterministic solvers. We modify these estimators to have lower variance, building on the works [Pilavcı et al., 2021] and [Pilavcı et al., 2022b].

We finally mention a proper generalization of Kirchhoff's network theorem (Chapter 1). *The following discussion will not be used in the remainder of this manuscript*, but helps to paint a more complete picture of how DPP-based estimators can be used in numerical linear algebra.

Mean Projection. Consider a linear operator $A : \mathbf{C}^p \to \mathbf{C}^n$, and the orthogonal projection P_A onto $\mathrm{im}A$. The mean-projection theorem considers the DPP with marginal kernel $K_A = A(A^*A)^{-1}A^*$ (*i.e.*, the *orthogonal* projection onto $\mathrm{im}A$), and reads as follows [Maurer, 1976, Lyons, 2003, Catanzaro et al., 2012, Kassel and Lévy, 2022b].

Theorem 3.3.5. *Suppose that $X \subseteq [n]$ follows the distribution $\mathrm{DPP}(K_A)$. Then, almost surely, the following Fredholm-alternative-like decomposition holds*[24]*:*

$$\mathbf{C}^n = \mathrm{im}A \oplus \left(\bigoplus_{x \in X} \mathbf{C}\delta_x \right)^{\perp}, \tag{3.39}$$

where $\{\delta_x\}_{x\in[n]}$ denotes a fixed orthonormal basis of $\mathbf{C}^n$ (used to express A and K_A). Further, denoting by P_X the projection onto $\mathrm{im}A$ parallel to[25] $\left(\bigoplus_{x\in X} \mathbf{C}\delta_x\right)^{\perp}$, *one has*

$$\mathbf{E}_{X\sim \mathrm{DPP}(K_A)}(\mathrm{P}_X) = K_A \tag{3.40}$$

or, as expressed in the usual basis of $\mathbf{C}^n = \bigoplus_{x\in[n]} \mathbf{C}\delta_x$, and for all $g \in \mathbf{C}^n$:

$$\mathbf{E}_{X\sim \mathrm{DPP}(K_A)} \left(A(A_{X,:})^{-1} c_{|X} \right) = K_A g. \tag{3.41}$$

That is, one can perform a DPP-based estimation of $K_A = A(A^*A)^{\dagger}A^*$, obtained as an average over non-orthogonal projections and, in particular, one can compute $K_A g$ for any $g \in \mathbf{C}^n$.
When A is full-rank, Theorem 3.0.2 is obtained as a corollary of Theorem 3.3.5. Indeed, one has

$$(A^*A)^{-1}A^*g = \mathbf{E}_{X\sim \mathrm{DPP}(K_A)}((A_{X,:})^{-1} g_{|X}) \tag{3.42}$$

23 For complex-valued random variables, one can apply this result to the real and imaginary parts respectively. Stronger results regarding convergence to the complex normal distribution $\mathcal{N}_{\mathbf{C}}(0, 1)$ also exist. See *e.g.* `https://terrytao.wordpress.com/2010/01/05/254a-notes-2-the-central-limit-theorem/`.

24 The (linear-algebraic) Fredholm altenative, named after Swedish mathematician Ivar Fredholm (1866−1927), states that $\mathbf{C}^n = \mathrm{im}A \oplus \ker A^*$.

25 A projection onto a subspace E_1 of $\mathbf{C}^n$ *parallel* to some supplementary E_2 (also *oblique* projection) is an operator $P : \mathbf{C}^n \to \mathbf{C}^n$ such that $P^2 = P$ and is defined, for all vectors $x = x_1 + x_2$ (with $x_1 \in E_1$ and $x_2 \in E_2$ following the decomposition $\mathbf{C}^n = E_1 \oplus E_2$), by $Px = x_1$. The difference with an *orthogonal* projection is that E_1 and E_2 need not be orthogonal. In particular, if B_1 denotes a matrix whose columns form a basis of E_1, and $B_2^{\perp}$ a matrix whose columns are a basis of the orthogonal complement of E_2, it is straightforward to show that $P = B_1((B_2^{\perp})^*B_1)^{-1}B_2^{\perp}$ (where $((B_2^{\perp})^*B_1)$ is always invertible by construction) [Hogben, 2006].

which, multiplied on the left by A, yields

$$K_A c = A\, \mathbf{E}_{X \sim \mathrm{DPP}(K_A)}\big((A_{X,:})^{-1} g_{|X}\big). \tag{3.43}$$

When A is not full-rank, Theorem 3.3.5 no longer a consequence of Theorem 3.0.2, but can still be applied, which is crucial in, *e.g.*, Kirchhoff's original network theorem [Nerode and Shank, 1961].

Remark 3.3.6. *In particular, one can show that Kirchhoff's network theorem (Chapter 1) is nothing but the corresponding estimation of the orthogonal projection of the imposed voltage g onto the kernel of ∇^*, that is, on the class of signals for which Kirchhoff's laws are satisfied. Recalling that* $\ker \nabla^* = \mathrm{im} \nabla$, *the corresponding kernel in this case is the uniform-spanning-tree kernel* $K_{\mathcal{T}} = \nabla(\nabla^* \nabla)^\dagger \nabla^*$.

3.4 SUMMARY

This concludes **Part** i of this manuscript, in which we have introduced a number of key concepts.

- We introduced in Chapter 1 a number of problems on graphs, and reviewed existing RandNLA methods for GSP problems, based on RSFs.
- We covered in Chapter 2 the case of graphs endowed with a connection, and specific associated problems. One important take-away from this chapter is that the spectrum of L_θ encodes useful information that can be used to solve smoothing and synchronization problems.
- Finally, we discussed determinantal point processes in Chapter 3. DPPs are nicely structured distributions, that implement a natural notion of repulsivity, and remain theoretically tractable. It turns out that the RSF distribution $\mathcal{D}_{\mathcal{F}}$ from Chapter 1 is a DPP, and that the RSF estimators from Theorems 1.2.1 and 1.2.2 follow from generic DPP properties. The computational applications of DPPs in RandNLA (specifically, Theorem 3.0.2) are unfortunately hindered by stringent restrictions, that we recall below.

  (C1) Sampling from a DPP is in general expensive.

  (C2) Computing solutions of the form $(A_{X,:})^{-1} y_{X,:}$ is also in general expensive.

  (C3) Even when both previous conditions can be alleviated, the approximation may remain of poor quality due to high variance.

  In the case of RSFs, Conditions (C1) and (C2) are not a problem, but condition (C3) typically is [Pilavcı et al., 2021]. It turns out that a slight variation on the variance-reduction technique introduced in [Pilavcı et al., 2022b] can largely improve the behavior of the estimators (see our development in Chapter 6).

Our main contribution in **Part** ii of this manuscript is to develop DPP-based RandNLA techniques for graphs endowed with a $U(\mathbf{C})$-connection. Specifically, we introduce a DPP over *multi-type spanning forests* (a variant of RSFs adapted to $U(\mathbf{C})$-connections), and develop forest-based estimators generalizing those of [Pilavcı et al., 2021]. We also develop improved variance-reduction techniques, and empirically show that our methods can outperform standard deterministic solvers for both the connection-aware Tikhonov smoothing problem, and the angular synchronization problem (from Chapter 2).

Part II

DETERMINANTAL POINT PROCESSES FOR $U(\mathbf{C})$-CONNECTIONS

4

RANDOM WALKS ON GRAPH BUNDLES

The most fruitful areas for the growth of the sciences were those which had been neglected as a no-man's land between the various established fields.
— Norbert Wiener[1]

Contributions in this chapter.

1. Introduction of novel random-walk-based estimators for connection-aware smoothing and interpolation. These estimators apply to $U(\mathbf{C})$-connections and to more general $O(\mathbf{R}^d)$-connections (Section 4.1).

2. In the case of $U(\mathbf{C})$-connections, discussion of a variance-reduction strategy (Section 4.2). This development allows to relate these simple estimators to the more involved ones we describe in Chapters 5 and 6 (this relation is made explicit in Chapter 7).

Random processes on $\mathbf{C}$-bundles, and on more general graph-bundles, have been the topic of a growing number of works in mathematics, the study of these phenomena often being motivated by applications in (statistical and mathematical) physics; see, *e.g.*, [Brydges et al., 1979, Forman,

1 See [Wiener, 2019].

1993, Kenyon, 2011, Kassel, 2015, Güneysu et al., 2016, Kassel and Lévy, 2021]. As a mean of contextualization, the reader may keep in mind that related models are classically used to study the behavior of electrons immersed in a magnetic field [Bellissard, 1992, Lieb and Loss, 1993].

To our knowledge, this line of research has yet to permeate into more computational sciences. This in clear contrast with the flurry of applications for well-behaved random processes defined on connection-free graphs (*e.g.*, [Avrachenkov et al., 2016] and references therein for random walks, or [Avena et al., 2018, Barthelmé et al., 2019, Avena et al., 2021, Pilavcı et al., 2021] for random forests), whereas we are only aware of two use-cases for non-trivial connections.

1. Ranking and community-detection in signed graphs [Jung et al., 2020, Neumann and Peng, 2022], leveraging connection-aware random walks and, more precisely, statistics of the form
$$\prod_{k=0}^{l-1} \sigma_{(u_k, u_{k+1})} \tag{4.1}$$
for some sampled paths $p = ((u_0, u_1), (u_1, u_2), ..., (u_{l-1}, u_l))$, and where we recall that $\sigma_e \in \{-1, 1\}$ is the sign of edge e.

2. Sparsification of (regularized) connection Laplacians [Fanuel and Bardenet, 2022], as obtained by drawing samples from a connection-aware forest-based distribution. Let us note that, in this work, the connection Laplacian is dubbed the *magnetic* Laplacian instead, following the tradition in mathematical physics.

Our work is related to both of these techniques.

More specifically, we develop throughout **Part** ii a number of RandNLA estimators based on *multi-type spanning forests*, that apply to, *e.g.*, connection-aware Tikhonov regularization and angular synchronization. Let us recall that a connection is a collection of maps $\Psi = (\psi_e)_{e \in \mathcal{E}}$ between isomorphic copies $\mathbf{C}_i$ of the complex plane $\mathbf{C}$ associated to each node $i \in \mathcal{V}$ (see Figure 4.1, a reproduction of Figure 2.2 from Chapter 2, that we will re-use for illustration purposes later in this chapter). Informally, a multi-type spanning forest is a subset of edges and nodes spanning the entire graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, the connected components of which form either rooted trees or unicycles (*i.e.*, connected subsets of edges containing exactly one cycle). We consider a distribution $\mathcal{D}_\mathcal{M}$ over the set of all multi-type spanning forests of $\mathcal{G}$ (see Chapter 5 for a more formal definition), which turns out to be a determinantal point process over $\mathcal{S} = \mathcal{V} \cup \mathcal{E}$. Our estimators are then based on the composition of connection maps along the branches of the trees, which takes the form of a transport map between the two endpoints of the path:
$$\prod_e \psi_e. \tag{4.2}$$
For the connection-aware Tikhonov smoothing problem (from Chapter 2), we are for instance able to show the following (which is proved in Chapter 6).

Theorem 4.0.1. *Let $\mathcal{G}$ be a graph endowed with a $U(\mathbf{C})$-connection, and $g \in \mathbf{C}^\mathcal{V}$ a noisy signal on $\mathcal{G}$. For a given multi-type spanning forest $\phi \subseteq \mathcal{V} \cup \mathcal{E}$ and two nodes a, b spanned by a rooted tree of ϕ, denote by $a \xrightarrow{\phi} b$ the set of edges of the unique path from a to b along edges of ϕ. If $v \in \mathcal{V}$ further belongs to a rooted tree of ϕ, denote by $r_\phi(v) \in \mathcal{V}$ the root of this tree.*
Consider then the statistics
$$\widetilde{f}(i, \phi, g) = \begin{cases} \left(\prod_{e \in a \xrightarrow{\phi} b} \psi_e\right) g(r_\phi(i)) \text{ if } i \text{ is spanned by a rooted tree,} \\ 0 \text{ if } i \text{ is spanned by a unicycle,} \end{cases} \tag{4.3}$$

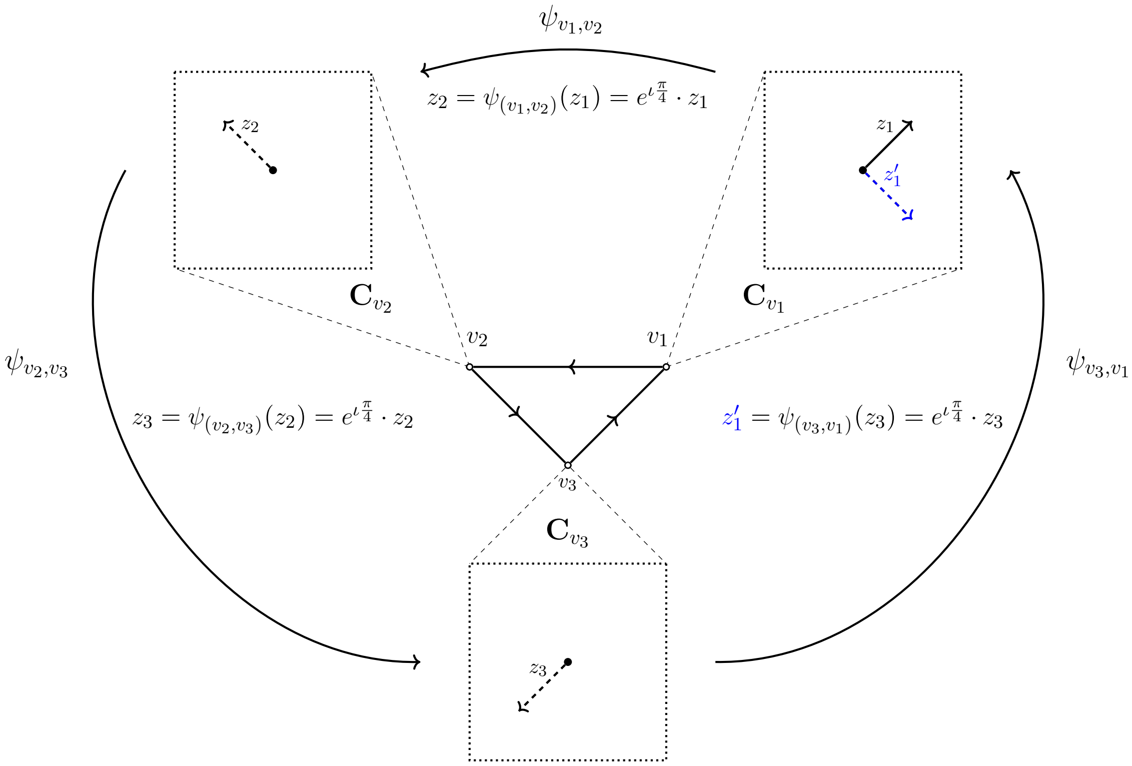


Figure 4.1: A $U(\mathbf{C})$-connection Ψ on the triangle graph K_3 (in the center). The connections maps ψ_e are rotations with associated angles $\theta_{(v_1,v_2)} = \theta_{(v_2,v_3)} = \theta_{(v_3,v_1)} = \frac{\pi}{4}$, and we represent their action on some $z_1 \in \mathbf{C}_{v_1}$. A cyclic path $C = ((v_1,v_2),(v_2,v_3),(v_3,v_1))$ is depicted using directed arrows, we denote by $\psi_C = \psi_{(v_3,v_1)} \circ \psi_{(v_2,v_3)} \circ \psi_{(v_1,v_2)}$ the composition of the connection maps along this cycle (a rotation of $\theta_C = \frac{3\pi}{4}$). Top right: $z_1 \in \mathbf{C}_{v_1}$ (bold arrow) and $z'_1 = \psi_C(z_1) \in \mathbf{C}_{v_1}$ (blue dotted arrow). Top left, bottom: $z_2 = \psi_{(v_1,v_2)}(z_2) \in \mathbf{C}_{v_2}$ and $z_3 = \psi_{(v_2,v_3)}(z_2) \in \mathbf{C}_{v_3}$ (dotted arrows).

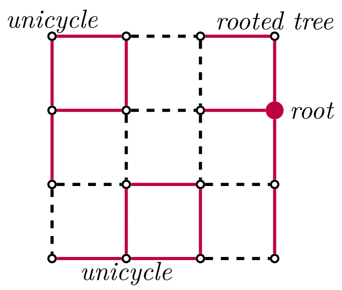
where a node i is spanned by a rooted tree (resp. a unicycle) of ϕ if it is an endpoint of some edge e belonging to a rooted tree (resp. a unicycle) in ϕ, or a root.
Then, the solution $f_* = (\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}g$ of the connection-aware Tikhonov smoothing problem reads

$$f_*(i) = \mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}(\widetilde{f}(i,\phi,g)), \tag{4.4}$$

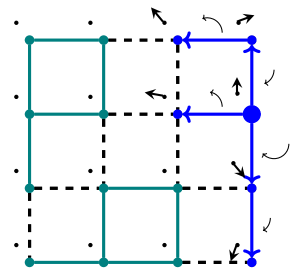
where the expectation is over the multi-type spanning forests ϕ sampled according to some probability distribution $\mathcal{D}_\mathcal{M}$*, that we introduce in Chapter 5.*

This translates to a straightforward estimation procedure.

1. Sample a MTSF ϕ distributed according to $\mathcal{D}_\mathcal{M}$.
2. Then, for each node $i \in \mathcal{V}$:



(a) A MTSF in purple, with two unicycles (top left, bottom) and one tree (right). The large purple node is the root of the tree.



(b) Estimation: we transport the values from the root (blue circle) to its tree via the blue arrows. The estimation is 0 on the unicycles (represented by arrowless dots at the top left of the nodes).

Figure 4.2: Theorem 4.0.1 illustrated on the 4×4 grid graph. Left: a MTSF. Right: estimation by propagations along the branches of the MTSF. The node-wise estimates (in $\mathbf{C}$) are represented as vectors, depicted as arrows (top left of the nodes), and the connection maps are represented as circular rotating arrows (only depicted along the edges of the unique tree in the MTSF we consider).

- if i is spanned by a rooted tree, transport the value $g(r_\phi(i))$ from its root $r_\phi(i)$ back to i, along the unique branch of ϕ linking the two:

$$\widetilde{f}(i, \phi, g) = \psi_{r_\phi(i) \xrightarrow{\phi} i}(g(r_\phi(i))); \tag{4.5}$$

- if i is spanned by a unicycle, set

$$\widetilde{f}(i, \phi, g) = 0. \tag{4.6}$$

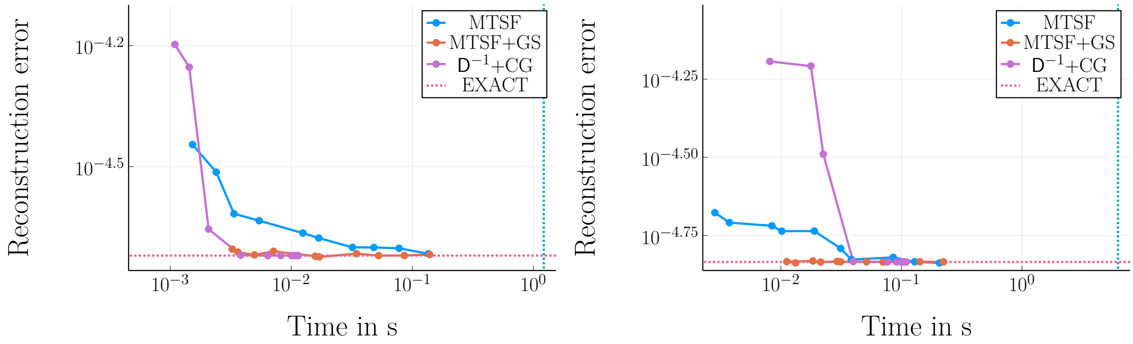
Theorem 4.0.1 ensures that we find f_* in expectation.
We illustrate this estimation procedure in Figure 4.2, on the 4×4 grid-graph.

One broader *leitmotif* throughout this second part of the manuscript is the development of randomized estimators for the entries of matrices $\left(\mathcal{L}_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}$, where $\mathcal{L}$ denotes some type of Laplacian operator, and $\mathcal{A} \subseteq \mathcal{V}$ is a subset of nodes in the graph.
For most of our results, $\mathcal{L} = \mathsf{L}_\theta$ is the connection Laplacian for a graph endowed with a $U(\mathbf{C})$-connection, and Theorem 4.0.1 is obtained from these techniques. In particular, when the connection is trivial, $\mathsf{L}_\theta = \mathsf{L}$ and we recover the forest-based estimator discussed in Chapter 1.
Some of our techniques also generalize to Laplacians $\mathcal{L}$ corresponding to situations where:

1. the graph is endowed with a more general connection, so that the connection maps ψ_e represent rotations in d-dimensional spaces;

2. the graph is replaced by a *simplicial complex*, which no longer encodes pairwise relationships (in the form of edges), but more general k-way relationships.

Let us remark that these more general connections appear in, *e.g.*, structure-from-motion problems [Tron et al., 2016], and that a number of signal-processing problems on cellular complexes have been emerging in the past few years [Barbarossa and Sardellitti, 2020, Schaub et al., 2021].



(a) DC-SBM with mean degree $\overline{d} \simeq 30$.

(b) DC-SBM with mean degree $\overline{d} \simeq 420$.

Figure 4.3: Runtime-precision comparisons for the graph Tikhonov smoothing problem, comparing two multi-type-spanning-forest-based estimators (MTSF and MTSF+GS) against a conjugate-gradient iteration with Jacobi preconditioner (D^{-1}+CG) on two different DC-SBM models, with $|\mathcal{V}| \simeq 10^4 = 10,000$. Results are averaged over 10 random-graph realizations for each of the two models, resulting in different average degrees $\overline{d}$: on the left, average degree $\overline{d} \simeq 30$; on the right, average degree $\overline{d} \simeq 420$. x-axis: average runtime. y-axis: average reconstruction error. Each data-point corresponds to a number $m \in \{1, 2, 3, 5, 8, 13, 22, 36, 60, 100\}$ of multi-type spanning forests (or of conjugate-gradient iterations) used for the estimation. The vertical (resp. horizontal) line is the computation-time (resp. error) obtained from an exact Cholesky-decomposition-based solver.

Our aim in this chapter is to introduce and study a class of simpler estimators, that will serve for finer analysis of those of Chapter 6 and Chapter 7. Specifically, we develop two Feynman-Kac estimators based on random walks[2], for both the problem of connection-aware smoothing and a connection-aware harmonic interpolation problem (Propositions 4.1.2 and 4.1.1). We focus on $U(\mathbf{C})$-connections, but discuss generalizations to more general connections along the way.
Our results in this chapter are related to those of [Brydges et al., 1979, Kassel, 2015, Kassel and Lévy, 2021]. Specifically, a different form of an identity we obtain in Section 4.1 (Equation (4.48)) has been previously established, *for continuous-time random walks*. Our developments are *inspired from this result*, but pertain to discrete-time walks, allow for simple proofs, and are tailored towards practical problems and applications, which are all novel contributions. Further, our proofs generalize to cellular complexes (a completely original setting, see Chapter 7 and Appendix E), and the discrete-time setting will allow to establish formal links between these Feynman-Kac estimators and Theorem 4.0.1 (which will follow from the discussions in Section 4.2, and are discussed in Chapter 7).

The remainder of **Part** ii specifically pertains to $U(\mathbf{C})$-connections, and is organized as follows.

- In Chapter 5, we establish a number of properties for the distribution $\mathcal{D}_{\mathcal{M}}$ over multi-type spanning forests. In particular, we show the following.

2 "The" Feynman-Kac formula, named after American physicist Richard Phillips Feynman (1918–1988) and Polish-American mathematician Mark Kac, expresses the solution of a specific partial differential equation as the expectation of a stochastic process. The term is more loosely used to describe results expressing the solution of variational problems as expectations, and can also be considered a generalized theoretical framework for walk-on-spheres algorithms.

1. $\mathcal{D}_{\mathcal{M}}$ is a determinantal point process, which we obtain from a strategy similar to the one we used in Chapter 1 to analyze the uniform-spanning-tree DPP[3].
2. Under a technical condition, samples of $\mathcal{D}_{\mathcal{M}}$ can be drawn using an efficient Wilson-like sampling algorithm (Algorithm 4, Chapter 4), with complexity linear in the size of the graph.

- We develop applications of this theory in Chapter 6. In particular, we prove a generalized version of Theorem 4.0.1, develop a number of variance-reduction techniques, empirically compare the resulting estimators to standard deterministic techniques, and further leverage these estimators to solve angular-synchronization problems.

- Our perspectives in Chapter 7 concern mathematical developments that pave the way to further applicability of our techniques. In particular, we discuss ways to lift, in some cases, the technical condition on the Wilson-like sampling algorithm, and the generalizations to cellular complexes. The determinantal theory of multi-type spanning forests does not generalize nicely to this setting, and we instead rely on observations laid-out in Section 4.2.

While our developments in these Chapters 4 and 5 are a bit lengthy, the reader shall be rewarded in Chapter 6. We tease in Figure 4.3 the resulting performance of two forest-based estimators derived from Theorem 4.0.1 (using different variance-reduction techniques), on two different graphs, as measured on instances of the connection-aware Tikhonov smoothing problem, and using our Julia implementation[4] (see Chapter 6 for details on the experimental setup). We observe that our methods have performance similar to that of conjugate-gradient-based solvers but, depending on the topology of the graph (specifically, if it is not *very sparse* and has heterogeneous degree-distribution) can reach solutions of quality equivalent to that of an exact solver much faster (the gain getting larger as the density increases).

3 Similar techniques are also employed in [Lyons, 2009] and [Kassel and Lévy, 2022a], but tailored towards a more mathematical audience.
4 https://gricad-gitlab.univ-grenoble-alpes.fr/gaia/synchromtsf

4.1 FEYNMAN-KAC ESTIMATORS

Elementary random processes can be used to express the solutions of complex problems, and we already mentioned some examples, such as the solution to the harmonic interpolation discussed in Chapter 1, or their "continuous-space"-counterpart, resulting in walk-on-spheres algorithms. It turns out that simple random-walk-based processes can also be leveraged to solve problems on graphs equipped with a connection. It is our goal in this chapter to describe two so-called Feynman-Kac formulas, for the connection-aware Tikhonov smoothing problem (from Chapter 2) and for a connection-aware harmonic interpolation problem (the solution of which we encountered in Equation (2.19) from Chapter 2). For us, a Feynman-Kac formula is nothing but the expression of the solution to variational problems as expectations of stochastic processes, that can then be used to perform Monte-Carlo estimations of the solutions.

4.1.1 *Definitions*

The specific stochastic processes we consider are based on random walks on graphs endowed with a set of "boundary" nodes, and the resulting estimation on a node i depends on the connection maps encountered along the edges of paths from i to this boundary. For clarity, let us preface our exposition by a few definitions.

- A *path* p in $\mathcal{G}$ is an ordered sequence of oriented edges in $\overrightarrow{\mathcal{E}}$. Formally, $p \in \overrightarrow{\mathcal{E}}^*$, where $\overrightarrow{\mathcal{E}}^* = \bigcup_{k\geq 0} \overrightarrow{\mathcal{E}}^k$, and $\overrightarrow{\mathcal{E}}^0 = \{\epsilon\}$ contains the "empty" path ϵ (for a digression on notations and terminology, see, *e.g.*, [Lothaire, 2005]).
- The *orientation-reversal* p^* of a path $p = ((u_0, u_1), (u_1, u_2), ..., (u_{l(p)-1}, u_{l(p)}))$ of length $l(p)$ is the path
$$p^* = ((u_{l(p)}, u_{l(p)-1}), ..., (u_1, u_0)). \tag{4.7}$$
- The *connection map* ψ_p associated to a path $p \in \overrightarrow{\mathcal{E}}^*$ is defined inductively: if $p = qr$ is the concatenation of two paths $q \in \overrightarrow{\mathcal{E}}^*$ from i to j and $r \in \overrightarrow{\mathcal{E}}^*$ from j to k, then qr is a path from i to k, and the resulting map $\psi_{qr} : \mathbf{C}_i \to \mathbf{C}_k$ (where we recall that $\mathbf{C}_v$ denotes an isomorphic copy of $\mathbf{C}$ associated to node i in the $\mathbf{C}$-bundle structure) is defined as
$$\psi_{qr} = \psi_q \circ \psi_r. \tag{4.8}$$
More concretely, for a path $p = ((u_0, u_1), (u_1, u_2), ..., (u_{l(p)-1}, u_{l(p)}))$, we end up with
$$\psi_p(z) = \left(\prod_{i=0}^{l(p)-1} \psi_{(u_i,u_{i+1})}\right)(z) \tag{4.9}$$
$$= e^{\left(\iota \sum_{i=0}^{l(p)-1} \theta_{u_i,u_{i+1}}\right)} \cdot z. \tag{4.10}$$
Note that, given these definitions, one has $\psi_p^* = \psi_{p^*}$ and, following our convention for connection maps on nodes, we often identify ψ_p with $e^{\left(\sum_{i=0}^{l(p)-1} \theta_{u_i,u_{i+1}}\right)}$ when there is no risk of misinterpretation.

In addition, we denote by $P_i^j \subseteq \overrightarrow{\mathcal{E}}^*$ the set of paths from node i to node j in a given graph and, more generally, $P_i^{\mathcal{A}} \subseteq \overrightarrow{\mathcal{E}}^*$ the set of paths from i to $\mathcal{A} \subseteq \mathcal{V}$. For consistency, we set $\psi_\epsilon = \mathrm{id}$.

4.1.2 *Description of the Estimators*

These definitions at hand, we are now able to state our results.

Connection-aware Interpolation. Let us first recall the connection-aware harmonic-interpolation problem which asks, given a $\mathbf{C}$-signal $g : \mathcal{A} \to \mathbf{C}$ defined on a subset of nodes $\mathcal{A} \subseteq \mathcal{V}$, to find a signal $f \in \mathbf{C}^{\mathcal{V}}$ such that

$$\begin{cases} f(i) = g(i) \text{ if } i \in \mathcal{A}, \\ (\mathsf{L}_\theta f)(i) = 0 \text{ if } i \in \mathcal{V} \setminus \mathcal{A}. \end{cases} \tag{4.11}$$

We denote by $\mathrm{h}_\theta(g) \in \mathbf{C}^{\mathcal{V}\setminus\mathcal{A}}$ the unique signal satisfying both conditions, which can be compactly expressed on $\mathcal{V} \setminus \mathcal{A}$ using the connection Laplacian[5]:

$$\mathrm{h}_\theta(g) = -((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}})^{-1}(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\, g. \tag{4.12}$$

This problem has appeared as a practical way to compute extensions of vector-fields (see Chapter 2), and could also be applied to, *e.g.*, angular synchronization with a few anchor nodes, for which the value ω_i is *a priori* known (as considered in, *e.g.*, [Boumal et al., 2014, Cucuringu, 2016]). For each node $i \in \mathcal{V}$, we then consider natural random walks on $\mathcal{G}$, that start from i and stop upon reaching a node $a \in \mathcal{A}$. More precisely, the probability of transitioning from node u to node v is given by

$$\frac{w_{u,v}}{d_u}, \tag{4.13}$$

which defines a probability measure $\nu_i^{\mathcal{A}}$ over $P_i^{\mathcal{A}}$:

$$\mathbf{P}_{\nu_i^{\mathcal{A}}}(p) = \mathbf{1}_{\{i\}}(u_0)\mathbf{1}_{\mathcal{A}}(u_l) \prod_{0\le k<l} \left(\mathbf{1}_{\mathcal{V}\setminus\mathcal{A}}(u_k)\frac{w_{u_k,u_{k+1}}}{d_{u_k}} \right), \tag{4.14}$$

where $p = ((u_0, u_1), ..., (u_{l-1}, u_l))$ is a path of length l, and we recall that $\mathbf{1}_S(i)$ denotes the indicator that $i \in S$, such that

$$\mathbf{1}_S(i) = \begin{cases} 1 \text{ if } i \in S, \\ 0 \text{ otherwise.} \end{cases} \tag{4.15}$$

We can then state our first Feynman-Kac estimator.

Proposition 4.1.1. *Let $i \in \mathcal{V}$ and $g : \mathcal{A} \to \mathbf{C}$ be a signal defined on some subset of nodes $\mathcal{A} \subseteq \mathcal{V}$ intersecting all the connected components of $\mathcal{G}$. For a path p sampled according to $\nu_i^{\mathcal{A}}$, denote by $a \in \mathcal{A}$ the endpoint of p. Then, for all $i \in \mathcal{V} \setminus \mathcal{A}$,*

$$(\mathrm{h}_\theta(g))(i) = \mathbf{E}_{p\sim\nu_i^{\mathcal{A}}}\left(\psi_{p^*}(g(a))\right) \tag{4.16}$$

In other words, one can estimate $\mathrm{h}_\theta(g)(i)$ as follows. See Algorithm 2 for a pseudo-code.

1. Run a random walk starting from i until reaching some node $a \in \mathcal{A}$ belonging to the boundary, and denote by p the resulting path.

5 Existence and uncity of the solution follow from elementary linear-algebraic manipulations. See Lemma 4.1.8 and Appendix B.1 for details.

Algorithm 2 Feynman-Kac smoothing estimator (Proposition 4.1.1).

$f_i \leftarrow 0$
Repeat m times
  Sample a path $p \in P_i^{\mathcal{A}}$ from $\nu_i^{\mathcal{A}}$ ▷ By running a random walk on $\mathcal{G}$
  $f_i \leftarrow f_i + \psi_{p^*}(g(a))$ ▷ a the first node of $\mathcal{A}$ encountered by p
Output $\frac{1}{m} f_i$

2. *Transport* the value $g(a)$ back to node i according to the connection map $\psi_p^* : \mathbf{C}_j \to \mathbf{C}_i$ associated to the path p^*; that is, set the estimation
$$f(i) = \psi_{p^*}(g(a)) \tag{4.17}$$
on node i.

In expectation, Proposition 4.1.1 precisely ensures that we find exactly $(\mathrm{h}_\theta(g))(i)$.

Let us note that Proposition 4.1.1 is but a slight variation on the classical estimator for harmonic interpolation in the trivial-connection case (Proposition 4.1.1), and accounting for the connection brings about a key difference:

- one no longer only considers the endpoint a, but the specific path p from i to a;
- in turn, it is necessary to perform parallel transport from a to i, along the path p.

Observations of this type actually hold true for all the results we discuss in this chapter and in Chapter 6.

Our second result is a variant of this strategy adapted to the connection-aware Tikhonov smoothing problem.

Connection-Aware Smoothing. Recall the connection-aware Tikhonov smoothing problem. For a noisy signal $g \in \mathbf{C}^{\mathcal{V}}$ (*e.g.*, of the form $g = f_\top + \varepsilon$ with $f_\top$ and underlying B-bandlimited signal an $\varepsilon \sim \mathcal{N}(0, \sigma^2 \mathsf{I})$), find the solution f_* of
$$\underset{f \in \mathbf{C}^{\mathcal{V}}}{\operatorname{argmin}} \; q\|f - g\|_2^2 + \langle f, \mathsf{L}_\theta f \rangle, \tag{4.18}$$
where q is a *positive* regularization parameter. The solution f_* to this problem reads
$$f_* = (\mathsf{L}_\theta + q\mathsf{I})^{-1} q\mathsf{I}\; g. \tag{4.19}$$

We consider for each node i a natural random walk on the extended graph $\mathcal{G}^\Gamma$ (which, we recall, is the graph with set of nodes $\mathcal{V}^\Gamma = \mathcal{V} \cup \{\Gamma\}$ and set of weighted edges $\mathcal{E}^\Gamma = \mathcal{E} \cup \bigcup_{v \in \mathcal{V}} \{\{v, \Gamma\}\}$, with weights $w_{v,\Gamma} = q$), starting from i and stopping when reaching Γ. The transition probability from node u to node v is given by
$$\frac{w_{u,v}}{d_u^\Gamma}, \tag{4.20}$$
where d_u^Γ denotes the degree of node u *in the extended graph* $\mathcal{G}^\Gamma$. Alternatively, and denoting by d_u the degree of u *in the graph* $\mathcal{G}$, transitions to another node v occur with probability
$$\begin{cases} \frac{w_{u,v}}{d_u + q} & \text{if } v \in \mathcal{V}, \\ \frac{q}{d_u + q} & \text{if } v = \Gamma. \end{cases} \tag{4.21}$$

Algorithm 3 Feynman-Kac smoothing estimator (Proposition 4.1.2).

1: $f_i \leftarrow 0$
2: **Repeat** m times
3: Sample a path $p \in P_i^\Gamma$ from ν_i^Γ ▷ By running an interrupted random walk on $\mathcal{G}$
4: $f_i \leftarrow f_i + \psi_{p_\Gamma^*}(g(j))$ ▷ j the last node before interruption of p
5: **Output** $\frac{1}{m} f_i$

Note that these two expressions for the transition probabilities are the same, since $d_u^\Gamma = d_u + q$ and $w_{u,\Gamma} = q$.
This process defines a probability measure ν_i^Γ over P_i^Γ:

$$\mathbf{P}_{\nu_i^\Gamma}(p) = \mathbf{1}_{\{i\}}(u_0)\mathbf{1}_{\{\Gamma\}}(u_l) \prod_{0 \leq k < l} \left(\mathbf{1}_{\mathcal{V}\setminus\{\Gamma\}(u_k)} \frac{w_{u_k,u_{k+1}}}{d_{u_k} + q_{u_k}} \right) \tag{4.22}$$

where $p = ((u_0, u_1), ..., (u_{l-1}, u_l))$ is a path of length l. Note that the resulting path $p \in P_i^\Gamma$ is always a concatenation of the form

$$p = p_\Gamma \ (j, \Gamma), \tag{4.23}$$

where $p_\Gamma \in P_i^j$, j is the last node in p before reaching the boundary Γ, and (j, Γ) is the last edge joining the two.
We then show the following Feynman-Kac formula.

Proposition 4.1.2. *Let $g \in \mathbf{C}^\mathcal{V}$ be a noisy signal signal on $\mathcal{G}$ and $i \in \mathcal{V}$. Writing $p = p_\Gamma \ (j, \Gamma)$ the same decomposition as above (for p a path sampled according to ν_i^Γ), we have*

$$f_*(i) = \mathbf{E}_{p\sim\nu_i^\Gamma} \left(\psi_{p_\Gamma^*}(g(j)) \right). \tag{4.24}$$

A simple estimation strategy ensues, which can be summarized as follows (see also Algorithm 3).

1. Run a random walk starting fron node i *on the extended graph* $\mathcal{G}^\Gamma$, until the boundary Γ is reached. Denote by p the resulting path, and by $p_\Gamma \in P_i^j$ its restriction to $\mathcal{G}$.
2. *Transport* the value $g(j)$ on the last node j reached before hitting Γ to node i:

$$f(i) = \psi_{p_\Gamma^*}(g(j)). \tag{4.25}$$

In expectation, Proposition 4.1.2 guarantees that we obtain $f_*(i)$.

One should note that Propositions 4.1.1 and 4.1.2 are similar. Before proceeding with the proof, let us point out an important underlying relation between the two problems.

Remark 4.1.3. *The similarity between the two Feynman-Kac formulas boils down to the fact that the solutions to the interpolation and smoothing problems share a common structure. Let L_θ^Γ denote the connection Laplacian of the extended graph $\mathcal{G}^\Gamma$, as considered in the connection-aware Tikhonov smoothing problem; note that we never defined the connection maps on the edges of the form $\{v, \Gamma\}$, and we can take the trivial connection $\psi_{v,\Gamma} = \mathrm{id}_{\mathbf{C}_v,\mathbf{C}_\Gamma}$ (which has no bearing on what follows). The harmonic interpolation and Tikhonov regularization read as follows.*

- *For the interpolation problem:*

$$\mathrm{h}_\theta(g) = \left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1} \left(-(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\right) g, \tag{4.26}$$

where $g \in \mathcal{A} \to \mathbf{C}$ is a signal to be extended.

- *For the smoothing problem, one should note that L_θ^Γ has the following block-structure:*

$$\mathsf{L}_\theta^\Gamma = \begin{bmatrix} \mathsf{L}_\theta + q\mathsf{I} & -q\mathbf{1} \\ -q\mathbf{1} & q|\mathcal{V}| \end{bmatrix}, \tag{4.27}$$

where $\mathbf{1}$ denotes the all-ones vector; in particular, we have

$$(\mathsf{L}_\theta^\Gamma)_{\mathcal{V}} = \mathsf{L}_\theta + q\mathsf{I}. \tag{4.28}$$

It follows that the solution to the smoothing problem reads

$$f_* = \left((\mathsf{L}_\theta^\Gamma)_{\mathcal{V}}\right)^{-1} q\mathsf{I}\ g_{TS} \tag{4.29}$$

for a noisy signal $g_{TS} \in \mathbf{C}^{\mathcal{V}}$

The similarity between the estimators of Propositions 4.1.1 and 4.1.2 is due to the common factor of the form $\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}$ (resp. $\left((\mathsf{L}_\theta^\Gamma)_{\mathcal{V}}\right)^{-1}$, in the specific case of the extended graph $\mathcal{G}^\Gamma$) appearing in both expressions, as shall become clear in the following proofs.

A second point we would like to raise is that our result actually hold in greater generality, in what concerns the nature of the connection. The following remark reviews the setting in which this generalization holds.

Remark 4.1.4. *The problems of interpolation and smoothing can also be extended to more general connections. Consider indeed a graph endowed with a connection $\Psi = (\psi_e)_{e\in\mathcal{E}}$ such that every map ψ_e acts by multiplication with a $O(\mathbf{R}^d)$-matrix, where $d \in \mathbf{N}$:*

$$\psi_e(x) = R_e \cdot x; \tag{4.30}$$

note that we do not a priori *restrict the space in which the signal x lives, as one may take,* e.g., *$x \in \mathbf{R}^d$ or $x \in O(\mathbf{R}^d)$. The case $d = 3$ is more common in applications, with the maps ψ_e's representing 3-dimensional rotations (with $R_e \in SO(\mathbf{R}^3)$ for each $e \in \mathcal{E}$); in that case, $x \in \mathbf{R}^3$ may represent the position of an object in space, whereas $x \in SO(\mathbf{R}^3)$ would represent another rotation (as encountered in synchronization problems; see* e.g.*, [Bandeira et al., 2013]). In concordance with our convention for $U(\mathbf{C})$-connections, we abuse notation and identify ψ_e with R_e.*
In that setting, one defines the (matrix of the) connection Laplacian L_Ψ by

$$(\mathsf{L}_\Psi)_e = \begin{cases} w_e R_{e^*} & \text{if } e \in \mathcal{E}, \\ 0 & \text{otherwise}, \end{cases} \tag{4.31}$$

which we regard as a $|\mathcal{V}| \times |\mathcal{V}|$ matrix with entries in the algebra of $d \times d$ matrices $M_d(\mathbf{C})$. Interpolation and smoothing are then defined as follows.

- Interpolation. *For a given signal g defined over a subset of nodes $\mathcal{A} \subseteq \mathcal{V}$, find a signal f such that*
$$\begin{cases} f(i) = g(i) \text{ if } i \in \mathcal{A}, \\ (\mathsf{L}_\Psi f)(i) = 0 \text{ if } i \in \mathcal{V} \setminus \mathcal{A}. \end{cases} \tag{4.32}$$

- Smoothing. *Given a noisy signal g, find the solution to the problem*
$$\underset{f}{\text{argmin}} \;\; q\|f - g\|^2_{HS^{\mathcal{V}}} + \frac{1}{2} \sum_{e \in \overrightarrow{\mathcal{E}}} w_e \|f(t_e) - R_e f(s_e)\|^2_{HS}, \tag{4.33}$$
where $\|.\|_{HS}$ is the norm associated to the Hilbert-Schmidt inner-product[6] $(x, y) \mapsto \langle x, y \rangle_{HS}$ *for $x, y \in M_d(\mathbf{C})$ (*i.e.*, the Frobenius norm*[7]*), and $\|.\|_{HS^{\mathcal{V}}}$ is the norm associated to the inner-product defined by $\langle f, g \rangle_{HS^{\mathcal{V}}} = \sum_{i \in \mathcal{V}} \langle f(i), g(i) \rangle_{HS}$. Note that we have*
$$\langle f, \mathsf{L}_\Psi f \rangle_{HS^{\mathcal{V}}} = \frac{1}{2} \sum_{e \in \overrightarrow{\mathcal{E}}} w_e \|f(t_e) - R_e f(s_e)\|^2_{HS}. \tag{4.34}$$

Let us stress that both problems make sense regardless of whether $g : \mathcal{A} \to \mathbf{R}^d$ (resp. $g \in (\mathbf{R}^d)^{\mathcal{V}}$) or $g : \mathcal{A} \to O(\mathbf{R}^d)$ (resp. $g \in O(\mathbf{R}^d)^{\mathcal{V}}$).
It is then possible to show that the solutions to these problems are unique, and respectively read
$$\mathrm{h}_\Psi(g) = -\left((\mathsf{L}_\Psi)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1} \left(\mathsf{L}_\Psi\right)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}} \; g \tag{4.35}$$
for the interpolation problem, and
$$f_* = (\mathsf{L}_\Psi + q\mathsf{I})^{-1} q\mathsf{I} \; g \tag{4.36}$$
for the smoothing problem[8]*.*
Propositions 4.1.1 and 4.1.2 and our proofs directly translate to this setting: one only needs to consider the corresponding connection maps ψ_p.

4.1.3 *Proofs*

We now focus on the proof of Proposition 4.1.1. The proof of Proposition 4.1.2 is obtained as a by-product (from Remark 4.1.3), along with the generalization to the setting of Remark 4.1.4.

Let us begin by recalling a well-known property of the random-walk Laplacian $\Delta = \mathsf{D}^{-1}\mathsf{L}$ (in the trivial-connection case) [Harary, 1966], which goes as follows.

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6 Named after German mathematician David Hilbert (1862−1943) and Baltic-German mathematician Erhard Schmidt (1876−1959).

7 Named after German mathematician Ferdinand Georg Frobenius, 1849−1917.

8 In order to show these results, one has to establish a number of properties. In particular, all the matrices involved are invertible, as follows from Remark 4.1.9 below. For the harmonic-interpolation solution, the argument for $U(\mathbf{C})$-connections then applies *verbatim* to general connections. For the smoothing problem, one should further check that L_Ψ is self-adjoint for the corresponding inner-product; this is easy to show, and we refer the reader to, *e.g.*, [Kassel and Lévy, 2021] for a general treatment for $\mathbf{V}$-bundles. This property established, the argument for $U(\mathbf{C})$-connections extends, but one no longer considers Fréchet-Wirtinger derivatives and instead relies on matrix derivatives [Petersen et al., 2008] (alternatively, the result also follows from a lengthy but direct computation, which would also apply to $U(\mathbf{C})$-connections).

Proposition 4.1.5. *Consider the natural random walk on $\mathcal{G}$ starting from node i, and let $j \in \mathcal{V}$ be some node of $\mathcal{G}$, and $l \in \mathbf{N}^*$.*
Then, $((\mathsf{I}-\Delta)^l)_{i,j}$ is the probability that a path of length l starting at node i ends up in node j:

$$((\mathsf{I}-\Delta)^l)_{i,j} = \sum_{\substack{p \in P_i^j \\ l(p)=l}} \left(\prod_{0 \leq k < l} \frac{w_{u_k,u_{k+1}}}{d_{u_k}} \right), \tag{4.37}$$

where $l(p)$ denotes the length of the path $p = ((i = u_0, u_1), (u_1, u_2), ..., (u_{l(p)-1}, u_{l(p)} = j))$, and we recall that P_i^j denotes the set of paths from i to j.

Note that $\mathsf{I}-\Delta$ instead is nothing but the normalized adjacency matrix $\mathsf{D}^{-1}\mathsf{A}$, and our notation reflects the form in which it appears in this is the form that naturally arises in the upcoming proofs.

Proposition 4.1.5 can be understood as a specialization of a more general lemma concerning graphs endowed with a $U(\mathbf{C})$-connection. In the following, $\Delta_\theta = \mathsf{D}^{-1}\mathsf{L}_\theta$ denotes the connection-aware random-walk Laplacian.

Lemma 4.1.6. *Consider the natural random walk on $\mathcal{G}$, two nodes $i, j \in \mathcal{V}$, and $l \in \mathbf{N}^*$.*
Then, one has:

$$((\mathsf{I}-\Delta_\theta)^l)_{i,j} = \sum_{\substack{p \in P_i^j \\ l(p)=l}} \left(\prod_{0 \leq k < l} \frac{w_{u_k,u_{k+1}}}{d_{u_k}} \right) \psi_{p^*}. \tag{4.38}$$

Lemma 4.1.6 is a straightforward generalization of Proposition 4.1.5, in which the connection is accounted for by the factors ψ_{p^*}. In particular, one should note that each p appearing in the right-hand-side on Equation (4.38) is associated to a map $\psi_{p^*} : \mathbf{C}_j \to \mathbf{C}_i$ from $\mathbf{C}_j$ to $\mathbf{C}_i$, which is in line with the action of $((\mathsf{I}-\Delta_\theta)^l)_{i,j}$ as an operator (for any $l \in \mathbf{N}^*$).
When $l = 0$, we simply have $((\mathsf{I}-\Delta_\theta)^l)_{i,j} = \psi_\epsilon = 1$, so that $(\mathsf{I}-\Delta_\theta)^l = \mathsf{I}$.

Proof of Lemma 4.1.6. First, note that the result holds by definition of Δ_θ for $l = 1$. We proceed by induction and assume the result true for some $l \in \mathbf{N}$. Then, recalling that $(\mathsf{I}-\Delta_\theta)_{v,j} = \frac{w_{v,j}}{d_v}$, we simply have:

$$\begin{aligned}
(\mathsf{I}-\Delta_\theta)^{l+1}_{i,j} &= \left((\mathsf{I}-\Delta_\theta)^l(\mathsf{I}-\Delta_\theta)\right)_{i,j} && (4.39)\\
&= \sum_{v \in \mathcal{V}} \sum_{\substack{p \in P_i^v \\ l(p)=l}} \left(\prod_{0 \leq k < l} \frac{w_{(u_k,u_{k+1})}}{d_{u_k}} \right) \frac{w_{v,j}}{d_v} \psi_{p^*} \psi_{(v,j)^*} && (4.40)\\
&= \sum_{\substack{p \in P_i^j \\ l(p)=l+1}} \left(\prod_{0 \leq k < l+1} \frac{w_{(u_k,u_{k+1})}}{d_{u_k}} \right) \psi_{p^*}. && (4.41)
\end{aligned}$$

■

Remark 4.1.7. *Lemma 4.1.6 also holds for more general connections. In that setting, denoting by* $\Delta_\Psi = \mathsf{D}^{-1}\mathsf{L}_\Psi$ *the connection-aware random-walk Laplacian*[9], *the same proof yields*

$$((\mathsf{I} - \Delta_\Psi)^l)_{i,j} = \sum_{\substack{p \in P_i^j \\ l(p)=l}} \left(\prod_{0 \le k < l} \frac{w_{u_k,u_{k+1}}}{d_{u_k}} \psi_{p^*} \right). \tag{4.42}$$

In addition to Lemma 4.1.6, the proof of Proposition 4.1.1 relies on another generic observation.

Lemma 4.1.8. *Let* $(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ *denote the restriction of the connection-aware random-walk Laplacian* Δ_θ *to its rows and columns indexed by* $\mathcal{V} \setminus \mathcal{A}$. *Suppose that* $\mathcal{A}$ *intersects all the connected components of* $\mathcal{G}$.
Then, $(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ *is invertible, and all its eigenvalues lie in* $(0, 2)$.

The proof of Lemma 4.1.8 is readily reduced to that of the connected case, which is then proved in two steps.

- First, notice that $(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ decomposes as:

$$(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}} = (\mathsf{D}^{-1}\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}} \tag{4.43}$$
$$= \mathsf{D}^{-1}_{\mathcal{V}\setminus\mathcal{A}}(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}, \tag{4.44}$$

  and that $(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ is invertible if and only if $(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ is. Since

$$\langle f, (\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}} f \rangle = \frac{1}{2} \sum_{e \in \overrightarrow{\mathcal{E}}} |f(t_e)\mathbf{1}_{\mathcal{V}\setminus\mathcal{A}}(t_e) - e^{\iota\theta_e} f(s_e)\mathbf{1}_{\mathcal{V}\setminus\mathcal{A}}(s_e)|^2 \tag{4.45}$$

  is positive for all non-identically-zero signals $f \in \mathbf{C}^{\mathcal{V}}$, it follows that all the eigenvalues of $(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ are also positive.

- Second, we further have the expression

$$\left((\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)_{i,j} = \begin{cases} 1 \text{ if } i = j, \\ -\frac{w_{i,j}}{d_i}\psi_{j,i} \text{ if } i \neq j, \end{cases} \tag{4.46}$$

  from which one should note that $\sum_{\substack{j \\ j \neq i}} \left| \left((\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)_{i,j} \right| < 1$.
  One then concludes from, *e.g.*, Gershgorin's circle theorem [Horn and Johnson, 2012][10], that all the eigenvalues of $(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ are smaller than 2.

We finally proceed with the proof of Proposition 4.1.1.

9 Note that, here, we could proceed with two equivalent choices of formalism: either take each entry $\mathsf{D}_{i,i} = d_i$, in which case it is understood as acting on the linear space of maps generated by the ψ_e's; or take $\mathsf{D}_{i,i} = d_i \mathsf{I}_{\mathbf{R}^d}$ with $\mathsf{I}_{\mathbf{R}^d}$ denoting the identity matrix in $\mathbf{R}^d$. For convenience, we adopt the former convention, which is in line with our definition of L_Ψ as a $|\mathcal{V}| \times |\mathcal{V}|$ matrix, whose entries are linear maps (instead of matrices).

10 Named after Soviet mathematician Semyon Aronovich Gershgorin, 1901−1933.

Proof of Proposition 4.1.1. It follows from Lemma 4.1.8 that the eigenvalues of $\mathsf{I} - (\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ are contained in $(-1, 1)$. Hence, for $i, j \in \mathcal{V}\setminus\mathcal{A}$, one may write

$$\left(((\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}})^{-1}\right)_{i,j} = \left[\sum_{l\geq 0}(\mathsf{I} - (\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}})^l\right]_{i,j} \tag{4.47}$$

$$= \sum_{p\in P_i^j}\left(\prod_{0\leq k<l}\frac{w_{u_k,u_{k+1}}}{d_{u_k}}\right)\psi_{p^*}, \tag{4.48}$$

where convergence of the right-hand-side (Equation (4.47)) is guaranteed by the bound on the spectrum, Equation (4.47) is the classical geometric-series expansion for matrix inverse[11], and Equation (4.48) is obtained the same computation as that of the proof of Lemma 4.1.6.
Multiplying Equation (4.48) by $(\mathsf{D}_{\mathcal{V}\setminus\mathcal{A}})^{-1}(-\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}$ on the right, with

$$\left((\mathsf{D}_{\mathcal{V}\setminus\mathcal{A}})^{-1}(-\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\right)_{i,j} = \frac{w_{i,j}}{d_i}\psi_{j,i} \tag{4.49}$$

accounting for the last transition from $\mathcal{V}\setminus\mathcal{A}$ to $\mathcal{A}$, we obtain for each $a \in \mathcal{A}$:

$$\left(-(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}(\mathsf{D}_{\mathcal{V}\setminus\mathcal{A}})^{-1}(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\right)_{i,a} = \sum_{p\in P_i^a}\left(\prod_{0\leq k<l}\frac{w_{u_k,u_{k+1}}}{d_{u_k}}\right)\psi_{p^*} \tag{4.50}$$

which, from Equation (4.44), yields

$$\mathrm{h}_\theta(g) = -\left(((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}})^{-1}\right)(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\ g. \tag{4.51}$$

■

Remark 4.1.9. *The convergence of the right-hand-side in Equation (4.47) can also be deduced from Equation (4.48), since*

$$\sum_{p\in P_i^j}\left|\left(\prod_{0\leq k<l}\frac{w_{u_k,u_{k+1}}}{d_{u_k}}\right)\psi_{p^*}\right| = \sum_{p\in P_i^j}\left(\prod_{0\leq k<l}\frac{w_{u_k,u_{k+1}}}{d_{u_k}}\right), \tag{4.52}$$

where we recognize on the total probability $\nu_i^j(P_i^j) = 1$.
This alternative does not rely on a spectral argument (as in Lemma 4.1.8), and does not use the commutativity of the connection maps. As such, it can be translated to $O(\mathbf{R}^d)$-connections, by replacing $|.|$ with an operator norm.

We now turn to the proof of Proposition 4.1.2. Our result is actually somewhat more general, and allows to estimate quantities of the form

$$(\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}\ g, \tag{4.53}$$

where Q is a diagonal matrix with non-negative coefficients $\mathsf{Q}_{i,i} = q_i$, such that at least one node *i in each connected component* of $\mathcal{G}$ is *positive*. The coefficients q_i's correspond to the weights of the edges $\{i, \Gamma\}$ in the extended graph $\mathcal{G}^\Gamma$, and we note that the expression in Equation (4.53) appears in regularization problems for signals degraded by heteroscedastic noise (as discussed in Chapter 2).

11 Also called a Neumann series, after German mathematician Karl Gottfried Neumann (1832–1925).

Proof of Proposition 4.1.2. From the same reasoning as in the proof of Proposition 4.1.1, and following Remark 4.1.3, we find that

$$(((\Delta_\theta^\Gamma)_\mathcal{V})^{-1})_{i,j} = \sum_{p \in P_i^j} \left(\prod_{0 \leq k < l(p)} \frac{w_{u_k, u_{k+1}}}{d_{u_k} + q_{u_k}} \right) \psi_{p^*}. \tag{4.54}$$

The conclusion follows by right-multiplication with $(\mathsf{D}+\mathsf{Q})^{-1}\mathsf{Q}$, where

$$\left((\mathsf{D}+\mathsf{Q})^{-1}\mathsf{Q}\right)_{j,j} = \frac{q_j}{d_{u_{l(p)-1}} + q_j} \tag{4.55}$$

accounts for the last transition from $u_{l(p)-1} = j$ to Γ. ■

Remark 4.1.10. *Somewhat tautologically, Proposition 4.1.2 also provides a nice probabilistic interpretation for connection-aware Tikhonov smoothing.*
As a smoothing operator, and in contrast to other operators such as, e.g., *the connection-aware adjacency matrix A_θ, Tikhonov regularization accounts for* long-range *interactions between nodes. In particular, the value of f_* on a given node i depends on the value on the nodes at which the random walk stops and, since information brought back along two different paths yields two different estimations, on the specific sampled path.*
For high values of q, the end-points of the paths tend to be localized near node i, on nodes for which, assuming the signal underlying degraded $f_\top$ is smooth, the noisy signal $g = f_\top + \varepsilon$ should take values close to that of i (up to rotations corresponding to the connection); in contrast, these end-points are spread-out across the graph for higher values of q, resulting in stronger smoothing.

Remark 4.1.11. *We recently noticed that a trivial-connection counterpart to Proposition 4.1.2 appears in [Wu et al., 2012, Wu, 2016]; this result is also similar to that of [Avrachenkov et al., 2017] (as discussed in Chapter 1). While not suitable for RandNLA purposes, this type of result is attractive when the estimation only needs to be performed on a few nodes, as may be the case for a few agents of interest in a very large network.*
Observations similar to Propositions 4.1.1 and 4.1.2 have also been laid out in a few works, such as [Güneysu et al., 2016, Kassel and Lévy, 2021] for continuous-time random walks, and in a number of other situations, where the connection often stems from mathematical-physics considerations [Brydges et al., 1979].

Remark 4.1.12. *The reader can check that, by the same arguments, the estimators of Propositions 4.1.1 and 4.1.2 carry over to $O(\mathbf{R}^d)$-connections. This is in stark contrast with the techniques we develop in Chapters 5 and 6, for which this more general setting seems to stump all our proof techniques.*

4.2 VARIANCE REDUCTION

It should become apparent that a practical way to account for the connection in Monte-Carlo estimators is to perform parallel transport along the sampled random paths, that is, according to $\psi_{p^*} : \mathbf{C}_j \to \mathbf{C}_i$. On the other, and by design, the simple estimators resulting from Propositions 4.1.1 and 4.1.2 suffer from two drawbacks when it comes to applications in RandNLA:

- the estimation is localized on a single node i;

- the variance of the estimation is high.

The second claim is not as clear as the first one, and we discuss it below.

Specifically, we consider the connection-aware smoothing estimator from Proposition 4.1.2, and discuss how its variance can be decreased by *neglecting cycles* in the sampled paths. Our goal is not to quantify this variance-reduction, but rather to introduce a nice construction that we use in Chapter 7 to relate Feynman-Kac estimators to the forest-based estimators of Chapters 5 and 6.

Let us begin by defining an equivalence relation on paths in P_i^Γ (recall that P_i^Γ denotes the set of paths from i to Γ in the extended graph $\mathcal{G}^\Gamma$): two paths $p, q \in P_i^\Gamma$ are equivalent, which we denote by $p \simeq q$, if the only difference between the two is the orientation of (at most) *one* cycle; that is, $p \simeq q$ if $p = aCb$ and $q = aC^*b$, with C a (possibly empty) cycle in $\mathcal{G}$. For convenience, we denote by $p(C)$ and $p(C^*)$ the two possible representatives of each of the induced equivalence classes, and by $\overline{p(C)}$ the equivalence class itself.
For any node $i \in \mathcal{V}$, we can write the expected value obtained at i by performing parallel transport, from which we obtain:

$$\sum_{p \in P_i^\Gamma} \mathbf{P}_{\nu_i^\Gamma}(p)\psi_{p_\Gamma^*} = \sum_{\overline{p(C)} \in P_i^\Gamma/\simeq} \left(\mathbf{P}_{\nu_i^\Gamma}(p(C))\psi_{p(C)_\Gamma^*} + \mathbf{P}_{\nu_i^\Gamma}(p(C^*))\psi_{p(C^*)_\Gamma^*}\right) \tag{4.56}$$

$$= \sum_{\overline{p(C)} \in P_i^\Gamma/\simeq} \mathbf{P}_{\nu_i^\Gamma}(\overline{p(C)})\left(\psi_{a^*} \circ (\psi_C + \psi_{C^*}) \circ \psi_{b^*}\right) \tag{4.57}$$

$$= \sum_{\overline{p(C)} \in P_i^\Gamma/\simeq} 2\mathbf{P}_{\nu_i^\Gamma}(\overline{p(C)})\left(\left(\psi_{a^*} \circ \left(\frac{\psi_C + \psi_{C^*}}{2}\right)\right) \circ \psi_{b^*}\right), \tag{4.58}$$

where $p(C) = aCb$, and we denote by $\mathbf{P}_{\nu_i^\Gamma}(\overline{p(C)}) = \mathbf{P}_{\nu_i^\Gamma}(p(C))$ in Equation 4.57 the probability of sampling either of the two representatives $p(C)$ or $p(C^*)$ (note that $\mathbf{P}_{\nu_i^\Gamma}(p_C) = \mathbf{P}_{\nu_i^\Gamma}(p_{C^*})$).

Let us re-phrase this observation in a more qualitative language: whereas each estimation in the Feynman-Kac estimator proceeds by parallel transport along a path $p(C)$, with the transport application *depending on the orientation of the cycle* C, we observe from Equation (4.58) that we can instead perform an *orientation-agnostic estimation* resulting in the same expectation, by applying the map $\frac{\psi_C+\psi_{C^*}}{2}$ instead of the connection map ψ_C; the factor 2 in Equation 4.58 in turn accounts for the possibility of sampling either $p(C)$ or $p(C^*)$. Here, θ_C is the argument of the holonomy ψ_C of C, as obtained from that of a path p *traversing* C *one single time*. Explicitly, for a cycle $C = \{\{v_1, v_2\}, \{v_2, v_3\}, ...\{v_{l-1}, v_l\}, \{v_l, v_1\}\}$, one has

$$\theta_C = \arg\left(\prod_{e \in p} \psi_e\right), \tag{4.59}$$

with ψ_p for $p = ((v_1, v_2), (v_2, v_3), ..., (v_{l-1}, v_l), (v_l, v_1))$ a path going through the cycle C. Note that, while θ_C depends on the orientation of the path p (in particular, $\psi_p = \psi_{p^*}^*$), $\cos(\theta_C)$ is agnostic to this choice of orientation, and the resulting probability $\mathbf{P}_{\mathcal{D}_\mathcal{M}}(\phi)$ is defined with no ambiguity. As a concrete example, one finds that $\theta_C = \frac{3\pi}{4}$ along the only cycle in Figure 4.1.

Provided that $\cos(\theta_C) \geq 0$ *for all cycles* C *in* $\mathcal{G}$, this allows to probabilistically interpret the maps $\frac{\psi_C+\psi_{C^*}}{2}$ in Equation (4.58), which act as multiplications by the factors $\cos(\theta_C)$:

$$\left(\frac{\psi_C+\psi_{C^*}}{2}\right)(z) = \cos(\theta_C)\cdot z. \tag{4.60}$$

Under the assumption that $\cos(\theta_C)$ is non-negative, we can understand Equation (4.58) as a variant of the Feynman-Kac estimator involving a Bernoulli trial (with success probability $\cos(\theta_C)$), going as follows. Suppose that the path p contains a unique cycle C, and takes the form $p = aCb\ (j, \Gamma)$, then:

1. if the trial succeeds, set
$$\psi_{(ab)^*}(g(j)) \tag{4.61}$$
as the estimation on node i.

2. If the trial fails, set 0 as the estimation.

Let us stress that, in expectation, this procedure still results in the smoothed signal f_*.

This argument extends to any number k of cycles, and translates to a significant reduction of variance for the Feynman-Kac estimator[12]: any path p with k cycles, sampled according to ν_i^Γ, belongs to an equivalence class of 2^k paths, for each of which the Feynman-Kac estimator results in a different value at node i, while the modified estimator assigns the same value to all 2^k paths.

One can even go a step further, and get rid of the non-negativity condition on $\cos(\theta_C)$. Indeed, for any cycle C such that $\cos(\theta_C) < 0$, one can perform a Bernoulli trial with success probability $-\cos(\theta_C) = (-1)\times\cos(\theta_C)$, and multiply the resulting estimate -1 each time such a cycle is constructed in the path p, thereby *extending the estimation to any* $U(\mathbf{C})$*-connection*.

Remark 4.2.1. *It is chiefly frustrating that this construction does not extend to higher-dimensional* $O(\mathbf{R}^d)$*-connections: in order to recover the factors* $\frac{\psi_C+\psi_{C^*}}{2}$*, one needs to rely on the commutativity of the connection maps for paths containing nested cycles*[13].

4.3 SUMMARY

We established in this chapter two Feynman-Kac formulas, allowing to estimate the solutions to the connection-aware Tikhonov regularization and harmonic-interpolation problems, simply by running random walks on the graph. Moreover, these results hold for general $O(\mathbf{R}^d)$-connections

12 Note that this requires to be careful in the definition of the successive cycles, as one cycle may be contained within another. Here, a convenient choice of definition is to define the set of cycles according to the following procedure: whenever a cycle is observed in the path p constructed in the random walk, add it to the set of cycles and remove it from p.

13 In the case of $U(\mathbf{C})$-connections (that can be understood as $O(\mathbf{R}^2)$-connections), the commutativity is immediate. The result would more generally hold for algebras such that $\psi_C + \psi_C^* = c\mathsf{I}$, where $c \in \mathbf{R}$ denotes some constant, which is for instance the case for the algebra of quaternions (hence, the arguments of this section generalize to this setting). Unfortunately, this property is very rare, and only holds for a restricted class of algebras, and does not even generalize to (the geometrically-motivated) Clifford algebras.
Clifford algebras, named after British mathematician and philosopher William Kingdon Clifford (1845−1879) find a number of applications in geometry and (regular and hyper-spectral) image processing, and formally generalize properties of the real, complex or quaternionic algebras.

and, from a mathematical perspective, boil down to the expression of the inverse of a matrix $A \in M_n(\mathbf{C})$ as

$$A^{-1} = \sum_{l \geq 0} (\mathsf{I} - A)^l, \tag{4.62}$$

when applicable. Unfortunately, the resulting estimators are ill-suited to RandNLA applications, due to both the local nature of the estimation and the high associated variance, and are not able to compete with standard linear-algebraic solvers for these problems.

In the next chapter, we introduce novel (random) forest-like decompositions for graphs endowed with a $U(\mathbf{C})$-connection that, exploiting this idea, we leverage in Chapter 6 to design global, efficient Monte-Carlo estimators. Let us spoil the surprise and proclaim that Theorem 4.0.1 will follow from expressing the entries of the inverse of a matrix $A \in M_n(\mathbf{C})$ using Cramer's rule (Equation 3.30):

$$(A^{-1})_{i,j} = (-1)^{i+j} \det(A)_{[n]\setminus\{j\},[n]\setminus\{i\}}, \tag{4.63}$$

and then computing the determinant using the Cauchy-Binet formula (much like in the proof of Theorem 3.0.2, for DPP-based estimation of solutions to linear least-squares problems).

Unfortunately, and in spite of the variety of tools we developed, we are not able to derive global estimators for $O(\mathbf{R}^d)$-connections. As such, Chapters 5 and 6 focus exclusively on $U(\mathbf{C})$-connections and the estimation of $\mathbf{C}$-signals, for smoothing, interpolation and angular-synchronization purposes.

5

MULTI-TYPE SPANNING FORESTS

The greatest reward lies in making the discovery; recognition can add little or nothing to that.

— Franz Ernst Neumann[1]

Contributions in this chapter.

1. Introduction of (multiple variants of) a determinantal point process over *multi-type spanning forests*.
2. Introduction of an efficient random-walk-based algorithm for this process.
3. Bound on the expected complexity of the sampling algorithm.
4. Efficient implementation of the sampling algorithm.

The first two points have also been independently developed in [Fanuel and Bardenet, 2022] (for one specific variant of the processes we consider), and our work resulted in a co-authored conference communication [Jaquard et al., 2023]. We also propose a significantly-improved implementation of the sampling algorithm [Jaquard et al., 2024], and discuss a number of algorithmic tricks in this chapter (our results are exposed in Chapter 6)[2].

1 See `https://mathshistory.st-andrews.ac.uk/Biographies/Neumann_Franz/quotations/`.
2 See `https://gricad-gitlab.univ-grenoble-alpes.fr/gaia/synchromtsf`.

The reader should by now be convinced that parallel transport is a powerful tool for Monte-Carlo estimators, and we are now ready to discuss *multi-type spanning forests*. Multi-type spanning forests are generalized spanning forests adapted specifically to graphs endowed with a connection, that we leverage in Chapter 6 to derive global Monte-Carlo estimators, based on parallel transport along the branches of these forests.

Our aim in this chapter is to establish the theoretical framework upon which our later analyses will be based. Specifically, our contribution is two-fold.

1. We define a probability distribution $\mathcal{D}_{\mathcal{M}}$ over the multi-type spanning forests of a graph, and show that this distribution is a determinantal point process. Our proof techniques are very similar to those used in Chapter 3 to study the uniform-spanning-tree distribution.

2. We derive an efficient random-walk-based sampling algorithm to draw samples from $\mathcal{D}_{\mathcal{M}}$, and bound the expected complexity of this algorithm.

In addition, we discuss important implementation details for the sampling algorithm. In Chapter 6, we will see that the resulting estimators are able to compete with standard deterministic algorithms.

As we found out, our developments have a significant overlap with those of the (independent) work in [Fanuel and Bardenet, 2022], with regards to the general theory of multi-type-spanning forests (as developed in this chapter), as indeed both works are based on this structure. Due to the similarity in both works, we co-authored a conference paper [Jaquard et al., 2023]. Finally, let us mention that the techniques of [Fanuel and Bardenet, 2022], which concern sparsifiers for connection-Laplacian-based systems, can be applied in conjunction to our own in some settings, as we discuss in Chapter 6.

Still, one of our aim in this work is to frame our results in as general a setting as possible, and to use streamlined arguments so as to make these techniques as accessible as can be. In turn, the reader will find a number of differences.

- The forests we consider have *rooted* trees, whereas those of [Fanuel and Bardenet, 2022] are unrooted. In particular, this allows for a simple, self-contained proof of Theorem 4.0.1 (concerning the forest-based estimator teased at the beginning of Chapter 4).

- We take a more hands-on approach in our construction of multi-type spanning forests, which allows to define more general forests *associated to a subset of nodes in the graph* (see Section 5.1 for the precise construction). This is nice in a number of proofs, and for instance yields instant generalizations of Theorem 4.0.1 to Tikhonov regularization in the presence of heteroscedastic noise, or to harmonic interpolation (as discussed in Chapter 6); in addition, we obtain a number of intermediate results that may be of independent interest.

- Our bound on the complexity of the sampling algorithm for multi-type spanning forests (Proposition 5.4.1) is completely novel.

Due to the similarity in both works, we co-authored a conference paper [Jaquard et al., 2023]. Finally, let us mention that the techniques of [Fanuel and Bardenet, 2022], which concern sparsifiers for connection-Laplacian-based systems, can be applied in conjunction to our own in some settings, as we discuss in Chapter 6.

5.1 MULTI-TYPE SPANNING FORESTS: DEFINITIONS

Given that simple random processes such as random walks only allow for local estimation, and do not result in efficient enough estimators, it is natural to ask whether techniques based on random-forest processes capturing properties of both the graph and the connection can be developed. Our goal in this section is precisely to propose such processes, as obtained from determinantal point processes. Key features of these processes are:

1. a dependence not only on the topology and edge-weights of the graph, but also on the incoherence of the connection;
2. an efficient Wilson-like sampling algorithm.

The end-result is a class of distributions over generalized forests called multi-type spanning forests, that can be leveraged for both smoothing and interpolation purposes (applications that are developed in Chapter 6). For simplicity of exposition, we focus on the smoothing problem, and point out the relevant generalizations for interpolation problems along the way.

Let us start from definitions.

Multi-Type Spanning Forests (MTSFs) in a graph $\mathcal{G}$ arise from solutions to a simple assignment problem: given the extended graph $\mathcal{G}^\Gamma$, associate to each node $i \in \mathcal{V}$ another node in its neighborhood *in the extended graph* $\mathcal{G}^\Gamma$. The solution to this problem is any map $\overrightarrow{\phi} : \mathcal{V} \to \mathcal{V}^\Gamma$, which corresponds to a subset of directed edges of $\mathcal{G}^\Gamma$. A Γ-*MTSF* ϕ^Γ is a subset of non-directed edges $\phi^\Gamma \subseteq \mathcal{V} \cup \mathcal{E}$ obtained by "forgetting" the orientation for some $\overrightarrow{\phi}$:

$$\phi^\Gamma = \left\{ \{i, \overrightarrow{\phi}(i)\};\ i \in \mathcal{V} \right\}. \tag{5.1}$$

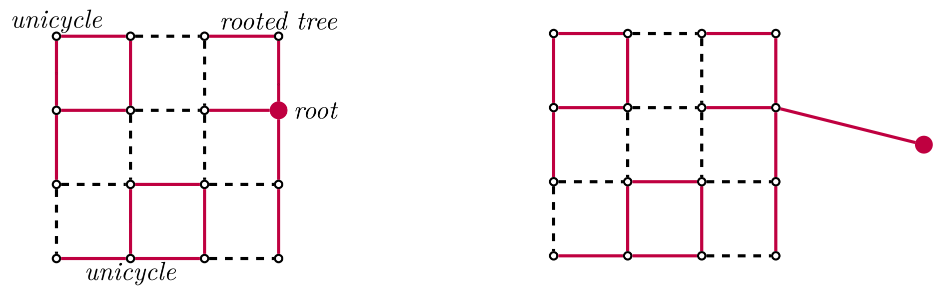
To each map $\overrightarrow{\phi}$ and Γ-MTSF ϕ, we associate a unique *Multi-Type Spanning Forest* $\phi \subseteq \mathcal{V} \cup \mathcal{E}$, which is obtained from ϕ^Γ by replacing the edges of the form $\{i, \Gamma\}$ by the nodes i:

$$\phi = \{i;\ \overrightarrow{\phi}(i) = \Gamma\} \cup \left\{ \{i, \overrightarrow{\phi}(i)\};\ i \in \mathcal{V} \right\}. \tag{5.2}$$

Let us emphasize that there is a one-to-one correspondence between ϕ and ϕ^Γ, which is completely analogous to the one between RSFs and spanning trees of $\mathcal{G}^\Gamma$ discussed in Chapter 1.
It immediately follows from the definition that multi-type spanning forests have a number of specific properties, and can be combinatorially characterized.

1. Since $\overrightarrow{\phi}$ assigns a neighbor to any node in $\mathcal{V}$, we have $|\phi| = |\mathcal{V}|$.
2. The edges of ϕ can be partitioned into connected components $c_\phi \subseteq \phi$, which must be one of the two following types.
    - Rooted trees $c_\phi \subseteq \mathcal{V} \cup \mathcal{E}$, such that $c_\phi \cap \mathcal{E}$ is a connected cycle-free subset of edges, and $c_\phi \cap \mathcal{V}$ is reduced to a signal node, connected to $c_\phi \cap \mathcal{E}$, and called a *root*.
    - Unicycles $c \subseteq \mathcal{E}$, that is, subsets of edges containing *exactly* one cycle.

We illustrate a MTSF and its corresponding Γ-MTSF in Figure 5.1, on the 4×4 grid-graph.
In the following, We denote by $\mathcal{M}(\mathcal{G})$ the set of all MTSFs over the graph $\mathcal{G}$. Let us also point-out that a unicycle-free MTSF is nothing but a rooted spanning forest [Avena and Gaudillière, 2013], whereas a tree-free MTSF is a spanning forest of unicycles [Kenyon, 2011].



(a) A multi-type spanning forest on $\mathcal{G}$. (b) Corresponding Γ-MTSF on $\mathcal{G}^\Gamma$.

Figure 5.1: A multi-type spanning forest on the 4×4 grid graph $\mathcal{G}$ (left), with two unicycles and one rooted tree, and the corresponding Γ-MTSF on $\mathcal{G}^\Gamma$ (right). Edges belonging to the (Γ-)MTSF are represented in purple, and roots as large purple nodes. We only represent the edges of the form (v, Γ) in $\mathcal{G}^\Gamma$ belonging to the Γ-MTSF.

Remark 5.1.1. *More generally, and for any subset of nodes $\mathcal{A} \subseteq \mathcal{V}$ in a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, one can define a $\mathcal{A}$-MTSF in a graph $\mathcal{V}$ as the subset of edges $\phi^{\mathcal{A}} \subseteq \mathcal{E}$ obtained from an assignment $\overrightarrow{\phi} : \mathcal{V} \setminus \mathcal{A} \to \mathcal{V}$. Precisely, and analogously to Γ-MTSFs:*

$$\phi^{\mathcal{A}} = \left\{ \{i, \overrightarrow{\phi}(i)\};\ i \in \mathcal{V} \setminus \mathcal{A} \right\}. \tag{5.3}$$

Similarly to MTSFs, the edges of $\mathcal{A}$-MTSFs (and of Γ-MTSFs) can be broken down in two classes:

1. *those edges that, in the graph where the edges of the form $\{i, a\}$, for $i \in \mathcal{V} \setminus \mathcal{A}$ and $a \in \mathcal{A}$, belong to (maximal) cycle-free subsets of edges, that we call a tree of $\phi^{\mathcal{A}}$;*
2. *those other that belong to unicycles of $\phi^{\mathcal{A}}$.*

We use $\mathcal{A}$-MTSFs in Chapter 6 to derive estimators for the connection-aware harmonic-interpolation problem. To that aim, and unlike MTSFs or RSFs, that are associated to Γ-MTSFs and for which it is implicit that every root $i \in \phi \cap \mathcal{A}$ corresponds to an edge $\{i, \Gamma\}$, it is important to remember the exact edges $\{i, a\}$ (for $a \in \mathcal{A}$).

For smoothing purposes, our interest lies in estimating quantities of the form

$$(\mathsf{L}_\theta + \mathsf{Q})^{-1} \mathsf{Q}\, g, \tag{5.4}$$

where $g \in \mathbf{C}^{\mathcal{V}}$ is some signal to be denoised; to that end, it is natural that the random processes we consider depend on the coefficients $\mathsf{Q}_{i,i} = q_i$. This is indeed the case, and we specifically consider the probability distribution $\mathcal{D}_{\mathcal{M}}$ defined over $\mathcal{M}(\mathcal{G})$ by

$$\mathbf{P}_{\mathcal{D}_{\mathcal{M}}}(\phi) \propto \prod_{r \in \phi \cap \mathcal{V}} q_r \prod_{e \in \phi \cap \mathcal{E}} w_e \prod_{C \in \mathcal{C}(\phi)} (2 - 2\cos(\theta_C)), \tag{5.5}$$

where $\mathcal{C}(\phi)$ denotes the set of cycles $C \subseteq \mathcal{E}$ belonging to the unicycles of ϕ. Recall from Section 4.2 that θ_C denotes the argument of the holonomy of ψ_C, as defined in Equation (4.59)

Importantly, let us remark that the dependence of $\mathcal{D}_{\mathcal{M}}$ on both the coefficients q_i's and the factors $\cos(\theta_C)$, which is a measure of the incoherence of the connection along the cycle C, can result in a complex interplay.

- The more incoherent a cycle is (the larger $\cos(\theta_C)$ is), the more likely it is to be sampled. In particular, a coherent cycle (with $\cos(\theta_C) = 1$) will never be sampled.
- Higher values of q_i's tend to favor sampling a larger number of trees, each spanning a smaller subset of nodes. This, in turn, makes it less likely for unicycles to be sampled, even for those containing a very incoherent cycles.
- On the flip side, lower values of q_i's favor sampling more unicycles.

Remark 5.1.2. *Distribution $\mathcal{D}_{\mathcal{M}}$ can equivalently be defined on Γ-MTSFs:*

$$\mathbf{P}_{\mathcal{D}_{\mathcal{M}}}(\phi^{\Gamma}) \propto \prod_{e\in\phi^{\Gamma}} w_e \prod_{C\in\mathcal{C}(\phi^{\Gamma})} \left(2 - 2\cos(\theta_C)\right), \tag{5.6}$$

with $\mathcal{C}(\phi^{\Gamma}) = \mathcal{C}(\phi)$.
More generally, one can define the distribution $\mathcal{D}_{M}^{\mathcal{A}}$ over $\mathcal{A}$-MTSFs by

$$\mathbf{P}_{\mathcal{D}_{\mathcal{M}}^{\mathcal{A}}}(\phi^{\mathcal{A}}) \propto \prod_{e\in\phi^{\mathcal{A}}} w_e \prod_{C\in\mathcal{C}(\phi^{\mathcal{A}})} \left(2 - 2\cos(\theta_C)\right), \tag{5.7}$$

where $\mathcal{C}(\phi^{\mathcal{A}})$ similarly denotes the set of cycles belonging to unicycles of $\phi^{\mathcal{A}}$.

One particularly salient feature of distribution $\mathcal{D}_{\mathcal{M}}$, that is useful to design MTSF-based estimators in Chapter 6, is that it defines a determinantal point process.

5.2 $\mathcal{D}_{\mathcal{M}}$ IS A DPP

Our goal in this section is to show that the distribution $\mathcal{D}_{\mathcal{M}}$ is a DPP over the space $\mathcal{S} = \mathcal{V} \cup \mathcal{E}$. This is a two-step endeavor, involving:

1. defining the kernel $K_{\mathcal{M}}$ of the DPP;
2. identifying the distribution $\mathrm{DPP}(K_{\mathcal{M}})$ with $\mathcal{D}_{\mathcal{M}}$.

Our proof techniques should be reminiscent of those we used to prove that the uniform-spanning-tree distribution is a DPP in Chapter 1, as well as those of, *e.g.*, [Lyons, 2009, Kassel and Lévy, 2022a] for other combinatorially-meaningful DPPs on graphs. We introduce along the way number of tools that are re-used in Chapter 6.

As our first order of business, we define the (projection) kernel associated to $\mathcal{D}_{\mathcal{M}}$.

Marginal Kernel. Recall from Chapter 1 the discrete differential operator $\nabla : \mathbf{C}^{\mathcal{V}} \to \mathbf{C}^{\mathcal{E}}$ associated to a graph $\mathcal{G}$, acting on a signal $f \in \mathbf{C}^{\mathcal{V}}$ in such a way that

$$(\nabla f)(e) = f(t_e) - f(s_e). \tag{5.8}$$

The combinatorial Laplacian L can be defined in terms of ∇, and one has

$$\mathsf{L} = \nabla^{*}\nabla \tag{5.9}$$

which, consequently and for any signal $f \in \mathbf{C}^\mathcal{V}$, yields

$$\|\nabla f\|_2^2 = \langle f, \mathsf{L} f\rangle \tag{5.10}$$
$$= \frac{1}{2}\sum_{e\in\overrightarrow{\mathcal{E}}} |f(t_e) - f(s_e)|^2. \tag{5.11}$$

One most important property, for our purposes, is that ∇ also serves to define the marginal kernel of the uniform-spanning-tree DPP $K_\mathcal{T}$:

$$K_\mathcal{T} = \nabla(\nabla^*\nabla)^\dagger\nabla^*, \tag{5.12}$$

which is nothing but the orthogonal projection onto im∇.
We follow a similar construction and, in order to define the marginal kernel of $\mathcal{D}_\mathcal{M}$, proceed in two steps.

- First, we need to define a discrete differential $\nabla_\theta : \mathbf{C}^\mathcal{V} \to \mathbf{C}^\mathcal{V}$ adapted to non-trivial connections, as proposed in [Kenyon, 2011]; in particular, this involves defining a "splitting" of the connection maps. To do so, we regard each entry $f(e)$ of an edge-signal $f \in \mathbf{C}^\mathcal{E}$ as belonging to a copy $\mathbf{C}_e$ of the complex plane $\mathbf{C}$, so that we have $\mathbf{C}^\mathcal{E} = \bigoplus_{e\in\mathcal{E}} \mathbf{C}_e$.
  We then rely on splittings of the form

  $$\psi_e : \mathbf{C}_{s_e} \xrightarrow{\psi_{s_e,e}} \mathbf{C}_e \xrightarrow{\psi_{e,t_e}} \mathbf{C}_{t_e}, \tag{5.13}$$

  such that $\psi_e = \psi_{e,t_e} \circ \psi_{s_e,e}$; note that it is always possible to obtain this type of decomposition, as one can take, *e.g.*, $\psi_{e,t_e} = \mathrm{id}_{\mathbf{C}_e,\mathbf{C}_{t_e}}$ and $\psi_{s_e,e}$ such that $\psi_{s_e,e}(z) = e^{\iota\theta_e}\cdot z$. The specific splittings will not be relevant for our purposes.
  We next define the *twisted discrete differential* ∇_θ as the operator satisfying

  $$(\nabla_\theta f)(e) = \sqrt{w_e}\psi_{t_e,e} f(t_e) - \sqrt{w_e}\psi_{e,s_e} f(s_e) \tag{5.14}$$

  which, expliciting its entries in the usual bases of $\mathbf{C}^\mathcal{V}$ and $\mathbf{C}^\mathcal{E}$, yields:

  $$\nabla_{e,v} = \begin{cases} -\sqrt{w_e}\psi_{s_e,e} & \text{if } v = s_e \\ \sqrt{w_e}\psi_{e,t_e} & \text{if } v = t_e \\ 0 & \text{otherwise.} \end{cases} \tag{5.15}$$

  It is then straightforward to show that $\mathsf{L}_\theta = \nabla_\theta^*\nabla_\theta$, and we recover

  $$\|\nabla_\theta f\|_2^2 = \langle f, \mathsf{L}_\theta f\rangle \tag{5.16}$$
  $$= \frac{1}{2}\sum_{e\in\overrightarrow{\mathcal{E}}} |f(t_e) - \psi_e f(s_e)|^2. \tag{5.17}$$

  In particular, Equation (5.17) is insensitive to the splittings of the ψ_e's.
  Let us remark that the DPP with marginal kernel

  $$\nabla_\theta(\nabla_\theta^*\nabla_\theta)^\dagger\nabla_\theta \tag{5.18}$$

  is known to sample spanning forests of unicycles in $\mathcal{G}$ [Kenyon, 2011].

(a) The 2×3 grid graph $\mathcal{G}$. (b) MTSF on $\mathcal{G}$.

Figure 5.2: The 2×3 grid graph (left), and a MTSF on this graph (right), with one unicycle (teal) and one rooted tree (blue).

- We then define a "rooted" variant of this process. We take inspiration from the kernel

$$K_{\mathcal{F}} = \begin{bmatrix} \nabla \\ \sqrt{q} \end{bmatrix} (\mathsf{L} + q\mathsf{I})^{-1} \begin{bmatrix} \nabla^* & \sqrt{q} \end{bmatrix}. \tag{5.19}$$

associated to the distribution $\mathcal{D}_{\mathcal{F}}$ over RSFs [Kulesza and Taskar, 2012], and consider the kernel

$$K_{\mathcal{M}} = \nabla_\theta^{\mathsf{Q}} (\mathsf{L}_\theta + \mathsf{Q})^{-1} (\nabla_\theta^{\mathsf{Q}})^*, \tag{5.20}$$

where $\nabla_\theta^{\mathsf{Q}} = \begin{bmatrix} \nabla_\theta \\ \sqrt{\mathsf{Q}} \end{bmatrix}$ is a $(|\mathcal{E}| + |\mathcal{V}|) \times |\mathcal{V}|$ matrix, with $\sqrt{Q}$ the diagonal matrix with entries $\sqrt{q_i}$'s. Note that $(\mathsf{L}_\theta + \mathsf{Q})^{-1}$ is always invertible according to Lemma 4.1.8, and that ∇_θ^Q is nothing but the restriction of the twisted discrete differential on $\mathcal{G}^\Gamma$ to those nodes indexed by $\mathcal{V}$.

Example 5.2.1. *For the 2×3 grid graph, as represented in Figure 5.2a, we have*

$$\nabla_\theta^Q = \begin{array}{cc} & \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{array}{c} (1,2) \\ (1,4) \\ (2,3) \\ (2,5) \\ (3,6) \\ (4,5) \\ (5,6) \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} & \begin{bmatrix} -\psi_{1,(1,2)} & \psi_{2,(1,2)} & 0 & 0 & 0 & 0 \\ -\psi_{1,(1,4)} & 0 & 0 & \psi_{4,(1,4)} & 0 & 0 \\ 0 & -\psi_{2,(2,3)} & \psi_{3,(2,3)} & 0 & 0 & 0 \\ 0 & -\psi_{2,(2,5)} & 0 & 0 & \psi_{5,(2,5)} & 0 \\ 0 & 0 & -\psi_{3,(3,6)} & 0 & 0 & \psi_{6,(3,6)} \\ 0 & 0 & 0 & -\psi_{(4,(4,5)} & \psi_{5,(4,5)} & 0 \\ 0 & 0 & 0 & 0 & -\psi_{5,(5,6)} & \psi_{6,(5,6)} \\ \sqrt{q_1} & 0 & 0 & 0 & 0 & 0 \\ 0 & \sqrt{q_2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \sqrt{q_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \sqrt{q_4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{q_5} & 0 \\ 0 & 0 & 0 & 0 & 0 & \sqrt{q_6} \end{bmatrix} \end{array}, \tag{5.21}$$

where the columns (resp. rows) are indexed by the nodes of $\mathcal{G}$ (resp. edges and nodes of $\mathcal{G}$). Equivalently, the columns of ∇_θ^Q can be indexed by the edges of $\mathcal{G}^\Gamma$, which corresponds to the Γ-MTSF convention.
Colored rows correspond to the edges and nodes of the MTSF in Figure 5.2b.

This allows to state our first theorem.

Theorem 5.2.2. *For a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, the distribution $\mathcal{D}_{\mathcal{M}}$ defined in Equation (5.5) is a DPP over the space $\mathcal{S} = \mathcal{V} \cup \mathcal{E}$, with marginal kernel given by $K_{\mathcal{M}}$. More precisely, $K_{\mathcal{M}}$ is a projection kernel and, for $\phi \in \mathcal{S}$,*

$$\det(K_{\mathcal{M}}) = \begin{cases} \frac{\prod_{r \in \phi \cap \mathcal{V}} q_r \prod_{e \in \phi \cap \mathcal{E}} w_e \prod_{C \in \mathcal{C}(\phi)} (2 - 2\cos(\theta_C))}{\det(\mathsf{L}_\theta + \mathsf{Q})} & \text{if } \phi \in \mathcal{M}(\mathcal{G}) \text{ is a MTSF,} \\ 0 \text{ otherwise,} \end{cases} \tag{5.22}$$

where $\det(\mathsf{L}_\theta + \mathsf{Q})$ is a normalization constant.

The proof of Theorem 5.2.2 relies on the following lemma.

Lemma 5.2.3. *Let $c_\phi \subseteq \mathcal{V} \cup \mathcal{E}$ be a connected component of a MTSF $\phi \in \mathcal{M}(\mathcal{G})$. Then, we have:*

$$\det((\nabla_\theta^{\mathsf{Q}})^* \nabla_\theta^{\mathsf{Q}})_{c_\phi, c_\phi} = \begin{cases} (2 - 2\cos(\theta_C)) \prod_{e \in c_\phi} w_e & \text{if } c_\phi \text{ is a unicycle with cycle } C, \\ q_r \prod_{e \in c_\phi \cap \mathcal{E}} w_e & \text{if } c_\phi \text{ is a tree rooted in } r. \end{cases} \tag{5.23}$$

Lemma 5.2.3 is proved in Appendix C.1. Let us point out that this is a simple extension of a result in [Kenyon, 2011].

Proof of Theorem 5.2.2. First, note that $K_{\mathcal{M}}$ is indeed a projection kernel, since one has

$$\mathsf{L}_\theta + \mathsf{Q} = (\nabla_\theta^{\mathsf{Q}})^* \nabla_\theta^{\mathsf{Q}}, \tag{5.24}$$

so that $K_{\mathcal{M}}^2 = K_{\mathcal{M}}$, and it then follows that the distribution $\mathrm{DPP}(K_{\mathcal{M}})$ associated to this kernel is a projection DPP.
The sample-size for projection DPPs is given by the rank of the kernel, in our case $\mathrm{rk}(K_{\mathcal{M}})$:

$$\mathrm{tr}(K_{\mathcal{M}}) = \mathrm{tr}((\mathsf{L}_\theta + \mathsf{Q})(\mathsf{L}_\theta + \mathsf{Q})^{-1}) = |\mathcal{V}|, \tag{5.25}$$

which, as a consequence, shows that $\mathbf{P}_{\mathrm{DPP}(K_{\mathcal{M}})}(\phi) = 0$ whenever $|\phi| \neq |\mathcal{V}|$, ensuring that $|\phi| = |\mathcal{V}|$ almost surely.
Now, let $\phi \subseteq \mathcal{V} \cup \mathcal{E}$ with $|\phi| = |\mathcal{V}|$. From Lemma 3.2.2 (*i.e.*, by multiplicativity of the determinant):

$$\det(K_{\mathcal{M}})_{\phi,\phi} = \frac{\det((\nabla_\theta^{\mathsf{Q}})^* \nabla_\theta^{\mathsf{Q}})_{\phi,\phi}}{\det(\mathsf{L}_\theta + \mathsf{Q})}, \tag{5.26}$$

and we are left with inspecting $\det((\nabla_\theta^{\mathsf{Q}})^* \nabla_\theta^{\mathsf{Q}})_{\phi,\phi} = \det(\nabla_\theta^{\mathsf{Q}})_{\phi,:} \det((\nabla_\theta^{\mathsf{Q}})^*)_{:,\phi}$ depending on ϕ. The two analyses are similar, and we start by inspecting $\det((\nabla_\theta^{\mathsf{Q}})^*)_{:,\phi}$. In this case, ϕ indexes the columns of $(\nabla_\theta^{\mathsf{Q}})^*$, and it is clear that two columns are linearly independent whenever they do not belong to the same component $c_\phi \subseteq \phi$ (see, *e.g.* Example 5.2.1).
If c_ϕ spans m nodes, $\det((\nabla_\theta^{\mathsf{Q}})^*)_{:,c_\phi}$ can only be non-zero if $|c_\phi| = m$ so that, almost surely, c_ϕ must be either a unicycle or a rooted tree (analogously to the definition of a MTSF as the solution to the assignment problem of a node or edge to each node of $\mathcal{G}$). It follows that ϕ must be a MTSF, and Lemma 5.2.3 expresses $\det((\nabla_\theta^{\mathsf{Q}})^* \nabla_\theta^{\mathsf{Q}})_{c_\phi, c_\phi}$ from $\det((\nabla_\theta^{\mathsf{Q}})^*)_{:,c_\phi}$ and $\det(\nabla_\theta^{\mathsf{Q}})_{c_\phi,:}$ explicitly for those connected components c_ϕ of ϕ. We finally obtain

$$\mathbf{P}_{\mathrm{DPP}(K)}(\phi) = \frac{\prod_{r \in \phi \cap \mathcal{V}} q_r \prod_{e \in \phi \cap \mathcal{E}} w_e \prod_{C \in \mathcal{C}(\phi)} ((2 - 2\cos(\theta_C))}{\det(\mathsf{L}_\theta + \mathsf{Q})} \tag{5.27}$$

whenever $\phi \in \mathcal{M}(\mathcal{G})$, which is exactly $\mathbf{P}_{\mathcal{D}_{\mathcal{M}}}(\phi)$. Note that we obtain the normalization constant $\det(\mathsf{L}_\theta + \mathsf{Q})$ as a by-product. ■

Remark 5.2.4. *When the connection is trivial, or perfectly coherent (*i.e., $\cos(\theta_C) = 1$ *for all cycles C in the graph), a number of simplifications occur.*

1. *All the components c_ϕ must be trees since, generalizing the analysis in,* e.g., *Remark 3.2.4,*

$$\det((\nabla_\theta^{\mathsf{Q}})^*)_{:,c_\phi} = 0 \tag{5.28}$$

whenever c_ϕ contains a cycle.

2. *Lemma 5.2.3 simply reads*

$$\det((\nabla_\theta^{\mathsf{Q}})^* \nabla_\theta^{\mathsf{Q}})_{c_\phi, c_\phi} = q_r \prod_{e \in c_\phi \cap \mathcal{E}} w_e. \tag{5.29}$$

In the end, we recover a proof that the distribution $\mathcal{D}_{\mathcal{F}}$ over RSFs is a DPP with kernel $K_{\mathcal{F}}$.

Remark 5.2.5. *Theorem 5.2.2 is readily translated to Γ-MTSFs. In this setting,* $\mathrm{DPP}(K_{\mathcal{M}})$ *is understood as sampling edges $\phi^\Gamma \subseteq \mathcal{E}^\Gamma$ in $\mathcal{G}^\Gamma$, and we simply have*

$$\mathbf{P}_{\mathrm{DPP}(K_{\mathcal{M}})}(\phi) = \begin{cases} \frac{\prod_{e \in \phi^\Gamma} w_e \prod_{C \in \mathcal{C}(\phi)} (2 - 2\cos(\theta_C))}{\det(\mathsf{L}_\theta^\Gamma)_{\mathcal{V}}} & \text{if } \phi \text{ is a } \Gamma\text{-MTSF,} \\ 0 & \text{otherwise.} \end{cases} \tag{5.30}$$

This is further generalized to $\mathcal{A}$-MTSFs in a graph $\mathcal{G}$. Consider indeed the restriction $(\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}}$ of the twisted differential operator ∇_θ to its columns indexed by $\mathcal{V} \setminus \mathcal{A}$, and define the marginal kernel

$$K_{\mathcal{M}}^{\mathcal{A}} = (\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}} \left(((\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}})^* (\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}} \right)^{-1} ((\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}})^*, \tag{5.31}$$

where the invertibility of $((\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}})^* (\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}} = (\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ follows from Lemma 4.1.8. Then, a completely analogous argument shows that

$$\mathbf{P}_{\mathrm{DPP}(K_{\mathcal{M}}^{\mathcal{A}})}(\phi) = \begin{cases} \frac{\prod_{e \in \phi \cap \mathcal{E}} w_e \prod_{C \in \mathcal{C}(\phi)} (2 - 2\cos(\theta_C))}{\det(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}} & \text{if } \phi \text{ is a } \mathcal{A}\text{-MTSF,} \\ 0 & \text{otherwise,} \end{cases} , \tag{5.32}$$

so that $\mathrm{DPP}(K_{\mathcal{M}}^{\mathcal{A}})$ *can be identified with $\mathcal{D}_{\mathcal{M}}^{\mathcal{A}}$.*

As a DPP, the distribution $\mathcal{D}_{\mathcal{M}}$ may be leveraged to estimate the solution of an associated linear-least-squares problem (following Theorem 3.0.2). In this case, this problem turns-out to correspond exactly to the Tikhonov regularization for heteroscedastic noise:

$$(\mathsf{L}_\theta + \mathsf{Q})^{-1} \mathsf{Q}\, g. \tag{5.33}$$

Whether this estimation is efficient or not depends on a number of factors and, in particular, on the time necessary to draw a sample from $\mathcal{D}_{\mathcal{M}}$. In particular, generic DPP samplers such as that of [Hough et al., 2006]) have $\mathcal{O}((|\mathcal{V}| + |\mathcal{E}|)^3)$ sampling-cost when sampling from $\mathcal{D}_{\mathcal{M}}$ (*i.e.*, from $\mathrm{DPP}(K_{\mathcal{M}})$), which is far too expensive in RandNLA applications. We detail in the next section an efficient Wilson-like algorithm for sampling MTSFs distributed according to $\mathcal{D}_{\mathcal{M}}$.

5.3 SAMPLING FROM $\mathcal{D}_{\mathcal{M}}$

Much like Wilson's algorithm, we use loop-erased random-walks to sample MTSFs from $\mathcal{D}_{\mathcal{M}}$. Much unlike Wilson's algorithm, the algorithm we derive depends on the incoherence of the connection, and the resulting MTSF will only be distributed according to $\mathcal{D}_{\mathcal{M}}$ under the following technical condition.

Condition 5.3.1 (Weak-Inconsistency). *For all cycles C in $\mathcal{G}$,* $\cos(\theta_C) \geq 0$.

Let us stress that this condition must hold *for all the cycles of the graph*, and not solely for the cycles $\mathcal{C}(\phi)$ of some MTSF ϕ. In the following, we refer to Condition 5.3.1 as "the" weak-inconsistency condition.

For the remainder of this section, we will focus on Γ-MTSFs instead of MTSFs, the reason being that our algorithm proceeds by running loop-erased random-walks on the extended graph $\mathcal{G}^\Gamma$. Specifically, the sampling goes as follows.

MTSF-Sampling Algorithm. Suppose that the nodes $\mathcal{V}$ of $\mathcal{G}$ are ordered in a queue, and start from $\phi^\Gamma = \emptyset$. The following is then repeated until the queue has been exhausted.

1. Start a random walk on the extended graph from the first node $i \in \mathcal{V}$ in the queue, and keep track of the resulting path p.
2. If p reaches Γ after l steps, add the $(l-1)$-th node to the set of roots $\phi \cap \mathcal{V}$.
3. If p self-intersects and forms a loop C, perform a Bernoulli trial with success probability $1 - \cos(\theta_C)$:
   - If the trial succeeds, add the edges in the path p to ϕ^Γ and stop the walk.
   - If the trial fails, remove the cycle from p and continue the walk.
4. If p reaches a node spanned by ϕ^Γ, add the edges in the path p to ϕ^Γ and stop the walk.

The corresponding pseudo-code is available in Algorithm 4, where `random_successor`(u) is a function which, at each call, randomly outputs either Γ with probability $\frac{q}{d_u+q}$, or some node $v \in \mathcal{V}$ with probability $\frac{w_{u,v}}{d_u+q}$.
Remark that the Bernoulli trial is only well-defined under the weak-inconsistency condition.

Proposition 5.3.2. *Suppose that the weak-inconsistency condition 5.3.1 holds, and that either:*

- *at least one of the q_i's is non-zero;*
- *at least one cycle C in $\mathcal{G}$ has non-trivial holonomy (*i.e.*, $\cos(\theta_C) \neq 1$).*

Then, Algorithm 4 outputs a Γ-MTSF ϕ^Γ distributed according to $\mathcal{D}_{\mathcal{M}}$.

Let us draw a few remarks concerning Algorithm 4. The first one concerns its relation to other Wilson-like sampling algorithms.

Remark 5.3.3. *Algorithm 4 was independently introduced in [Fanuel and Bardenet, 2022] to sample non-rooted MTSFs. Moreover, it is but a simple mixture of*

2 We abuse here notation, and denote by p the set of *non-oriented* edges in the path.

Algorithm 4 MTSF sampling algorithm.

$\phi \leftarrow \emptyset$
while ϕ not spanning **do**
  Let $i \in \mathcal{V}$ be the first node in the queue not spanned by ϕ
  $u \leftarrow i$ ▷ u is the current node of the random walk
  $p \leftarrow \epsilon$ ▷ ϵ the empty path
  while $(u \neq \Gamma)$ and (p does not intersect ϕ or contain a cycle) **do**
    $u' \leftarrow$ `random_successor`(u) ▷ Move to next node
    $e \leftarrow (u, u')$, $p \leftarrow pe$ ▷ Add e to the path p
    if p contains a cycle C **then**
      Remove this cycle from p with probability $\cos(\theta_C)$
    end if
    $u \leftarrow u'$
  end while
  $\phi \leftarrow \phi \cup p$
end while
Output ϕ

- *Wilson's algorithm for rooted spanning forests [Wilson, 1996, Avena and Gaudillière, 2013];*
- *the algorithm described in [Kassel and Kenyon, 2017] to sample spanning forests of unicycle distributed according to the DPP with marginal kernel described in Equation* (5.18)*.*

The algorithms in both [Kassel and Kenyon, 2017] and [Fanuel and Bardenet, 2022] are only guaranteed to sample from the desired distribution under the weak-inconsistency condition 5.3.1. Further, this condition also appears, in a different guise, in [Kassel and Lévy, 2021], where it allows to simplify technical statements concerning Gaussian free fields on graphs endowed with a unitary connection.
*Note that there is also a number of practical situations for which weak-inconsistency can always be satisfied: for instance, when the incoherence of the connection is controlled by a scaled parameter (*e.g.*, for GSP over directed graphs), or in applications to angular synchronization such as,* e.g.*, perceived-luminance reconstruction [Yu, 2009] or ranking [Cucuringu, 2016].*

The second remark is more practically relevant as it concerns a variation of Algorithm 4, and generalizes existing observations for those similar algorithms we discussed in Remark 5.3.3.

Remark 5.3.4. *Algorithm 4 can be modified to sample from any distribution over* Γ*-MTSFs of the form*

$$\mathbf{P}(\phi^\Gamma) \propto \prod_{e \in \phi^\Gamma} w_e \prod_{C \in \mathcal{C}(\phi^\Gamma)} \alpha(C), \tag{5.34}$$

where $\alpha : \mathcal{C}(\phi) \to [0, 2]$ *is invariant under the orientation of the cycle* C*.*
The corresponding modification amounts to replacing the success-probability $\cos(\theta_C)$ *of the Bernoulli trial by* $\frac{\alpha(C)}{2}$*. The same proof will carry over to this setting.*

Our last remark is about the generalization to $\mathcal{A}$-MTSFs, and is crucial to applications in harmonic-interpolation problems.

Remark 5.3.5. *In order to sample from the distribution $\mathcal{D}_M^{\mathcal{A}}$ over $\mathcal{A}$-MTSFs such that*

$$\mathbf{P}_{\mathcal{D}_{\mathcal{M}}^{\mathcal{A}}}(\phi^{\mathcal{A}}) \propto \prod_{e \in \phi^{\mathcal{A}}} w_e \prod_{C \in \mathcal{C}(\phi^{\mathcal{A}})} (2 - 2\cos(\theta_C)), \tag{5.35}$$

the walks in Algorithm 4 should be stopped whenever some node $a \in \mathcal{A}$ is reached.
Much naturally, Remark 5.3.4 also applies to this setting, and one can sample $\mathcal{A}$-MTSFs from

$$\mathbf{P}(\phi^{\mathcal{A}}) \propto \prod_{e \in \phi^{\mathcal{A}}} w_e \prod_{C \in \mathcal{C}(\phi^{\mathcal{A}})} \alpha(C) \tag{5.36}$$

as well.

It turns out that much of the combinatorial complexity in Algorithm 4 can be abstracted away, and we preface the proof of Proposition 5.3.2 with an equivalent formulation obtained from the *stack representation* introduced in [Diaconis and Fulton, 1991] (and classically used to analyze the algorithm of [Wilson, 1996], used to sample spanning trees of a graph uniformly).

More precisely, we consider an equivalent description of Algorithm 4 (originally developed for Algorithm 1), which goes as follows.

- As an initialization step, associate to each node $i \in \mathcal{V}$ an infinite stack of neighboring nodes $u_i = (u_i^1, u_i^2, \dots) \in (\mathcal{V}^\Gamma)^\omega$, where each u_i^k is a neighbor of i in $\mathcal{G}^\Gamma$ independently drawn with probability $\frac{w_{i,u_i^k}}{d_i}$.

  Note that this association of nodes describes a subset $\overrightarrow{\phi} \subseteq \overrightarrow{\mathcal{E}^\Gamma}$ of directed edges in the extended graph $\mathcal{G}^\Gamma$, with
  $$\overrightarrow{\phi} = \bigcup_{i \in \mathcal{V}} \{(i, u_i^1)\}. \tag{5.37}$$

- Then, repeat the following.
  1. Detect subsets of edges in $\overrightarrow{\phi}$ that form cycles C_k in the graph $\mathcal{G}$ and, for each cycle, perform a Bernoulli trial with success probability $\cos(\theta_{C_k})$.
  2. If the trial is successful for a cycle C_k then, for all nodes i spanned by C_k, remove the top node of the stack u_i:
  $$(u_i^1, u_i^2, \dots) \mapsto (u_i^2, u_i^3, \dots). \tag{5.38}$$
  3. If the trial fails, "accept" cycle C_k, and never perform the trial again for this cycle in the subsequent iterations.

  The process ends once all cycles have been "accepted", or there are no cycles in ϕ.

Upon termination, "forgetting" the orientation of the edges in $\overrightarrow{\phi}$ results in a Γ-MTSF ϕ.
Further, and as is classically observed for Wilson's algorithm [Wilson, 1996], the order in which the cycles are processed does not influence the output $\overrightarrow{\phi}$. For completeness, we recall the argument of [Wilson, 1996] in Appendix C.2.

A consequence of this observation is that Algorithm 4 is nothing but an implementation of the previous procedure[3], and we are now ready to proceed with the proof of Proposition 5.3.2. Note

3 In addition, adopting the stack representation frames Algorithm 4 as a *partial rejection sampling* algorithm [Guo et al., 2019].

that our proof pertains to the stack-representation-based formulation, and let us point out that we closely follow the corresponding argument for spanning forests of unicycles exposed in [Kassel and Kenyon, 2017].

Proof of Proposition 5.3.2. First, it is clear that the algorithm terminates (with probability 1) since, for at least one node i, there is a non-zero probability that $u_i^k = \Gamma$ for some k (one may alternatively reason about the probability of cycles being kept). Moreover, any given Γ-MTSF ϕ^Γ such that $\mathbf{P}_{\mathcal{D}_\mathcal{M}}(\phi^\Gamma) > 0$ can be output by this procedure (in form of some $\overrightarrow{\phi}$), as it suffices that the edges of ϕ^Γ are reflected in the first layers of the stacks.
Consider then some $\overrightarrow{\phi}$ output as a result of:

- all the edges of $\overrightarrow{\phi}$ appearing at the top of the stacks;
- a finite sequence of cycles $\overline{C} = (C_{l_i})_{i\in I}$ being removed;
- a finite number of cycles C_k being "accepted".

In particular, conditioning over the removed cycles $\overline{C}$, $\overrightarrow{\phi}$ is output with probability

$$\mathbf{P}(\overrightarrow{\phi}) = \sum_{\overline{C}} \mathbf{P}(\overrightarrow{\phi} \mid \overline{C} \text{ is removed})\ \mathbf{P}(\overline{C} \text{ is removed}) \tag{5.39}$$

$$= \sum_{\overline{C}} \mathbf{P}(\overrightarrow{\phi} \text{ appears and has its cycles accepted } \wedge \overline{C} \text{ is removed}) \tag{5.40}$$

$$= \left(\sum_{\overline{C}} \mathbf{P}(\overline{C} \text{ is removed})\right) \prod_{e\in\overrightarrow{\phi}} w_e \prod_k (1-\cos(\theta_{C_k})), \tag{5.41}$$

where the Equation (5.41) follows from the independence of the nodes sampled in the stacks.
Noting that each cycle C in a Γ-MTSF ϕ^Γ is observed with two possible orientations in some $\overrightarrow{\phi}$, we finally obtain

$$\mathbf{P}(\phi) \propto \prod_{e\in\phi^\Gamma} w_e \prod_{C\in\mathcal{C}(\phi)} (2-2\cos(\theta_C)), \tag{5.42}$$

where the set of cycles $\mathcal{C}(\phi)$ corresponds to the set of cycles C_k, the orientations of which have been discarded. ■

This shows the correctness of Algorithm 4, and provides an alternative mean to sample from $\mathcal{D}_\mathcal{M}$. What remains to show is that this sampling algorithm is far more efficient than a generic DPP sampler.

5.4 RUNTIME ANALYSIS

We conclude this chapter by an analysis of the computational cost of Algorithm 4. Specifically, our final result is the following.

Proposition 5.4.1. *Algorithm 4 can be implemented to have* $\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right)$ *expected time-complexity, where* $q_{\min} = \min_i\{q_i;\ i\in\mathcal{V}\}$, *and* $\mathrm{Vol}(\mathcal{G}) = 2\sum_{e\in\mathcal{E}} w_e$.

Our proof of Proposition 5.4.1 proceeds in two steps.

- First, we need to compute (and bound) the expected number of random-walk steps in Algorithm 4.
- Second, we argue that this bound indeed translates to a bound on the expected complexity.

5.4.1 *Expected Number of Steps*

It turns out that the expected number of steps was recently established in [Fanuel and Bardenet, 2024], as described in Proposition 5.4.2 below.

Proposition 5.4.2 (From [Fanuel and Bardenet, 2024])**.** *Denote by $T_{\mathcal{M}}$ the number of calls to* `random_successor` *in a run of Algorithm 4. Then,*

$$\mathbf{E}(T_{\mathcal{M}}) = \operatorname{tr}\left((\mathsf{L}_\theta + \mathsf{Q})^{-1}(\mathsf{D} + \mathsf{Q})\right). \tag{5.43}$$

We refer the reader to [Fanuel and Bardenet, 2024] for the proof. Let us now simplify the right-hand-side of Equation (5.43).

In order to get a more intuitive bound, notice that we have

$$\begin{aligned}
\operatorname{tr}\left((\mathsf{L}_\theta + \mathsf{Q})^{-1}(\mathsf{D} + \mathsf{Q})\right) &= \sum_{i\in\mathcal{V}} (\mathsf{L} + \mathsf{Q})^{-1}_{i,i}(q_i + d_i) &(5.44)\\
&= \sum_{i\in\mathcal{V}} \frac{\left((\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}\right)_{i,i}}{q_i}(q_i + d_i), &(5.45)
\end{aligned}$$

where we know that (from, *e.g.*, Equation (3.7))

$$\left((\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}\right)_{i,i} = \mathbf{P}_{\phi\sim\mathcal{D}_{\mathcal{M}}}(\{i,\Gamma\} \in \phi \in \phi \cap \mathcal{V}) \tag{5.46}$$

and, hence, lies in $[0, 1]$. It follows that $\mathbf{E}(T_{\mathcal{M}})$ is in $\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right)$ and, more precisely:

$$\mathbf{E}(T_{\mathcal{M}}) \leq |\mathcal{V}| + 2\frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}, \tag{5.47}$$

where we recall that the volume of $\mathcal{G}$ is defined by $\mathrm{Vol}(\mathcal{G}) = 2\sum_{e\in\mathcal{E}} w_e$ which, when the graph is unweighted, is simply $2|\mathcal{E}|$.
Further, in the specific case that $\mathsf{Q} = q\mathsf{D}$, the bound is linear in $|\mathcal{V}|$:

$$\mathbf{E}(T_{\mathcal{M}}) \leq \left(1 + \frac{2}{q_{\min}}\right)|\mathcal{V}|. \tag{5.48}$$

Let us remark that this is much faster than the generic DPP-sampling algorithms of Chapter 3, which result in $\mathcal{O}\left((|\mathcal{V}| + |\mathcal{E}|)^3\right)$ complexity in this case.

Remark 5.4.3. *Though we rely here on the characterization from [Fanuel and Bardenet, 2024], we first obtained the bounds of Equations (5.47) and (5.48) from an independent argument. In particular, we can show that*

$$\mathbf{E}(T_{\mathcal{M}}) \leq \operatorname{tr}\left((\mathsf{L} + \mathsf{Q})^{-1}(\mathsf{D} + \mathsf{Q})\right), \tag{5.49}$$

where the right-hand-side is known to correspond exactly to the expected number of steps of Algorithm 4 for trivial connections. *Equations* (5.47) *and* (5.48) *can be directly derived from this proposition as well (from a completely analogous argument). Interestingly, the proof does not rely on the algebraic expression of Proposition 5.4.2, but on a simple probabilistic argument. See Appendix C.3*[4].

5.4.2 *Implementation Strategies*

As the complexity of Algorithm 4 does not only depend on the number of random-walk steps, but also on the cost of the cycle-detection procedure, we propose in the following a number of implementation strategies to be used in practice. In particular, one of them achieves the

$$\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right) \tag{5.50}$$

expected complexity.

One-counter detection. The first strategy consists in dynamically assigning a numerical value $\mathrm{c}_v \in \mathbf{N}$ to each node $v \in \mathcal{V}$ encountered during the random walk (initialized to $\mathrm{c}_v = 0$), based on a global counter $\mathrm{c} \in \mathbf{N}$. This counter is updated as follows.

- c is incremented each time a new random walk begins (instruction at line 3 of Algorithm 4), and we set $\mathrm{c}_{u'} = \mathrm{c}$ whenever the random walk reaches u' on the instruction at line 7 of Algorithm 4 (if not already spanned by a MTSF).
- If a cycle C is created in node u' and discarded, we reset $\mathrm{c}_v = 0$ for all nodes v spanned by C *except* u'.

A cycle can then be detected in $\mathcal{O}(1)$ time by checking the value of $\mathrm{c}_{u'}$ at each step of the random walk: there is a cycle if $\mathrm{c}_{u'} = \mathrm{c}$, and no cycle if $\mathrm{c}_{u'} < \mathrm{c}$.

In a run of Algorithm 4, resetting the values c_v when cycles are discarded requires going through at most $T_{\mathcal{M}}$ nodes, which results in $\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right)$ overall expected *time complexity*, thereby proving Proposition 5.4.2.

In practice, the c_v's need to be reset because, whenever a cycle is discarded, nodes spanned by this cycle should no longer be remembered as spanned by the path p. This is expensive and we found it faster in our experiments to use an implementation of the following strategy.

Multiple-counters detection. In order to bypass the expensive resetting step, we consider *multiple* counters, in addition to the c_v's (indexed by a global c).
More specifically, we keep track of couples of values $(\mathrm{id}_v, \mathrm{val}_v) \in \mathbf{N}^2$ for each node $v \in \mathcal{V}$,

4 In particular, the upper-bound in Equation (5.49) describes one of many situations where a phenomenon of domination is observed as the result of adding a connection or, in mathematical-physics terminology, a magnetic field (see, *e.g.*, [Güneysu et al., 2016] for examples), and may be of independent interest. In our case, this domination concerns the expected number of steps, which decreases in the presence of a connection; in mathematical physics, these phenomena may relate to, *e.g.*, ground-state energy levels of particles (which take the form of spectral bounds on related operators) [Güneysu, 2012].
One should note though that in our case, and as obtained from the proof, this results holds more generally for any map $\alpha : \mathcal{C}(\phi) \to [0, 2]$, and does not necessarily stem from a $U(\mathbf{C})$-connection.

initialized to $(0,0)$, which are updated based on the values of two global counters $\mathrm{id}, \mathrm{val} \in \mathbf{N}^2$, and of values $\mathrm{cap}(\mathrm{id})$ *associated to each value of* id.
The id_v's should be thought of as the IDs of counters associated to each v (amongst multiple others), and the val_v's as the values of these counters. These values val_v's are reset to 0 whenever c is incremented (and a new random walk is initiated), and are otherwise updated by applying the following rules for all nodes u' reached by the random walk (if not already spanned by the MTSF).

- If no cycle is created and discarded at u', set $\mathrm{id}_{u'} = \mathrm{id}$, $\mathrm{val}_{u'} = \mathrm{val}$, and increment val.
- If a cycle is created at node u' and discarded, store the value $\mathrm{cap}(\mathrm{id}) = \mathrm{val}$ (the maximum value for the id^{th} counter), set $\mathrm{cap}(k) = 0$ for all $k > \mathrm{id}_{u'}$, $\mathrm{val}_{u'} = \mathrm{val}$, and increment id.

Using these counters, a cycle is detected at some base node u' if $\mathrm{val}_{u'} \leq \mathrm{cap}(\mathrm{id}_{u'})$.

The number id of counters used is unbounded, and we did not derive an expected time complexity for this multiple-counters strategy but consistently obtained better performance in our measurements when using it.

Remark 5.4.4. *The reader interested in implementation details will find a number of other algorithmic tricks leveraged in our Julia implementation*[5]*. Examples include:*

- *implementing the random walker using the alias method (for weighted graphs, which allows for $O(1)$ sampling time of each new random neighbor, modulo some $O(n)$ pre-processing) [Walker, 1977];*
- *only keeping track of the value val_v associated to the largest id_v;*
- *merging multiple counters (both ID's and values) in order to decrease the number of book-keeping operations.*

Remark 5.4.5. *The $\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right)$ complexity can be daunting for very small values of q_i's, but one should remember that this is only an upper bound and, in particular, does not account for the presence of unicycles in the sample.*

5.5 SUMMARY

The main contribution and novelty of the work we exposed in this chapter is certainly an elementary theory of ($\mathcal{A}$-)MTSFs.

1. ($\mathcal{A}$-)MTSFs are spanning subsets of edges in the graph $\mathcal{G}$, with connected components being either rooted trees of unicycles. Further, we showed that these structures naturally arise as generalizations of rooted spanning forests, through the lens of determinantal point processes. From this perspective, $\mathcal{A}$-MTSFs are naturally associated to the distribution $\mathcal{D}_{\mathcal{M}}^{\mathcal{A}}$ defined by
$$\mathbf{P}_{\mathcal{D}_{\mathcal{M}}^{\mathcal{A}}}(\phi^{\mathcal{A}}) \propto \prod_{e \in \phi^{\mathcal{A}}} w_e \prod_{C \in \mathcal{C}(\phi^{\mathcal{A}})} (2 - 2\cos(\theta_C)) \,. \tag{5.51}$$

5 https://gricad-gitlab.univ-grenoble-alpes.fr/gaia/synchromtsf

2. What is especially remarkable about these structures is that we are able to derive an efficient Wilson-like sampling algorithm from $\mathcal{D}_{\mathcal{M}}^{\mathcal{A}}$, *under a condition on the connection*; namely, if it is weakly incoherent. In the specific case that $\mathcal{A} = \{\Gamma\}$ (with Γ connected to all the other nodes i's by edges of weight q_i), our algorithm has

$$\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right) \tag{5.52}$$

expected complexity.

Another important contribution, though mostly invisible in these pages, is our Julia implementation of Algorithm 4, which is *more than an order of magnitude faster than that of [Fanuel and Bardenet, 2022]*. Apart from the implementation tricks we already mentioned, this mainly owes to a careful low-level implementation, and the use of more efficient data representations. The reader interested in those details can consult this implementation directly[6].

Our next step, in Chapter 6, is to develop global counterparts to these estimators, by leveraging the properties of $\mathcal{D}_{\mathcal{M}}^{\mathcal{A}}$ as a DPP.

6 At `https://gricad-gitlab.univ-grenoble-alpes.fr/gaia/synchromtsf`.

6

PROCESSING **C**-SIGNALS WITH MTSFS

All the evolution we know of proceeds from the vague to the definite.
— Charles Sanders Peirce[1]

Contributions in this chapter.

1. We introduce novel forest-based estimators for connection-aware smoothing and interpolation, based on multi-type spanning forests. We show that these estimators are unbiased, and characterize the variance of the smoothing estimator (Section 6.1)

2. Development of two variance-reduction techniques for the smoothing estimators. We characterize the variance of one of the resulting estimators, and provide a concentration bound on the error (Section 6.2).

1 See [Peirce, 1974].

3. Thorough experimental analysis of the improved smoothing estimators, which are able to outperform standard solvers depending on the topology of the graph.

4. Application of these estimators in iterative schemes for angular synchronization. We analyze the performance of this strategy and show that we are able to outperform existing solvers depending on the topology of the graph (Section 6.3.3).

While multi-type spanning forests can be formally obtained as a generalization of rooted spanning forests to graphs endowed with a $U(\mathbf{C})$-connection, it is a stretch, based on our developments in Chapter 4, to dub that generalization “intuitive”.
And indeed, we have so far left aside an important part of the theory: multi-type spanning forests can be more naturally understood as forming sets of edges along which it is natural to perform parallel transport, which in turn allows to design efficient, global estimators for a number of problems. Our aim in this chapter is to describe the associated phenomena[2] (Section 6.1), and leverage them in practical estimations (Sections 6.2 and 6.3).

We recall in the next few pages a number of applicative considerations that came up in **Part** i. This is nothing but a reminder, and the reader may comfortably skip to Section 6.1.

Let us first recall an important feature of the connection Laplacian, namely, that its quadratic form acts as a measure of coherence of a signal $f \in \mathbf{C}^{\mathcal{V}}$ with respect to the connection:

$$\langle f, \mathsf{L}_\theta f \rangle = \frac{1}{2} \sum_{e \in \overrightarrow{\mathcal{E}}} |f(t_e) - e^{\iota\theta_e} f(s_e)|^2; \tag{6.1}$$

a property that allows for convenient formulations of different problems.

(1) Angular synchronization, which asks to recover a set of unknown phase factors $x_i = e^{\iota\omega_i}$ from degraded pairwise offset-measurements $\theta_{i,j} = \omega_j - \omega_i + \varepsilon_{i,j}$, by solving

$$\underset{f \in U(\mathbf{C})^{\mathcal{V}}}{\operatorname{argmin}} \ \langle f, \mathsf{L}_\theta f \rangle, \tag{6.2}$$

with the connection given by $\psi_e = e^{\iota\theta_e}$ for each edge e. The spectrum of the connection Laplacian has remarkable properties with respect to the angular-synchronization problem:

- the eigenvector of L_θ associated to its smallest eigenvalue λ_1 yields a robust approximation to the solution of Problem (6.2), and λ_1 is a measure of the quality of the synchronization itself;
- the spectral gap between the first two smallest eigenvalues of L_θ is a measure of the difficulty of Problem (6.2), which accounts for both the incoherence of the connection, and the topology of the graph.

(2) Connection-aware Tikhonov smoothing of a signal $g \in \mathbf{C}^{\mathcal{V}}$, which is one strategy that can be used to recover a signal $f_\top$ from a measurement $g = f_\top + \varepsilon$ degraded with noise (with $f_\top$

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2 For the sake of clarity, we postpone the discussion of the formal links with our Feynman-Kac formulas to Chapter 7.

a "low-frequency" signal and noise generated according to, *e.g.*, $\varepsilon \sim \mathcal{N}_{\mathcal{C}}(0, \sigma^2 \mathsf{I})$), proceeds by applying a soft low-pass filter, which amounts to computing the optimal solution to:

$$\underset{f \in \mathbf{C}^{\mathcal{V}}}{\operatorname{argmin}} \; q\|f - g\|_2^2 + \langle f, \mathsf{L}_\theta f \rangle, \tag{6.3}$$

where $q \in \mathbf{R}_+^*$ is a positive regularization parameter. The solution $f_* \in \mathbf{C}^{\mathcal{V}}$ to this problem multiplies the "high-frequency" components of the signal g by a damping coefficient, and reads

$$f_* = (\mathsf{L}_\theta + q\mathsf{I})^{-1} q\mathsf{I} \; g \tag{6.4}$$

which, on appropriate graphs, should allow to recover $f_\top$ faithfully. This formulation is adapted to signals degraded by homoscedastic noise, and one may introduce another estimator in the presence of heteroscedastic noise to better deal with this type of degradation, in which case one considers estimators of the form $f_* = (\mathsf{L}_\theta + \mathsf{Q})^{-1} \mathsf{Q} \; g$, where Q is a diagonal matrix with non-negative coefficients $\mathsf{Q}_{i,i} = q_i$, representing node-wise regularization parameters.

(2') Connection-aware harmonic interpolation is a companion problem to the connection-aware Tikhonov smoothing problem and asks, given a **C**-signal $g : \mathcal{A} \to \mathbf{C}$ defined on some subset of nodes $\mathcal{A} \subseteq \mathcal{V}$, to find a signal $f \in \mathbf{R}^{\mathcal{V}}$ satisfying

$$\begin{cases} f(a) = g(a) \text{ for all } a \in \mathcal{A}, \\ (\mathsf{L}_\theta f)(i) = 0 \text{ for all } i \in \mathcal{V} \setminus \mathcal{A}, \end{cases} \tag{6.5}$$

the solution of which can be compactly expressed, on $\mathcal{V} \setminus \mathcal{A}$ using the connection Laplacian:

$$-((\mathsf{L}_\theta)_{\mathcal{V} \setminus \mathcal{A}})^{-1} (\mathsf{L}_\theta)_{\mathcal{V} \setminus \mathcal{A}} \; g. \tag{6.6}$$

This problem has appeared as a practical way to compute extensions of vector-fields, and could also be applied to, *e.g.*, angular synchronization with a few anchor nodes for which the the value ω_i is *a priori* known.

Let us stress that, in all of the above applications, the connection is known *a priori* (*e.g.*, as a result of some pre-processing steps), and the estimation only pertains to the graph signals. The next remark introduces a simple denoising pipeline that we use to illustrate our work later in this chapter.

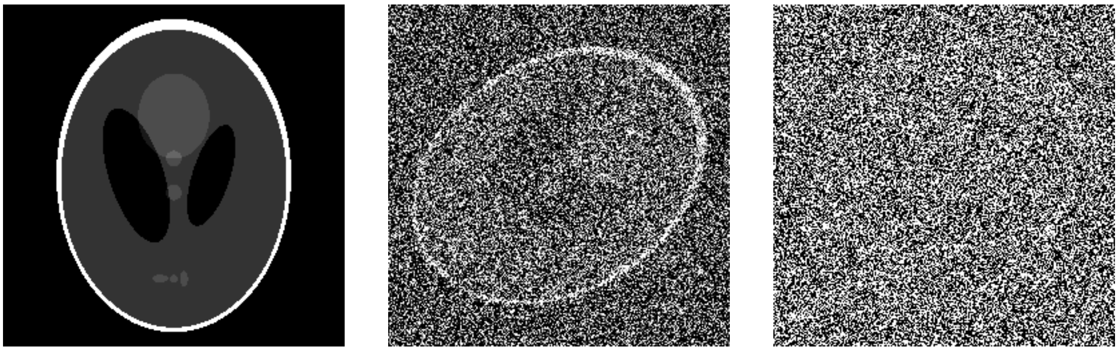
Remark 6.0.1. *Consider the following pipeline, inspired from the problem of denoising by intra-class averaging in,* e.g.*, cryo-EM pipelines.*

- *We are given n copies $(I_i)_{i=1}^n$ of a same image I_* that have been rotated and degraded, and are of the form*

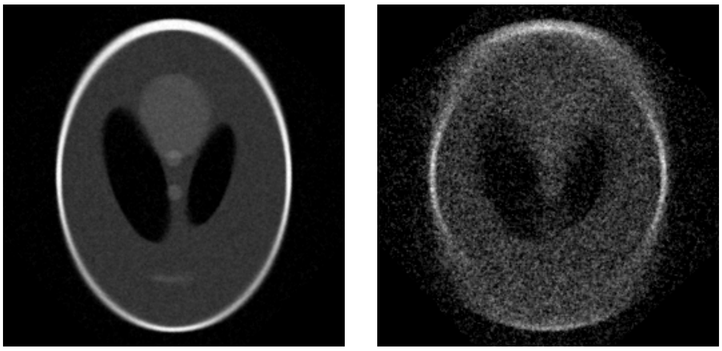
$$I_i = r_i(I_*) + \varepsilon, \tag{6.7}$$

where r_i is some unknown rotation and ε some (zero-mean) noise on the image.

- *Given the set of images $(I_i)_{i=1}^n$, estimate the rotations $\widetilde{r}_{i,j}$ that best register pairs of images I_i and I_j. This can be achieved using different types of registration tools and, typically, is only performed for a subset of pairs $\{i, j\}$, which correspond to the edges of an underlying graph with n nodes.*



(a) Left: Shepp-Logan phantom (I_*). Center: rotated image degraded by a mild amount of noise ($\varepsilon \sim \mathcal{N}(0, \mathsf{I})$). Right: rotated image degraded by a higher amount of noise ($\varepsilon \sim \mathcal{N}(0, 5\mathsf{I})$).



(b) Reconstructed image $\widetilde{I}$. Left: mild noise-level. Right: higher noise-level.

Figure 6.1: Top row: reference image I_* and degraded images I_i for two different noise levels. Bottom row: reconstructed images from $n = 10^3 = 1000$ copies of I_* degraded by a (fixed-level) noise.

- *From the data of the approximate rotations $\widetilde{r}_{i,j}$, estimate the underlying rotations $\widetilde{r}_i \simeq r_i$. This is the core problem of the processing, and is an instance of the angular-synchronization problem (recalled below).*
- *Finally, compute an estimation of the ground-truth image I_* as the mean of the rotated images:*

$$\widetilde{I} = \frac{1}{n} \sum_{i=1}^{n} \widetilde{r}_i^{-1}(I_i). \tag{6.8}$$

The connection here is estimated from the data as a pre-processing step, and the non-trivial part of the processing lies in determining the estimated rotations $\widetilde{r}_i$.
We illustrate this process in Figure 6.1, using the Shepp-Logan phantom as a base image[3]; finer details on the experimental setup and methods used are available in Section 6.3.

3 The Shepp-Logan phantom, named after American mathematician and statistician Lawrence Alan Shepp (1936−2013) and American electrical-engineer Benjamin Franklin "Tex" Logan, Jr. (1927−2015), is a classical test image for image-reconstruction algorithms, representing an approximation of a human head as the superposition of 10 ellipses.

From a computational perspective, the problems of connection-aware smoothing, interpolation and angular synchronization are similar: computing a solution requires numerically solving a linear-algebra problem (matrix inversion, eigenvalue decomposition), and the generic cost of exact, practical algorithms for these problems is in $\mathcal{O}(|\mathcal{V}|^3)$ (resp., $\mathcal{O}(|\mathcal{V} \setminus \mathcal{A}|^3)$ for interpolation)[4]. As we mentionned in the beginning of Chapter 1, this complexity prohibits the computation of *exact* solutions even for moderately large graphs ($|\mathcal{V}| \simeq 10^4 = 10,000$), and high-quality approximations are typically used instead. A common strategy consists in using Krylov-subspaces-based methods, that are based on successive matrix-vector multiplications [Saad, 2003, Saad, 2011], and result in standard deterministic algorithms used across a wide range of applications. There are two main drawbacks to these methods: their rate of convergence largely depends on the problem at hand, and the time of each iteration depends on the density of the matrix considered; as an illustration, recall the following theorem (Chapter 1), which pertains to the conjugate-gradient descent algorithm (and applies, in particular, to the connection-aware Tikhonov smoothing problem).

Theorem 6.0.2. *Given a positive definite $n \times n$ symmetric matrix A and a vector b, consider the linear-least-square problem (which is equivalent to solving a system $Aw = y$)*

$$\underset{w}{\operatorname{argmin}} \|Aw - y\|_2^2, \tag{6.9}$$

where $\|x\|_2 = \langle x, x \rangle$ denotes the usual squared l_2-norm. Denoting by w_* the optimal solution to this problem, the conjugate-gradient algorithm outputs an approximation $\widetilde{w}$ of w_* satisfying

$$\frac{\|\widetilde{w} - w_*\|_A^2}{\|w_*\|_A^2} \leq \varepsilon, \tag{6.10}$$

where $\varepsilon > 0$, and $\|x\|_A^2 = \langle x, Ax \rangle$, in time $\mathcal{O}\left(t_A \sqrt{\kappa_A} \log\left(\frac{1}{\varepsilon}\right)\right)$, with κ_A the condition number of A, and t_A the cost of a matrix-vector product involving A.

In practice, for the conjugate-gradient algorithm for instance, one may choose to deal with poor conditioning by building a *preconditioner* [Ferronato, 2012], that is, by solving a system of the form $\mathsf{P}^{-1}Aw = \mathsf{P}^{-1}y$ instead which, for a good choice of P, could largely improve the condition number of the system; the problem here is that building a good-quality preconditioner P is itself *"a combination of art and science"* [Saad, 2003], and a notoriously difficult problem. Further, note that the two issues of Krylov-subspaces methods may very well come up on a same instance of the problem: denser systems can be ill-conditioned, and classical preconditioners are mostly suitable for sparse, structured systems, making the problem all the more challenging (for example of well-known challenging systems, see, *e.g.*, [Chen et al., 2024]).

Monte-Carlo estimators follow from a different strategy, in which one proceeds to estimate an expectation

$$\mathbf{E}_{X \sim \mathcal{D}}(f(X)) \tag{6.11}$$

by drawing some number m of i.i.d. realizations $X_1, ..., X_m \sim \mathcal{D}$ (for the estimators we develop in this chapter, the X_i are multi-type spanning forests sampled according to the distribution $\mathcal{D}_{\mathcal{M}}$), and computing:

$$\mu(X_1, ..., X_m) = \frac{1}{m} \sum_{X_i} f(X_i) \tag{6.12}$$

4 The theoretically optimal time-complexity for virtually all numerical-linear-algebra problems is the same as that of matrix-matrix matrix multiplications [Demmel et al., 2007], in $\mathcal{O}(n^\omega)$, where it is known that $\omega \leq 2.371552$ [Williams et al., 2024]. However, these algorithms are impractical due to large constants in the complexity.

which, by virtue of the law of large numbers, converges to $\mathbf{E}_{X\sim\mathcal{D}}(f(X))$. Practically, the $\mathcal{O}\left(\frac{\sigma}{\sqrt{m}}\right)$ convergence rate of Monte-Carlo estimators, where σ^2 denotes the variance of the estimator $f(X)$, is often quite slow, as compared to other algorithms (*e.g.*, when $\mathbf{E}_{X\sim\mathcal{D}}(f(X))$ is the solution to a linear-least-square problem), but randomized estimators may enjoy a number of chief advantages:

1. Monte-Carlo estimators are not suited for high-quality approximations, but can be used to obtain sensible approximations at moderate cost, especially for large-scale problems;

2. algorithms formulated *via* Monte-Carlo estimators may largely differ from deterministic ones, allowing for flexible schemes of computations, and to further exploit parallelism or distributed computing;

3. depending on the problem, deterministic algorithms with reasonable complexity may not even exist[5], in which case Monte-Carlo estimators are the only tool to obtain a solution;

Speaking to the growing success of randomized algorithms in handling very-large-scale problems, Monte-Carlo estimators have found their way into standard textbooks and mature toolboxes for numerical linear algebra, together with well-established deterministic techniques, under the label of Randomized Numerical Linear Algebra (RandNLA); see [Martinsson and Tropp, 2020] for a review[6]. The solutions to the connection-aware graph problems above can all be expressed using linear-algebraic expressions involving large matrices and, as such, our MTSF-based Monte-Carlo estimators fall into this challenging framework.

5 For a striking historical example, see [Lovász and Simonovits, 1993].

6 Beyond Monte-Carlo estimators, an important line of research in RandNLA consists in the randomized construction of preconditioners; an illustrative contribution in this sense is that of [Kyng and Song, 2018], which proposes to build preconditioners for systems of the form $\mathsf{L}w = y$ involving the combinatorial graph Laplacian by computing a sparse approximate inverse of L, as obtained by sampling spanning trees of $\mathcal{G}$ uniformly.

6.1 ESTIMATING $\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}$

Similarly to the Feynman-Kac formula we developed in Chapter 4, we show in the following that parallel transport along branches of (random) multi-type spanning forests allows to express the entries of matrices of the form $\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}$. Phenomena of this type allow to design simple MTSF-based estimators, and to show that, for instance, the global estimator for connection-aware Tikhonov smoothing teased in Chapter 4 (Theorem 4.0.1) is indeed unbiased. The underlying property is captured, from a determinantal perspective[7], in the following theorem.

Theorem 6.1.1. *Let $\mathcal{G}$ be a graph endowed with a $U(\mathbf{C})$-connection. For a given multi-type spanning forest ϕ and two nodes u, v spanned by the same component of ϕ, denote by $u \xrightarrow{\phi} v$ the unique path from u to v in ϕ. Note that $\left(u \xrightarrow{\phi} v\right) \subseteq \mathcal{E}$.*
The entries of $(\mathsf{L}_\theta + \mathsf{Q})^{-1}$ can then be expressed as

$$\left((\mathsf{L}_\theta + \mathsf{Q})^{-1}\right)_{i,j} = \sum_{\phi \in \mathcal{M}(\mathcal{G})} \left(\mathbf{1}_{c_\phi(j)}(i) \prod_{r \in \phi \cap \mathcal{V}} q_r \prod_{e \in \phi \cap \mathcal{E}} w_e \prod_{C \in \mathcal{C}(\phi)} (2 - 2\cos(\theta_C)) \right) \psi_{j \xrightarrow{\phi} i}, \tag{6.13}$$

where $\mathbf{1}_{c_\phi(j)}(i)$ is the indicator that i belongs to the set of nodes $c_\phi(j)$ spanned by a tree of ϕ rooted in node j.
In particular, when $q_j \neq 0$, we have

$$\left((\mathsf{L}_\theta + \mathsf{Q})^{-1}\right)_{i,j} = \frac{1}{q_i} \mathbf{E}_{\phi \sim \mathcal{D}_\mathcal{M}} \left(\mathbf{1}_{c_\phi(j)}(i) \psi_{j \xrightarrow{\phi} i} \right). \tag{6.14}$$

Before proving Theorem 6.1.1, let us remark that *a completely analogous argument* allows to show a generalization to $\mathcal{A}$-forests, that we later use to derive an MTSF-based estimator for connection-aware harmonic interpolation. The statement of this result is a bit cumbersome, and left to Appendix D.1.
Our proof of Theorem 6.1.1 is straightforward, and only relies on the Cauchy-Binet formula and a variant of Lemma 5.2.3.

Proof of Theorem 6.1.1. We begin by rewriting $(\mathsf{L}_\theta + \mathsf{Q})^{-1}_{i,j}$ as an expectation, starting from Cramer's rule (Equation (3.30)):

$$(\mathsf{L}_\theta + \mathsf{Q})^{-1}_{i,j} = (-1)^{i+j} \frac{\det(\mathsf{L}_\theta + \mathsf{Q})_{\mathcal{V}\setminus\{j\},\mathcal{V}\setminus\{i\}}}{\det(\mathsf{L}_\theta + \mathsf{Q})}. \tag{6.15}$$

We can rewrite the numerator using the Cauchy-Binet formula:

$$\det(\mathsf{L}_\theta + \mathsf{Q})_{\mathcal{V}\setminus\{j\},\mathcal{V}\setminus\{i\}} = \sum_{\substack{\phi \subseteq \mathcal{V} \cup \mathcal{E} \\ |\phi| = |\mathcal{V}|-1}} (-1)^{i+j} \det\left(\left[((\nabla_\theta^{\mathsf{Q}})^*)_{:,\phi} \ \delta_j\right]\right) \det\left(\begin{bmatrix} (\nabla_\theta^{\mathsf{Q}})_{\phi,:} \\ \delta_i \end{bmatrix}\right), \tag{6.16}$$

7 As we will discuss in Chapter 7, it is likely that similar results holds for multi-type spanning forests drawn according to a larger class of probability distributions.

where δ_i is the i-th vector in the usual basis of $\mathbf{C}^{\mathcal{V}}$. An argument analogous to that Lemma 5.2.3 then shows that the product of determinants

$$\det\left(\left[((\nabla_\theta^{\mathsf{Q}})^*)_{:,\phi}\ \delta_j\right]\right)\det\left(\begin{bmatrix}(\nabla_\theta^{\mathsf{Q}})_{\phi,:}\\ \delta_i\end{bmatrix}\right) \tag{6.17}$$

can only be non-zero if ϕ contains a tree $T_i^j \subseteq \mathcal{E}$ spanning both i and j (with no root), with contribution $\psi_{j\xrightarrow{\phi}i}\prod_{e\in T_i^j} w_e$ to the product, and if *all* the other components are rooted trees or unicycles, with associated contributions described in Lemma 5.2.3. We thus obtain

$$\begin{aligned}(\mathsf{L}_\theta+\mathsf{Q})^{-1}_{i,j} &= \frac{1}{q_j}\sum_{\phi\in\mathcal{M}(\mathcal{G})}\mathbf{P}_{\phi\sim\mathcal{D}_\mathcal{M}}(\phi)\mathbf{1}_{c_\phi(j)}(i)\psi_{j\xrightarrow{\phi}i} &(6.18)\\ &= \frac{1}{q_j}\mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}\left(\mathbf{1}_{c_\phi(j)}(i)\psi_{j\xrightarrow{\phi}i}\right), &(6.19)\end{aligned}$$

where $\mathbf{1}_{c_\phi(j)}(i)$ is the indicator that i belongs to the set of nodes $c_\phi(j)$ spanned by the tree rooted in j. ■

Remark 6.1.2. *Note that the proof of Theorem 6.1.1 does not use the fact that $\mathcal{D}_\mathcal{M}^\mathcal{A}$ is a determinantal point processes* explicitly, *but uses much of the same tools we used to prove that it is indeed one. This can serve as an alternative motivation for the definition of $\mathcal{A}$-MTSFs. Further, and though we did not try to derive it explicitly, it is likely that a general version of Theorem 6.1.1 holds for any projective DPP (similarly to Theorems 3.0.2 and 3.3.5).*

We directly obtain Theorem 4.0.1 as a corollary of Theorem 6.1.1, recalled below. Note that the following statement pertains to the more general variant of connection-aware Tikhonov smoothing adapted to heteroscedastic noise, in which case the solution reads:

$$(\mathsf{L}_\theta+\mathsf{Q})^{-1}\mathsf{Q}g. \tag{6.20}$$

Corollary 6.1.3. *Let $\mathcal{G}$ be a graph endowed with a $U(\mathbf{C})$-connection and, for a given multi-type spanning forest ϕ and node $u\in\mathcal{V}$ spanned by a rooted tree of ϕ, denote by $r_\phi(u)\in\mathcal{V}$ the root of this tree. Assume that there exists either some node v such that $q_v>0$, or some cycle C with non-trivial holonomy ($\cos(\theta_C)\neq 1$), and consider some $i\in\mathcal{V}$. We define the estimator*

$$\widetilde{f}(i,\phi,g) = \begin{cases}\psi_{r_\phi(i)\xrightarrow{\phi}i}\left(g(r_\phi(i))\right) & \text{if } i \text{ is spanned by a rooted tree,}\\ 0 & \text{if } i \text{ is spanned by a unicycle,}\end{cases} \tag{6.21}$$

which propagates the value from $r_\phi(i)$ to i if $i\in\mathcal{V}$ belongs to a rooted tree, *and sets $\widetilde{f}(i,\phi,g)=0$ otherwise. Then, we have:*

$$f_*(i) = \mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}(\widetilde{f}(i,\phi,g)). \tag{6.22}$$

Algorithm 5 MTSF-based estimator (Theorem 6.1.3).

$f_i \leftarrow 0\ \forall i \in \mathcal{V}$
Repeat m times
  Sample $\phi \in \mathcal{M}(\mathcal{G})$ from $\mathcal{D}_{\mathcal{M}}$ ▷ Algorithm 4 can be used
  for $i \in \mathcal{V}$ **do**
    if i belongs to a rooted tree of ϕ **then**
      $f_i \leftarrow f_i + \psi_{r_\phi(i) \xrightarrow{\phi} i}\left(g(r_\phi(i))\right)$
    end if
  end for
Output $\frac{1}{m}(f_1, ..., f_{|\mathcal{V}|})$

Proof. Equation (6.22) is obtained by a direct computation:

$$\left((\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}g\right)(i) = \langle \delta_i, (\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}g \rangle \tag{6.23}$$

$$= \sum_{j \in \mathcal{V}} q_j (\mathsf{L}_\theta + \mathsf{Q})^{-1}_{i,j} g(j) \tag{6.24}$$

$$= \sum_{j \in \mathcal{V}} \mathbf{E}_{\phi \sim \mathcal{D}_{\mathcal{M}}} \left(\psi_{j \xrightarrow{\phi} i} (g(j)) \, \mathbf{1}_{c_\phi(j)}(i) \right) \tag{6.25}$$

$$= \mathbf{E}_{\phi \sim \mathcal{D}_{\mathcal{M}}} (\widetilde{f}(i, \phi, g)), \tag{6.26}$$

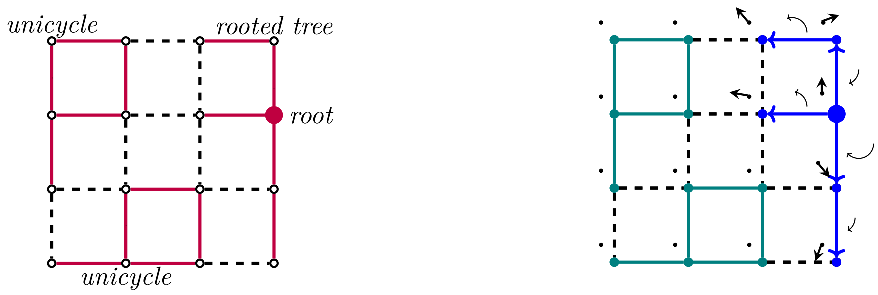
where Equation (6.25) follows Equation (6.14) (in particular, it holds whenever $q_j \neq 0$; and in case $q_j = 0$ we have $\mathbf{P}_{\mathcal{D}_{\mathcal{M}}}(\phi \wedge j \in \phi \cap \mathcal{V}) = 0$). ∎

Let us stress that Theorem 6.1.1 (and Corollary 6.1.3) holds *for any connection*, with no regard for the incoherence of the connection.
Corollary 6.1.3 can be understood as a specialization of Theorem 3.0.2, for estimating the solution of linear-least-squares systems using DPPs, but is much more efficient than in the general case.

- Assuming weak-inconsistency, Algorithm 4 allows to sample multi-type spanning forests in $\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right)$ time.
- The estimation for a given MTSF ϕ is a simple locally-coherent signal $\widetilde{f}$, with each value $\widetilde{f}(i)$ obtained by transporting the value at the root of the tree, which can be computed in $\mathcal{O}(|\mathcal{V}|)$ time. Intuitively, one may break this estimation down into two parts: first, finding a set of well-distributed points in the graph (the roots of ϕ, distributed according to a DPP, which is a repulsive point process); second, building a low-variance, *global* approximation of the smooth signal by transporting the value of the noisy signal at the root to a given neighborhood (the corresponding tree).

The resulting algorithm is summarized in Algorithm 5 and results, for m multi-type spanning forests, in $\mathcal{O}\left(m\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right)\right)$ complexity. See Figure 6.2 for an illustration of the estimation process.

In order to build more intuition about $\widetilde{f}$, one may consider the following.
As a smoothing operator, and in contrast to other operators such as the adjacency matrix A_θ,



(a) A MTSF in purple, with two unicycles (top left, bottom) and one tree (right). The large purple node is the root of the tree.

(b) Estimation: we transport the values from the root (blue circle) to its tree via the blue arrows. The estimation is 0 on the unicycles (represented by arrowless dots at the top left of the nodes).

Figure 6.2: Reproduction of Figure 4.2. Corollary 6.1.3 illustrated on the 4×4 grid graph. Left: a MTSF. Right: estimation by propagations along the branches of the MTSF. The node-wise estimates (in $\mathbf{C}$) are represented as vectors, depicted as arrows (top left of the nodes), and the connection maps are represented as circular rotating arrows (only depicted along the edges of the unique tree in the MTSF we consider).

Tikhonov regularization accounts for *long-range* interactions between nodes; this can for instance be understood in light of the probabilistic interpretation of the Feynman-Kac formula from Proposition 4.1.2. These long-range interactions, as a result of incoherence in the connection, tend to bias the estimation on a given node (since information brought back along two different paths yields two different estimations).
Unicycles help to account for this phenomenon: the more incoherent a cycle is (*i.e.*, the smaller $\cos(\theta_C)$ is), the more likely it is to be sampled (according to Equation (5.5)) and, hence, not to be accounted for in the estimation (0 estimate along the spanned nodes).

It is also possible to characterize the variance of the estimator $\tilde{f}$. In the following, we denote by $\mathrm{var}(\tilde{z}) = \mathbf{E}(|\tilde{z}|^2) - \left|\mathbf{E}(\tilde{z})^2\right|$ the variance of a complex-valued random variable $\tilde{z}$, and quantify the weighted expected squared error

$$\mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}\left(\|\widetilde{f} - f_*\|^2_{\mathsf{Q}}\right) = \sum_{i\in\mathcal{V}} q_i \mathrm{var}\left(\widetilde{f}(i,\phi,g)\right), \tag{6.27}$$

where $\|f\|^2_{\mathsf{Q}} = \sum_{i\in\mathcal{V}} q_i |f(i)|^2$ (note that the equality in Equation (6.27) follows from the linearity of the expectation).

Proposition 6.1.4. *Under the same assumptions as in Corollary 6.1.3, we have*

$$\mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}\left(\|\widetilde{f}(\phi,g) - f_*\|^2_{\mathsf{Q}}\right) = \|g\|^2_{\mathsf{Q}} - \|f_*\|^2_{\mathsf{Q}}. \tag{6.28}$$

The proof relies on the observation that $\widetilde{f}(\phi,g)$ is *linear in* g, so that we can write $\widetilde{\mathsf{S}}_\phi g = \widetilde{f}(\phi,g)$ for some linear operator/matrix $\widetilde{\mathsf{S}}_\phi$, and otherwise consists in a simple computation. Note that we have $\mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}\left(\widetilde{\mathsf{S}}_\phi g\right) = f_*$ and

$$\mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}\left(\widetilde{\mathsf{S}}_\phi\right) = (\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}. \tag{6.29}$$

Proof of Proposition 6.1.4. First, notice that $\widetilde{\mathsf{S}}_\phi^* \mathsf{Q} \widetilde{\mathsf{S}}_\phi = \mathsf{Q}$. We then have

$$\mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}\left(\|\widetilde{S}(\phi,g)-f_*\|_\mathsf{Q}^2\right) = \sum_{i\in\mathcal{V}} q_i\left(\mathbf{E}\left(\left|\left(\widetilde{S}_\phi g\right)(i)\right|^2\right) - \left|\mathbf{E}\left(\left(\widetilde{S}_\phi g\right)(i)\right)^2\right|\right) \tag{6.30}$$

$$= \left(\sum_{i\in\mathcal{V}}\langle g, \mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}\left(\widetilde{\mathsf{S}}_\phi^*(q_i\delta_i\delta_i^*)\widetilde{\mathsf{S}}_\phi\right) g\rangle\right) - \|f_*\|_\mathsf{Q}^2 \tag{6.31}$$

$$= \|g\|_\mathsf{Q}^2 - \|f_*\|_\mathsf{Q}^2, \tag{6.32}$$

where δ_i is usual basis vector of $\mathbf{C}^\mathcal{V}$ with non-zero entry $\delta_i(i) = 1$, Equation (6.31) follows from the linearity of the expectation, and Equation (6.32) from the previous observation that $\widetilde{\mathsf{S}}_\phi^* \mathsf{Q} \widetilde{\mathsf{S}}_\phi = \mathsf{Q}$. ■

Remark 6.1.5. *In actual implementations, and under weak-inconsistency, the propagation maps used in Algorithm 5 can be computed as a by-product of Algorithm 4. Indeed, one only requires knowledge of: the root of each tree, whether or not a node i belongs to a rooted tree and, if it does, the propagation map from the root $r_\phi(i)$ to i (which can be derived from the loop-erased random path during the sampling process). The actual multi-type spanning forest ϕ need never be used in the estimation.*

Remark 6.1.6. *In case the weak-inconsistency condition 5.3.1 is not satisfied, Algorithm 4 no longer samples multi-type spanning forests* according to $\mathcal{D}_\mathcal{M}$, *but samples multi-type spanning forests nonetheless, according to some other distribution. One may still be able to use the estimator $\widetilde{f}$ to estimate f_*, by proceeding in a number of alternative ways.*

1. *One may simply decide to "keep" all the cycles that are too incoherent during an execution of Algorithm 4 (*i.e.*, those cycles C such that* $\cos(\theta_C) < 0$*). Provided that either (or both) of the following holds:*
   - *there are few inconsistent cycles (with* $\cos(\theta_C) \leq 0$*);*
   - *the q_i's are large enough;*

   *this strategy should still result in a sensible (though biased) estimator. Let us note that there are some applications for which we expect such conditions to hold (*e.g.*, the q_i's are typically large in [Sharp et al., 2019] and, further, inconsistency often results from noise in real-data, which translates to mostly weakly-incoherent cycles; see Section 6.3.2 below for an example where this strategy is successful).*

2. *One may build upon the previous strategy, and perform importance sampling in order to estimate af_*, for some constant $a \in \mathbf{R}_+^*$ [Tokdar and Kass, 2010]. Indeed, the above sampling strategy results in MTSFs distributed according to*

$$\mathbf{P}_{IS}(\phi) \propto q^{|\phi\cap\mathcal{V}|}\prod_{C\in\mathcal{C}(\phi)} \min(2, 2-2\cos(\theta_C)), \tag{6.33}$$

   so that one may use the importance weights[8]

$$w(\phi) = \prod_{C\in\mathcal{C}(\phi)} \max(1, 1-\cos(\theta_C)). \tag{6.34}$$

8 These specific importance weights were originally proposed in the context of connection-Laplacian sparsification [Fanuel and Bardenet, 2022].

3. *Going further, one can consider a self-normalized importance-sampling estimator [Owen, 2013]. For a set of m multi-type spanning forests $\{\phi_k\}_{k=1}^m$ sampled according to the distribution of Equation (6.34), this estimator reads*

$$\widetilde{f}_{IS} = \frac{1}{\sum_{k=1}^m w(\phi_k)} \sum_{k=1}^m \tilde{f}(\phi_k, g) \tag{6.35}$$

and, as $m \to \infty$, tends to f_*. Note that this results holds regardless of the incoherence of the connection, but only yields an unbiased estimator in the limit.

Remark 6.1.7. *In particular, our approach is flexible enough to estimate quantities of the form $(\mathsf{L}_\theta + q\mathsf{D})^{-1}(qD)g'$, which in turn allows to estimate $(\widetilde{\mathsf{L}_\theta} + q\mathsf{I})^{-1}q\mathsf{I}g$ by setting $g' = \mathsf{D}^{-\frac{1}{2}}g$ and leveraging the decomposition*

$$q(\widetilde{\mathsf{L}_\theta} + q\mathsf{I})^{-1}g = \mathsf{D}^{\frac{1}{2}}((\mathsf{L}_\theta + q\mathsf{D})^{-1}(q\mathsf{D}))g', \tag{6.36}$$

where we recall that $\widetilde{\mathsf{L}_\theta} = \mathsf{D}^{-\frac{1}{2}}\mathsf{L}_\theta\mathsf{D}^{-\frac{1}{2}}$ is the normalized connection Laplacian. In this case, and under weak-inconsistency, Algorithm 4 has $\mathcal{O}(|\mathcal{V}|)$ complexity (from Equation (5.48)), and so does the resulting estimator $\widetilde{f}$.

Theorem 6.1.1 also translates to a simple MTSF-based estimator for the connection-aware interpolation problem.

Corollary 6.1.8. *Let $\mathcal{G}$ be a graph endowed with a $U(\mathbf{C})$-connection, and $\mathcal{A} \subseteq \mathcal{V}$ a subset of nodes intersecting all the connected components of $\mathcal{G}$. For a given $\mathcal{A}$-MTSF $\phi^{\mathcal{A}}$ on $\mathcal{G}$ and node $u \in \mathcal{V}$ spanned by a tree of $\phi^{\mathcal{A}}$, denote by $r_{\phi^{\mathcal{A}}}(u)$ the unique node of the tree connected to some node $a \in \mathcal{A}$ by an edge of $\phi^{\mathcal{A}}$, and by $a \xrightarrow{\phi^{\mathcal{A}}} v$ the unique path in $\phi^{\mathcal{A}}$ from a to v. For $i \in \mathcal{V}$, we define the estimator*

$$\widetilde{f}_{\mathcal{A}}(i, \phi^{\mathcal{A}}, g) = \begin{cases} \psi_{r_{\phi^{\mathcal{A}}}(i) \xrightarrow{\phi^{\mathcal{A}}} i} \left(\psi_{a, r_{\phi^{\mathcal{A}}}(i)}\left(g(a)\right)\right) & \text{if } i \text{ is spanned by a tree of } \phi^{\mathcal{A}}, \\ 0 & \text{if } i \text{ is spanned by a unicycle,} \end{cases} \tag{6.37}$$

which we can simply re-write as:

$$\widetilde{f}_{\mathcal{A}}(i, \phi^{\mathcal{A}}, g) = \begin{cases} \psi_{a \xrightarrow{\phi^{\mathcal{A}}} i} \left(g(a)\right) & \text{if } i \text{ is spanned by a tree of } \phi^{\mathcal{A}}, \\ 0 & \text{if } i \text{ is spanned by a unicycle,} \end{cases} \tag{6.38}$$

Then, for all $i \in \mathcal{V} \setminus \mathcal{A}$, we have:

$$(\mathrm{h}_\theta(g))(i) = \mathbf{E}_{\phi^{\mathcal{A}} \sim \mathcal{D}_{\mathcal{M}}^{\mathcal{A}}} \left(\widetilde{f}_{\mathcal{A}}(i, \phi^{\mathcal{A}}, g)\right) \tag{6.39}$$

We omit the proof of Corollary 6.1.8, which is identical to that of Corollary 6.1.3, whereby we simply need to replace the factor $\mathsf{Q}g$ with the estimation of $(-\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\, g$ obtained from the generalization of Theorem 6.1.1 to $\mathcal{A}$-MTSFs (as detailed in Appendix D.1).

Let us conclude this theoretical section with two remarks.

Remark 6.1.9. *It has been observed (for trivial connections [Pilavcı et al., 2021], but this holds in more generality as well) that the interpolation estimator $\widetilde{f}_{\mathcal{A}}$ can also be derived from $\widetilde{f}$, by considering weights $q_i = q_a$ for all i's and a fixed $q_a \in \mathbf{R}_+^*$ if $i \in \mathcal{A}$, and $q_i = 0$ otherwise. In particular, a straightforward computation shows that, for all $i \in \mathcal{V} \setminus \mathcal{A}$*

$$\lim_{q_a \to \infty} \widetilde{f}(i) = (\mathrm{h}_\theta(g))(i), \tag{6.40}$$

with the limiting estimator corresponding exactly to Algorithm 5.
Note that this construction is as the one we considered in Chapter 1 to relate semi-supervised node-classification and Tikhonov regularization.

Remark 6.1.10. *Since $\mathcal{D}_{\mathcal{M}}^{\mathcal{A}}$ is a DPP, we also recover a generalization of the efficient "number of roots" estimator from Chapter 1. In the specific case of MTSFs, the resulting expression is easy to write, and we find*

$$\mathbf{E}_{\phi \sim \mathcal{D}_{\mathcal{M}}} \left(|\phi \cap \mathcal{V}| \right) = \mathrm{tr} \left((\mathsf{L}_\theta + \mathsf{Q})^{-1} \mathsf{Q} \right). \tag{6.41}$$

We are now ready to evaluate our estimators.

6.2 MTSF-BASED TIKHONOV SMOOTHING: EXPERIMENTAL VALIDATION AND FURTHER IMPROVEMENTS

We investigate in this section the performance of MTSF-based estimators for smoothing[9], as compared to classical conjugate-gradient-based techniques. All empirical results are based on our Julia implementation[10].
For convenience, and since we aim at analyzing the behavior of our estimators with respect to different paremeters, we mostly focus on synthetically generated data and, in particular, the connections we consider are generated according to the following model.

Connection Model. For a given graph $\mathcal{G}$ with n nodes, we associate to each of its nodes v an angle ω_v chosen uniformly in $[0, 2\pi)$, and we set

$$\theta_e = \omega_{t_e} - \omega_{s_e} + \eta \varepsilon_e \tag{6.42}$$

for all edges e, with ε_e a perturbation uniformly distributed in $[-1, 1]$, and $\eta \in \mathbf{R}_+^*$ a scaling constant. Unless otherwise specified, we take $\eta = \frac{\pi}{2n}$ so as to ensure that the weak-inconsistency condition (Condition 5.3.1) is satisfied and that, consequently, efficient sampling of multi-type spanning forests from the the distribution $\mathcal{D}_{\mathcal{M}}$ is possible based on Algorithm 4.

Let us first discuss a fundamental issue: in this (favorable) weakly-incoherent setting, is the MTSF-based estimator $\widetilde{f}$ (for *one* MTSF) fast to compute, as compared to, *e.g.*, one iteration of a standard conjugate-gradient estimator?

6.2.1 *Runtime of $\widetilde{f}$ as a function of the density, in Erdös-Rényi graphs*

We empirically analyze the runtime of our estimator $\widetilde{f}$ with respect to the two following parameters.

9 Similar results are expected for connection-aware harmonic interpolation.
10 Available at https://gricad-gitlab.univ-grenoble-alpes.fr/gaia/synchromtsf.

- The mean degree $\overline{d}$ of the graph, which controls the graph's density (which parametrizes both the sampling-time of MTSFs and the cost of matrix-vector multplications). In particular, we consider the values $\overline{d} \in \{50, 100, 150, 200\}$.

- The choice of regularization parameter q which, in this specific experiment[11], only impacts the sampling-time of multi-type spanning forests. In our experiment, we q take values equal to $q'\overline{d}$, with $q' \in \{10^{-3}, 10^{-1}, 1\}$.
In this regime, the complexity bound of Proposition (5.4.1) effectively becomes *linear in* $|\mathcal{V}|$.

Experimental Setup. We generate Erdös-Rényi random graphs $\mathcal{G}$ of size $n = 10000$ for varying $p \in [0, 1]$, so that each edge e independently appears with probability p. To control the density, we set $p = \frac{\overline{d}}{n-1}$ so that the expected mean degree of these random graphs is $\overline{d}$. For each such graph, we generate a random signal $f \in \mathbf{C}^{\mathcal{V}}$ with independent complex Gaussian entries $f_v \sim \mathcal{N}_{\mathbf{C}}(0, 1)$.
We then measure the running time of:

1. sampling *one* multi-type spanning forest ϕ and applying the estimator $\widetilde{f}$;

2. *one* matrix-vector multiplication $\mathsf{L}_\theta f$, where L_θ is implemented as a sparse Hermitian matrix in CSC format.

Note that this second operation is the most expensive part of an iteration of the conjugate-gradient algorithm, and serves as a first simple baseline to compare computational costs of MTSF-based and conjugate-gradient-based estimators.
The results, depicted in Figure 6.3, are the average over 10000 time measurements (to account for runtime variability on any given graph), themselves averaged over 10 realizations of the graph G (in order to neglect potential graph-specific phenomena).

Analysis. Two trends emerge in Figure 6.3.

- First, as q' (hence q) increases, computing $\widetilde{f}$ becomes less expensive, since the probability of the random walk transitioning to Γ at each step increases, which reduced the time needed for each walk.

- Second, in contrast to the matrix-vector product $\mathsf{L}_\theta f$, our estimator is less sensitive to the density of the graph: computing $\widetilde{f}$ is more expensive than computing $\mathsf{L}_\theta f$ for $\overline{d} = 50$, but systematically faster for $\overline{d} = 200$. This reflects the $\mathcal{O}(|\mathcal{V}|)$ expected complexity for $\widetilde{f}$ (obtained with the parametrization $q = q'\overline{d}$), as compared to the $\mathcal{O}(|\mathcal{E}|)$ complexity of matrix-vector multiplication.

Remark 6.2.1. *We obtain similar results (not shown) for smaller/larger graphs as well, as indeed the resulting complexities remain unchanged.*

Remark 6.2.2. *The $\mathcal{O}(|\mathcal{V}|)$ computational profile resulting from choosing $\mathsf{Q} = q'\mathsf{D}$ may seem specific, but also corresponds to an important applicative setting for angular synchronization, as it is the same as that obtained when estimating $(\widetilde{\mathsf{L}}_\theta + q'\mathsf{I})^{-1}(q'\mathsf{I})g$. See Section 6.3 for more context.*

11 In our later experiments from Section 6.2.3, this parameter also affects the conditioning of the system, hence the speed of conjugate-gradient algorithms.

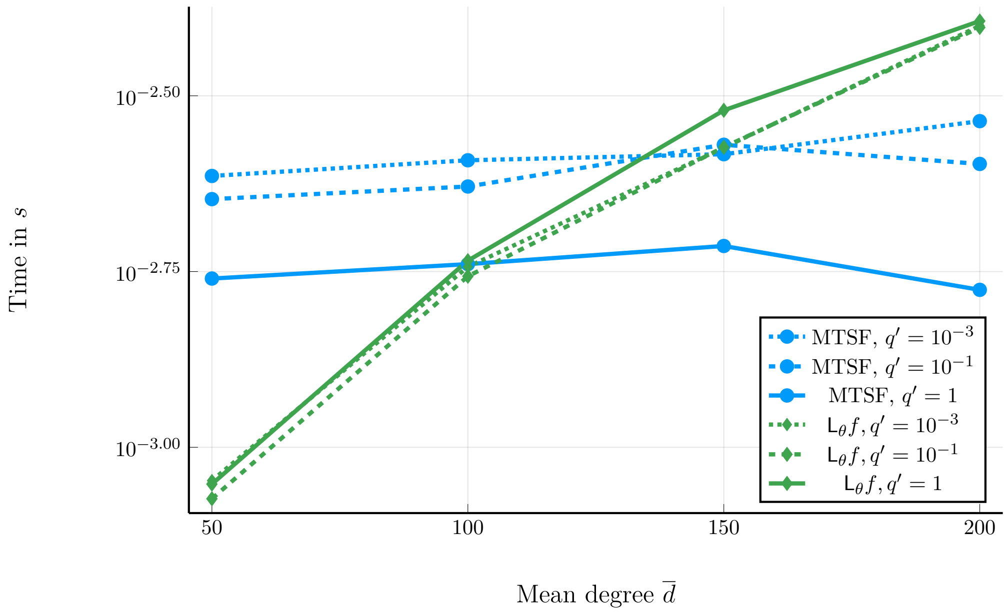


Figure 6.3: Runtime when varying $\overline{d}$, for different values of q.

In addition to showcasing the role of our two parameters, this also answers our first question: from a computational perspective, sampling one multi-type spanning forest and applying $\widetilde{f}$ *can* be faster than a matrix-vector multiplication, *depending on the choice of parameter*.

Unfortunately, the number of forests m required to get a high-quality approximation of f_* may be very high, as the $\mathcal{O}\left(\frac{\sigma}{\sqrt{m}}\right)$ convergence rate of Monte-Carlo estimators, where σ^2 denotes the variance of the estimator, is typically too slow for $\widetilde{f}$ to compete with conjugate-gradient algorithms estimators, unless the graph results in a poorly-conditioned, dense system, so that conjugate-gradient methods converge very poorly (which is not typical of real-world graphs, that most often exhibit sparsity, but may happen in some specific settings).

This brings about our next question: can we design variance-reduction techniques that result in significant variance-reduction and minimal computational overhead, to make MTSF-based estimators competitive?

6.2.2 *Variance reduction*

Let us insist once more on three important properties of Monte-Carlo estimators:

- being unbiased;
- having low variance;
- being cheaply-computable.

As such, Monte-Carlo estimators can be largely improved when used in conjunction with variance-reduction techniques, but it remains necessary to ascertain unbiasedness.

We propose two such improvements over $\widetilde{f}$, based on the two (classical) approaches of *Rao-Blackwellization* (named after the works [Blackwell, 1947, Rao, 1992]) and *control variates* [Kroese et al., 2013] respectively, generalizing the variance reduction techniques introduced in [Pilavcı et al., 2021, Pilavcı et al., 2022b] for estimators based on rooted spanning forests.

Remark 6.2.3. *Similar control-variates-based variance-reduction techniques have been proposed in [Pilavcı et al., 2022b] (improving on the RSF-based estimator from Chapter 1), but never properly benchmarked; we explore in the following generalizations of these techniques to our setting, and obtain with these techniques significant speed-ups* when applied to graphs with homogeneous degree-distribution*, but not on general graphs. We further design a refinement of this technique that is effective* regardless of the degree distribution*.*

Rao-Blackwellization

Rao-Blackwellization leverages the two laws of total expectation and variance which, roughly, state that conditioning an estimator using another statistic extracted from the same sample still results in an unbiased estimator, with lower variance (see, *e.g.*, [Weiss et al., 2006]). Here, our technique consists in conditioning on the set of edges in ϕ. This yields the following estimator:

$$\overline{f}(i,\phi,g) = \begin{cases} \psi_{r_\phi(i) \xrightarrow{\phi} i} \left(h_\phi(r_\phi(i), g)\right) & \text{if } i \text{ is spanned by a rooted tree,} \\ 0 \text{ if } i \text{ is spanned by a unicycle.} \end{cases} \tag{6.43}$$

where h_ϕ is defined as

$$h_\phi(r,g) = \frac{\sum_{j \in c_\phi(r)} q_j \psi_{j \xrightarrow{\phi} r} g(j)}{\sum_{j \in c_\phi(r)} q_j}, \tag{6.44}$$

and $c_\phi(r)$ is the *set of nodes* spanned by the tree containing r.
This estimator translates to Algorithm 6, which amounts to:

- Computing at the root r of each tree the average $h_\phi(r,g)$ of the values of g over the tree (obtained by transporting the values $g(i)$ from each node i in the tree to the node r according to the connection maps).
- Transporting this average back to the other nodes of the tree.

As compared to Algorithm 5, this incurs little additional cost, and still results in $\mathcal{O}\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right)$ complexity per MTSF[12]. Moreover, we are still able to prove that the estimation is unbiased.

Proposition 6.2.4. *Assuming that either:*

- *at least one of the q_i's is positive;*
- *there exists a cycle with non-trivial holonomy;*

we have

$$\mathbf{E}_{\phi \sim \mathcal{D}_\mathcal{M}} \left(\overline{f}(i,\phi,g)\right) = f_*(i). \tag{6.45}$$

12 It is also interesting to note the similarity with the Belief Propagation algorithm over trees [Mezard and Montanari, 2009].

Algorithm 6 Rao-Blackwellized MTSF-based estimator (Proposition 6.2.4).

$f_i = 0\ \forall i \in \mathcal{V}$, $h_r = 0\ \forall r \in \mathcal{V}$
Repeat m times
  Sample $\phi \in \mathcal{M}(\mathcal{G})$ from $\mathcal{D}_\mathcal{M}$ ▷ Via Algorithm 4
  for $r \in \phi \cap \mathcal{V}$ **do**
    for $j \in c_\phi(r)$ **do**
      $h_r = h_r + \psi_{j \xrightarrow{\phi} r}\, g(j)$ ▷ Propagate and average at the root
    end for
  end for
  for $i \in \mathcal{V}$ belonging to a rooted tree of ϕ **do**
    $f_i = f_i + \psi_{r_\phi(i) \xrightarrow{\phi} i} \left(h_{r_\phi(i)}\right)$ ▷ Propagate back
  end for
Output $\frac{1}{m}(f_1, ..., f_{|\mathcal{V}|})$

Proof. We will express $\overline{f}$ as a conditional expectation, the result will follow from the law of total expectation. Here, we choose a connected subset of edges $\pi \subseteq \mathcal{E}$, and condition on $\phi \cap \mathcal{E}$ containing π as one of its connected components (that is, as a rooted tree or unicycle), which we denote by $\pi \sqsubseteq_\mathcal{E} \phi$. For i belonging to the connected component spanned by π, we have:

$$\begin{aligned}
\mathbf{E}_{\phi \sim \mathcal{D}_\mathcal{M}}(\widetilde{f}(i, \phi, g) \mid \pi \sqsubseteq_\mathcal{E} \phi) &= \sum_{\phi \in \mathcal{M}(\mathcal{G})} \mathbf{P}_{\mathcal{D}_\mathcal{M}}(\phi \mid \pi \sqsubseteq_\mathcal{E} \phi)\ \widetilde{f}(i, \phi, g) && (6.46)\\
&= \sum_{\substack{\phi \in \mathcal{M}(\mathcal{G}) \\ \pi \sqsubseteq_\mathcal{E} \phi}} \frac{q_r \mathbf{1}_{\phi \cap c_\phi(i)}(r)}{\sum_{r' \in c_\phi(i)} q_{r'}}\ \widetilde{f}(i, \phi, g) && (6.47)\\
&= \overline{f}(i, \phi, g). && (6.48)
\end{aligned}$$

■

The variance of improved estimator can be derived similarly to that of $\widetilde{f}$. In particular, denoting by $\overline{\mathsf{S}}_\phi$ the operator such that $\overline{\mathsf{S}}_\phi g = \overline{f}(\phi, g)$, which is such that $\overline{\mathsf{S}}_\phi^* \mathsf{Q} \overline{\mathsf{S}}_\phi = \mathsf{Q} \overline{\mathsf{S}}_\phi$ and $\mathbf{E}_{\phi \sim \mathcal{D}_\mathcal{M}}(\overline{\mathsf{S}}_\phi) = (\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}$, we obtain the following.

Proposition 6.2.5. *Under the same assumptions as in Proposition 6.2.4, we have*

$$\mathbf{E}_{\phi \sim \mathcal{D}_\mathcal{M}} \left(\|\overline{f}(\phi, g) - f_*\|_\mathsf{Q}^2 \right) = \langle g, \mathsf{Q}(\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q} g\rangle - \|f_*\|_\mathsf{Q}^2. \tag{6.49}$$

For homogeneous q_i's, $\overline{\mathsf{S}}_\phi$ is Hermitian, and amenable to existing concentration inequalities. In particular, we obtain from the matrix-Bernstein inequality of [Tropp, 2012] the following result.

Proposition 6.2.6. *Let $\varepsilon, \delta \in (0,1)^2$, and assume that the assumptions of Proposition 6.2.4 hold. Consider sampling m multi-type spanning forests $\{\phi_k\}_{k=1}^m$, with*

$$m \geq 2\frac{6}{\varepsilon^2} \log\left(\frac{|\mathcal{V}|}{\delta}\right). \tag{6.50}$$

Then, for any signal $g \in \mathbf{C}^{\mathcal{V}}$,

$$\mathbf{P}\left(\left\|\left(\frac{1}{m}\sum_{k=1}^{m}\overline{\mathsf{S}}_{\phi_k}\right)g - f_*\right\|_2 \leq \varepsilon\|g\|_2\right) \geq 1-\delta. \tag{6.51}$$

We defer the detail of the proof to Appendix D.2. In words, this result states that, in order for the estimation $\overline{f}$ to be close to f_* *at a given threshold, and with a given probability*, the number of multi-type spanning forests m should grow *logarithmically* with $|\mathcal{V}|$.

Remark 6.2.7. *We obtain the bound in Equation (6.51) from a more explicit bound, which takes the form*

$$m \geq 2\frac{\varepsilon + \frac{q}{q+\lambda_1} + \frac{q^2}{(1+q+\max_{v\in\mathcal{V}} d_v)^2}}{\varepsilon^2}\log\left(\frac{|\mathcal{V}|}{\delta}\right). \tag{6.52}$$

In particular, it is interesting to note the explicit dependency on the incoherence of the connection (as quantified by λ_1), the maximum degree of the graph, and the parameter q.

Control variates

Our second variance-reduction technique consists in the introduction of control variates: an addition of another random quantity to $\overline{f}$, designed to have zero mean (so that the expectation remains unchanged), but resulting in an estimator with lower variance when designed properly. We propose to use a single *gradient-descent*[13] step with parametrized step-size α:

$$\hat{f}(\phi,g) = \overline{f}(\phi,g) - \alpha\mathsf{P}(\mathsf{Q}^{-1}(\mathsf{L}_\theta+\mathsf{Q})\overline{f}(\phi,g) - g), \tag{6.53}$$

where $\mathsf{P} = \left(\mathsf{Q}^{-1}\mathsf{D}+\mathsf{I}\right)^{-1}$ is a diagonal Jacobi preconditioner for the system $\mathsf{Q}^{-1}(\mathsf{L}_\theta+\mathsf{Q})f = g$, and $\alpha \in \mathbf{R}_+^*$. Note that this operation, which acts as an additional smoothing step, is only well-defined when *all* the q_i's are *positive*.
This additional gradient-descent step is easy to implement, and only entails entails *a single* matrix-vector multiplication, at $\mathcal{O}(|\mathcal{E}|)$ cost. For an estimation leveraging m MTSFs, this results in $\mathcal{O}\left(m\left(|\mathcal{V}| + \frac{\mathrm{Vol}(\mathcal{G})}{q_{\min}}\right) + |\mathcal{E}|\right)$ complexity. Further, the estimator remains unbiased.

Proposition 6.2.8. *Assuming that either:*

- *at least one of the q_i's is positive;*
- *there exists a cycle with non-trivial holonomy;*

we have

$$\mathbf{E}_{\mathcal{D}_\mathcal{M}}(\hat{f}(i,\phi,g)) = f_*(i). \tag{6.54}$$

Proof. Recalling that $f_* = \mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}(\hat{f}(\phi,g)) = (\mathsf{L}_\theta+\mathsf{Q})^{-1}\mathsf{Q}g$, we obtain by linearity of the expectation that:

$$\mathbf{E}_{\phi\sim\mathcal{D}_\mathcal{M}}(\hat{f}(\phi,g)) = f_* - \alpha\mathsf{P}(g-g) = f_*. \tag{6.55}$$

■

13 Note that solving the linear system arising from the Tikhonov smoothing problem indeed amounts to minimizing a quadratic functional of the form $z \mapsto \frac{1}{2}\langle z, (\mathsf{L}_\theta+\mathsf{Q})^{-1}\mathsf{Q}z\rangle + \langle z, g\rangle$. A similar observation holds when trying to solve a preconditioned system as well.

Let us finally note that a good choice of step-size α is crucial in order to obtain a significant reduction of variance. We simply take $\alpha = 1$ in the following, as it results in meaningful variance reduction in our other experiments, and seems close to optimal in this respect[14]; see Appendix D.3 for empirical results backing this choice.

Remark 6.2.9. *The reader familiar with those techniques may interpret the estimator $\hat{f}$ as a randomized variant on multi-grid method (for a general reference on multi-grid methods, see,* e.g.*, [Trottenberg et al., 2000]):*

- *each multi-type spanning forest defines a coarsening of the graph, in which the problem is easy to solve (*e.g.*, by applying $\widetilde{f}$ or, in our case, $\overline{f}$), and allows for an approximation with small low-frequency error;*
- *the additional smoothing* via *gradient-descent then reduces the high-frequency error.*

This perspective is in line with existing observations about rooted spanning forests, in the trivial-connection setting, that have also been used to build coarsening operators for multi-resolution analysis of graph signals [Avena et al., 2021, Avena et al., 2020].
Further, one may use any high-frequency smoother in place of the gradient-descent step such as, e.g.*, one iteration of the Jacobi method [Saad, 2003], resulting in (here, for a non-preconditioned system, but the same techniques also apply to preconditioned systems):*

$$\hat{f}_J = \mathsf{Q}(\mathsf{D} + \mathsf{Q})^{-1}(g + \mathsf{Q}^{-1}\mathsf{A}_\theta \hat{f}). \tag{6.56}$$

In fact, a short calculation shows that $\hat{f}_J = \hat{f}$, but the unbiasedness also follows from a fixed point argument and the convergence of the Jacobi method for systems involving strictly diagonally dominant matrices, and a similar argument would apply to, e.g.*, Gauss-Seidel[15] or (Symmetric) Successive Over-Relaxation methods as well [Saad, 2003].*

It remains to compare the resulting estimators with conjugate-gradient methods.

6.2.3 *Runtime-Precision trade-offs for denoising* **C**-*signals on varying graph topologies*

We compare the performance of our improved estimators $\overline{f}$ and $\hat{f}$ with conjugate-gradient methods (see, *e.g.*, [Saad, 2003]). The objective we consider is the reconstruction a signal $f_\top \in \mathbf{C}^\mathcal{V}$ given a noisy degradation $g = f_\top + \varepsilon$.

Let us recall that, in the specific instance that $f_\top \in \mathbf{R}^\mathcal{V}$ and the connection on $\mathcal{G}$ is trivial ($\psi_e = \mathrm{id}_{\mathbf{C}_{se},\mathbf{C}_{te}}$ for all $e \in \mathcal{E}$), a common assumption in the graph signal processing literature is that $f_\top$ is a smooth, band-limited signal signal (as discussed in Chapter 1), that is, a linear combination of the first (low-frequency) eigenvectors of L (associated to the smallest eigenvalues) [Puy et al., 2018, Lorenzo et al., 2018].
We here similarly assume that $f_\top \in \mathbf{C}^\mathcal{V}$ is B-bandlimited, *i.e.* $f_\top \in \mathrm{span}(u_1, ..., u_B)$[16], with u_i *the i-th eigenvector of* L_θ. Solving the connection-aware Tikhonov smoothing problem

$$\underset{f \in \mathbf{C}^\mathcal{V}}{\operatorname{argmin}} \; q\|f - g\|_2^2 + \langle f, \mathsf{L}_\theta f\rangle \tag{6.57}$$

14 This contrasts with the similar variance reduction technique proposed in [Pilavcı et al., 2022b], developed for trivial connections, and for which a good choice of step-size is less straightforward).
15 Named after German mathematician, physician and astronomer Carl Friedrich Gauss (1777−1855), and German mathematician Philipp Ludwig von Seidel (1821−1896).
16 The linear subspace generated by the vectors u_1,...,u_B.

should then allow to faithfully recover $f_\top$ from g, by penalizing high-frequency components in the optimal solution $f_* = (\mathsf{L}_\theta + q\mathsf{I})^{-1}(q\mathsf{I})\ g$ (this is not a classical consideration, as it is not *a priori* clear that the eigenvectors associated to the smallest eigenvalues of L_θ represent smooth signals, and we argue for its well-foundedness in Appendix D.4).

There is a number of classical ways to solve the corresponding system, among which:

- using a Cholesky-decomposition-based solver[17] to obtain the exact solution f_*;
- using a conjugate-gradient-based iteration to obtain a high quality approximation of f_*.

We compare our estimators with these methods, and evaluate the performance of each estimator against the two following cost functions.

1. The reconstruction error
$$e_r(f) = \frac{\|f - f_\top\|_2}{n}, \tag{6.58}$$
which measures the quality of the denoising of the signal g.

2. The approximation error
$$e_a(f) = \frac{\|f - f_*\|_2}{n}, \tag{6.59}$$
which measures the quality of the approximation of f_*.

Our estimators and the conjugate-gradient algorithms are respectively parametrized by the number of MTSFs used and the number of gradient steps, that we both denote by m.

Experimental Setup. For each graph, we generate a B-bandlimited signal $f_\top$, each u_i being weighted by a random complex Gaussian value $a_i \sim \mathcal{N}_{\mathbf{C}}(0,1)$. We then degrade $f_\top$ with some additive Gaussian noise $\varepsilon \sim \mathcal{N}_{\mathbf{C}}(0,\sigma^2)$ (independently on each entry, with σ^2 such that the SNR is equal to 2 [18], and determine the optimal parameter q_* for which $e_r(f_*) = e_r(q_*(\mathsf{L}_\theta + q_*\mathsf{I})^{-1}g)$ is minimized[19], before measuring the errors and running time associated to 10 different values of m, logarithmically-spaced values from 1 to 100:
$$m \in \{1, 2, 3, 5, 8, 13, 22, 36, 60, 100\}. \tag{6.60}$$

The iterative algorithms are initialized at the noisy signal g.
We compare the following.

1. The MTSF-based estimators $\overline{f}$ and $\hat{f}$.

2. The conjugate-gradient descent algorithm with no preconditioner, with a simple diagonal Jacobi preconditioner $\mathsf{P} = (q^{-1}\mathsf{D} + \mathsf{I})^{-1}$, and with a CROUT ILU preconditioner[20]. Computing this high-quality ILU preconditioner is too expensive to be competitive with the other methods, and we only include these results on one type of graph, for illustration purposes.

17 Named after French engineer and mathematician André-Louis Cholesky, 1875−1918.
18 Precisely, we set $\sigma^2 = \frac{1}{\mathrm{SNR}\times n}$, with SNR = 2 the desired signal-to-noise ratio.
19 We perform our search in $(0, 30)$.
20 We use the implementations from the libraries: https://github.com/JuliaLinearAlgebra/IterativeSolvers.jl, https://github.com/JuliaLinearAlgebra/Preconditioners.jl and https://github.com/haampie/IncompleteLU.jl.
For the ILU preconditioner, we fix the drop-threshold to 0.1.

Each runtime measurement is averaged over 100 runs, and the results are averaged over 5 realizations of the noise ε (to account for signal-dependency in the resulting errors). These results are performed and averaged over 10 realizations each of the random graph models detailed below, and we plot the overall mean results in Figure 6.4.

Graphs used. We use the following graph models.

- An ε-graph obtained by sampling $|\mathcal{V}| = 10000$ i.i.d. points x_i's in $[0,1]^3$, with an edge between x_i and x_j whenever $\|x_i - x_j\|_2 < 0.1$.
- A graph generated from a Stochastic Block Model (SBM), with $|\mathcal{V}| = 10000$ nodes, each belonging to one of two communities C_1 and C_2 of size 5000 each. In this model, an edge is drawn randomly between two nodes i, j with probability equal to $\frac{c_{k,l}}{n}$, where k, l denote respectively the community label of i and j. Here, we take $c_{1,1} = c_{2,2} = 36$ and $c_{1,2} = c_{2,1} = 4$, resulting in average degree $\overline{d} = 40$.
- A graph generated from a related Degree-Corrected Stochastic Block Model (DC-SBM 1), with two communities of size 5000 and an edge between nodes $i \in C_k$ and $j \in C_l$ present with probability $\xi_i \xi_j \frac{c_{k,l}}{n}$. ξ_i is a randomly sampled positive real value, representing the intrinsic *connectivity* of node i, with $\mathbf{E}(\xi_i) = 1$ and finite second moment (as defined in Chapter 1). Here, we take $c_{1,1} = c_{2,2} = 36$, $c_{1,2} = 4$, and the p_i's are drawn from a normalized mixture of Gaussian distributions[21], resulting in a graph with mean degree $\overline{d}$ close to 40. The objective of adding this model is to illustrate, at constant density, how the degree distribution affects the results.
- Another DC-SBM-graph (DC-SBM 2), with higher density (typically with an average degree more than 10 times that of the previous DC-SBM model). We take two communities of size 5000, $c_{1,1} = c_{2,2} = 480$, $c_{1,2} = 20$, and the p_i's are drawn from another mixture of Gaussian distributions[22]. The objective of this second DC-SBM model is to illustrate how density affects the results.
- We also illustrate the results on a real-world graph: a relationship graph for internet Autonomous Systems (AS), recorded by the Center for Applied Internet Data Analysis (CAIDA) and provided in the SNAP datasets [Leskovec and Krevl, 2014][23].

In the event of a randomly generated graph containing isolated nodes, we remove these nodes, so that the graph remains connected (to make the interpretations simpler). We summarize in Table 6.1 the information concerning the graphs generated, and the associated optimal q_*'s (averaged over all realizations of the noise ε). All those graphs are endowed with a synthetic connection as specified in Equation (6.42). The parameters used for the (DC) SBM models ensure that the resulting graphs have a strong community structure[24].
For the ε-graph and the AS-graph, we arbitrarily set $B = 5$ when generating the bandlimited signal $f_\top$. For (DC) SBMs, we take $B = 2$; recall that, for weakly-incoherent connections, the

21 Specifically, the connectivity parameters are independently sampled from a mixture of $\mathcal{N}(50, 20)$, $\mathcal{N}(500, 100)$ and $\mathcal{N}(10000, 100)$, with weights of 0.59, 0.4 and 0.01 respectively, and then normalized.
22 Here from the mixture of $\mathcal{N}(50, 20)$, $\mathcal{N}(1000, 50)$, $\mathcal{N}(5000, 100)$ and $\mathcal{N}(10000, 100)$, with weights of 0.45, 0.1, 0.44 and 0.01.
23 We use the library available at `https://github.com/JuliaGraphs/SNAPDatasets.jl`.
24 Specifically, this ensures that the community-detectability threshold from Equation (1.34) is (largely) satisfied for each model.

Graph	$\lvert\mathcal{V}\rvert$	$\overline{d}$	d_{min}	d_{max}	q_*
ε-graph	10000	37.3	5.2	67.5	6.518
SBM	10000	39.87	17.7	66.6	21.04
DC-SBM 1	9833.7	33.4	1	906.1	2.634
DC-SBM 2	9932.8	418.8	1	1513.7	4.31
AS CAIDA	26475	4.032	1	2628	0.4808

Table 6.1: Experimental parameters associated to each graph. Values are averaged over all 10 realizations for random graph models.

eigenvectors of the first two connection Laplacian L_θ reflect the community structure of these graphs and, in particular, taking $B = 2$ here results in signals coherent with the community structure.

Let us note that, for all random graph models, Tikhonov regularization is expected to result in meaningful smoothing of the signal (as argued in Appendix D.4).

Discussion. Let us first comment on the approximation error for the ε-graph (Figure 6.4a, left). Both of our estimators converge linearly in log-log-scale with a mild slope, as expected for Monte-Carlo estimators, and which is in overall too slow, as indeed any of the other three conjugate-gradient-based algorithms achieves a better approximation error within 3 steps than our estimators in 100 steps (as expected for Monte-Carlo convergence rates), and our methods are not competitive with respect to this measure. However, from a graph-signal-processing perspective (as introduced in Chapter 2 for **C**-signals), and in this denoising setting, the quality of the denoising is reflected in the reconstruction error (depicted in all the plots of Figure 6.4). Overall, we observe that:

(1) $\hat{f}$ consistently performs better than $\overline{f}$, but how much better depends on the graph (see *e.g.* SBM and DC-SBM 1), and improvements are smaller for graphs with more heterogeneous degree distributions (DC-SBM graphs).

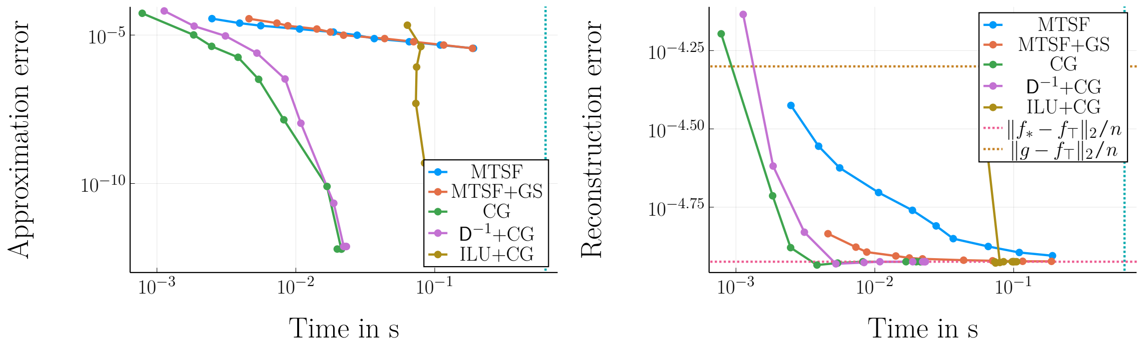
(2) Compared with MTSF-based estimators, the conjugate-gradient algorithms perform *worse* on graphs with heterogeneous degree distributions.

(3) Conjugate-gradient solvers perform worse than our methods when the density increases.

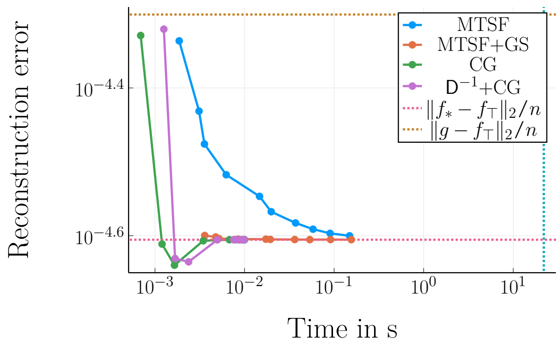
The relatively poor performance of conjugate-gradient estimators on DC-SBM 1 (bullet point (2)) is partly explained by the conditioning of the system considered: indeed, a bound on the condition number κ is given by

$$\kappa = \frac{q_* + \lambda_n}{q_* + \lambda_1} \leq \frac{q_* + 2\max_{v\in V} d_v}{q_* + \lambda_1}, \tag{6.61}$$

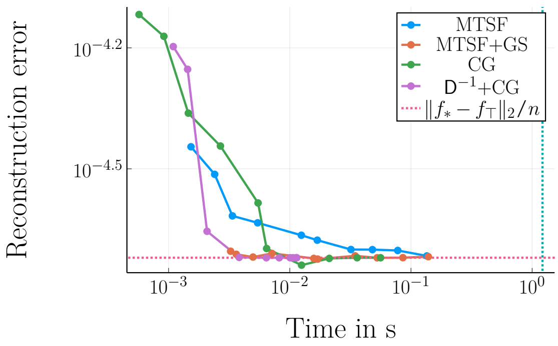
where $\lambda_1 \leq ... \leq \lambda_n$ denote the eigenvalues of L_θ, and where we recall that λ_1 is a measure of the quality of the optimal angular assignment (the lower λ_1 is, the better the quality of the synchronization), which is small for our synthetic connection (good synchronization due to low incoherence), and results in an ill-conditioned system (that the diagonal Jacobi preconditioner is not able to correct for). This suggests that conjugate-gradient algorithms require more iterations in the presence of high-degree nodes. In contrast, the bound of Proposition 6.2.6 suggests that



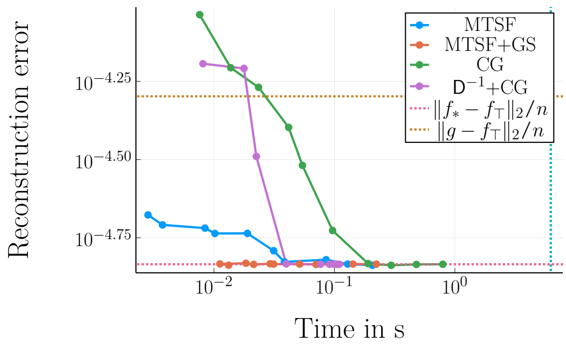
(a) ε-graph. Left: Approximation error. Right: Reconstruction error.



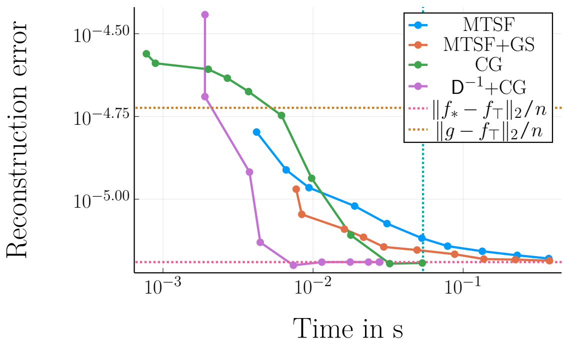
(b) SBM Reconstruction error.



(c) DC-SBM 1 Reconstruction error.



(d) DC-SBM 2 Reconstruction error.



(e) AS CAIDA Reconstruction error.

Figure 6.4: Runtime-precision trade-offs for Tikhonov smoothing with optimal q. Each datapoint corresponds to a value of m. The vertical blue line records the average runtime of the Cholesky-based solver, the horizontal red line the average reconstruction error of the exact solution to the Tikhonov problem, and the horizontal yellow line the average reconstruction error of the noisy signal. Vertical bars represent standard deviations.

the presence of high-degree nodes tends to decrease the number of forests required (though we stress again that his dependency is quite weak).
Conjugate-gradient iterations are slower for high-density graphs (with $\mathcal{O}(|\mathcal{E}|)$ time *per iteration*, and as observed in Section 6.2, and $\overline{f}$ and $\hat{f}$ reach near-optimal reconstruction around 10 *times faster* than (preconditioned) conjugate-gradient methods on DC-SBM 2 (with expected overall complexities in $\mathcal{O}\left(m\left(\frac{|\mathcal{E}|}{q}+|\mathcal{V}|\right)\right)$ and $\mathcal{O}\left(m\left(|\mathcal{V}|+\frac{|\mathcal{E}|}{q}\right)+|\mathcal{E}|\right)$ respectively).

Our observations so far show that our estimators are competitive with standard methods, especially for low-accuracy solutions, and in the following situations:

- On graphs resulting in poorly conditioned systems (with high-degree nodes and for low values of q_*).
- On graphs with higher density (in which case MTSF-based methods seem to slow-down at slower rates).

Note though that our experimental setup, and in particular our random connection model, is only representative of connections encountered in a subset of applications (*e.g.*, those considered in [Yu, 2012], [Cucuringu, 2016] or [Furutani et al., 2019]).

We conclude this section by a number of remarks that should be kept in mind when comparing MTSF-based estimators with standard strategies.

Remark 6.2.10. *Our comparisons are performed on a* single thread *of a commercial laptop. In particular, we use existing conjugate-gradient implementations that only leverage instruction-level parallelism (SIMD) and neither thread/core-level parallelism, nor GPU accelerations, which is known to be much more time-efficient for very-dense systems (see,* e.g.*, [Krüger and Westermann, 2005], amongst many others), nor any kind of distributed implementation.*
In a similar manner, our MTSF-sampling implementation does, at this point, not allow for parallel sampling, which is usually a very attractive feature of Monte-Carlo estimators (see, e.g.*, [Rosenthal, 2000]).*
The respective trade-offs between these more efficient implementations remain to be explored.

Remark 6.2.11. *The reader may wonder why we do not use a high-quality preconditioner instead of a diagonal Jacobi preconditioner. In our experiments, we found that computing a high-quality incomplete-factorization-based preconditioners indeed allows for much faster convergence, but is also too expensive to compute, and not able to compete with simpler non-preconditioned conjugate-gradient iteration and MTSF-based estimators. Further, it is known that the convergence of the conjugate-gradient algorithm preconditioned with this type of preconditioner slows down as the size of the system increases [Ferronato, 2012]. Similar observation were drawn for algebraic-multigrid preconditioners in [Pilavcı et al., 2021], in the trivial-connection case.*
*State-of-the-art preconditioners are nowadays built using RandNLA techniques (*e.g.*, [Frangella et al., 2023]). In particular, for combinatorial-Laplacian-based systems, uniform-spanning-tree-based preconditioners are the preferred option [Kyng and Song, 2018, Kaufman et al., 2022] . In this spirit, a MTSF-based preconditioner tailored to the connection-aware Tikhonov smoothing problem has also been proposed in [Fanuel and Bardenet, 2022]. This preconditioner could also be used, in conjunction with a conjugate-gradient iteration, as a post-processing step for our Monte-Carlo estimator, if one wishes to further reduce the approximation error.*

Remark 6.2.12. *Further profiling of our implementation shows that the most expensive operations during the computation of $\overline{f}$ (and $\hat{f}$) are the instantiation of the data structures we use, and* *the successive memory accesses to the different θ_e. These issues would likely benefit from further democratization of randomized computational schemes, with more efficient compiler optimizations and architectures specialized for these types of computations.*

Let us now turn to applications of multi-type spanning forests to angular synchronization.

6.3 ANGULAR SYNCHRONIZATION

We describe in this section how our MTSF-based estimators for connection-aware Tikhonov smoothing can be applied to angular-synchronization solvers. Recall from Chapter 2 that, to perform angular synchronization, we aim to minimize the incoherence

$$\underset{f \in U(\mathbf{C})^n}{\operatorname{argmin}} \ \langle f, \mathsf{L}_\theta f \rangle, \tag{6.62}$$

which is NP-hard in general [Zhang and Huang, 2006], and prohibitively expensive to compute.

Our MTSF-based synchronization scheme (described in Section 6.3.1 below) approximates the solutions to spectral relaxations of the minimization problem from Equation (6.62) taking the form[25]:

$$\underset{\|f\|_2^2 = n}{\operatorname{argmin}} \ \langle f, \mathsf{L}_\theta f \rangle. \tag{6.63}$$

As discussed in Chapter 2, the solution to this problem is given by the eigenvector u_1 associated to the *smallest* eigenvalue λ_1 of L_θ and, unless the connection is coherent, differs from that of Equation (6.62).
From a theoretical front, relaxations of this type are known to enjoy two types guarantees (as discussed in Chapter 2):

1. as the incoherence of the connection decreases, the solution to the problem in Equation (6.63) approaches that of Equation (6.62);

2. for Gaussian noise (of the form in Equation (2.27)), the solution to the problem in Equation (6.63) is correlated to the ground-truth angular assignment x *for any noise levels* (this is a very weak guarantee but has, to our knowledge, not been established for other methods).

In addition, there exist standard techniques to compute u_1, such as, *e.g.*, an inverse power method, or a Lanczos iteration[26].

We preface the exposition of our methods with a brief review of existing approaches, as compared to spectral relaxations.

- *Semi-definite relaxations* [Singer, 2011], which re-frame the problem of Equation (6.62) as a semi-definite programming problem before relaxing it to obtain a more efficiently-solvable expression. These techniques are known to yield results similar to those of the spectral relaxation, but are classically considered too computationally impractical past mid-sized instances ($n \simeq 10^4$)[27].

---

25 Alternatively, one could replace $\mathcal{L}_\theta$ with the normalized connection Laplacian $\widetilde{\mathcal{L}_\theta} = \mathsf{D}^{\frac{1}{2}} \mathsf{L}_\theta \mathsf{D}^{\frac{1}{2}}$, to which our methods also apply (see Remark 6.1.7.

26 Named after Hungarian mathematician and physicist Kornél Lánczos, 1893−1974.

27 This is also a very active area of research, and modern methods and solvers are much less sensitive to these issues; see, *e.g.*, [Boumal, 2023] and references therein.

- *Message-passing algorithms*, implementing heuristics from statistical physics. For connections degraded by Gaussian noise (taking the form of Equation (2.27)), and on the complete graph[28], *approximate-message-passing* algorithms have been conjecture to be statistically optimal among polynomial time algorithms [Perry et al., 2018], even at higher noise levels, but cannot be applied to arbitrary topologies. *cycle-edge message-passing*, as developed in [Lerman and Shi, 2022], allows exact recovery under a theoretical corruption model but is more computationally-expensive than spectral relaxations.

- *The generalized power method* of [Boumal, 2016], which iteratively applies a power-method-like iteration
$$(f)_i \mapsto \frac{(A_\theta f)_i}{|(A_\theta f)_i|}, \tag{6.64}$$
to some signal f_0, and provably recovers the optimal solution of Problem (6.62) in the presence of (low) Gaussian noise, on complete graphs. The performance of this method in benchmarks is similar to that of spectral relaxations (when using the normalized connection Laplacian $\widetilde{\mathsf{L}_\theta}$) [He et al., 2023].

- *Descent techniques* (*e.g.*, [Maunu and Lerman, 2023, Liu et al., 2023]), which perform a form of Riemannian gradient-descent (on the manifold $U(\mathbf{C})^{\mathcal{V}}$) and allow, in some cases and for low-enough corruption, for exact recovery (in [Maunu and Lerman, 2023], for instance, under a specific corruption model).

- *Graph Neural Networks*, that can reach performance comparable to that of spectral-relaxation methods for low-noise regimes, and often obtain state-of-the-art synchronization-quality in high-noise regimes [He et al., 2023, Janco and Bendory, 2023]. From an engineering perspective, these methods are difficult to compare with training-free alternatives; in a sense, graph neural networks and our MSTF-based estimators for the spectral relaxation are antipodal, as one aims for maximal synchronization-quality (at the cost of training), and the other at faster runtime (at the cost of an approximation).

Note that message-passing algorithms and descent techniques (along with others we did not mention as well) can be applied to synchronization problems over more general groups[29], but are, to our knowledge, not competitive on benchmarks for *angular* synchronization [He et al., 2023].

We now discuss how to use our randomized estimators for eigenvector-computation.

6.3.1 *Proposed Approach*

One way to solve the spectral relaxation of Equation (6.63) is to perform an inverse power iteration, by setting $f_0 \in \mathbf{C}^{\mathcal{V}}$ and iterating:
$$f_{r+1} = \frac{\mathsf{L}_\theta^{-1} f_r}{\|\mathsf{L}_\theta^{-1} f_r\|_2} \tag{6.65}$$
until convergence. Note that for this iteration to be well-defined, L_θ needs to be invertible, *i.e.*, the angular synchronization problems needs to be non-trivial.

28 Alternatively, on (asymptotically) very-dense Erdös-Rényi models.

29 Let us mention that, while we have not encountered this idea in the literature, it is also possible to generalize the generalized power method to this setting, whereby the $O(\mathbf{R}^d)$-factors are readily obtained from a polar decomposition (thus generalizing the component-wise normalization) [Horn and Johnson, 2012].

Our approach stems from the observation that L_θ and $q^{-1}(\mathsf{L}_\theta + q\mathsf{I})$ share the same eigenvectors. The power method applied to this regularized matrix consists in computing iterations of the form:

$$f_{r+1} = \frac{(\mathsf{L}_\theta + q\mathsf{I})^{-1}(q\mathsf{I})f_r}{\|(\mathsf{L}_\theta + q\mathsf{I})^{-1}(q\mathsf{I})f_r\|_2}, \tag{6.66}$$

where an estimation of $q(\mathsf{L}_\theta + q\mathsf{I})^{-1}f_r$ can be obtained by our estimators $\overline{f}$ and $\hat{f}$ (or other deterministic algorithms). The convergence of this iteration is geometric with ratio

$$\frac{\mu_1}{\mu_2} = \frac{\lambda_2 + q}{\lambda_1 + q}, \tag{6.67}$$

where $\mu_1 \leq \mu_2$ and $\lambda_1 \leq \lambda_2$ denote respectively the two smallest eigenvectors of $q(\mathsf{L}_\theta + q\mathsf{I})^{-1}$ and L_θ [Saad, 2011]. The choice of q is then a trade-off between:

- Low values of q that induce smaller $\frac{\lambda_2+q}{\lambda_1+q}$ ratio, favoring faster convergence of the power iteration.
- Large values of q that enable a faster sampling time of MTSFs (in $\mathcal{O}\left(\frac{|\mathcal{E}|}{q}\right)$).

Remark 6.3.1. *One could instead maximize $\langle f, \mathsf{A}_\theta f\rangle$ in Equation (6.62). This formulation is common in theoretical works, that mostly focus on (often dense) Erdös-Rényi graphs, and applying the power method to A_θ is often efficient on these graphs. For other topologies, more involved estimators may be necessary, in case the ratio α_1/α_2 of the top two eigenvalues α_1 and α_2 of A_θ is large.*
For trivial connections, such cases include:

- *regular grids (with α_1/α_2 growing worse with the dimension);*
- *graphs with homogeneous spatial correlations (*e.g.*, ε-graphs, nearest-neighbors-graphs), or graphs with homogeneous degree distributions* and *bottlenecks such as in SBMs (in the specific case of regular graphs, this phenomenon is described by the Cheeger inequality).*

*Our experiments suggest that the same behavior arises for non-trivial connections, and it is interesting to note that these situations co-incide with those for which the angular-synchronization problem is statistically difficult (*e.g.*, from the Cramér-Rao bounds of [Boumal et al., 2014], as discussed in Chapter 2), and also correspond to larger $\frac{\lambda_1}{\lambda_2}$ ratios. In turn, quenchings of this spectral ratio taking the form $\frac{\mu_1}{\mu_2}$ allow for faster convergence.*

6.3.2 *Illustration: Intra-Class Denoising*

Before delving into precise computation time comparisons with state-of-the-art methods, we first illustrate our proposed method on a toy denoising problem, inspired from an application in cryo-EM [Scheres, 2012, Zhao and Singer, 2014], following Remark 6.0.1.

Recall that we consider n copies $(I_i)_{i=1}^n$ of some image I_* that have been rotated and degraded:

$$I_i = r_i(I_*) + \varepsilon, \tag{6.68}$$

with r_i a rotation and $\varepsilon \sim \mathcal{N}(0, \sigma^2\mathsf{I})$ some Gaussian noise. The rotations r_i are unknown, and the goal is to recover image I_*. This is a simplified version of the intra-class denoising problem in cryo-EM, where each (2D) image corresponds to a noisy projection of a (3D) molecule observed in an unknown orientation.

In the absence of rotations r_i, a solution consists in averaging the I_i's, trivially recovering I_* as $n \to \infty$. This strategy fails in our setting (due to the rotations), and we need to estimate the r_i's before denoising. To do that, we first estimate angles $\theta_{i,j}$ between a subset of pair of images $\mathcal{E}$ (representing the edges of a graph) using image moments [Gonzalez, 2009][30], and perform angular synchronization to estimate the rotations r_i's. We then rotate and average the images I_i's.

Experimental Setup. We take for I_* the 256×256 Shepp-Logan phantom [Shepp and Logan, 1974], $n = 1000$, $q = 10^{-3}$, $k = 20$ iterations of the power method, and randomly choose an underlying graph $\mathcal{G}$ from an Erdös-Rényi model edge-inclusion probability $p = \frac{5}{n}$. We uniformly sample rotations r_i with angles in $(-\frac{\pi}{2}, \frac{\pi}{2})$, that we try to estimate[31].
We display the recovered images in Figure 6.5 (solving the spectral relaxation (6.63) using both our method, with $m = 10$ MTSFs for the estimator $\overline{f}$ at each step, and an exact solver), for different noise levels $\sigma^2 \in \{1, 5\}$. We also plot the image reconstruction error

$$e_I(x) = \|I_* - x\|_2 \tag{6.69}$$

for $\sigma^2 = 5$ for varying values of m.
In this specific setting, we have no guarantee that the connection resulting from the image-moment-based estimation is weakly-incoherent and, following Remark 6.1.6, we always accept a cycle when encountered in the sampling process if $\cos(\theta_C) < 0$ (Algorithm 4), and apply estimator $\overline{f}$ on the resulting MTSF. Note that we do not perform importance sampling.

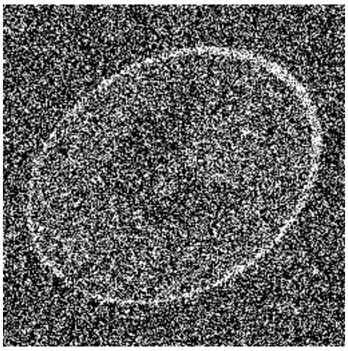
Even though the connection may not be weakly-incoherent, the MTSF-based synchronization allows to obtain results of a quality very similar to exact synchronization on this problem, even at higher noise levels. What the reader should take away from this illustration is that multi-type spanning forests can indeed be used in practical settings, as the quality of the synchronization is sufficient to fit into processing pipelines (at least, those pipelines for which the spectral relaxation of Equation (6.63) results in a good enough quality of synchronization). The novelty of our work though does not lie solely proposing in good-quality estimators, but mostly in developing *fast* estimators. As such, we also explore, in the following section, performance-runtime trade-offs similar to those of Section 6.2 for the connection-aware Tikhonov smoothing problem. The comparison to deterministic methods in this case is more involved, as one needs to account for a number of additional parameters.

1. *The incoherence of the connection*, which is an important parameter that did not appear in our smoothing experiments, for which all connections were very coherent and satisfied the weak-inconsistency condition. For our randomized synchronization scheme though, we only need to estimate quantities of the form

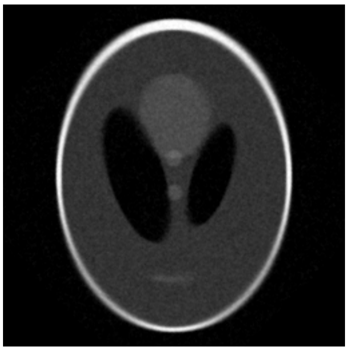
$$\frac{(\mathsf{L}_\theta + q\mathsf{I})^{-1}(q\mathsf{I})f_r}{\|(\mathsf{L}_\theta + q\mathsf{I})^{-1}(q\mathsf{I})f_r\|_2}, \tag{6.70}$$

which remain invariant when each estimate $(\mathsf{L} + q\mathsf{I})^{-1}q\mathsf{I}\, f_r$ is scaled by a multiplicative factor a. As such, we can (and do) rely on the importance sampling strategy from Remark 6.1.6.

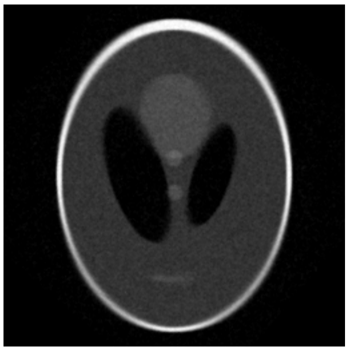
30 Specifically, we estimate the orientation of each of the two images using the eigenvectors of the matrix of its covariant moments, and take $\theta_{i,j}$ the angle between these two sets of eigenvectors
31 We restrict the possible rotations so that we can use the simple image-moment-based registration, other methods could be used instead.



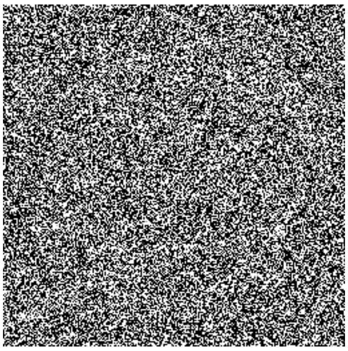
(a) $r(I_*) + \varepsilon, \sigma^2 = 1$



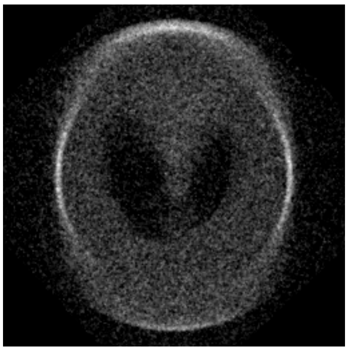
(b) $\sigma^2 = 1$, exact



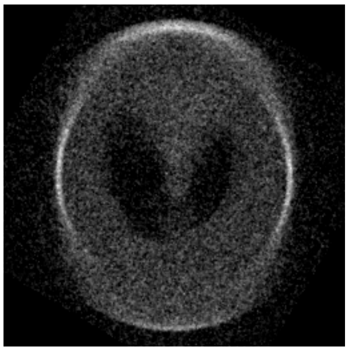
(c) $\sigma^2 = 1$, MTSF



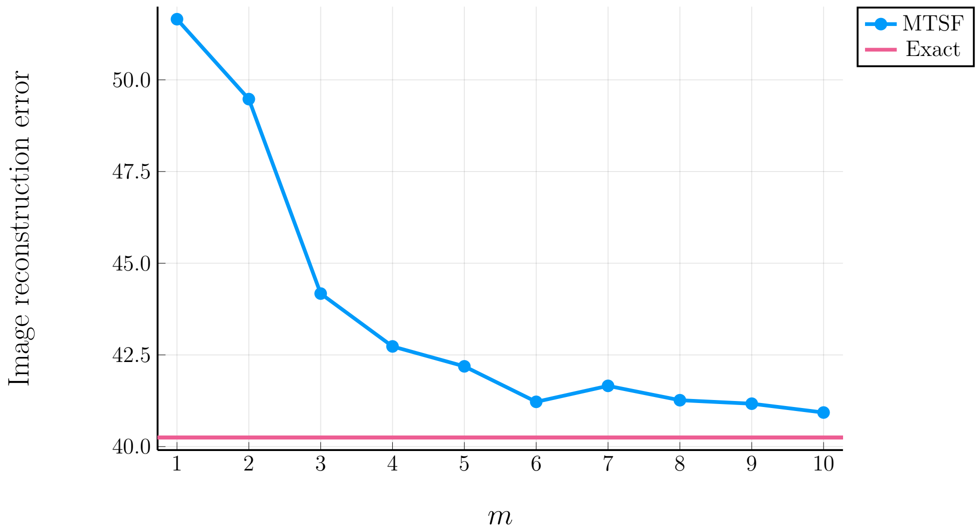
(d) $r(I_*) + \varepsilon, \sigma^2 = 5$



(e) $\sigma^2 = 5$, exact



(f) $\sigma^2 = 5$, MTSF



(g) Error for $\sigma^2 = 5$, $m \in \{1, ..., 10\}$,

Figure 6.5: Examples of recovery from noisy images, with different noise levels, using both the exact solution to the spectral relaxation from Equation (6.63) and the approach from Section 6.3.1.

.

2. *The choice of hyper-parameters q, k and m*, none of which is fixed *a priori*: all of those parameters impact both the quality and the computation time of the estimation, and interact in a complicated way.

Of course, the topology of the graph remains a major factor in deciding which method to use.

6.3.3 *Numerical Evaluation*

We compare the performance of the method from Section 6.3.1 on synthetic graph data using four different iterative solvers: $\overline{f}$, $\hat{f}$ and a conjugate-gradient descent (with and without diagonal preconditioning).

Experimental Setup. We work with the SBM and DC-SBM graph models discussed in Section 6.2.1. We endow each graph G with a random connection generated according to the model described in Section 6.2.3, for both coherent ($\eta = \frac{\pi}{2n}$) and incoherent connections ($\eta = \frac{\pi}{10}$), and aim to recover the vector x with $x_i = e^{\iota \omega_i}$.
Starting from f_0 taken uniformly in $U(\mathbf{C})^n$, we perform k iterations of the power iteration of Equation (6.66), estimating $(\mathsf{L}_\theta + q\mathsf{I})^{-1}(q\mathsf{I})\, f_r$ with m MTSFs (resp. conjugate-gradient iterations) at each step, for both of our methods. We use $m = 3$ for coherent connections, and $m = 10$ for incoherent ones. For incoherent connections, we *do* use the importance sampling strategy from Remark 6.1.6.
We average the synchronization errors

$$e_s(f) = \min_{r \in U(\mathbf{C})} \frac{\|f - rx\|_2}{n} \tag{6.71}$$

over 20 executions, and measure the mean runtimes over 100 runs. We set $q = 10^{-2} \times \overline{d}$ and take measurements for different values of k (10 logarithmically-spaced values in $\{1, ..., 100\}$). The results are averaged over 10 realizations of the random graphs.
We also take measurements for a Lanczos-iteration-based computation[32] (for the matrix L_θ). Finally, we consider two simple tree-based synchronization algorithm, going as follows.

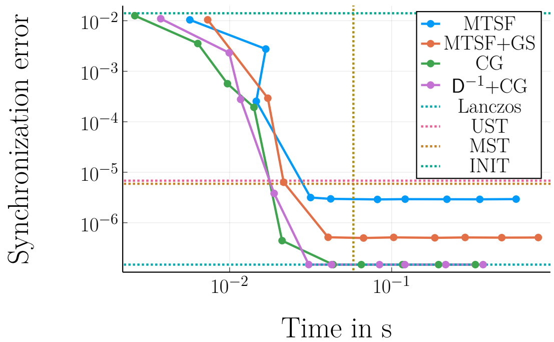
1. First, fix a root node $v \in V$, then:
   - sample a spanning tree of G uniformly (a "UST", which is sampled using Wilson's algorithm [Wilson, 1996]),
   - propagate the value from v to the other nodes (taking into account the offsets).

   Note that this procedure does *not* depend on k or m.

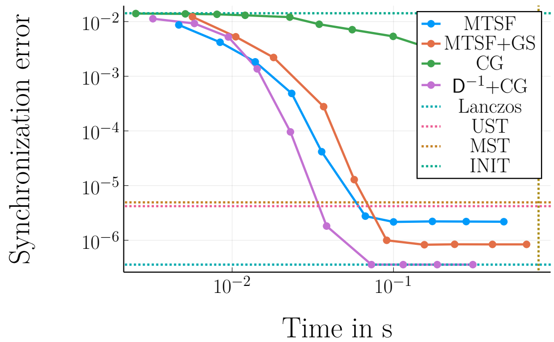
2. A completely analogous strategy, propagating along a *maximum spanning tree* (MST), with respect to the edge-weights $w_{i,j} = |\cos(\theta_{i,j})|$. This strategy is inspired from surface reconstruction techniques such as that of [Hoppe et al., 1992], and expected to perform better than the simpler uniform-spanning-tree strategy depending on the connection (in particular, when said connection stems from the discretization of a surface).

All results are depicted in Figure 6.6.

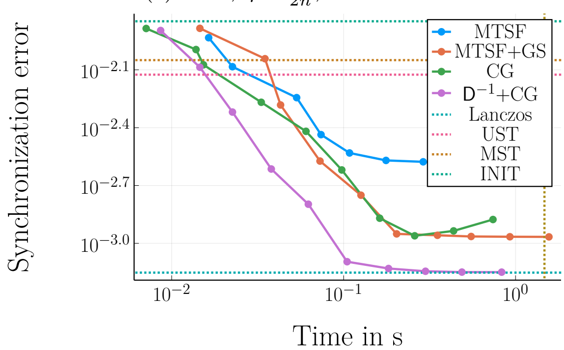
32 Precisely, we use a Krylov-Schur algorithm [Stewart, 2002], as implemented in `https://jutho.github.io/KrylovKit.jl/stable/man/eig/#KrylovKit.eigsolve`. For a concise introduction to Krylov-Schur methods, see, *e.g.* [Hernández et al., 2007].



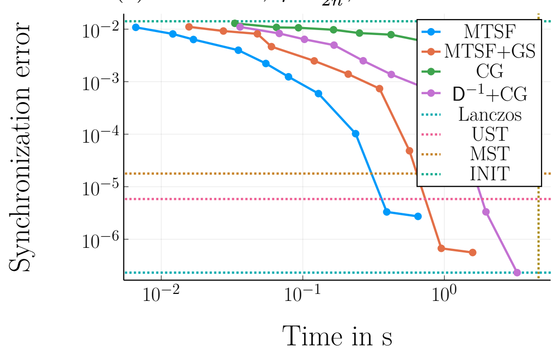
(a) SBM, $\eta = \frac{\pi}{2n}$, $m = 3$.



(b) DC-SBM 1, $\eta = \frac{\pi}{2n}$, $m = 3$.



(c) DC-SBM 1, $\eta = \frac{\pi}{10}$, $m = 10$.



(d) DC-SBM 2, $\eta = \frac{\pi}{2n}$, $m = 3$.

Figure 6.6: Runtime-precision trade-offs for angular synchronization for $m = 10$. Each data point corresponds to a value of k. The vertical yellow line records the average runtime of the Lanczos-based computation, the horizontal blue line the synchronization error for the Lanczos-based computation. The horizontal red (resp. orange) line is the mean synchronization error for the UST-based (resp. MST-based) synchronization, and the horizontal teal line the error for the random initialization.

Comments. Results vary greatly depending on the graph and connection. For coherent connections ($\eta = \frac{\pi}{2n}$), we observe the following:

(1) Krylov-subspaces-based methods are sensitive to : all methods perform similarly on the SBM, but the conjugate-gradient-based power-methods and Lanczos iteration converge more slowly on the (less well-conditioned) DC-SBM 1 graph.

(2) Higher density results in significant slow-downs for Krylov-subspaces methods, and MTSF-based iterations are roughly 10 times faster than the preconditioned-conjugate-gradient iteration for equivalent precision on the DC-SBM 1.

For these coherent connections, the tree-based methods also achieve good synchronization quality, and are cheap to compute.
This is no longer the case for incoherent connections (DC-SBM 1 with $\eta = \frac{\pi}{10}$): in this case, our estimators require more MTSFs (resp. conjugate-gradient iterations) to converge ($m = 10$), and achieve a synchronization error around 10 times smaller than tree-based methods.

For spectral relaxations, our results suggest that MTSF-based solutions perform no-worse than deterministic solutions, and get comparatively better as the density of the graph increases. These techniques are only suitable for approximations, which meshes well with spectral-relaxation-based techniques being mostly primarily interesting in two situations:

- at low noise-levels, in which case the quality of the solution is good, and a small loss due to the MTSF-based approximant is acceptable (*e.g.*, the image-denoising pipeline from above);
- at high noise-levels, for which spectral methods are not able to provide good-quality solutions, and in which case a (fast) rough approximation of the solution may be desirable (which would be interesting for, *e.g.*, computational geometry pipelines).

We conclude this chapter with two complementary remarks.

Remark 6.3.2. *We also experimented with two other estimation strategies (not shown): the spectral relaxation using the normalized Laplacian $\widetilde{\mathsf{L}}_\theta$, and a generalized-power-method-like algorithm, replacing the global normalization in Equation (6.66) by a component-wise normalization, so that each f_r belongs to $U(\mathbf{C})^n$ (resulting in an algorithm similar to that of [Boumal, 2016]). In our experiments, we did not observe any qualitative differences with the previously-described strategy.*

Remark 6.3.3. *We kept our synchronization strategy as straightforward as possible, but could improve it in a number of different ways.*

1. *An important drawback of our strategy is that it requires to set a value for q, which is the subject of a trade-off in the comparison with conjugate-gradient-based methods. Our strategy can be extended to estimate Rayleigh-quotient iterations [Trefethen and Bau, 2022], such that q no longer needs to be fixed, and that better convergence rates may be obtained. In particular, this would require sampling from the distribution $\mathcal{D}_\mathcal{M}$ for a decreasing sequences of values of q, which can be achieved by a variation of the "coupled forest" algorithm of [Avena and Gaudillière, 2013, Avena and Gaudillière, 2018] (originally developed for rooted spanning forests, in the trivial-connection setting).*
2. *Our empirical results are based on an implementation in which a new set of m multi-type spanning forests is sampled at each of the k iterations of the power method. This is expensive,*

and results in a somewhat unfair comparison with conjugate-gradient methods (for which, e.g.*, a preconditioner is only computed once), and performing the k estimations based on the same set of m multi-type spanning forests would result in a significant speed-up. The question that remains to be investigated is how much this would impact the resulting synchronization error.*

3. *Our strategy is based on the eigenvectors of the matrix L_θ, but considering the eigenvectors of the normalized connection Laplacian $\widetilde{\mathsf{L}}_\theta$ instead consistently results in better synchronization-quality in a variety of benchmarks. Our techniques extend to this situation, and enjoy a more advantageous asymptotic complexity in this case: sampling multi-type spanning forests has $\mathcal{O}(|\mathcal{V}|)$ cost in this setting (from Equation (5.48)).*

6.4 SUMMARY

This work should serve as a proof-of-concept that multi-type spanning forests *can* indeed be leveraged to build efficient, competitive RandNLA routines:

- for connection-aware Tikhonov smoothing (and interpolation);
- for angular synchronization.

We believe that an important originality of our work lies in exhibiting the interaction between the runtime-performance trade-offs and *the topology of the graph* which, to our knowledge, has never been considered in the literature before. To put this statement in better perspective, let us note that:

- *in the case of the connection-aware Tikhonov smoothing problem*, we are not aware of any systematic study of any property;
- *in the case of angular synchronization*, most theoretical studies focus on (very) dense Erdös-Rényi models, and principled experimental comparisons between different methods remain scarce.

Given that, in contrast to classical problems on graphs (for which the graph often represents a real-world network), many problems involving connections build *both the graph and the connection* as a part of the pre-processing, this type of study should be valuable to practitioners in diverse fields.

As compared to Krylov-subspaces-based methods, our MTSF-based estimators can reach competitive performance, especially in low-precision regimes, and even *without implementing parallel sampling.* In addition, MTSF-based methods have another noteworthy advantage and, in contrast to conjugate-gradient methods, are very sensitive to both the density of the graph and the conditioning of the resulting systems, are less sensitive to both parameters.

This is rather remarkable, as:

1. MTSF-based smoothing is a specific instanciation of the estimator of Theorem 3.0.2 (for general projective DPPs), which is in general much too inefficient for RandNLA purposes;
2. our Julia implementation of MTSF-based estimators is comparatively not as optimized as those of the other methods it is compared to.

In addition, our empirical evaluations all pertain to sparse graphs, and comparatively better results are to be expected on dense graphs (though it is not clear how relevant this setting is to practical applications).

Finally, let us not that the results of our analyses also applies to the RSF-based estimators of [Pilavcı et al., 2021], and that MTSF-based strategies can be also be employed in distributed systems (where each node represents, *e.g.*, an autonomous entity), in which case all complexity-bounds translate to bounds on the communication-complexity of the resulting distributed algorithm.

7

CONCLUSION AND PERSPECTIVES

Un exemple n'est pas forcément un exemple à suivre.

— Albert Camus[1]

Making sense of graph-supported signals is a challenging task, owing to different factors.

1. *Irregularity of the domain.* Graphs represent complex situations, stemming from different kinds of networks or geometrical situations, and developing processing tools adapted to the variety of problems defined over these domains is a very dynamic area of research.

2. *Computational challenges.* In most applications, the underlying graph that serves as the basis for the processing is large, and typically much too large for the different steps of the processing to be handled by exact algorithms.

A major trend consists in designing processing pipelines, based on variational properties of the signals, that, in particular, boil down to solving linear-algebraic problems. From this perspective, the thesis exposed in this manuscript is in line with a classical idea: random processes can help to understand and design processing pipelines, and are valuable tools towards efficient implementations of these pipelines.

1 See [Camus, 2021].

7.1 ON WHAT HAS BEEN DONE...

The novelty and specificity of our work lies in the setting we consider. Whereas most works in graph signal processing consider real-valued signals, we are concerned with *complex-valued signals* that, what is more, are supported on graphs *endowed with a connection.*
This situation is not novel *per se*, as it can be encountered in different works from geometry and (mathematical) physics, and neither are the inference problems we consider on those graphs, but adopting a signal-processing point of view on these problems is, to our knowledge, original, and can be surprisingly fruitful.

This allows for the implementation of radically different estimation strategies. In place of high-quality approximations to the solutions of problems involved in a processing pipeline (obtained from classical numerical-linear-algebraic tools), one sometimes only needs low-quality approximations, that can be obtained very efficiently from RandNLA estimators and, down the line, yield equivalent results.
Our simplest result in this line pertains to Feynman-Kac estimators for the problems of smoothing and interpolation, and is similar to a number of results in mathematical physics. An important feature of these results is that Feynman-Kac formulas are easily-interpretable; in particular, a smoothed out signal is nothing but a aggregation of noisy values on nodes reached by an interrupted random walk, modulo transport by the connection maps.

There are two important contributions in our work.
First is the introduction of multi-type spanning forests, which are a generalization of spanning forest that naturally arise on graphs endowed with a connection, for (at least) two different reasons:

1. multi-type spanning forests are samples from a determinantal point process generalizing that associated to the classical uniform-spanning-tree distribution;
2. and also arise as natural structures upon which transport maps allow to estimate the solution to smoothing and interpolation problems.

Second is the fact that multi-type spanning forests can be used to perform Monte-Carlo estimation. From our Julia implementation, and as compared to conjugate-gradient-based algorithms, we observe that MTSF-based techniques become more efficient:

- as the conditioning of the problem worsens;
- as the density of the graph increases.

Importantly, both criteria pertain to the topology of the underlying graph, and the choice of solver should depend on this topology. Another convenient observation is that practical use-cases for MTSF-based estimators mostly agree with applicative hypotheses.

- Weak-inconstistency is a required condition for fast sampling of multi-type spanning forests, but connection-aware Tikhonov smoothing similarly mostly makes sense for connections that are not too incoherent (as argued in Appendix D.4).
- Incoherent connections frequently come up in angular-synchronization problems, and MTSF-based estimators can be applied seamlessly to these problems, thanks to importance sampling.

Let us also recall that, using self-normalized importance sampling, we can *always* benefit from a fast sampling algorithm, but that the resulting estimation is only unbiased when the number of forests m goes to ∞.
Further, and though we did not precisely discuss or benchmark the associated estimators, one is able to estimate the trace of various Laplacian-like operator by counting the number of roots of these forests (as follows from Proposition 3.7, and similarly to Theorem 1.2.2), such as

$$\mathrm{tr}\left((\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}\right) \tag{7.1}$$

These techniques are known to rival state-of-the-art methods for trivial connections [Pilavcı et al., 2022a], and we expect similar result in this case.

Remarkably, MTSF-based estimations can be tied back to more classical tools, inspired by underlying variational properties. In particular, one of our estimators can be understood as a type of elementary randomization scheme for *geometric* multi-grid algorithms [Xu and Zikatanov, 2017], whereby multi-type spanning forests induce coarsenings of the graph allowing to reduce the "low-frequency" error. What the relevant hypotheses enabling this scheme are have not yet cristallized in our work, but this observation is, *to our knowledge*, the first link between the randomized-numerical-linear-algebra and geometric-multi-grid literatures, and is likely to warrant some more interest [Pilavcı et al., 2022a].

From a computational perspective, there is a number of different ways our results could be improved upon.

1. First, by designing faster sampling algorithms for multi-type spanning forests. In the case of the related uniform-spanning-tree distribution, the search for algorithms more efficient than Wilson's is an active area of research, exploring both exact and Markov-Chain-Monte-Carlo algorithms (see, *e.g.* [Wilson, 1996, Schild, 2018, Anari et al., 2022]).

2. Second, by awaiting more specialized hardware and optimizations. Let us especially stress this point: MTSF-based solutions do not, at the moment, benefit nearly as much of existing architectures and compiler optimizations as more classical methods do. Indeed, the runtime of high-performance algorithms is often dominated by the time spent fetching data from memory (this is in particular the case for our estimators); memory accesses of standard deterministic algorithms are very well-studied, and benefit from numerous optimizations, whereas those of randomized algorithms do not (but could, *in expectation*!). It would not be surprising to obtain two-times speed-ups for forest-based estimators from this type of improvements alone (for an illustration of comparative benchmarks depending on the underlying architecture, as applied to level 3 BLAS operations, see, [Kågström et al., 1998]).

3. Third, by leveraging more efficient variance-reduction techniques. We only explored a few avenues, but many generic methods may also applied to MTSF-based estimators [Owen, 2013]. Specializing these techniques to obtain efficient algorithms for our specific purposes is an important avenue for further research.

While alternative sampling algorithms (resp. variance-reduction techniques) *may* be more difficult to implement than Wilson's algorithm (resp. control-variates), it is important to keep in mind that modern Krylov-subspaces methods benefit from very mature and high-quality implementations, which allow to subtly deal with inherent issues of these methods such as numerical instability.

In spite of promising first results for MTSF-based estimators, a frustrating theoretical limitation of our strategies is that it is hard to generalize them to other settings of interest. In particular, our Wilson-like sampling algorithm does not extend to highly-incoherent connections (though, as recalled, this situation can be handled with importance sampling), and we discuss below another natural generalization to which our techniques do not translate: simplicial complexes.

Determinantal point processes are a very convenient tool for our developments in Chapters 5 and 6 but, on the other hand, this formalism is poorly adapted to these more general settings. Instead, it may be more convenient to rely on Feynman-Kac estimators to *build* forest-based estimators. This idea is further explored in Sections 7.2.1 and 7.2.2 below.

7.2 ... AND REMAINS TO BE EXPLORED

Chiefly missing from our work are applications and a comparative study of different estimators on real-world problems: multi-type spanning forests allow for *approximate solutions*, and whether or not such approximations are suitable in a given pipeline is extremely problem-dependent.

Our work did not focus of these important applications but, instead, on further theoretical developments, aiming at broader generalizations of forest-based estimators. We sketch below a few preliminary results, that we had thus far left aside.

7.2.1 *Towards Highly-Incoherent Connections*

In general, there is no reason for $U(\mathbf{C})$-connections to be weakly-inconsistent and, in fact, connections are often highly incoherent in practical angular-synchronization problems. While we are able to handle this situation thanks to an importance-sampling trick, the fact remains that we are unable to sample multi-type spanning forests from our Wilson-like algorithm and, in turn, to estimate f_* in this case. On the other hand, we *are* in position to use our Feynman-Kac estimator which, we already noted, relies on a similar estimation strategy: parallel transport along paths in the graph.

We sketch below a formal (but partial) relationship between the two types of estimators and obtain. More precisely, we exhibit the way in which the loop-erasure procedure considered in Algorithm 4 can be understood as stemming from this variance-reduction technique.

Recall from the discussion of Section 4.2 that, in expectation, the Feynman-Kac estimators can be modified to neglect the loops. More specifically, each time a loop C is encountered when running the random walk started at node i, we can perform a Bernoulli trial with success probability $\cos(\theta_C)$, and, set decide on the the estimation at node i depending on these trials.

- If the path contains k loops and the k trials are successful, estimate the signal at node i while neglecting these loops.
- If one of these trials fails, set the estimation at node i to be 0.

The crucial observation follows, as one can identify:

- the modified Feynman-Kac estimator, and in particular the Bernoulli trial for each cycle;
- the cycle-acceptance trial of our Wilson-like sampling algorithm for multi-type spanning forests.

More specifically, consider the first node i in the queue of the MTSF-sampling algorithm, at which the first loop-erased random walk starts: this walk constructs a path p and, at each encountered cycle C in $\mathcal{G}$, may be interrupted due to cycle-acceptance, with probability $1 - \cos(\theta_C)$, in which case the ensuing estimation at node i is $\widetilde{f}(i) = 0$.
Similarly, consider the modified Feynman-Kac estimator, which performs successive Bernoulli trials with success probability $\cos(\theta_C)$, and notice that the result of the estimation at node i is decided *as soon as a trial fails*, in which case it is going to be 0. Effectively, the resulting path p upon which the parallel transport will be applied describes a part of a multi-type spanning forest: the branch on a unicycle, and the associated cycle.
In both cases, if no cycle is accepted (resp. no Bernoulli trial fails), the walk ends upon reaching node Γ, and the estimation at node i is described by the parallel transport from Γ to i along the path $p \subseteq \mathcal{E}$.

The question is then the following. Can such a construction be extended to *all* the nodes in $\mathcal{G}$, so as to recover the (unbiasedness of the) MTSF-based estimator $\widetilde{f}$? This question is important for a few reasons.

- First and foremost, this construction would both allow to extend MTSF-based estimators to highly-incoherent connections, and provide a much more intuitive construction of multi-type spanning forests.
- Second, it is likely to be more flexible: instead of interpreting the factors $\frac{\psi_C + \psi_{C^*}}{2}$ probabilistically (hence conjecturally resulting in MTSF-based estimators), one could, *e.g.*, systematically apply the map $\frac{\psi_C + \psi_{C^*}}{2}$, which would (conjecturally) result in connection-aware RSF-based estimators.

It is humbling that we still lack an answer to this question, and I personally believe that it is crucial to the broader development of forest-based estimators.

Remark 7.2.1. *A similar cycle-equivalence argument has also been used to prove the determinantality the spanning-forest-of-unicycles measure of [Forman, 1993, Kenyon, 2011] in [Kassel, 2015] (in the weakly-inconsistent setting), and to derive the exact expected number of steps for the MTSF-sampling algorithm in [Fanuel and Bardenet, 2024].*
These results crucially rely on determinantal arguments inspired from those of [Marchal, 2000] (see also [Avena and Gaudillière, 2013] for an application to the determinantality of the rooted-spanning-forest distribution), and it is also conceivable that we could actually reconstruct MTSF-based estimators using such a Marchal-like argument.

Note that these techniques would only apply to $U(\mathbf{C})$-connections, and constructing forest-based estimators for general connections remains completely open.

7.2.2 *Simplicial Complexes*

Despite the large number of applications that can be developed based on (random processes over) graphs, this formalism only allows to model pairwise connections, as described by edges, which can be restrictive.

- From a system-analysis perspective, many real-world systems exhibit complex multi-way relationships, owing to, *e.g.*, dynamic behaviors in a system [Torres et al., 2021].

Figure 7.1: A simplicial complex and an arbitrary orientation of its simplices. Left: original non-oriented complex, with six 0-simplices, nine 1-simplices and three 2-simplices (represented as yellow triangles). Right: the same complex endowed with an orientation, represented as directed arrows for 1-simplices, and circular arrows for 2-simplices.

- From a geometrical and physical-modeling perspective, graphs only allow to reason about (discretizations of) functions and co-vector fields over manifolds, while the description of many important phenomena (in, *e.g.*, electromagnetism and fluid mechanics) involves higher-order fields (that is, differential forms or other tensors) [Sontz, 2015].

We sketch in the following a generalization of our work to a classical mathematical model of higher-order systems, capable of accommodating both perspectives: simplicial complexes[2].

Formal developments involving simplicial complexes require a more involved mathematical apparatus, and the remainder of this section is mostly informal. The reader interested in the details can consult Appendix E (which is mostly self-contained, but challenging).

A *simplicial complex* is a couple $\mathcal{X} = (\mathcal{V}, \mathcal{S})$ of a (finite) set of vertices $\mathcal{V}$, and of a downward-closed set of *simplices* $\mathcal{S} \subseteq 2^{\mathcal{V}}$. The simplices of $\mathcal{X}$ can be graded according to their cardinality, which translates to a geometrical interpretation of simplicial complexes. More specifically, we can write

$$\mathcal{S} = \bigcup_{k \geq 0} \mathcal{S}_k, \tag{7.2}$$

where $S_k = \{\sigma \in \mathcal{S};\ \#\sigma = k+1\}$ denotes the set of simplices of size $k+1$. In particular, $\mathcal{S}_1$ is (in bijection with) the set of vertices $\mathcal{V}$, and S_2 describes a set of edges between these nodes. In turn, $\mathcal{S}_k$ describes k-way relationships between nodes, called k-simplices, which we can graphically represent as (geometrical) k-simplices. We illustrate this gradation in Figure 7.1 on a simple simplicial complex, which contains 0-simplices (vertices), 1-simplices (edges), and 2-simplices (triangles).

An important property of simplicial complexes is that their set of simplices is downward-closed, that is, for every simplex $\sigma \in \mathcal{S}$ and $\tau \subseteq \sigma$, it holds that $\tau \in \mathcal{S}$. In other words, this is equivalent to asking that the *faces* $\tau \in \mathcal{S}_{k-1}$ of each simplex $\sigma \in \mathcal{S}_k$ also belong to the simplicial complex[3]; for instance, if a triangle $\{v_1, v_2, v_3\} \in \mathcal{S}_3$ belongs to $\mathcal{S}$, then the faces of this triangle $\{v_1, v_2\}, \{v_1, v_3\}, \{v_2, v_3\} \in \mathcal{S}_2$, and so do the faces of these edges, so that $\{v_1\}, \{v_2\}, \{v_3\} \in \mathcal{S}_1$. In the case of graphs, this amounts to requiring that edges be defined between pairs of vertices, and that there are no "half-edges".

2 All our developments generalize to regular cell complexes as well.
3 Relaxing this condition results in an hypergraph [Torres et al., 2021].

In this generalized setting, one considers signals that are defined on *oriented* simplices. An edge $\{v_0, v_1\} \in \mathcal{S}_1$ has two possible orientations: (v_0, v_1) or (v_1, v_0). A triangle $\{v_0, v_1, v_2\} \in \mathcal{S}_2$ also has two possible orientations, which represent an a circular order on its constituting vertices: either (v_0, v_1, v_2), or (v_0, v_2, v_1). We denote by $\overrightarrow{\mathcal{S}_k}$ the set of oriented k-simplices, which contains both oriented versions of each simplex. In the following, we fix a reference orientation for each simplex σ, and denote by $\overline{\sigma}$ the reverse orientation.
A k-signal is then nothing but a function $f : \overrightarrow{\mathcal{S}_k} \to \mathbf{C}$ such that $f(\sigma) = -f(\overline{\sigma})$. These signals may represent, *e.g.*, measurements of a (directed) electrical current is a circuit (Chapter 1), of a velocity flow, or of a magnetic field [Warnick, 2021].

k-signals have been the object of recent interest from the graph-signal-processing community. In particular, the question of smoothing k-signals has emerged, and several variants of Tikhonov smoothing have been considered. To this end, one defines different discrete differential operators acting on k signals (generalizing the combinatorial Laplacian), called the upper, lower, and (full) Hodge Laplacians, and respectively denoted by $\mathsf{L}_k^{\uparrow}$, $\mathsf{L}_k^{\downarrow}$, and $\mathsf{L}_k = \mathsf{L}_k^{\uparrow} + \mathsf{L}_k^{\downarrow}$ (see Appendix E for a precise definition). The k-signal smoothing problems may then take the form

$$\underset{f \in C^k(\mathcal{X}, \mathbf{C})}{\operatorname{argmin}} \quad q\|f - g\|^2 + \langle f, \mathsf{L}_k^{\uparrow} f\rangle, \tag{7.3}$$

or any variant involving the other Hodge Laplacians in the regularization term. In the following, we denote by f_* the solution to this problem.
This type of problem has appeared in a number of applications, involving the analysis of velocity-field of ribonucleic acid from noisy measurements [Barbarossa and Sardellitti, 2020], of oceanic flows from measured displacements of buoys drifting in the ocean [Schaub et al., 2020], or of political-books co-purchasing data using edge-centrality measures [Schaub et al., 2020].

Can forest-based estimators be designed for problems involving simplicial complexes? We are not able to answer this question at this point, but can obtain a number of preliminary results, which we briefly discuss below (see Appendix E for details).

1. We are able to derive a a family of generalizations of the distribution $\mathcal{D}_{\mathcal{F}}$, that sample a specific type of generalized forests over simplicial complexes (for each value of k). These distributions can be used to perform Monte-Carlo estimation of f_* (or of other variants), but we were not able to derive an efficient sampling algorithm.

2. *More interestingly*, we are able to develop a simple Feynman-Kac formula for f_*. The resulting estimator can be computed by running a (lazy, interrupted) random walk on the k-simplices of the complex, which goes as follows:
   - With probability $\frac{k}{2k+1}$, stay at σ and do not interrupt the walk (lazy step).
   - Otherwise, with probability $\frac{q}{q+d_\sigma^{\uparrow}}$, interrupt the walk at σ.
   - Otherwise, select a "neighbor" of σ uniformly at random, that is, new simplex $\sigma' \in \mathcal{S}_k \setminus \{\sigma\}$ such that $\sigma \cap \sigma' \in \mathcal{S}_{k-1}$.

   Upon the walk being interrupted at some node j, this results in a random path $p = (i, \sigma_1, ..., \sigma_{l-1}, j)$, and we are able to show that

$$f_*(i) = \mathbf{E}_p\left(\psi_{p^*}(g(j))\right), \tag{7.4}$$

where $\psi_{p^*} \in \{-1, 1\}$ is a multiplicative factor depending on the orientations of the simplices encountered in the path p.

If we were able to extend the strategy of Section 7.2.1 to build forest-based estimators from Feynman-Kac formulas, the same strategy may apply to this setting as well.

7.3 ON TEACHING

In the last three years, I also devoted a number of hours to teaching, and had the opportunity to supervise students during exercise and practical sessions for the following courses.

- *Introduction to Artificial Intelligence and Machine Learning*, at the Ense3 engineering school (Grenoble), second semester of 2023−2024. This is an optional course for first-years engineering students (three years past high-school, in the french system), which covered a number of classical tools for statistical learning including, *e.g.*: basics on classification and regression problems, bias-variance trade-offs and model selection issues, nearest-neighbors, linear and logistic regression, regularization issues, decision trees and forests, neural networks. I supervised computer-based practical sessions.
- *Continuous Optimization*, at Université Grenoble Alpes (Grenoble), second semester of 2022−2023. A course for third-year mathematics and computer-science majors. The course starts from an introduction to multi-variate analysis (in the general setting of Banach spaces[4]), and explores theoretical properties of classical optimization strategies such as Newton's method or (conjugate) gradient descents, from a mathematical-engineering perspective. I supervised paper-based exercise sessions.
- *Systems and Environments for Development*, at Université Grenoble Alpes (Grenoble), second semester of 2021−2022. An introductory class for first-year university students, familiarizing them with UNIX-like environments, shell tools and Bash scripting, the C programming language, finite automata, and interactions between those topics. I supervised both paper-based exercise and computer-based practical sessions.

The first two courses listed above are obviously related to theory developed in this thesis, but what about the third one? Automata theory takes many forms, and it's nice to notice that a variant called *quantum finite automata* are actually described by graphs with edge-associated unitary maps [Kondacs and Watrous, 1997]. Would cross-fertilization of our work with the (probabilistic) monitoring/model-checking of those systems be possible?

7.4 LIST OF PUBLICATIONS

The content of this thesis is the object of one conference-proceedings publication [Jaquard et al., 2023] (for a preliminary version), and of one preprint submitted to a refereed journal [Jaquard et al., 2024] (currently in revision for *SIAM Journal on Mathematics of Data Science*).

[Jaquard et al., 2023]. Jaquard, H., Fanuel, M., Amblard, P.-O., Bardenet, R., Barthelmé, S., and Tremblay, N. (2023). Smoothing Complex-Valued Signals on Graphs with Monte-Carlo. In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*

4 Named after Polish mathematician Stefan Banach, 1892−1945.

[**Jaquard et al., 2024**]. Jaquard, H., Amblard, P.-O., Barthelmé, S., and Tremblay, N. (2024). Random Multi-Type Spanning Forests for Synchronization on Sparse Graphs

Our Julia implementation of MTSF-based estimators is also freely available: `https://gricad-gitlab.univ-grenoble-alpes.fr/gaia/synchromtsf`.

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RÉSUMÉ SUBSTANTIEL EN FRANÇAIS

Les quelques pages qui suivent visent à exposer de manière brève et introductive le travail réalisé dans cette thèse, et sont certainement destinées aux curieux plutôt qu'aux experts. Le lecteur intéressé par les détails techniques leur préférera le reste du document, rédigé en langue anglaise.

Un *graphe* $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ est la donnée d'un ensemble fini de *sommets* $\mathcal{V}$, et de paires de sommets $\mathcal{E}$, appelées ses *arêtes*. Un dessin vaut mille mots, et nous représentons ci-dessous un graphe à la fois typique et exceptionnel [Holton and Sheehan, 1993].

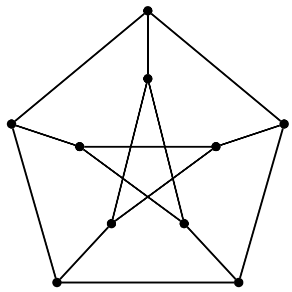


Figure 7.2: Le graphe « de Petersen ».

La théorie des graphes, vaste et dynamique, vise à étudier et exploiter les propriétés de ces graphes. À ce titre, de nombreux points de vue peuvent être adoptés, et nous en listons quelques-uns ci-dessous.

- Les théories combinatoires des graphes, qui, par exemple, étudient et comptent des structures combinatoires présentent dans les graphes.
- Les théories algébriques des graphes, qui étudient des objets algébriques définis à partir de graphes, et leurs liens avec la structure du graphe.
- Le traitement du signal sur les graphes, qui étudie le traitement de signaux mesurés sur des graphes, et non plus sur des grilles régulières.
- Les probabilités sur les graphes, qui étudient des processus aléatoires définis sur les graphes, comme par exemple des marches aléatoires.
- Les théories « physiques » des graphes, qui étudient les propriétés statistiques de réseaux.
- Les théories « géométriques » des graphes, qui étudient des graphes comme des objets géométriques discrétisés, par exemple des surfaces.

Une distinction stricte de ces théories a bien peu de sens : leurs interactions riches et complexes constituent l'un des grands charmes de la théorie des graphes.

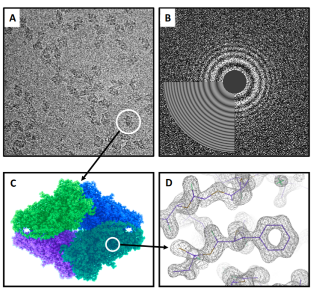
Un second charme indéniable de la théorie des graphes est son utilité pratique. Les graphes permettent de modéliser de très nombreuses situations tant, en physique qu'en (télé-)communications, en biologie qu'en informatique. Ces modèles sont associés à des « problèmes » sur graphes que l'on cherche à résoudre, typiquement par une procédure algorithmique. Dans le cadre de cette thèse, ces problèmes impliques des fonctions $f : \mathcal{V} \to \mathbf{C}$ à valeurs complexes définies sur les sommets du graphes, souvent appelés *signaux* sur graphes. Nous décrivons ci-dessous trois exemples, qui nous l'espérons surprendront le lecteur, de situations dans lesquels de tels problèmes apparaissent, et auxquelles les outils que nous développons peuvent s'appliquer.

1. *La cryomicroscopie électronique* (cryo-em). Une technologie d'imagerie qui vise à obtenir des modèles 3D à haute résolution de molécules, et pour laquelle les graphes sont utilisés de plusieurs manières. Les techniques de cryo-em se basent sur la mesure de très nombreuses prises de vues 2D, extrêmement bruitées et à des angles inconnus, de la molécule (en 3D). Il est nécessaire de débruiter ces mesures, et une approche consiste à construire un graphe dont : les sommets représentent les images bruitées, les arêtes représentent des paires d'images pour lequelles on a calculé un alignement « au mieux » de ces images. Sur la base de cette information, on cherche à retrouver une rotation pour chaque image, de sorte à pouvoir les « moyenner » pour éliminer le bruit. Ces rotations se représentent par des nombres complexes : le signal f dans cet exemple.

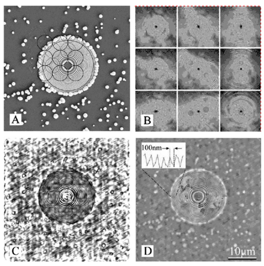
2. La *ptychographie*. Une méthode d'imagerie électronique par diffraction, qui trouve des applications dans de nombreux domaines tels que la cristallographie ou l'imagerie cellulaire. La ptychographie implique un problème d'estimation de phase d'un signal $f \in \mathbf{C}^n$ (inconnu) à partir de « patterns » de diffractions obtenus en illuminant différentes régions d'un objet d'intérêt. Les méthodes modernes d'estimation de phase procèdent par une approximation de différents décalages de phases $f_i(f_j)^*$, pour certaines paires d'indices (i, j), qui décrivent un graphe, et sont ensuite utilisées pour estimer f.

3. *Le grooming*. Une technologie utilisée par les artistes graphiques pour générer automatiquement de la fourure sur un modèle 3D, à partir de quelques brins qui servent de guide. Dans cet exemple, le graphe est décrit par la surface discrète (*e.g.*, triangulée), et le signal f correspond à une direction dans le plan tangent à chaque sommet, qui représente la direction des différents brins de fourures. Dans cet exemple, la géométrie du modèle doit être prise en compte, ce qui est typiquement encodé grâce à une rotation associée à chaque arête.

Nous illustrons ces trois applications dans la figure ci-dessous.



(a) Imagerie 3D d'une enzyme de β-galactosidase enzyme par cryo-em. Image de [Bendory et al., 2020].



(b) Imagerie par ptychographie d'une plaque à zone de Fresnel en or d'épaisseur 1μm. Image de [Rodenburg et al., 2007, Rodenburg, 2008].

(c) Modèle de Licorne du film Pixar *Onward*, dont le pelage et les ailes sont générées automatiquement à partir de quelques brins guides. Image de [De Goes et al., 2020].

Dans tous ces exemples, le graphe est équipé d'une structure supplémentaire : une rotation associée à chaque arête orientée $e = (v, v')$, qui sera en général représentée par un nombre complexe ψ_e (de module $|\psi_e| = 1$, et telle que $\psi_{(v,v')} = \psi^*_{(v',v)}$). C'est ce que l'on appelle une *connection unitaire* (dans la suite, connection). Un graphe équipé d'une connection peut se représenter par le *Laplacian de connection* L_θ du graphe, qui est une matrice définie par

$$\mathsf{L}_\theta = \mathsf{D} - \mathsf{A}_\theta, \tag{7.5}$$

avec A_θ une matrice d'adjacence, telle que $(\mathsf{A}_\theta)_{i,j} = e^{-\iota\theta_{i,j}}$ si $\{i,j\} \in \mathcal{E}$ et $(\mathsf{A}_\theta)_{i,j} = 0$ sinon, et D la matrice diagonale des « degrés » du graphe, avec $\mathsf{D}_{i,i} = \sum_{i \neq j} |(\mathsf{A}_\theta)_{i,j}|$.
Le Laplacien de connection a une propriété remarquable : il permet de quantifier la « cohérence » d'un signal sur un graphe, vis-à-vis de la connection. Plus précisément, on a

$$\langle f, \mathsf{L}_\theta f \rangle = \sum_{\{i,j\} \in \mathcal{E}} |f(i) - e^{\iota\theta_{j,i}} f(j)|^2, \tag{7.6}$$

et on observe en particulier que, pour un signal f donné, $\langle f, \mathsf{L}_\theta f \rangle = 0$ *si et seulement si* $f(i) = e^{\iota\theta_{j,i}} f(j)$ pour toute paire de sommet $\{i,j\} \in \mathcal{E}$.

Quantifier la cohérence d'un signal est utile : les trois problèmes d'estimation présentés plus haut peuvent se ramener à déterminer un signal lisse, et plus précisément aux deux problèmes suivant.

1. Le problème de la *synchronisation angulaire*, qui consiste à déterminer résoudre le problème

$$\underset{f}{\operatorname{argmin}} \ \langle f, \mathsf{L}_\theta f \rangle \tag{7.7}$$

   sous la contrainte que tous les $f(i)$ aient module 1.

2. Le problème de la *régularisation de Tikhonov*, qui consiste à lisser un signal g en résolvant

$$\underset{f}{\operatorname{argmin}} \ q\|f - g\|_2^2 + \langle f, \mathsf{L}_\theta f \rangle, \tag{7.8}$$

   pour un certain paramètre de régularisation $q \in \mathbf{R}_+^*$.

Le premier problème est compliqué, en général NP-difficile[5], et sa solution est typiquement approchée par celle d'un autre problème (par exemple, obtenue par lissage itératif d'un autre signal). La solution f_* du second problème est plus facile à calculer, et prend la forme

$$f_* = q(\mathsf{L}_\theta + q\mathsf{I})^{-1} \ g, \tag{7.9}$$

qui est donc obtenue par résolution d'un système linéaire.

Bien que le calcul de f_* se ramène à un problème bien connu, ce problème peut être long à résoudre dans de nombreuses situations. Cette observation tient en particulier à deux raisons.

1. Si le graphe est grand, le calcul *exact* de f_* devient trop cher, et ne peut plus être effectué en temps raisonnable.

5 C'est-à-dire, de manière informelle, qu'il *n'existe pas* d'algorithme pour le résoudre impliquant un nombre d'opérations polynomial en la taille du graphe.

2. Les méthodes de calcul de f_* *approchées*, qui sont en général des méthodes itératives, peuvent nécessiter de (trop) nombreuses itérations pour converger vers une solution de bonne qualité. Ces méthodes ont historiquement été développées principalement pour des graphes « grille », mais ne sont plus aussi efficaces sur des graphes arbitraires.

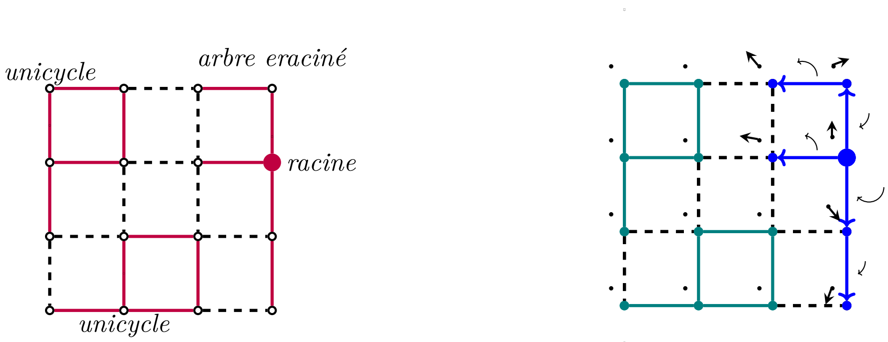
Comme la majorité des graphes rencontrés dans les problèmes de recherche actuels sont à la fois grands et très irréguliers, développer des méthodes de résolution efficaces est une problématique importante.

Notre travail consiste précisément à proposer de nouvelles techniques de résolution pour les problèmes de régularisation de Tikhonov et de synchronisation angulaire. Nos développements s'inscrivent dans le domaine de l'*algèbre linéaire numérique randomisée*, qui vise à accélérer les algorithmes d'algèbre linéaire numérique en introduisant de l'aléatoire dans les calcul; plus précisément, nous introduisons des estimateurs de Monte-Carlo. Une estimation de Monte-Carlo d'une quantité $\hat{x}$ consiste en un tirage de variables aléatoires qui sont ensuite moyennées pour approcher $\hat{x}$. Dans notre cas, nous estimons $\hat{x} = f_*$, et nous nous basons sur cette estimation pour estimer la solution de problèmes de synchronisation angulaire.

À ces fins, nous introduisons un nouveau type de décomposition (aléatoire) du graphe, appelées *forêts couvrantes multi-types*. De manière informelle, une forêt couvrante multi-types est un sous-ensemble ϕ de sommets et d'arêtes du graphe qui satisfait les propriétés suivantes.

- Le graphe entier doit être couvert par la forêt : chaque sommet v doit soit appartenir à la forêt, soit être connecté à une arête de la forêt.
- Les composantes connexes de la forêt doivent décrire :
  - soit un arbre enraciné ;
  - soit un unicycle, c'est à dire un sous-ensemble d'arêtes contenant *exactement* un cycle.

Nous illustrons une forêt couvrante multi-types dans la Figure 7.4, sur un graphe grille.



(a) Une forêt couvrante multi-types comprenant deux unicycles (en haut à gauche et en bas) et un arbre enraciné (en haut à droite).

(b) Estimation : la valeur à la racine est transportée aux autres sommets de l'arbre, en appliquant la rotation associée à chaque arête. L'estimation 0 sur les unicycles est représentée par un point (sans flèche).

Figure 7.4: Illustration d'une forêt couvrante multi-types (à gauche), et de notre estimateur $\widetilde{f}$ (à droite). L'estimation $\widetilde{f}(i) \in \mathbf{C}$ sur chaque sommet est représenté par un vecteur, et la connection par des flèches courbées (seulement pour les arêtes appartenants à des arbres enracinés).

Nous considérons alors une distribution de probabilités $\mathcal{D}_{\mathcal{M}}$ sur les forêts couvrantes multi-types d'un graphe, admettant une définition technique qui dépend à la fois :

- du paramètre q ;
- du nombre d'arbres enracinés dans ϕ ;
- de la connection sur le graphe.

Pour cette distribution, nous prouvons alors le résultat suivant.

Theorem 7.4.1. *Soit g un signal bruité, et ϕ une forêt tirée selon la loi $\mathcal{D}_{\mathcal{M}}$.*
Nous définissons l'estimateur $\widetilde{f}$ pour chaque sommet $i \in \mathcal{V}$ par :

$$\widetilde{f}(i) = \begin{cases} \left(\prod_{e \in a \xrightarrow{\phi} b} \psi_e\right) g(r_\phi(i)) & \text{si } i \text{ is couvert par un arbre enraciné,} \\ 0 & \text{si } i \text{ est couvert par un unicycle,} \end{cases} \tag{7.10}$$

où $r_\phi(i)$ dénote le sommet racine de l'arbre auquel appartient i, et $a \xrightarrow{\phi} b$ la suite d'arêtes reliant un sommet a à un sommet b.
Alors, en moyenne sur les $\phi \sim \mathcal{D}_{\mathcal{M}}$, nous avons

$$\mathbf{E}\left(\widetilde{f}(i)\right) = f_*(i). \tag{7.11}$$

Ce théorème se traduit en un procédé d'estimation simple de f_*, que nous illustrons en Figure 7.4. En quelques mots, ce processus est le suivant.

1. Tirer une forêt ϕ selon $\mathcal{D}_{\mathcal{M}}$.
2. Pour chaque sommet $i \in \mathcal{V}$:
   - si i est couvert par un arbre enraciné, propager la valeur en la racine vers i selon l'unique chemin les reliant dans ϕ, en prenant en compte les rotations rencontrées ;
   - si i est couvert par un unicycle, estimer 0 sur ce sommet.

Le théorème précédent assure que l'on retrouvera bien le signal f_* en moyenne.

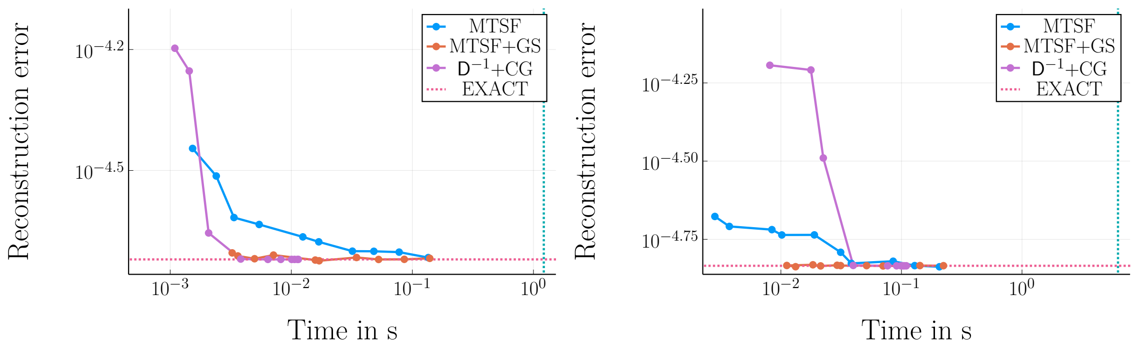
Cette description est malheureusement réductrice, et cache un bon nombre de complexités. Nous adressons en particulier les questions suivantes.

- *Comment faire, en pratique, pour tirer ϕ selon $\mathcal{D}_{\mathcal{M}}$?* C'est une étape nécessaire à l'estimation, et il n'est pas *a priori* évident que ce tirage puisse être réalisé de manière efficace. Nous proposons un algorithme simple et rapide pour résoudre ce problème, et bornons son temps de calcul théoriquement.

- *La moyenne empirique calculée sur m forêts est-elle une bonne approximation de f_* ?* Si l'on compare notre estimateur aux autres méthodes de calcul de f_*, notre estimation n'est pas d'assez bonne qualité. Nous proposons donc plusieurs variantes de l'estimateur $\widetilde{f}$ qui permettent d'obtenir une bien meilleure estimation avec le même nombre de forêts, et établissons diverses garanties regardant ces estimateurs.

- *Nos estimateurs atteignent-ils de meilleures performances que les autres méthodes de calcul de f_*?* Cela dépend du graphe, et nous proposons un jeu de tests structuré pour analyser le comportement de nos estimateurs sur différents graphes, et pour différents choix de paramètres. Plus précisément, nous considérons que le signal g est obtenu comme dégradation d'un signal sous-jacent $f_\top$, et comparons la capacité des différents algorithmes à estimer $f_\top$. Il en ressort que nos algorithmes peuvent rivaliser avec les techniques bien établies et permettent dans certains cas, selon la topologie du graphe, une accélération significative. Nous illustrons nos résultats dans la Figure 7.5.

- *Comment ces estimateurs peuvent-ils être utilisés pour résoudre des problèmes de synchronisation angulaire?* Il se trouve que lisser itérativement un signal de départ f_0 *via* régularisation de Tikhonov converge vers un estimateur bien établi de la solution du problème de synchronisation angulaire, et nous pouvons appliquer nos estimateurs pour approcher chacun des lissages successifs. Les comparaisons avec les méthodes plus classiques sont similaires au problème précédent.



(a) DC-SBM with mean degree $\overline{d} \simeq 30$.

(b) DC-SBM with mean degree $\overline{d} \simeq 420$.

Figure 7.5: Comparaisons en temps de calcul et en qualité de reconstruction de $f_\top$ de différentes méthodes de calcul de la régularisation de Tikhonov sur deux modèles de graphes aléatoires : par deux de nos estimateurs (MTSF et MTSF+GS) et par une méthode classique (D^{-1}+CG : gradient conjugué préconditionné diagonalement). Chaque point de donnée représente un nombre $m \in \{1, 2, 3, 5, 8, 13, 22, 36, 60, 100\}$ de forêts utilisées (resp. d'itérations de gradient conjugué). La ligne verticale représente le temps de calcul de f_* via une méthode exacte (décomposition de Cholesky); la ligne verticale la qualité de reconstruction associée.

La recherche future permettra sans doute d'améliorer nos résultats, en proposant :

- des algorithmes d'échantillonnage plus efficaces;
- des variantes de nos estimateurs plus efficaces;
- des implémentations plus efficaces.

Il est important de garder en tête que les méthodes auxquelles nous nous comparons bénéficient de nombreuses années de recherche et d'implémentations de qualité.

Concluons enfin par une remarque qui ravira certainement le lecteur : les résultats que nous développons ont attrait à toutes les « théories » mentionnées à la première page de ce résumé.

APPENDICES

A

SMOOTHING WITH THE CONNECTION LAPLACIAN

A.1 PROOF OF PROPOSITION 2.1.1

Denote by $C : \mathbf{C}^{\mathcal{V}} \to \mathbf{R}$ the cost function

$$C(f) = q\|f - g\|_2^2 + \langle f, \mathsf{L}_\theta f\rangle. \tag{A.1}$$

We use a standard argument from **CR**-calculus, and seek the zeros of the Fréchet-Wirtinger derivatives of C (see *e.g.* [Bouboulis, 2010]). Let us first compute these derivatives: for $f, h \in \mathbf{C}^{\mathcal{V}}$, we have

$$C(f+h) = q\langle (f+h) - g, (f+h) - g\rangle + \langle (f+h), \mathsf{L}_\theta(f+h)\rangle, \tag{A.2}$$

$$= C(f) + D_f^W C(h) + D_f^{W^*} C(h) + \left(q\langle h, h\rangle + \langle h, \mathsf{L}_\theta h\rangle\right), \tag{A.3}$$

where $(h \mapsto q\langle h, h\rangle + \langle h, \mathsf{L}_\theta h\rangle) \in o(h)$, and the associated (conjugate) Fréchet Wirtinger derivatives of C (at f) $D_f^W C$ and $D_f^{W^*} C : \mathbf{C}^{\mathcal{V}} \to \mathbf{C}$ are given by:

$$D_f^W C(h) = \langle q(f-g) + \mathsf{L}_\theta, h\rangle \tag{A.4}$$

$$D_f^{W^*} C(h) = \langle h, q(f-g) + \mathsf{L}_\theta\rangle. \tag{A.5}$$

The result follows from the fact that $D_f^W C = 0$ (resp. $D_f^{W^*} C = 0$) identically if and only if $f = q(\mathsf{L}_\theta + q\mathsf{I})^{-1} g$.

B

CONNECTION-AWARE HARMONIC INTERPOLATION

B.1 SOLUTION OF THE CONNECTION-AWARE HARMONIC-INTERPOLATION PROBLEM

We proceed identically to the case of real-valued signals.

Let us consider the following decomposition of L_θ:

$$\mathsf{L}_\theta = \begin{bmatrix} (\mathsf{L}_\theta)_{\mathcal{A}} & (\mathsf{L}_\theta)_{\mathcal{A},\mathcal{V}\setminus\mathcal{A}} \\ (\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}} & (\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}} \end{bmatrix}. \tag{B.1}$$

Then, $f \in \mathbf{C}^{\mathcal{V}}$ is a solution to the connection-aware interpolation problem if and only if

$$\begin{bmatrix} \mathsf{I} & 0 \\ (\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}} & (\mathsf{L}_\theta)\mathcal{V}\setminus\mathcal{A} \end{bmatrix} f = \begin{bmatrix} g \\ 0 \end{bmatrix} \tag{B.2}$$

and, assuming invertibility of $(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ (given by Lemma 4.1.8 when $\mathcal{A}$ intersects all the connected components of $\mathcal{G}$), it follows that the (hence unique) solution f satisfies

$$f_{\mathcal{V}\setminus\mathcal{A}} = -((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}})^{-1}(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A},\mathcal{A}}\, g. \tag{B.3}$$

C

MISSING PROOFS FROM CHAPTERS 5

C.1 PROOF OF LEMMA 5.2.3

Suppose first that c_ϕ is a unicycle (this is the case treated in [Kenyon, 2011]). We fix an orientation of the edges of c_ϕ such that all edges are oriented *towards* the cycle C, and that the edges belonging to C form a directed cycle (there are only two such orientations, we choose one arbitrarily). The only non-zero permutations in the determinant

$$\det(\Delta^*_{\mathsf{Q}})_{:,c_\phi} = \sum_{\sigma\in\mathcal{S}_m}\prod_{1\le i\le m}\operatorname{sgn}(\sigma)(\Delta^*_{\mathsf{Q}})_{i,\sigma(i)}, \tag{C.1}$$

where $\operatorname{sgn}(\sigma)$ denotes the signature of permutation (bijection from nodes to edges) σ, correspond to these orientations, and map vertices s_e to edges e ($\sigma(s_e)=e$). These two bijections, denoted σ_C and σ_{C^*}, differ only on nodes adjacent to edges in C, and correspond to the two possible orientations of the cycle. We then obtain:

$$\det(\Delta^*_{\mathsf{Q}})_{:,c_\phi} = (-1)^m\operatorname{sgn}(\sigma_C)\prod_{e\in c_\phi\setminus C}\sqrt{w_e}\psi_{s_e,e}\left(\prod_{e'\in C}\sqrt{w_{e'}}\psi_{s_{e'},e'} - \prod_{e'\in C}\sqrt{w_{e'}}\psi_{t_{e'},e'}\right). \tag{C.2}$$

Performing a similar computation for $\det(\Delta_{\mathsf{Q}})_{c_\phi,:}$ and multiplying the two resulting expressions then allows to show that (this computation also appears in [Kenyon, 2011]):

$$\det(\nabla^*_{\mathsf{Q}}\nabla_{\mathsf{Q}})_{c_\phi,c_\phi} = (2-2\cos(\theta_C))\prod_{e\in c_\phi} w_e, \tag{C.3}$$

where we used $\psi_e = \psi_{e,t_e}\circ\psi_{s_e,e}$ and $\psi_C+\psi_C^* = 2\cos(\theta_C)$.
Similarly, for c_ϕ a rooted tree with root $r\in c_\phi\cap\mathcal{V}$, we have

$$\det(\Delta^*_{\mathsf{Q}})_{:,c_\phi} = \sqrt{q_r}\times(-1)^m\operatorname{sgn}(\sigma_C)\prod_{e\in(c_\phi\cap])\setminus C}\sqrt{w_e}\psi_{s_e,e}, \tag{C.4}$$

which results in

$$\det(\nabla_{\mathsf{Q}})_{c_\phi,c_\phi} = q_r\prod_{e\in c_\phi\cap\mathcal{E}} w_e. \tag{C.5}$$

C.2 WILSON'S ARGUMENT FOR THE INDEPENDENCE OF CYCLES IN THE STACK REPRESENTATION

We translate the argument of [Wilson, 1996] to show that the order in which the cycles are detected in the MTSF-sampling algorithm does not influence the result.

As a first step, let us encapsulate all remaining randomness in the algorithm as a pre-processing step. In addition to the $|\mathcal{V}|$ stacks $u_i \in (\mathcal{V}^\Gamma)^\omega$ associated to each node $i \in \mathcal{V}$, we shall consider that all the detected cycles of $\overrightarrow{\phi}$ are "colored", that is, that we keep track, for all nodes i belonging to a cycle C, of the depth of the stack u_i at which this cycle appears[1] and that, for all "colored" cycles, the (random) output of the Bernoulli trial has been drawn in advance.

Consider then a cycle C that could be "detected", that is, for which there exists a sequence of "colored" cycles $(C_1, C_2, ..., C_{l-1}, C_l)$ (as ordered by successive chronological detection in an execution), and a second "colored" cycle C_1' that could have been detected in place of C_1.

- *If C_1' shares vertices with other "colored" cycles in the sequence* $(C_1, C_2, ..., C_{l-1}, C_l)$, let i denote the first "colored" cycle in the sequence such that C_1' and C_i share a vertex.
  If C_1' and C_i are not equal as non-colored cycles, then there exists some node w such that the associated maps $\overrightarrow{\phi_1'}$ and $\overrightarrow{\phi_i}$ point to two different nodes: $\overrightarrow{\phi_1'}(w) \neq \overrightarrow{\phi_i}(w)$. But w is never spanned in the sequence of detected "colored" cycles $(C_1, C_2, ..., C_{l-1}, C_l)$ and, hence, one must have $\overrightarrow{\phi_1'}(w) = \overrightarrow{\phi_i}(w)$; a contradiction.
  It follows that C_1' and C_i *are* equal as "colored" cycles as well, and that the order of detection does not matter, as both sequences $(C_1, C_2, ..., C_{l-1}, C_l)$ and $(C_i, C_2, ..., C_{i-1}, C_{i+1}, ..., C_{l-1}, C_l)$ result in the same state for the $|\mathcal{V}|$ stacks.
- *If C_1' does not share any vertex with the sequence* $(C_1, C_2, ..., C_{l-1}, C_l)$, the same conclusion holds trivially.

Finally, note that if a cycle C_i were to be "kept" (resp. "discarded") in a given order of detection, it would also be "kept" (resp. "discarded") in any other, as the randomness is part of the pre-processing.

C.3 UPPER-BOUNDING THE NUMBER OF STEPS

We will show that the upper bound

$$\mathbf{E}(T_{\mathcal{M}}) \leq \operatorname{tr}\left((\mathsf{L}+\mathsf{Q})^{-1}(\mathsf{D}+\mathsf{Q})\right). \tag{C.6}$$

can be obtained as a simple corollary of a more precise result in the trivial-connection case, for which it is known that:

$$\mathbf{E}(T_{\mathcal{M}}) = \operatorname{tr}\left((\mathsf{L}+\mathsf{Q})^{-1}(\mathsf{D}+\mathsf{Q})\right). \tag{C.7}$$

Let us briefly argue for the correctness of this claim. We assume the connection to be trivial through the following points.

- First, let us consider the stack-representation-based formulation of Algorithm 4. The crux of the argument is to recall that the order in which the cycles are removed does not matter, and that we can consequently assume that the random walk starts from any node $i \in \mathcal{V}^\Gamma$. The expected number of calls to random_successor(i) is then the expected number of times a random walk starting from node i returns to i before reaching Γ, which is classically given by $\pi_i(t_{i,\Gamma} + t_{\Gamma,i})$, where [Aldous and Fill, 2002]:
  - π_i is the stationary distribution of the random-walk at node i;

1 A formalist may hence regard the following argument as pertaining to "colored" subsets of edges $\overrightarrow{\phi} = \bigcup_{i\in\mathcal{V}} \left(\{(i, u_i^1\}, j_i\right)$, with j_i denoting the current depth of the stack at node i.

– $t_{i,j}$ is the hitting time of j starting from node i (*i.e.*, the expected number of steps to reach j).

Summing over all i's, we find

$$\mathbf{E}(T_{\mathcal{M}}) = \sum_{i \in \mathcal{V}} \pi_i (t_{i,\Gamma} + t_{\Gamma,i}), \tag{C.8}$$

which is nothing but the *mean commute time* μ_Γ to Γ in $\mathcal{G}^\Gamma$.

- Second, from classical identities regarding the mean commute time [Aldous and Fill, 2002, Marchal, 2000], it holds that

$$\mathbf{E}(T_{\mathcal{M}}) = \operatorname{tr}\left(\mathsf{I} - \mathsf{P}_{\mathcal{V}}\right), \tag{C.9}$$

where P is the transition matrix of the random walk; in our case, $\mathsf{P} = \mathsf{I} - \Delta^\Gamma$ (where Δ^Γ is the random-walk Laplacian associated to the extended graph $\mathcal{G}^\Gamma$), and we have $\mathsf{P}_{i,j} = \frac{w_{i,j}}{d_i^\Gamma}$ on the off-diagonal.

- Finally, expliciting the entries of P, we find

$$\mathbf{E}(T_{\mathcal{M}}) = \operatorname{tr}\left((\mathsf{L} + \mathsf{Q})^{-1} (\mathsf{D} + \mathsf{Q})\right). \tag{C.10}$$

For non-trivial connections, the number of calls to each `random_successor`(i) is in general *less* than $\pi_i(t_{i,\Gamma} + t_{\Gamma,i})$, since a cycle may be accepted, which directly translates to

$$\mathbf{E}(T_{\mathcal{M}}) \leq \operatorname{tr}\left((\mathsf{L} + \mathsf{Q})^{-1} (\mathsf{D} + \mathsf{Q})\right). \tag{C.11}$$

D

COMPLEMENTARY MATERIAL FOR CHAPTER 6

D.1 $\mathcal{A}$-FORESTS

We show the following generalization of Theorem 6.1.1 to $\mathcal{A}$-forests.

Theorem D.1.1. *Let $\mathcal{A} \subseteq \mathcal{V}$ be a subset of nodes in a graph $\mathcal{G}$ endowed with a $U(\mathbf{C})$-connection, such that $\mathcal{A}$ intersects all the connected components of $\mathcal{G}$ and, for a given $\mathcal{A}$-MTSF $\phi^{\mathcal{A}}$, denote by $u \xrightarrow{\phi^{\mathcal{A}}} v$ the unique path from node u to node v in $\phi^{\mathcal{A}}$ in the graph $\mathcal{G}_{\mathcal{V}\setminus\mathcal{A}}$ with nodes $\mathcal{V}\setminus\mathcal{A}$ and edges $\mathcal{E}_{\mathcal{V}\setminus\mathcal{A}} = \{\{i,j\} \in \mathcal{E};\ i,j \in \mathcal{V}\setminus\mathcal{A}\}$. Note that $\left(u \xrightarrow{\phi^{\mathcal{A}}} v\right) \in \mathcal{E}_{\mathcal{V}\setminus\mathcal{A}}$.*

The entries of $\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}$ can then be expressed as

$$\left(\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}\right)_{i,j} = \sum_{\phi^{\mathcal{A}} \in \mathcal{M}^{\mathcal{A}}(\mathcal{G})} \left(\mathbf{1}_{c^a_\phi(j)}(i) \prod_{e \in \phi^{\mathcal{A}}\setminus\{j,a\}} w_e \prod_{C \in \mathcal{C}(\phi^{\mathcal{A}})} (2 - 2\cos(\theta_C))\right) \psi_{j \xrightarrow{\phi^{\mathcal{A}}} i}, \tag{D.1}$$

*where $\mathbf{1}_{c^a_\phi(j)}(i)$ is the indicator that i is spanned by an $\mathcal{A}$-tree of $\phi^{\mathcal{A}}$ (as defined in Remark 5.1.1) such that $\{j,a\} \in \phi^{\mathcal{A}}$ (for a given $a \in \mathcal{A}$), and $\mathcal{M}^{\mathcal{A}}(\mathcal{G})$ denotes the set of all $\mathcal{A}$-MTSFs over $\mathcal{G}$. In particular, when $\sum_{a\in\mathcal{A}} w_{j,a}$ is non-zero (*i.e.*, if j is adjacent to some node in $\mathcal{A}$), Equation (D.1) takes the form:*

$$\left(\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}\right)_{i,j} = \mathbf{E}_{\phi^{\mathcal{A}} \sim \mathcal{D}^{\mathcal{A}}_{\mathcal{M}}} \left(\mathbf{1}_{c^a_\phi(j)}(i) \frac{1}{w_{j,a}} \psi_{j \xrightarrow{\phi^{\mathcal{A}}} i}\right) \tag{D.2}$$

Proof. We begin by re-writing the entries of $\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}$ from Cramer's rule (Equation (3.30)):

$$\left(\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}\right)_{i,j} = (-1)^{i+j} \frac{\det(\mathsf{L}_\theta)_{(\mathcal{V}\setminus\mathcal{A})\setminus\{j\},(\mathcal{V}\setminus\mathcal{A})\setminus\{i\}}}{\det(\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}}. \tag{D.3}$$

We can then re-write the numerator using the Cauchy-Binet formula (Equation (3.13)):

$$\det(\mathsf{L}_\theta)_{(\mathcal{V}\setminus\mathcal{A})\setminus\{j\},(\mathcal{V}\setminus\mathcal{A})\setminus\{i\}} = \sum_{\substack{\phi \subseteq \mathcal{E} \\ |\phi| = |\mathcal{V}\setminus\mathcal{A}|-1}} (-1)^{i+j} \det\left(\left[\left(((\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}})^*\right)_{:,\phi^{\mathcal{A}}} \ \ \delta_j\right]\right) \det\left(\begin{bmatrix} \left((\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}}\right)_{\phi^{\mathcal{A}},:} \\ \delta_i \end{bmatrix}\right), \tag{D.4}$$

where δ_i is the i-th vector in the usual basis of $\mathbf{C}^{\mathcal{V}}$. From an argument completely analogous to that of Lemma 5.2.3, we obtain that the product of determinants

$$\det\left(\left[\left(((\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}})^*\right)_{:,\phi^{\mathcal{A}}} \;\; \delta_j\right]\right) \det\left(\begin{bmatrix} \left((\nabla_\theta)_{:,\mathcal{V}\setminus\mathcal{A}}\right)_{\phi^{\mathcal{A}},:} \\ \delta_i \end{bmatrix}\right) \tag{D.5}$$

can only be non-zero if $\phi^{\mathcal{A}}$ contains a tree $T_i^j \subseteq \mathcal{E}_{\mathcal{V}\setminus\mathcal{A}}$ (which is *not* an $\mathcal{A}$-tree) spanning both i and j, with contribution $\psi_{j\xrightarrow{\phi^{\mathcal{A}}} i} \prod_{e\in T_i^j} w_e$ to the product, and if *all* the other components are $\mathcal{A}$-trees or unicycles, with associated contributions generalizing those of Lemma 5.2.3. We thus obtain the expected expression

$$\left(\left((\mathsf{L}_\theta)_{\mathcal{V}\setminus\mathcal{A}}\right)^{-1}\right)_{i,j} = \sum_{\phi^{\mathcal{A}}\in\mathcal{M}^{\mathcal{A}}(\mathcal{G})} \left(\mathbf{1}_{c_\phi^a(j)}(i) \prod_{e\in\phi^{\mathcal{A}}\setminus\{j,a\}} w_e \prod_{C\in\mathcal{C}(\phi^{\mathcal{A}})} (2-2\cos(\theta_C))\right) \psi_{j\xrightarrow{\phi^{\mathcal{A}}} i}. \tag{D.6}$$

■

D.2 MATRIX CONCENTRATION BOUNDS

Let us begin by stating the result from [Tropp, 2012].

Theorem D.2.1 (1.4 from [Tropp, 2012])**.** *Consider a set of* m self-adjoint *matrices* $\{\mathsf{X}_k\}_{k=1}^m$ *in* $M_n(\mathbf{C})$*, sampled independently. Assume that, almost surely and for each such matrix, both the conditions*

$$\mathbf{E}(\mathsf{X}_k) = 0 \tag{D.7}$$

$$\|\mathsf{X}_k\| \leq R \tag{D.8}$$

are satisfied, where $\|.\|$ *denotes the operator norm*[1]*.*
Then, for all $\varepsilon \geq 0$*,*

$$\mathbf{P}\left(\left\|\sum_{k=1}^m \mathsf{X}_k\right\| \geq \varepsilon\right) \leq n e^{\frac{\frac{-\varepsilon^2}{2}}{\sigma^2 + R\frac{\varepsilon}{3}}}, \tag{D.9}$$

where we define

$$\sigma^2 = \left\|\sum_{k=1}^m \mathbf{E}\left(\mathsf{X}_k^2\right)\right\|. \tag{D.10}$$

Note that this bound is quite powerful and, in particular, that the matrices X_k need not be sampled from the same distribution. Our setting is quite simple in comparison, as all the matrices we consider *are* sampled from the same distribution.

Specifically, we consider the matrices $\mathsf{X}_k = \frac{1}{m}(\overline{\mathsf{S}}_{\phi_k} - \mathsf{K})$ associated to a set of m multi-type spanning forests $\{\phi_k\}_{k=1}^m$, with $\mathsf{K} = (\mathsf{L}_\theta + \mathsf{Q})^{-1}\mathsf{Q}$ denoting the “matrix” solution to the connection-aware Tikhonov smoothing problem. We assume that all m multi-type spanning forests are sampled according to $\mathcal{D}_{\mathcal{M}}$, and forget the subscripts in the following. Note that we have

$$\mathbf{E}(\mathsf{X}_k) = 0 \tag{D.11}$$

1 Associated to the usual l_2-norm on $\mathbf{C}^n$.

for all k.

In order to determine R, we bound the spectrum of each X_k:

$$\|\mathsf{X}_k\| \leq \frac{1}{m}\left(\|\overline{\mathsf{S}}_{\phi_k}\| + \|\mathsf{K}\|\right). \tag{D.12}$$

Since we have both $\|\mathsf{K}\| = \frac{q}{q+\lambda_1}$ (where λ_1 is the smallest eigenvalue of L_θ) and $\|\overline{\mathsf{S}}_{\phi_k}\| \leq 1$ (by, *e.g.*, Gershgorin's circle theorem), we can take

$$R = \frac{1}{m}\left(1 + \frac{q}{q+\lambda_1}\right). \tag{D.13}$$

In order to simplify our bounds later-on, we can notice that

$$R \leq \frac{3}{m}. \tag{D.14}$$

As for the value of σ^2, we obtain from the triangle inequality that

$$\sigma^2 = \frac{1}{m}\|\mathsf{K} - \mathsf{K}^2\| \tag{D.15}$$

$$\leq \frac{1}{m}\left(\frac{q}{q+\lambda_1} + \frac{q^2}{(q+\lambda_{\max})^2}\right). \tag{D.16}$$

This expression can be made more interpetable by lower-bounding $\lambda_{\max}$, and we find that:

$$\sigma^2 \leq \frac{1}{m}\left(\frac{q}{q+\lambda_1} + \frac{q^2}{(1+q+\max_{v\in\mathcal{V}} d_v)^2}\right), \tag{D.17}$$

which is obtained from the following lemma.

Lemma D.2.2. *For a graph $\mathcal{G}$ endowed with a $U(\mathbf{C})$-connection, we have*

$$1 + \max_{v\in\mathcal{V}} d_v \leq \lambda_{\max} \leq 2\max_{v\in\mathcal{V}} d_v. \tag{D.18}$$

Proof. The right-hand-side inequality immediately follows from Gershgorin circle's theorem. To obtain the left-hand-side inequality, let $v_{\max} \in \mathcal{V}$ denote one vertex of $\mathcal{G}$ with maximum degree $d_{v_{\max}} = \max_{v\in\mathcal{V}} d_v$, and consider the signal $f \in \mathbf{C}^{\mathcal{V}}$ defined by

$$f(i) = \begin{cases} d_{v_{\max}} \text{ if } i = v_{\max} \\ -e^{\iota\theta_{v,i}} \text{ if } i \text{ is adjacent to } v_{\max} \\ 0 \text{ otherwise.} \end{cases} \tag{D.19}$$

Clearly, $\frac{\langle f, \mathsf{L}_\theta f\rangle}{\langle f, f\rangle} = 1 + \max_{v\in\mathcal{V}} d_v$, from which the lower-bound on $\lambda_{\max}$ ensues. ∎

It follows from our previous bounds on R and σ^2 that

$$\left(\sigma^2 + R\frac{\varepsilon}{3}\right) \leq \frac{1}{m}\left(\varepsilon + \frac{q}{q+\lambda_1} + \frac{q^2}{(1+q+\max_{v\in\mathcal{V}} d_v)^2}\right). \tag{D.20}$$

Finally, $\| \sum_{k=1}^m \mathsf{X}_k \| \leq \varepsilon$ implies by definition that

$$\left\| \sum_{k=1}^{m} \mathsf{X}_k g \right\|_2 \leq \varepsilon \|g\|_2 \tag{D.21}$$

for any signal $g \in \mathbf{C}^{\mathcal{V}}$, so that putting all the previous results together results in:

$$\mathbf{P}\left(\left\| \left(\frac{1}{m} \sum_{k=1}^{m} \overline{\mathsf{S}}_{\phi_k} \right) g - f_* \right\|_2 \leq \varepsilon \|g\|_2 \right) \geq 1 - \delta \tag{D.22}$$

whenever we have

$$m \geq 2 \frac{\varepsilon + \frac{q}{q+\lambda_1} + \frac{q^2}{(1+q+\max_{v\in\mathcal{V}} d_v)^2}}{\varepsilon^2} \log\left(\frac{|\mathcal{V}|}{\delta} \right). \tag{D.23}$$

D.3 CHOICE OF α FOR CONTROL VARIATES

We plot in Figure D.1 the average of the errors

$$\|\overline{f} - f_*\|_2 \tag{D.24}$$

and

$$\|\hat{f} - f_*\|_2 \tag{D.25}$$

over 5 trial estimations, for different values values of α. We always take $m = 20$ MTSFs for each estimation, and set $q = 10^{-2} \times \overline{d}$. We use one realization of each of the random graph models from Section 6.2.

The extent of the error reduction depends on the graph, but the optimal choice of α seems to always be close to 1. This behavior can be observed across order-of-magnitude variations of q (not shown), and setting $\alpha = 1$ results in meaningful variance reduction for our experiments in Chapter 6.

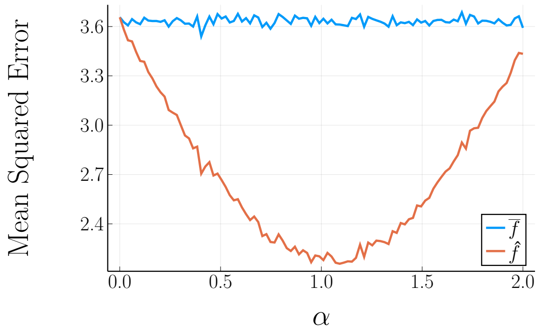
D.4 COMPARING THE SPECTRUM OF L AND L_θ

Our goal in this section is to shed some light on the structure captured by the spectrum of L_θ, and reassure the reader that, *when the connection is sufficiently coherent*, similar information can be extracted from the eigenvalues and eigenvectors of L and L_θ respectively. As such, it is possible to leverage them in a similar manner in, *e.g.*, graph-signal-processing applications, such as the connection-aware Tikhonov smoothing problem from Chapter 2, or to analyze properties of the graph (for instance, to interpret λ_2 as a measure of the connectivity of $\mathcal{G}$).

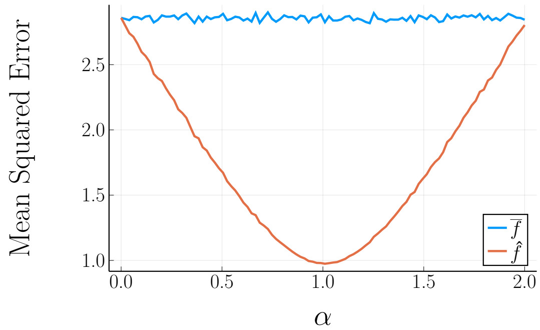
Coherent connection. Let us first consider a connection of the form

$$\theta_{i,j} = \omega_j - \omega_i \tag{D.26}$$

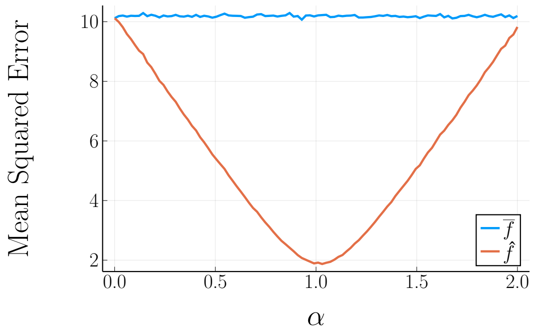
for all edges $\{i, j\} \in \mathcal{E}$, with given angles $\omega_i \in [0, 2\pi)$. Similar situations frequently come up in applications, where the angles ω_i represent some unknown data, and $\theta_{i,j}$ are the observed offsets (see Section 2.3 for more applicative context). This situation represents what we call a *coherent* connection, obtained from an exact difference of angles associated to each node in $\mathcal{V}$: in this



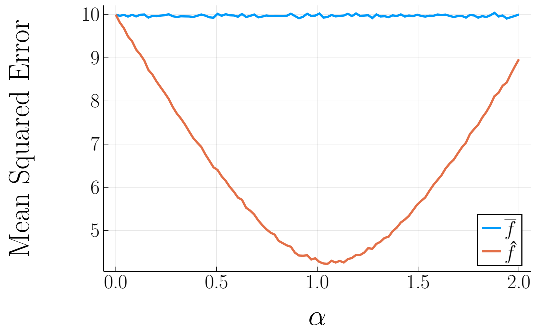
(a) ε-graph mean error.



(b) SBM mean error.



(c) DC-SBM 1 mean error.



(d) AS CAIDA mean error.

Figure D.1: Mean error as a function of α.

case, the holonomy is trivial for all cycles C, *i.e.*, we always have $\left(\prod_{e \in \mathbf{C}} \psi_e\right) = 1$. Here, it is straightforward to show that the eigendecomposition of L_θ can be easily recovered from that of L. Indeed, consider the eigendecomposition $\mathsf{L} = \mathsf{U}\Lambda\mathsf{U}^*$ of L, where we recall that $(\mathsf{L}_\theta)_{i,j} = e^{\iota(\omega_i - \omega_j)}$ on the off-diagonal, and denote respectively by $x \in \mathbf{C}^{\mathcal{V}}$ the ground-truth vector for the angular-synchronization problem, with entries $x_i = e^{\iota\omega_i}$, and by D_x the diagonal matrix with $(\mathsf{D}_x)_{i,i} = x_i$. In this situation, we have

$$\mathsf{L}_\theta = \mathsf{D}_x \mathsf{L} \mathsf{D}_x^* = (\mathsf{D}_x\mathsf{U})\Lambda(\mathsf{D}_x\mathsf{U})^*, \tag{D.27}$$

where the first equality follows from applying the diagonal scaling to the off-diagonal entries: $e^{\iota\omega_i}(\mathsf{L})_{i,j}e^{-\iota\omega_j} = e^{\iota(w_i - \omega_j)} = (\mathsf{L}_\theta)_{i,j}$. Consequently, L and L_θ have the same eigenvalues, and the eigenvectors v_k of L_θ are given by $\mathsf{D}_x u_k$, with u_k the eigenvectors of L. In particular, $v_1 = \mathsf{D}_x \frac{\mathbf{1}}{\sqrt{|\mathcal{V}|}} = \frac{x}{\sqrt{|\mathcal{V}|}}$, with $\mathbf{1}$ the all-ones vector, and we recover the exact spectral solution for coherent connections.
Going one step further, and denoting by f^h and f_θ^h the signals obtained by applying a filter $h : \mathbf{R} \to \mathbf{R}$ based on the eigendecomposition of L and L_θ respectively, it follows that

$$f_\theta^h = (\mathsf{D}_x f)^h, \tag{D.28}$$

so that applying a filter based on either L_θ or L is completely equivalent.

Let us now describe the more realistic case of connections that are *not* coherent.

Incoherent connection. In most situations of interest, Equation (D.26) no longer holds, and the diagonal re-scaling from Equation (D.27) can no longer be applied to recover v_k from u_k. Consider for instance a model of the form

$$\theta_{i,j} = \omega_j - \omega_i + \eta\varepsilon_{i,j}, \tag{D.29}$$

where each $\epsilon_{i,j}$ represents a possible source of incoherence, and η is a scale parameter controlling the overall magnitude of this incoherence. Noisy connections of the form in Equation (D.29) may be considered for a number of reasons.

- Because of an intrinsic property of the model, such as in, *e.g.*, the connection used to encode directed graphs (Equation (2.20)), for which $\omega_i = 0$ for all $i \in \mathcal{V}$). In this case, the scale parameters η and $\gamma 2\pi$ play a similar role.
- Because the measured offsets $\theta_{i,j}$ are not measured exactly but are subject to, *e.g.*, noise or some other type of corruption [El Karoui and Wu, 2016], as is commonly the case in angular-synchronization problems.

Importantly, one should note that when $\eta_{i,j} = 0$ for all edges $\{i, j\}$, we recover exactly the coherent connection from Equation (D.26). This serves as a basis to interpret the eigenvectors v_k of L_θ for small values of η, *i.e.*, for weakly incoherent connections. Indeed, L_θ can be understood as an analytic perturbation of L in η [Kato, 2013]:

$$\mathsf{L}_\theta(\eta) = \mathsf{L} + \sum_{k\geq 1} \eta^k \frac{\mathsf{M}_k}{k!}, \tag{D.30}$$

with $(\mathsf{M}_k)_{i,j} = (\iota\theta_{i,j})^k$ for $i \neq j$, and $(\mathsf{M}_k)_{i,i} = 0$ on the diagonal. A general theorem of Rellich[2], applicable to analytic perturbations of Hermitian operators, then states that, provided that the eigenvalues of L are separated, the eigenvalues (resp. eigenvectors) of $\mathsf{L}_\theta(\eta)$ can also be expressed as power series in η, with leading term corresponding to the eigenvalue (resp. eigenvector) of L. Thus, for η low enough, the spectrum of L is similar to that of L_θ.

These conclusions are moreover borne out empirically and we now empirically compare the eigenvectors of L and L_θ on different types of graphs. The results are elementary, and should not be surprising, but we are not aware of another qualitative analysis of this type appearing in the literature.

Building on the previously-described situation, we consider connections of the form

$$\theta_{i,j} = \omega_j - \omega_i + \eta\varepsilon_{i,j}, \tag{D.31}$$

where each $\epsilon_{i,j}$ represents uniformly distributed noise in $[-1, 1]$, and η is a scale parameter controlling the overall magnitude of the noise.

We represent in Figures D.2 and D.3 the second and third eigenvectors u_k, v_k (for $k \in \{2, 3\}$) of L and L_θ respectively, along with the re-scaled eigenvectors $\mathsf{D}_x^* v_k$ (with D_x the diagonal matrix with entries $x_i = e^{\iota\omega_i}$ the ground-truth solution to the angular-synchronization problem, so that D_x^* should approximate u_k), for different values of η ($\eta_1 \simeq 0.11$ and $\eta_2 \simeq 0.75$), and two different randomly generated graphs.

- An ε-graph built from $n = 100$ points uniformly sampled in $[0, 1]^3$, with $\varepsilon = 0.3$ (Figure D.2).

2 Franz Rellich, Austrian-German mathematician, 1906–1955.

- A SBM with two communities on size 50 ($n = 100$), with $c_{1,1} = c_{2,2} = 19$ and $c_{1,2} = 1$ (Figure D.3). This is the same model we considered in our illustration from Section 1.1.3.

For each graph, we plot the real and imaginary parts of the entries of the corresponding eigenvectors. We only use a single realization of these graphs, equipped with a single realization of the connection. Note that, for both of these graphs, the classical GSP pipelines can be safely applied (Section 1.1), and recall that the entries of the eigenvectors v_k of L_θ are defined up to a global rotation r. We further measure the difference between the vectors u_k and $\mathsf{D}_x^* v_k$ as a function of η using the normalized error

$$\min_{r \in U(\mathbf{C})} \frac{\|u_k - r(\mathsf{D}_x^* v_k)\|_2}{n}, \tag{D.32}$$

which are computed across a linear range of values of η in $[0, 1]$. The global rotation $r \in U(\mathbf{C})$ serves to find the rotation that best registers the two eigenvectors u_k and $\mathsf{D}_x^* v_k$. The errors corresponding to $\eta = \eta_1, \eta_2$ are highlighted, along with another value $\eta_0 = \frac{\pi}{2n}$, that will play a special role in **Part** ii, and serves to analyse a forest-based estimator for the connection-aware Tikhonov smoothing. **We encourage the reader to come back to this development at after reading through Chapter 6**.
We can summarize the resulting observations as follows.

(O1) For low noise levels ($\eta = \eta_1$), $\mathsf{D}_x^* v_k$ provides a good approximation of u_k, as can be observed in Figure D.2.

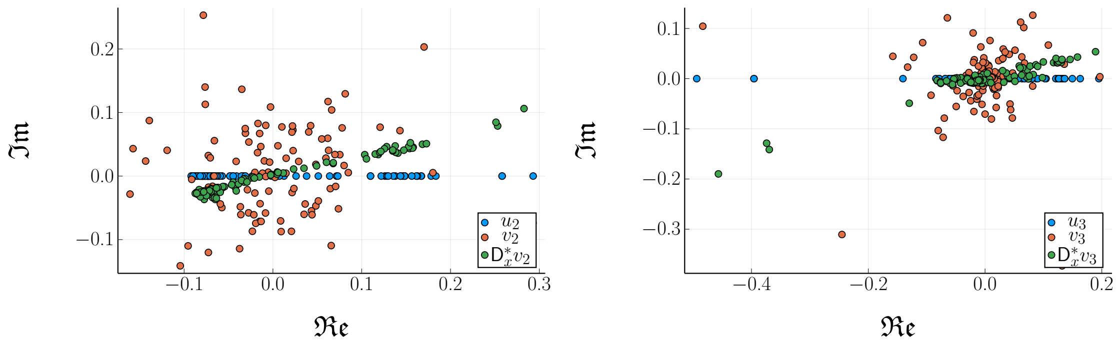
(O2) This is no longer the case for higher noise levels ($\eta = \eta_2$): in Figure D.2, $\mathsf{D}_x^* v_2$ seems to roughly recover the shape of u_2, but $\mathsf{D}_x^* v_3$ appears completely unrelated to u_3.

(O3) Important structural properties captured by the combinatorial Laplacian L seem to also be captured by L_θ, even for high noise levels: for the SBM, $\mathsf{D}_x^* v_2$ clearly partitions the graph into the same communities as u_2, which is a spectral descriptor of the community-structure of the graph, discussed in relation with the spectral clustering strategy in Chapter 1 (Figure D.3). This behavior persists even for higher noise-levers (Figure D.4).

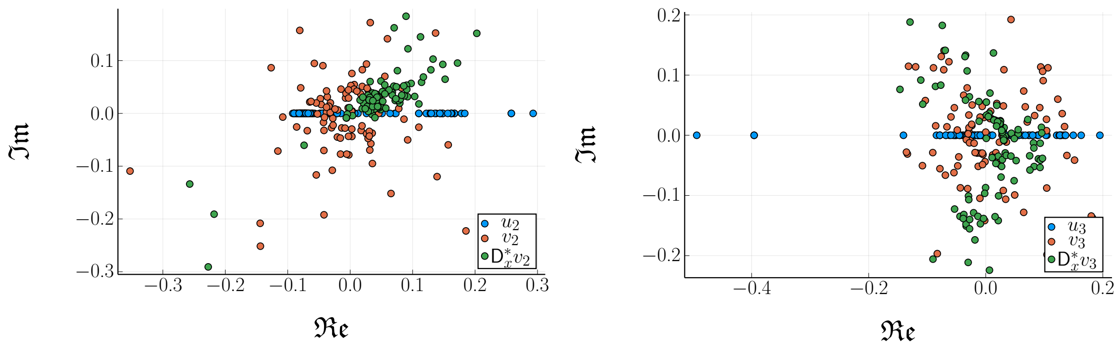
(O4) These behaviors are reflected in the normalized-error plots (Figure D.5): a greater error is associated with higher values of η, and the error decays smoothly as η goes to 0. For the SBM (not shown), the error decays linearly in η.

Overall, these results provide weak evidence that the eigenvectors of L and L_θ behave similarly in low-noise regimes (on the connection). In particular, for these regimes, we expect B-bandlimited signals to be meaningfully smooth, and the connection-aware Tikhonov smoothing problem to output a smooth signal.

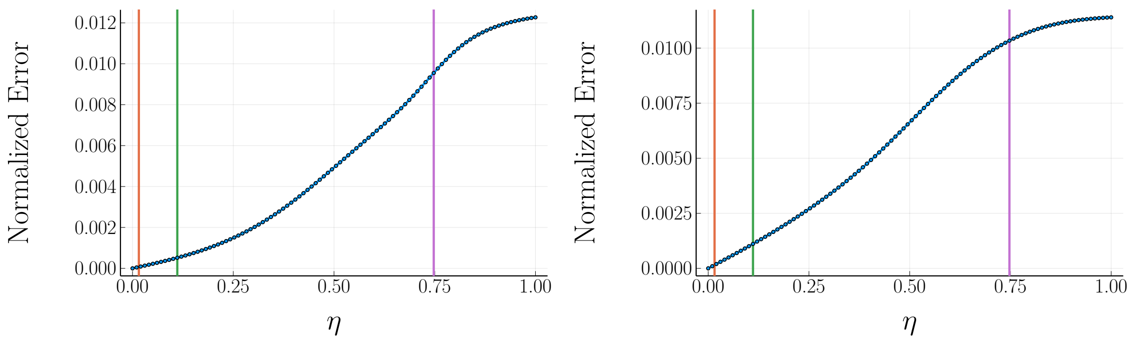
Notwithstanding, let us finally stress that, depending on the graph and the choice of connection, even small values of η may result in very incoherent connections. We illustrate this phenomenon on a cyclic graph of size $n = 100$, in a more controlled setup: the values of $\varepsilon_{i,j}$ are no longer randomly sampled, and we instead set $\varepsilon_{i,j} = 1$ for all edges i, j along a consistent orientation (*i.e.*, no two directed edges pointing at the same node). The whole graph consists of one large, incoherent cycle, and the incoherence grows larger with η. In this setting the value $\eta = \eta_0$ corresponds to the following thresholding behavior, as encountered in Chapter 5 (in the form of the weak-inconsistency condition).



(a) Representation on u_k, v_k and $\mathsf{D}_x^* v_k$ in the complex plane. $\eta_1 \simeq 0.11$.



(b) Representation on u_k, v_k and $\mathsf{D}_x^* v_k$ in the complex plane. $\eta_2 \simeq 0.75$.



(c) Normalized errors between the eigenvectors u_k and $\mathsf{D}_x^* v_k$ for the ε-graph, as a function of the incoherence, varying with η. The colored vertical lines represent the values of η_0 (orange), η_1 (green) and η_2 (purple).

Figure D.2: Results of the comparisons between u_k, v_k and $\mathsf{D}_x^* v_k$ for the ε-graph. Left: second eigenvector. Right: third eigenvector.

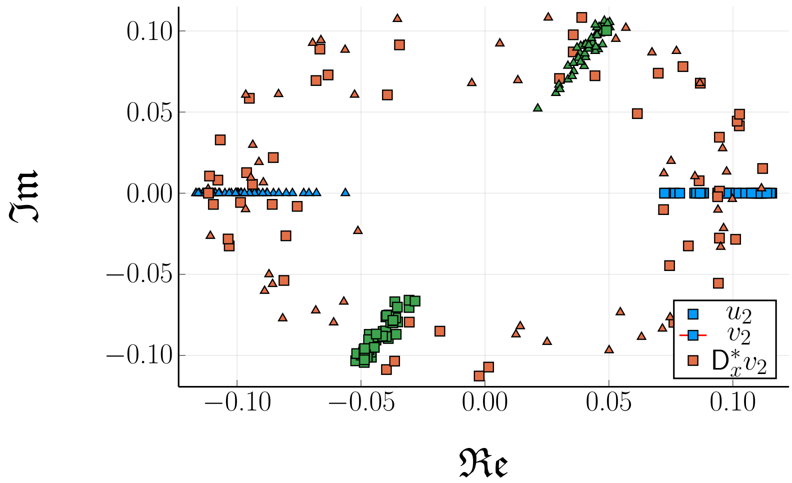


Figure D.3: Community-structure indicators using u_2 (blue), v_2 (orange) and $\mathsf{D}_x^* v_2$ (green) for the SBM-graph with $\eta = \eta_1 \simeq 0.11$, as represented in the complex plane. The two communities are depicted as triangles and squares respectively. The entries of v_2 are not informative of the community-structure, whereas those of $\mathsf{D}_x^* v_2$ describe the same classes as those of u_2 (including some mis-classified nodes).

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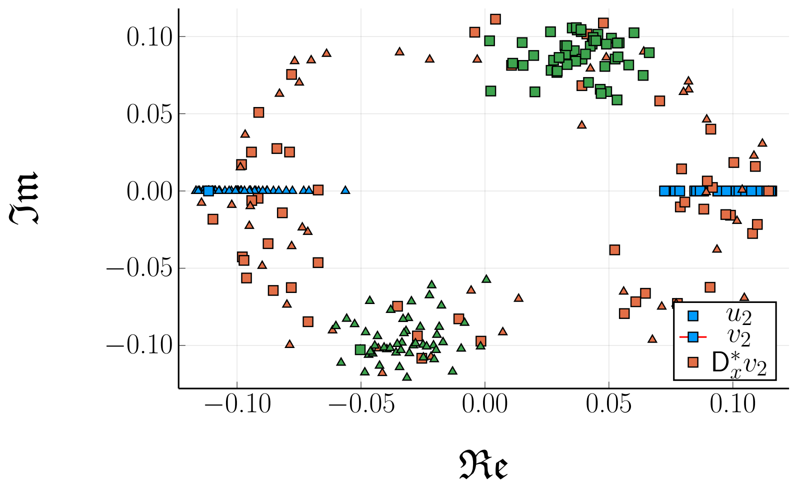

Figure D.4: Community-structure indicators using u_2 (blue), v_2 (orange) and $\mathsf{D}_x^* v_2$ (green) for the SBM-graph with $\eta = \eta_2 \simeq 0.75$. The entries of u_2 and $\mathsf{D}_x^* v_2$ remain informative even at higher noise-levels.

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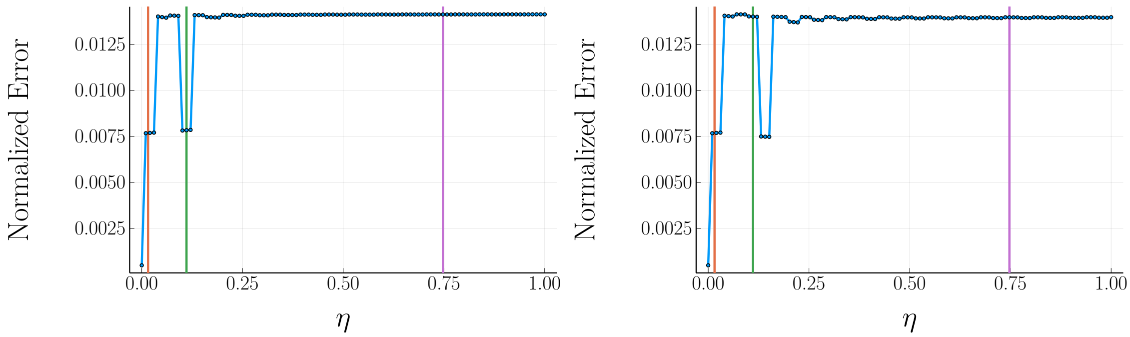


Figure D.5: Normalized errors between the eigenvectors u_k and $\mathsf{D}_x^* v_k$ for the cycle-graph, as a function of the incoherence, varying with η. Left: second eigenvector. Right: third eigenvector. Vertical lines represent η_0 (orange), η_1 (green) and η_2 respectively (purple).

- When $\eta < \eta_0$, parallel transport of a complex $z \in \mathbf{C}$ (over some node v) all the way along the unique loop C (back to v) results in multiplication of z by a factor $e^{\iota \eta n}$, which always has a *positive* real part.
- When $\eta > \eta_0$, the factor $e^{\iota \eta n}$ may have *negative* real part.

This thresholds represents a sharp transition in the behavior of sampling algorithms for multi-type spanning forests, as discussed in **Part** ii (Chapter 5). We plot the corresponding normalized errors in Figure D.5.
The approximationconnections may exhibit additional structure, resulting in connection-Laplacian-eigenvectors that largely differ from our of understanding in the trivial-connection case (recall from Chapter 1 that, in the case of the cycle, the Laplacian-eigenvectors are nothing but the usual discrete Fourier modes).

Remark D.4.1. *Note that, when $\omega_i = \omega$ for all $i \in \mathcal{V}$ and a fixed ω, the $\theta_{i,j}$'s take the form*

$$\theta_{i,j} = \eta \varepsilon_{i,j}, \tag{D.33}$$

which is analogous to the connection in Equation (2.20).
In fact, an argument similar to ours was also proposed in [Furutani et al., 2019], in the case of the connection Laplacian associated to a directed graph with scale-parameter γ. Inspecting the power-series decompositions of the eigenvalues and eigenvectors of L_θ would be a natural way to shed light on what a good choice of γ should be.

Connection-aware graph-signal-processing is still a very nascent research topic, and techniques making sense of more incoherent connections may emerge in the future. Nevertheless, it is clear that, just like for the angular synchronization problem, the connection Laplacian is a practical modeling tool for problems on $\mathbf{C}$-bundles.

E

IN ALL GENERALITY: FORESTS AND RANDOM WALKS ON SIMPLICIAL COMPLEXES

... un auteur ne nuit jamais tant à ses lecteurs que quand il dissimule une difficulté.

— Évariste Galois[1]

We begin with careful definitions.

An orientation of a k-simplex $\{v_0, ..., v_k\}$ is defined as any ordering of its constituting vertices, up to an *even* permutation. The motivation for this definition is that, for a given permutation, each simplex can only have two different orientations, and requiring that this permutation be even ensures that inverting the order of two vertices in the orientation results in the opposite orientation of the same simplex, from which we for instance recover the familiar $(s_e, t_e)^* = (t_e, s_e)$ identity for 1-simplices.
In the following, *we fix an orientation of all simplices*, and denote by σ (resp. $\overline{\sigma}$) the *oriented* simplex (resp., its reversely-oriented counterpart) corresponding to a non-oriented simplex $\sigma \in \mathcal{S}$. The set of all oriented k-simplices, which contains both σ and $\overline{\sigma}$ for each k-simplex $\sigma \in \mathcal{S}_k$, is denoted by $\overrightarrow{\mathcal{S}_k}$.

In order to define signals of simplicial complexes, it remains to account for an intuitive requirement: as is observed for electrical currents, a signal on k-simplices should be such that $f(\sigma) = -f(\overline{\sigma})$. This motivates the two definitions that we introduce below. In the following, $\mathbf{A}$ denotes a ring (*e.g.*, $\mathbf{Z}$, $\mathbf{R}$, $\mathbf{C}$, or some matrix algebra).

- *Chains*, which are defined, for each k, as $\mathbf{A}$-linear combinations of oriented k-simplices, modulo an equivalence relation $\sim$. More specifically, we define the relation $\sigma \sim \tau$ whenever $\tau = \overline{\sigma}$, and define the set of k-chains on the simplicial complex $\mathcal{X}$ as

$$C_k(\mathcal{X}, \mathbf{A}) = \begin{cases} \mathbf{A}^{\mathcal{V}} \text{ if } k = 0, \\ \mathbf{A}^{\overrightarrow{\mathcal{S}_k}} /\!\sim \text{ if } k \geq 1 \\ \{0\} \text{ if } k < 0. \end{cases} \tag{E.1}$$

1 See [Galois, 1908].

In particular, quotienting by the equivalence relation for $k \geq 1$ translates to the relation $\sigma + \overline{\sigma} = 0$ in $C_k(\mathcal{X}, \mathbf{A})$, which is similar to the property we want signals to satisfy[2].

- *Co-chains* (or "k-forms", for a given k), which are nothing but linear forms on $C_k(\mathcal{X}, \mathbf{A})$, *i.e.*, linear maps of the form $f : C_k(\mathcal{X}, \mathbf{A}) \to \mathbf{A}$, and we denote by $C^k(\mathcal{X}, \mathbf{A})$ the set of k-co-chains over $\mathcal{X}$. Co-chains describe linear measurements over chains: if $\sigma \in C_k(\mathcal{X}, \mathbf{A})$ decomposes as $\sigma = a + b$, we have

$$f(\sigma) = f(a) + f(b), \tag{E.2}$$

which translates to $f(\sigma) = -f(\overline{\sigma})$. Further, in case $\mathbf{A}$ is a field, we can decompose co-chains according to a linear-basis decomposition and, in particular, if $\sigma = \sum_{\tau \in \mathcal{S}_k} a_\tau \tau$ (with $a_\tau \in \mathbf{A}$), we obtain for any k-co-chain f that

$$f(\sigma) = \sum_{\tau \in \mathcal{S}_k} a_\tau f(\tau), \tag{E.3}$$

so that a co-chain is completely determined by its values on k-simplices.

In the following, we call k-signal a k-co-chain $f \in C^k(\mathcal{X}, \mathbf{A})$ (for given simplicial complex $\mathcal{X}$ and ring of coefficients $\mathbf{A}$). Let us also note that, when $\mathbf{A}$ is a field, $C_k(\mathcal{X}, \mathbf{A})$ is isomorphic to $C^k(\mathcal{X}, \mathbf{A})$, and that most works define k-signals as chains. The reason we are especially careful in our definitions is that we will need to consider chains with coefficients in $\mathbf{A} = \mathbf{Z}$. Note also that we already encountered 1-chains in Chapter 1 (Remark 3.2.4), and that we introduced an operator $\mathsf{B} : C_1(\mathcal{X}, \mathbf{A}) \to C_0(\mathcal{X}, \mathbf{A})$ mapping 1-chains to their *boundary*. This operator is nothing but the dual of the discrete differential operator ∇, and our way to generalizations on higher-order simplicial complexes.

Equipped with both the definition of a chain and an adequate notion of signals on simplicial complexes, we are finally able to define generalized differential operators relating (co-)chains of different order. The *boundary operator* $\mathsf{B}_k : C_k(\mathcal{X}, \mathbf{A}) \to C_{k-1}(\mathcal{X}, \mathbf{A})$ is a linear map defined on oriented k-simplices, which are of the form $\sigma = (v_0, ..., v_k)$, by

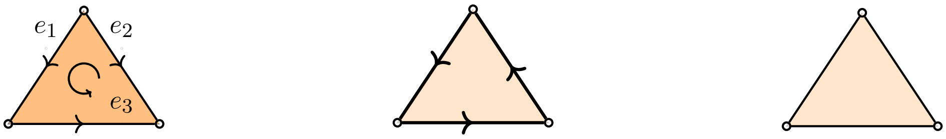
$$\mathsf{B}_k \sigma = \sum_{i=0}^{k} (-1)^i (\sigma \setminus v_i), \tag{E.4}$$

where $(\sigma \setminus v_i) = (v_1, ..., v_{i-1}, v_{i+1}, ..., v_k)$ denotes the (oriented) face of the simplex σ obtained from σ by removing node v_i from σ. Note that this operation is well-defined, since $\mathcal{S}$ is downward-closed. In low dimensions, this operation agrees with an intuitive definition, and we represent in Figure E.1 the boundary of a simple 2-chain containing a single (oriented) triangle.
In turn, the adjoint $\mathsf{B}_k^* : C^{k-1}(\mathcal{X}, \mathbf{A}) \to C^k(\mathcal{X}, \mathbf{A})$ of B_k defines a discrete differential operator for k-signals $f \in C^k(\mathcal{X}, \mathbf{A})$, which acts on k-chains $\sigma = (v_0, ..., v_k)$ as follows[3]:

$$(\mathsf{B}_k^* f)(\sigma) = \sum_{i=0}^{k} (-1)^i f(\sigma \setminus v_i). \tag{E.5}$$

2 Depending on their affinities, the reader may be more comfortable with the equivalent formalism of quotienting a ring by an ideal (here, generated by elements of the form $\sigma + \overline{\sigma}$), and remark that $\mathbf{A}^{\overrightarrow{\mathcal{S}_k}} / \sim \; = \mathbf{A}^{\overrightarrow{\mathcal{S}_k}} / \langle \sigma + \overline{\sigma};\ \sigma \in \overrightarrow{\mathcal{S}_k} \rangle$.

3 The adjunction is here defined with respect to the regular l_2 inner-product, which *only holds for* $\mathbf{A} = \mathbf{Z}$*,* $\mathbf{R}$ *or* $\mathbf{C}$. Otherwise, one can take the resulting expression as the definition (or use a systematic adjuction theorem).



(a) Oriented 2-simplex τ. (b) Boundary $\mathsf{B}_2\tau = e_1 - e_2 + e_3$. (c) Null boundary $\mathsf{B}_1\mathsf{B}_2\tau = 0$.

Figure E.1: Successive applications of the boundary operators, starting from a chain τ supported on a 2-simplex. Left: original 2-simplex, represented as an oriented orange triangle, with reference orientations for the edges. Center: boundary of this 2-simplex (which is a 1-chain), represented as thick directed edges. Right: The resulting null 0-chain (obtained from $\mathsf{B}_1 e_1 - \mathsf{B}_1 e_2 + \mathsf{B}_1 e_3 = (t_{e_1} - s_{e_1}) - (t_{e_2} - s_{e_2}) + (t_{e_3} - s_{e_3}) = 0$).

As a special case, we recover $\nabla = \mathsf{B}_1^*$ on 0-signals.

This allows to define a family of Laplacian-like operators, called the upper, lower, and (full) Hodge Laplacians[4], and respectively defined as:

$$\mathsf{L}_k^{\uparrow} = \mathsf{B}_k \mathsf{B}_k^*, \tag{E.6}$$

$$\mathsf{L}_k^{\downarrow} = \mathsf{B}_k^* \mathsf{B}_k, \tag{E.7}$$

$$\mathsf{L}_k = \mathsf{B}_k \mathsf{B}_k^* + \mathsf{B}_k^* \mathsf{B}_k, \tag{E.8}$$

which are all positive semi-definite positive linear operators mapping k-signals to other k-signals. What is remarkable is that these operators allow for variational interpretations of k-signals: much like the quadratic form associated to the combinatorial Laplacian $\mathsf{L} = \mathsf{L}_0^{\uparrow} = \mathsf{L}_0$ can be understood as measuring how much a given 0-signal f differs from a (connected-component-wise) constant signal, those associated to the Hodge Laplacians measure how much k-signals differ from other reference signals. We describe in the remark below the reference signals associated to the full Hodge Laplacian, which are perhaps the most spectacular of the three.

Remark E.0.1. *A central property of boundary operators is to define a so-called* chain complex*, which boils down to the fact that we always have*

$$\mathsf{B}_{k-1}\mathsf{B}_k = 0. \tag{E.9}$$

This seemingly innocent property is suprisingly useful, as it implies that $\mathrm{im}\mathsf{B}_k \subseteq \ker \mathsf{B}_{k-1}$*, and we illustrate it in Figure E.1. As a particular example, it appears that the boundary of triangles are given by three-edges cylces, the boundary of which is in turn trivial.*
This allows to define the (simplicial) homology groups[5]

$$H_k(\mathcal{X}, \mathbf{A}) = {}^{\ker \mathsf{B}_k}\!\big/_{\mathrm{im}\mathsf{B}_{k+1}}, \tag{E.10}$$

the elements of which $H_k(\mathcal{X}, \mathbf{A})$ *are equivalence classes of* k*-chains. Homology groups are of wide-spread usage in algebraic topology, for two reasons.*

1. *Remarkably, the group* $H_k(\mathcal{X}, \mathbf{A})$ *does not depend on the precise combinatorial structure of the complex, but only on its* topology[6].

4 Named after British mathematician William Vallance Douglas Hodge, 1903−1975, in reference to his work on the de Rham co-homology of smooth manifolds).
5 That also happen to be **A**-modules.
6 Rather, on the topology of its *geometrical realization*, which is a representation of simplicial complexes into space.

2. *Furthermore, homology groups enjoy convenient algebraic properties, which allow to perform calculations* pertaining to topological features of the complex.

As an example of property captured by the homology groups, we have already shown that $H_0(\mathcal{X}, \mathbf{C}) \simeq \mathbf{C}^{\beta_0}$*, with* β_0 *the number of connected components of the graph* $(\mathcal{V}, \mathcal{S}_1)$ *(since* $\ker \mathsf{B}_1 \simeq \ker \mathsf{L}$*), and that, if* $\mathcal{S} = \mathcal{S}_0 \cup \mathcal{S}_1$, $H_1(\mathcal{X}, \mathbf{C}) \simeq \mathbf{C}^{\beta_1'}$*, with* β_1' *the number of* algebraic cycles *of the graph* $(\mathcal{V}, \mathcal{S}_1)$ *(Remark 3.2.4). A similar intuition can be obtained in the presence of higher-order simplices, in which case one shows that* $H_1(\mathcal{X}, \mathbf{C}) \simeq \mathbf{C}^{\beta_1}$ *is generated by the* β_1 *chains bounding "holes" in the complex (for the complex of Figure 7.1, we have* $\beta_1 = 1$*)*[7]*. See [Hatcher, 2005] for a general reference on algebraic topology.*
Of interest in signal-processing applications are the following (natural, linear) isomorphisms between homology groups, the kernel of L_k*, and* co-homology groups*, which are defined as*

$$H^k(\mathcal{X}, \mathbf{A}) = {}^{\ker \mathsf{B}_{k+1}^*}\!/\!_{\mathrm{im} \mathsf{B}_k^*}. \tag{E.11}$$

More specifically, if $\mathbf{A}$ *is a field, we have [Lim, 2020]:*

$$H^k(\mathcal{X}, \mathbf{A}) \simeq \ker \mathsf{L}_k \simeq H_k(\mathcal{X}, \mathbf{A}) \tag{E.12}$$

and, as is further obtained from the proof, those k-signals in the kernel of L_k are supported on chains co-linear with the generators of $H_k(\mathcal{X}, \mathbf{A})$ (resp. $H^k(\mathcal{X}, \mathbf{A})$). *For* 0*-signals, those are connected-component-wise constant signals and, for* 1*-signals, those are cycle-wise constant-up-to-sign signals supported on cycles surrounding "holes" in the complex.*
Let us finally recall a classical structural property for groups of the form $H_k(\mathcal{X}, \mathbf{A})$*, namely, a weak form of the structure theorem for finitely generated abelian groups, which states that* $H_k(\mathcal{X}, \mathbf{A})$ *takes the form*

$$H_k(\mathcal{X}, \mathbf{A}) = \mathbf{A}^{\beta_k} \oplus [H_k(\mathcal{X}, \mathbf{A})], \tag{E.13}$$

where $[H_k(\mathcal{X}, \mathbf{A})]$ *is a* finite *group called the* torsion *subgroup of* $H_k(\mathcal{X}, \mathbf{A})$*; though its discussion is outside the scope of this section,* $[H_k(\mathcal{X}, \mathbf{A})]$ *does carry meaningful topological information about* $\mathcal{X}$*. The decomposition of Equation (E.13) will be important for our later developments (see,* e.g.*, Equation (E.22) below).*

Aside from this technical digression, it is clear upon finer inspection that the quadratic forms associated to the Hodge Laplacians penalize signals in very specific ways. *We focus in the remainder of this section on* $\mathbf{C}$*-valued signals.*
Let us first remark that, as can be obtained from the proof of Equation (E.12), any k-signal can be broken-up into three parts, according to the so-called Hodge decomposition:

$$C^k(\mathcal{X}, \mathbf{C}) = \mathrm{im} \mathsf{B}_{k+1} \oplus \mathrm{im} \mathsf{B}_k^* \oplus \ker \mathsf{L}_k. \tag{E.14}$$

For real-valued 1-signals, this corresponds to the well-known vector-calculus decomposition of flows into:

- a gradient flow of the form $\mathsf{B}_1^* f_0 = \nabla f_0$, that arise from, *e.g.*, potential differences in electrical networks;

7 The numbers β_k are known as the *Betti numbers*, and are named after Italian mathematician Enrico Betti (1823−1892).

- a curl flow of the form $\mathsf{B}_2 f_2$, that account for circulations that can be observed as a result of, *e.g.*, fluid flows surrounding an object;
- an *harmonic* flow $f_1 \in \ker \mathsf{L}_k$.

In particular, both the gradient and curl flows can be penalized, thanks to the lower and upper Hodge Laplacians respectively, or both at once using the full Hodge Laplacian.
If one measures a noisy k-signal g that, for modelling reasons, is expected to have small non-harmonic components, it for instance makes sense to perform a Tikhonov-like regularization of the form

$$\underset{f \in C^k(\mathcal{X}, \mathbf{C})}{\operatorname{argmin}} \quad q\|f-g\|^2 + \langle f, \mathsf{L}_k f \rangle \tag{E.15}$$

in order to estimate the underlying signal [Schaub et al., 2021]. Let us stress though that this type of hypothesis is very much application-dependent, and that this operation may only make sense for specific topologies of the complex.

In spite of the study of simplicial-complexes-based tools still being in its infancy, it is the subject of increasing interest, and finds applications in a number of practical situations, such as the following (which are early examples amongst a growing body of works).

- Estimating the velocity-field of ribonucleic acid from noisy measurements [Barbarossa and Sardellitti, 2020].
- Analyzing oceanic flows from measured displacements of buoys drifting in the ocean [Schaub et al., 2020].
- Analysing political-books co-purchasing data using edge-centrality measures [Schaub et al., 2020].

Note that it is not yet clear whether such techniques allow to reach clear-cut state-of-the-art performance but, much like graphs, these emerging, geometrically-inspired tools allow for a simple modelling of different problems.

Let us now sketch our preliminary results concerning forest-based estimators for problems on simplicial complexes. For simplicity, we restrict our exposition to a Tikhonov smoothing problem for k-signals involving the upper-Laplacian $\mathsf{L}_k^{\uparrow}$, which reads

$$\underset{f \in C^k(\mathcal{X}, \mathbf{C})}{\operatorname{argmin}} \quad q\|f-g\|^2 + \langle f, \mathsf{L}_k^{\uparrow} f \rangle, \tag{E.16}$$

and with solution $f_* = (\mathsf{L}_k^{\uparrow} + q\mathsf{I})^{-1}(q\mathsf{I})\, g$. All our results translate to (smoothing and interpolation) problems involving the lower and (full) Hodge Laplacians as well.
We further omit the proofs of our statements, that can be obtained using arguments similar to those of Chapters 4 and 6, and only strive to frame the problems in this familiar setting, while pointing out the relevant differences.

Determinantal Approach. Let us first adopt a determinantal perspective, and build a generalization of the distribution $\mathcal{D}_{\mathcal{F}}$ over rooted spanning forests. Let us decsribe our construction in two steps.

1. First, we consider a generalization of the uniform-spanning-tree process, as introduced in [Lyons, 2009]. More specifically, we consider a projective DPP with marginal kernel

$$K_k = \mathsf{B}_k^*(\mathsf{B}_k\mathsf{B}_k^*)^\dagger\mathsf{B}_k, \tag{E.17}$$

which is nothing but the orthogonal projection from $C^k(\mathcal{X},\mathbf{C})$ onto $\mathrm{im}\mathsf{B}_k^* \subseteq C^k(\mathcal{X},\mathbf{C})$, and it follows that $\mathrm{rk}(K_k) = |\mathcal{S}_k| - \beta_k$ (note that, when $k=1$, we recover $\mathrm{rk}(K_1) = |\mathcal{E}| - \beta_1 = |\mathcal{V}| - \beta_0$, as we already established in Chapter 1 for the uniform-spanning-tree process).
We then need to determine the structure of the samples $\mathrm{t}_k \subseteq \mathcal{S}_k$ from $\mathrm{DPP}(K_k)$. From Lemma 3.2.2, we know that $\mathbf{P}_{\mathrm{DPP}(K_k)}(\mathrm{t}_k)$ is non-zero if and only if $\det(\mathsf{B}_k^*\mathsf{B}_k) > 0$, that is, if $\ker(\mathsf{B}_k)_{:,\mathrm{t}_k}) = \{0\}$. In order to further interpret this condition, let us note that each t_k induces a sub-complex $\mathcal{X}_{\mathrm{t}_k} = (\mathcal{V}, \mathrm{t}_k \cup \bigcup_{i\leq k-1}\mathcal{S}_i)$ of $\mathcal{X}$, and that $(\mathsf{B}_k)_{:,\mathrm{t}_k}$ is nothing but the boundary operator associated to this complex. We already encountered this construction in Chapter 1 (Remark 3.2.4), in the context of the uniform-spanning-tree process, in which setting it follows from the condition $\ker((\mathsf{B}_k)_{:,\mathrm{t}_k}) = \{0\}$ that, almost surely, samples of $\mathrm{DPP}(K_1)$ are cycle-free subsets of edges, *i.e.*, trees. More generally, we now obtain that, almost surely, the samples $\mathrm{t}_k \subseteq \mathcal{S}_k$ of $\mathrm{DPP}(K_k)$ define sub-complexes such that

$$H_k(\mathcal{X}_{\mathrm{t}_k},\mathbf{C}) = \{0\}. \tag{E.18}$$

In particular, for low-dimensional complexes, this translates to the absence of k-dimensional "voids" in the complex $\mathcal{X}_{\mathrm{t}_k}$.
By a combination of Lemma 3.2.2 and a classical identity[8], we finally obtain that, for all $\mathrm{t}_k \subseteq \mathcal{S}_k$ of size $|\mathrm{t}_k| = |\mathcal{S}_k| - \beta_k$ and such that $\ker((\mathsf{B}_k)_{:,\mathrm{t}_k}) = \{0\}$,

$$\mathbf{P}_{\mathrm{DPP}(K_k)}(\mathrm{t}_k) \propto |[H_{k-1}(\mathcal{X}_{\mathrm{t}_k},\mathbf{Z})]|^2, \tag{E.19}$$

thereby characterizing the distribution $\mathrm{DPP}(K_k)$ completely.

2. Second, we consider an associated "forest"-process, defined as a projective DPP with kernel

$$K_{\mathcal{F}_k} = \begin{bmatrix}\mathsf{B}_k^*\\ \sqrt{q}\mathsf{I}\end{bmatrix}(\mathsf{L}_k^\uparrow + q\mathsf{I})^{-1}\begin{bmatrix}\mathsf{B}_k & \sqrt{q}\mathsf{I}\end{bmatrix}. \tag{E.20}$$

From arguments similar to those of our precedent point, we are able to derive that:

- the samples of $\mathrm{DPP}(K_{\mathcal{F}_k})$ are subsets $\phi \subseteq \mathcal{S}_k \cup \mathcal{S}_{k-1}$ of k- and $(k-1)$-simplices, with cardinality $|\mathcal{S}_{k-1}|$ (which is typically much larger than $|\mathcal{S}_k| - \beta_k$);
- each such ϕ should further satisfy $H_k(\mathcal{X}_{\phi\cap\mathcal{S}_k},\mathbf{C}) = \{0\}$.

For $k=1$, we exactly recover the rooted-spanning-forest structure associated to $\mathcal{D}_{\mathcal{F}}$, but an important difference occurs at higher order: whereas "connected" subsets of 1-simplices (*i.e.*, trees) are always incident to exactly one 0-simplex (the root of the tree), in general, connected subsets of k-simplices in ϕ are incident to *multiple* $(k-1)$-simplices; in turn, the structure of those $(k-1)$-simplices must also be constrained. Unfortunately, this constraint does not translate to a simple combinatorial condition, and boils down to the fact that, almost surely, we have

$$\ker\left(\begin{bmatrix}\mathsf{B}_k & \sqrt{q}\mathsf{I}\end{bmatrix}\right)_{:,\phi} = \{0\}, \tag{E.21}$$

8 Precisely, that the number of integer points in the fundamental domain of a lattice of a lattice determined by a system of integer-valued vectors is (up to sign) the determinant of this system of vectors.

from which the two previous conditions also follow. Those ϕ that satisfy this condition are called *k-forests* in the following.

We are then in position to explicitly compute the probability of drawing each k-forest ϕ as

$$\mathbf{P}_{\mathrm{DPP}(K_{\mathcal{F}_k})}(\phi) \propto q^{|\phi \cap \mathcal{S}_{k-1}|} \times |[H_{k-1}(\mathcal{X}_{\mathrm{t}_k}, \mathbf{Z})]|^2. \tag{E.22}$$

What is remarkable is that, in spite of the precise combinatorial structure of k-forests being hard to characterize, these structures are still amenable to a theoretical analysis. In particular, it is possible to derive an higher-order analogue of Theorem 6.1.1 to (theoretically) estimate the values of $\left((\mathsf{L}_k^{\uparrow} + q\mathsf{I})^{-1}\right)_{i,j}$. Unfortunately, this does not translate to useful RandNLA algorithms, as we are *not* able to derive an efficient sampling algorithm for k-forests. Even worse: finding such algorithms to sample from the (simpler) distribution $\mathrm{DPP}(K_k)$ has been open for fifteen years[9].

All in all, determinantal point process do not result in satisfying estimators for Tikhonov smoothing of k-signals, and involve some uncomfortable theoretical subtleties. As a breath of fresh air, it turns out that we can derive *simple Feynman-Kac estimators* for this same problem.

Random-Walk Approach. In order to design Feynman-Kac estimators, one first has to settle on an appropriate definition of random walks on simplicial complexes. What such a process should look like is neither obvious, nor a classical problem with well-established solutions, and indeed a number of different definitions have been proposed in recent years.

1. For theoretical purposes, with works such as those of [Parzanchevski and Rosenthal, 2017, Mukherjee and Steenbergen, 2016,Rosenthal, 2014], in which case the overarching motivation is to introduce processes that can help to define and study *expansivity* properties of the complex by way of precise relationships between asymptotic properties of the process, spectral properties of Hodge Laplacians, and the homological properties of the complex.

2. For practical purposes, with, notably, the work of [Schaub et al., 2020], which introduces a simple process that, importantly, is easier to define, and can be leveraged to build intuition about some heuristic processing pipelines.

In both cases, an important difficulty that comes up is that the entries of the Hodge Laplacians depend on the reference orientations chosen for the simplices, which is in turn reflected in the studied random processes. This issue can already be observed when looking at the action of these Laplacians. For instance, and for a given k-signal f and $\sigma \in \mathcal{S}_k$, we have

$$(\mathsf{L}_k^{\uparrow} f)(\sigma) = d_\sigma^{\uparrow} f(\sigma) - \sum_{\substack{\sigma' \in \mathcal{S}_k \setminus \{\sigma\} \\ \sigma \cap \sigma' \in \mathcal{S}_{k-1}}} \psi_{\sigma,\sigma'} f(\sigma'), \tag{E.23}$$

where $\{\sigma' \in \mathcal{S}_k \setminus \{\sigma\};\ \sigma \cap \sigma' \in \mathcal{S}_{k-1}\}$ is the set of k-simplices sharing a face with σ, with the "upper-degree" $d_\sigma^{\uparrow}$ of a k-simplex σ being defined as the number of $(k+1)$-simplices containing it[10], and $\psi_{\sigma,\sigma'}$ denoting some coefficient $\{-1, 1\}$ that depends on the choice of orientation (the precise value of which can be computed explicitely).

For our purposes, it is possible to completely disregard this difficulty by, as our notation suggest,

9 We did try to derive such algorithms, using partial-rejection-sampling strategies, but the resulting strategies turn out to be even more expensive than generic DPP-samplers.

10 Note that $|\{\sigma' \in \mathcal{S}_k \setminus \{\sigma\};\ \sigma \cap \sigma' \in \mathcal{S}_{k-1}\}| = (k+1) d_\sigma^{\uparrow}$.

interpreting the factors $\psi_{\sigma,\sigma'}$ as connection maps $x \mapsto \psi_{\sigma,\sigma'} x$. This enables the definition of a very simple process, for which we are able to obtain crisp results.
More specifically, we consider an interrupted, lazy random walk over the k-simplices $\mathcal{S}_k$ which, starting from a given $i \in \mathcal{S}_k$, transitions from a given simplex σ as follows.

- With probability $\frac{k}{2k+1}$, stay at σ and do not interrupt the walk (lazy step).
- Otherwise, with probability $\frac{q}{q+d_\sigma^\uparrow}$, interrupt the walk at σ.
- Otherwise, select a "neighbor" of σ uniformly at random, that is, new simplex $\sigma' \in \mathcal{S}_k \setminus \{\sigma\}$ such that $\sigma \cap \sigma' \in \mathcal{S}_{k-1}$.

Upon the walk being interrupted at some node j, this results in a random path $p = (i, \sigma_1, ..., \sigma_{l-1}, j)$, and we are able to show that[11]

$$f_*(i) = \mathbf{E}_p\left(\psi_{p^*}(g(j))\right). \tag{E.24}$$

The proof of Equation (E.24) is analogous to that of Proposition 4.1.2, with a slight subtlety.

1. In place of the restricted random-walk Laplacian $(\Delta_\theta)_{\mathcal{V}\setminus\mathcal{A}}$ (as considered in the extended graph $\mathcal{G}^\Gamma$), we have to consider a matrix of the form

$$\left((k+1)\mathsf{D}_k^\uparrow + q\mathsf{I}\right)^{-1}\mathsf{L}_k^\uparrow, \tag{E.25}$$

   where $\mathsf{D}_k^\uparrow$ is the diagonal matrix of upper-degrees. The diagonal of this matrix is non-zero, which we account for by considering a lazy random walk, the probability $\frac{k}{2k+1}$ following from an explicit calculation.

2. In turn, we have to ensure that the spectrum of the matrix

$$\mathsf{M} = \mathsf{I} - \left((k+1)\mathsf{D}_k^\uparrow + q\mathsf{I}\right)^{-1}\mathsf{L}_k^\uparrow \tag{E.26}$$

   remains bounded in $(-1, 1)$, so that a series-development argument is still applicable. This is indeed the case, as we can show from an argument analogous to that of Lemma 4.1.8 that all the eigenvalues of M lie in $(1 - \frac{k+2}{k+1}, 1)$.

In the end, we remain unable to derive forest-based estimators, but our Feynman-Kac estimator is amenable to the variance-reduction strategy described in Section 7.2.1 and, if this strategy can indeed be leveraged to build forest-based estimators, may be benefit from it as well.

Remark E.0.2. *Our Feynman-Kac estimators further extend to models involving more complicated connection maps between simplices. This is the discrete counterpart to the more general setting in which physical laws are formulated [Sontz, 2015], and we refer the reader to [Hansen and Ghrist, 2019, Hansen, 2020] for a general introduction on these models.*

11 Since the connection maps only encode a multiplication by ± 1, we have $\psi_p = \psi_{p^*}$.